



PES
UNIVERSITY

CELEBRATING 50 YEARS

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Introduction

Mamatha.H.R

Department of Computer Science and
Engineering

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Introduction

Mamatha H R

Department of Computer Science and Engineering

Course content

Unit 1: Descriptive Statistics and Sampling Distributions

14 Hours

Sampling and Descriptive Statistics: Introduction, Motivating Examples and Scope. Statistics: Introduction, Types of Data, Types of Experiments—Controlled and Observational Study, Types of Statistics, Descriptive and Inferential Statistics, Sampling: Sampling Methods, Sampling Errors.

Sampling Distributions: Chebyshev's inequality, Linear function of random variables, Normal Probability Plots, Introduction to Generation of Random Variates and mention the types, Acceptance-Rejection method, Sampling Distribution, The Central Limit Theorem and Applications.

Self Learning: Mean, Median, Mode

Unit 2: Point Estimation, MLE, Confidence Intervals

14 Hours

Point Estimation, MLE: Principles of Point Estimation – Mean Squared Error for Bernoulli, Binomial, Poisson, Normal, Maximum Likelihood Estimate for Bernoulli, Binomial, Poisson, Normal. Introduction to Multivariate Normal Distribution, MAP distribution.

Confidence Intervals: Interval Estimates for Mean of Large and Small Samples, Student's t Distribution, Interval Estimates for Proportion of Large and Small Samples, Confidence Intervals for the Difference between Two Means, Confidence interval for difference between Two Proportions, and Interval Estimates for Paired Data. Factors affecting Margin of Error.

Course content

Unit 3: Hypothesis Testing and Simple Linear Regression

14 Hours

Hypothesis Testing: Introduction, Large-Sample tests for a Population Mean, Drawing conclusions from the results of Hypothesis tests, the relationship between confidence interval and hypothesis tests, Tests for population proportion, Small-Sample Tests for a Population Mean. Distribution-Free Tests, Chi-squared Test, Fixed Level Testing, Type I and Type II Errors, Power of a Test, Factors Affecting Power of a Test.

Simple Linear Regression: Introduction, Correlation, Fitting Least Squares Line, Checking Assumption

Unit 4: Engineering Optimization

14 Hours

Introduction to Optimization, Types of Optimization, Mathematical concepts of Objective Function, Constraints and Decisions, Constrained Linear Optimization- LPP – Formulation, Graphical Method, The Simplex Method, Discrete Variable Optimization – Integer Programming Problem, Branch and Bound Method, Constrained Non-linear Optimization – Lagrangian Multipliers method, Kuhn Tucker Conditions, Unconstrained non-linear Optimization – Gradient Descent Method, Evolutionary Computation - Genetic Algorithms.

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Course content: Applications

Unit 1: Applications:

1. Poisson distribution: Calculation of number of calls received in a specified time duration in call centers, Modeling the Arrival of Patients in an Emergency Room
2. Variance, Standard Deviation: Monitoring Transaction Variability in E-Commerce, Identifying the customer satisfaction in online shopping
3. Central Limit Theorem: Estimating Average Waiting Time in Public Transport, Load Balancing in Distributed Systems and Internet Traffic Prediction

Unit 2: Applications:

1. Application of Point Estimation (MLE) in Email Spam Detection
2. Application of Mean Squared Error (MSE) in Sensor Calibration
3. Application of Confidence Intervals in Clinical Trials

Unit 3: Applications:

1. Hypothesis Testing in E-commerce Conversion Rates.
2. Chi-Square Test for analyzing the association between vaccination and recovery of the patients considering COVID data, Analyzing the relationship between gender and preference for a product purchase.
3. Identifying Type 1 and Type 2 Errors in Spam Mail Classification.
4. Application of Simple Linear Regression in Real Estate Price Prediction, Stock Price Prediction

Unit 4: Applications:

1. Minimize a Loss functions in Neural Networks using Batch gradient descent (Unconstrained Optimization)
2. Lagrange Multipliers to find local maxima and minima of a function subject to equations constraints (Constrained Optimization)
3. Case study on Bayesian Optimization with Discrete Variables (Discrete Variable optimization)
4. Use Genetic Algorithms to optimize Production Scheduling in a manufacturing environment, focusing on minimizing total production costs while meeting job deadlines and machine constraints. Evaluate the GA's effectiveness against traditional scheduling methods.

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Tools and Textbooks



Tools / Languages/Libraries: Jupyter Notebook, Python, Pandas, Matplotlib, Scipy, Seaborn, BeautifulSoup, Numpy, Scikit learn.

Text Book(s):

1. “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2024.
2. “Engineering Optimization Theory and Practice”, Singiresu S. Rao, John Wiley & Sons, Inc, 5th Edition, 2020

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Evaluation Policy

	Conduction	Reduced to
ISA 1	40	20
ISA 2	40	20
Assignment	20	10
ESA	100	50

Assignment Components

Banana Level Problem	2 Marks (Conducted for 4 marks)
Orange Level Problem	3 Marks (Conducted for 6 marks)
Jackfruit level problem (Datathon)	5 Marks (Conducted for 10 marks)

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Data Science



- Have you ever wondered how YouTube recommends videos of your liking?
- How Google's autocomplete works?
- How Gmail filters your emails into spam and non-spam categories?

Data Science enables us to solve real-world problems by leveraging the power of data.

Many everyday applications—such as recommendation systems, fraud detection, and predictive analytics—would not be possible without access to relevant data. In essence,

Data Science is the discipline of extracting meaningful insights and solutions from data.

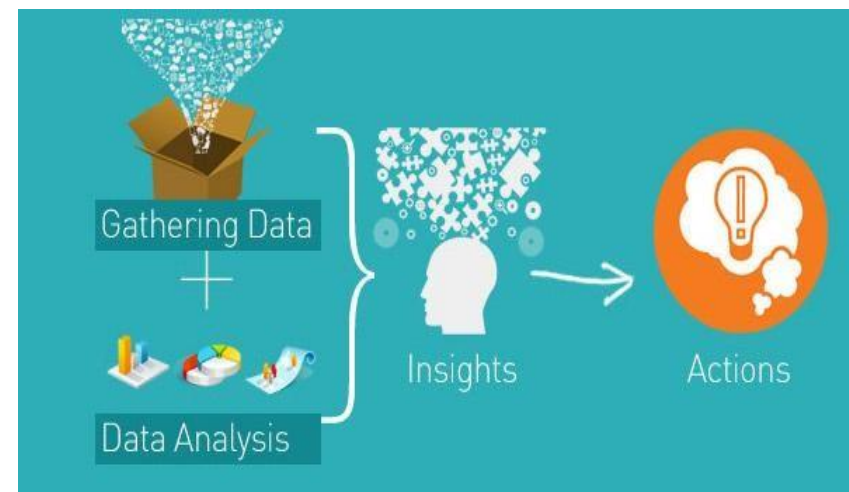


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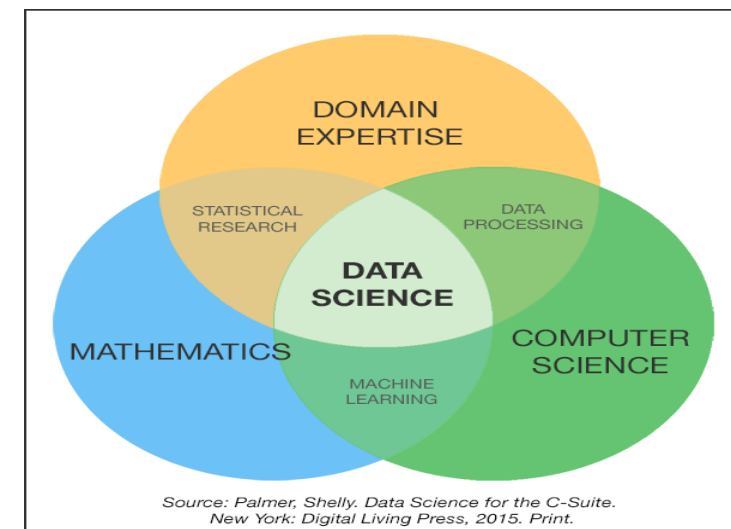
Data Science

Data Science is an interdisciplinary field dedicated to extracting knowledge and actionable insights from data.

- By combining techniques from Statistics, Computer Science, Mathematics, and Domain-specific knowledge (e.g., Business or Healthcare), data science enables informed decision-making and problem-solving across a wide range of sectors.
- The insights derived from data are applied to address complex challenges, optimize processes, and drive innovation in fields such as finance, healthcare, marketing, education, and more.



Source: theblog.adobe..com



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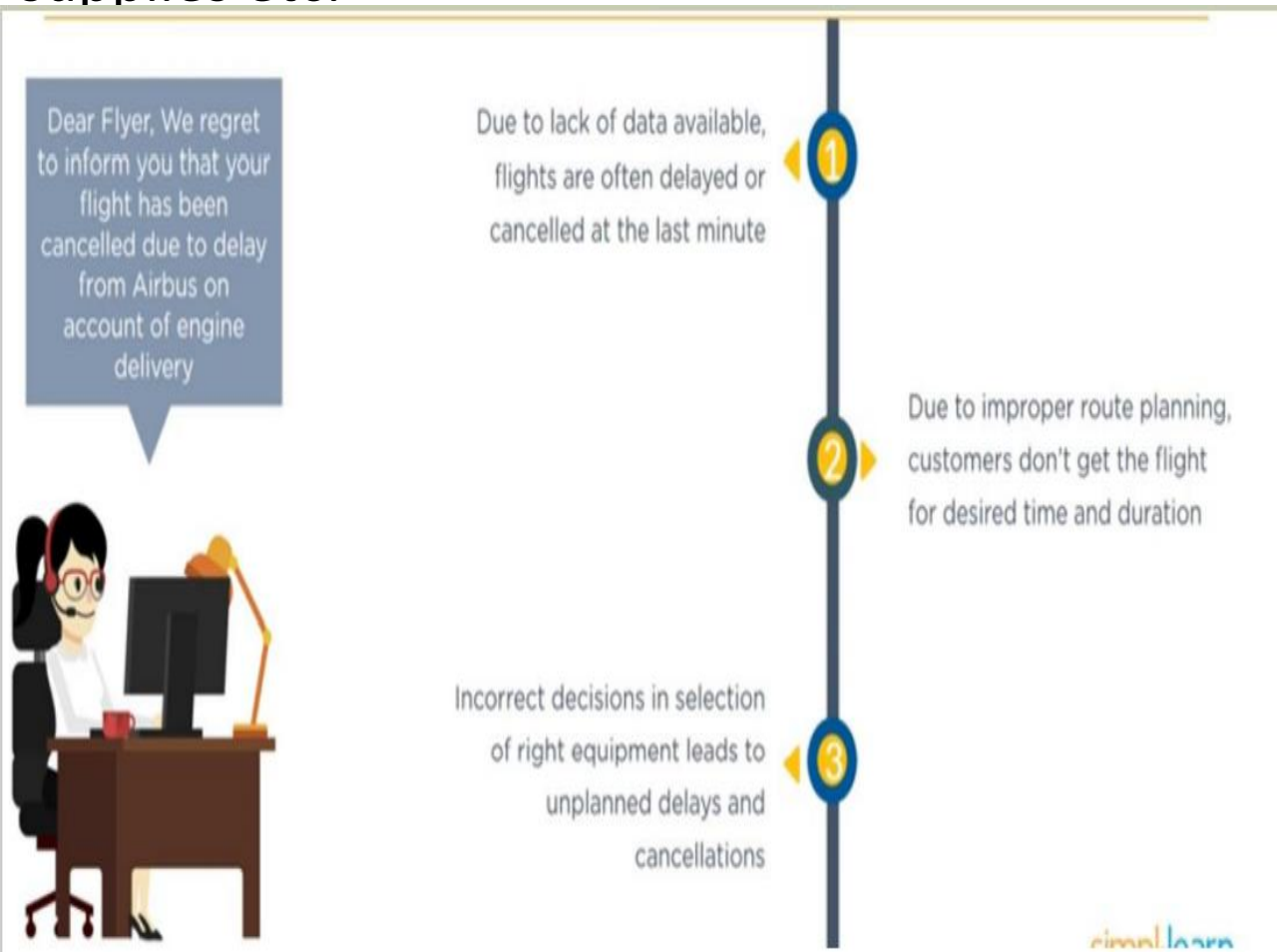
Applications of Data Science



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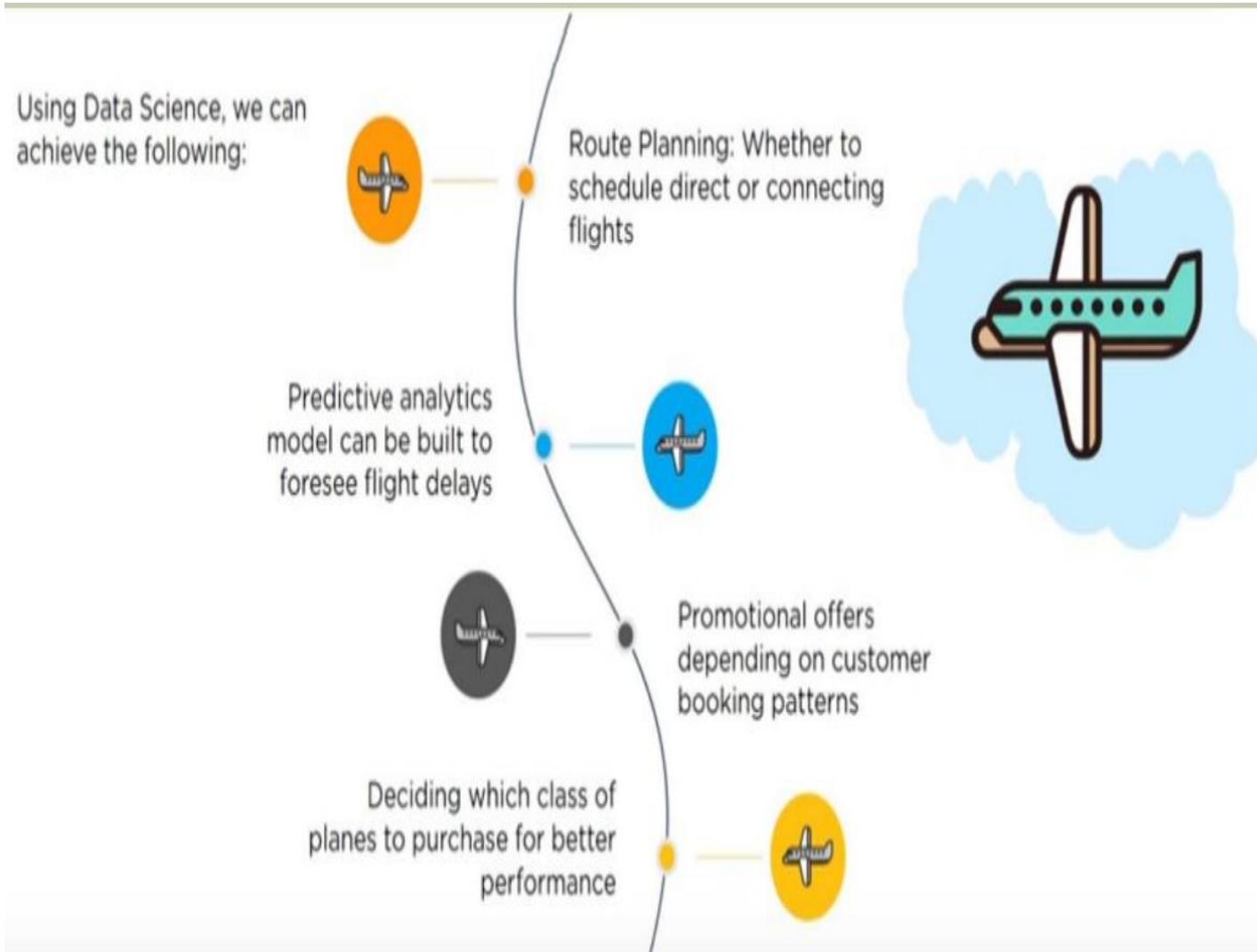
Applications of Data Science : Data Science in Airlines Industry

Data Science is used for various purposes like: route planning, revenue management, prediction on in-flight sales and food supplies etc.



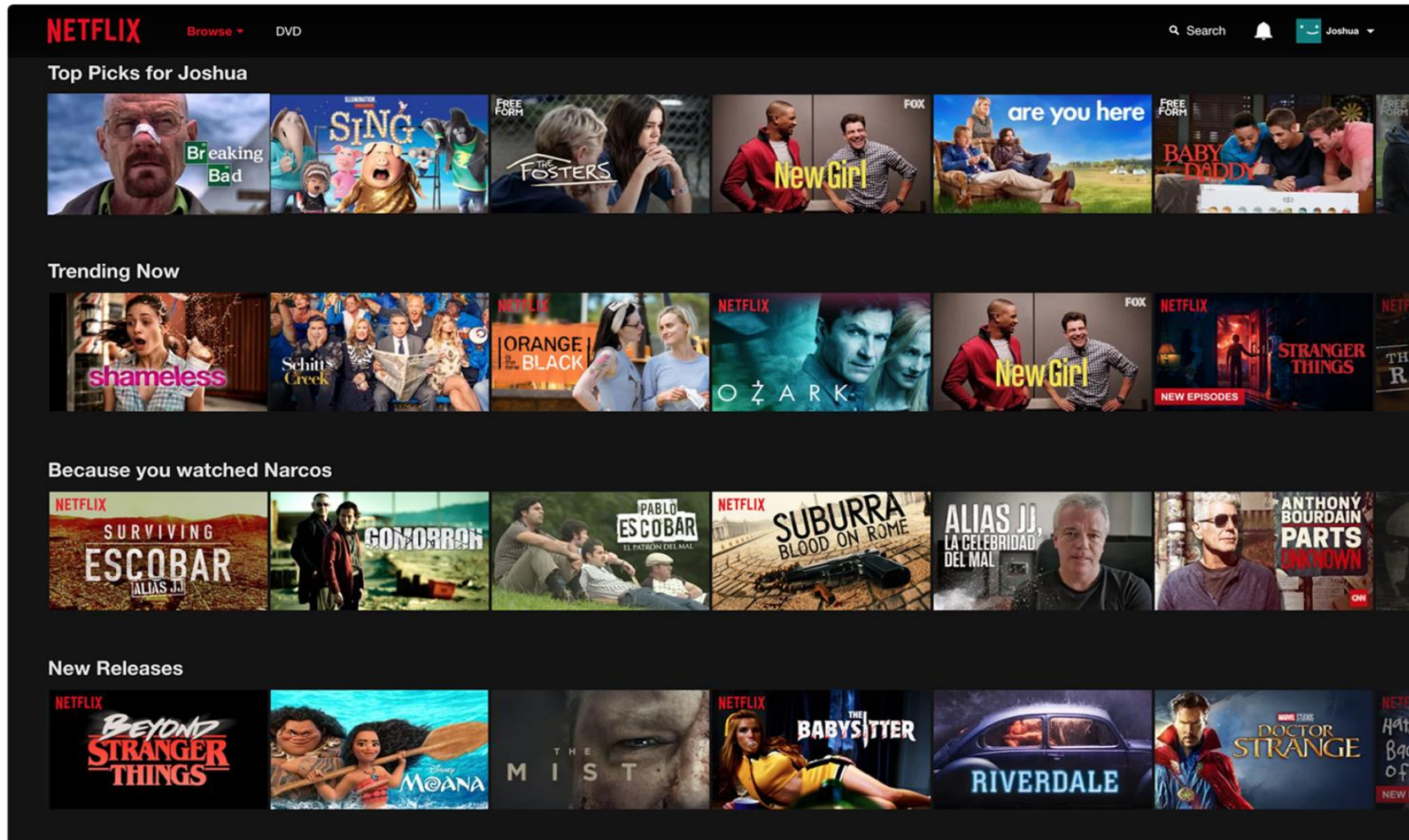
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Applications of Data Science : Data Science in Airlines Industry



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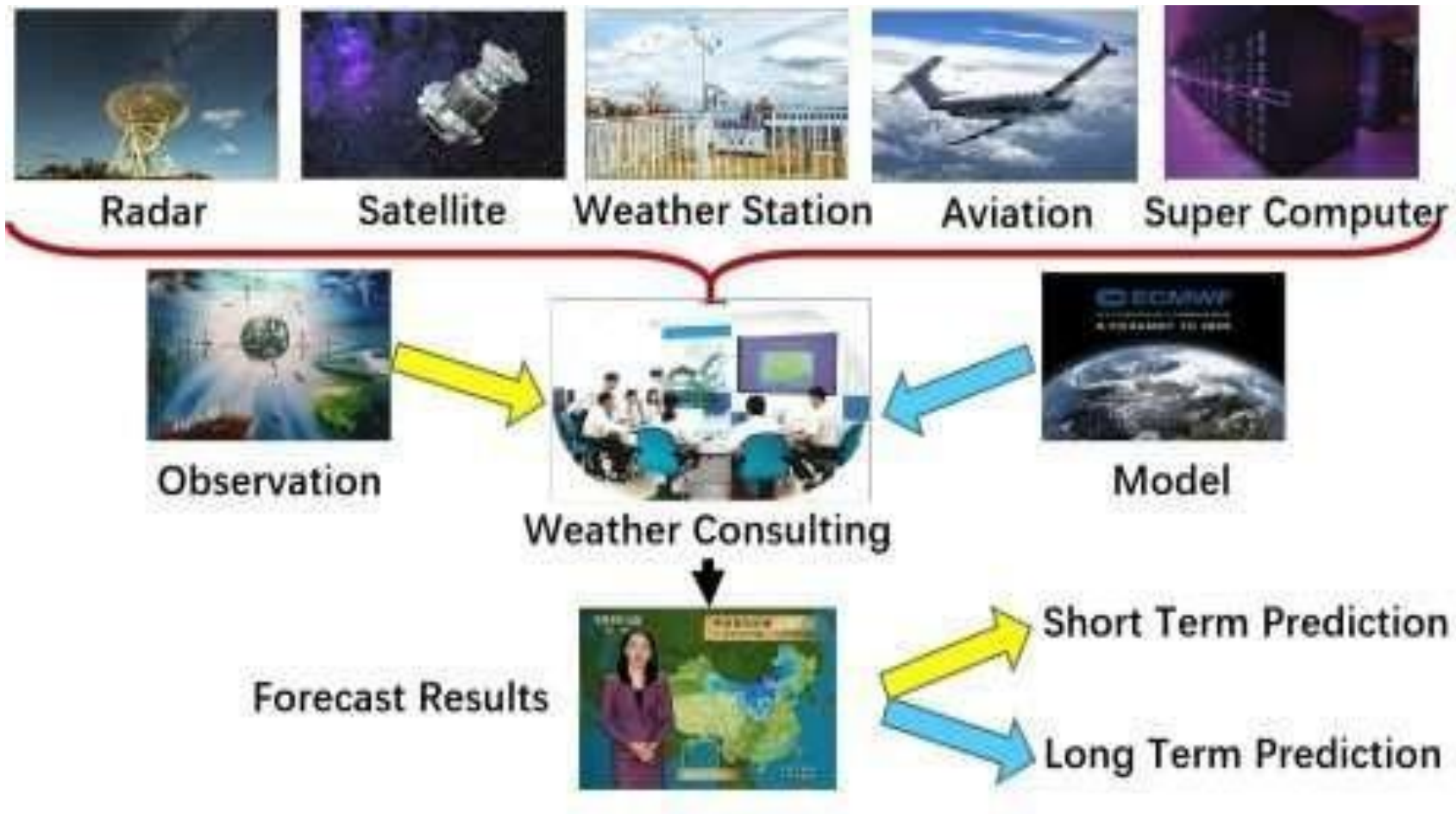
Applications of Data Science : Data Science in Recommender systems



Source: <https://medium.com/swlh/recommendations-in-time-context-93b32f73d98d>

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Applications of Data Science : Data Science in Weather Forecasting



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Applications of Data Science : Data Science in Sports



Source: <https://arstechnica.com/information-technology/2015/10/big-data-an-it-buzzword-that-is-actually-producing-results/>

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Applications of Data Science : Data Science in Politics



Political parties and their strategists have realized the **importance of mining real-time demographic and polling data**.

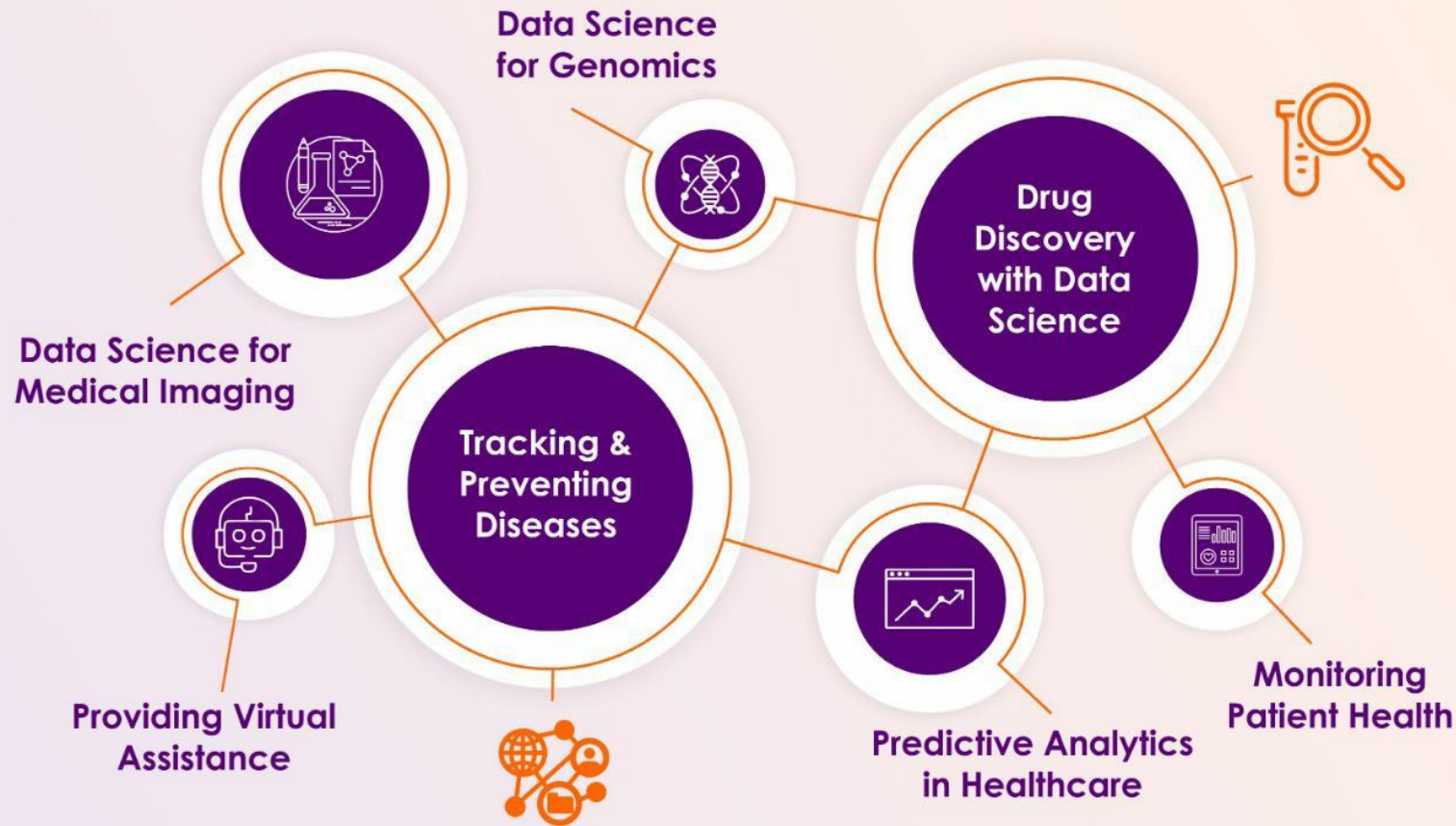
The various data points may include voter sentiment, mass emotions, citizen concerns in different constituencies, popular outlooks in various states, etc. Political parties can use these insights to,

- pull voter donations
- convert undecided voters
- enroll young volunteers
- organize resources
- social media campaigns
- improve effectiveness of electioneering activities etc.

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Applications of Data Science : Data Science in Healthcare & Medicine

The Need for Data Science in Healthcare Industry



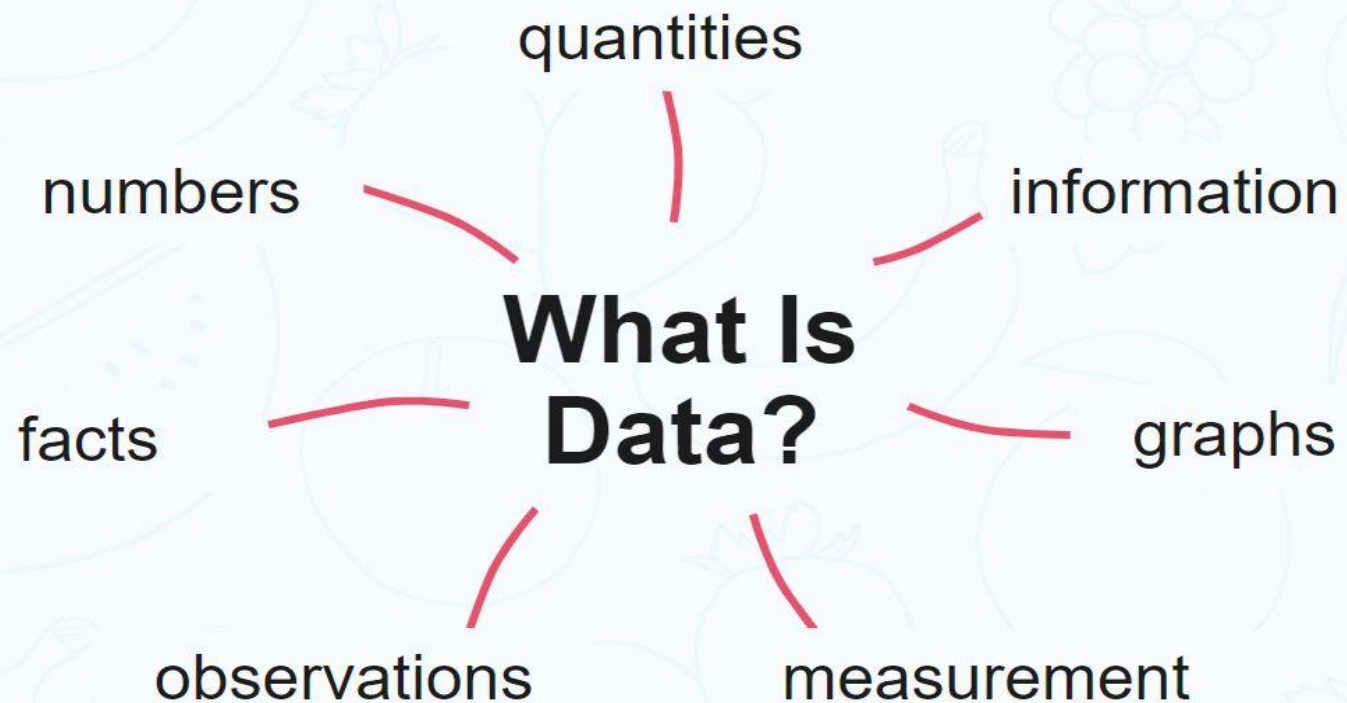
Source:

<http://www.primeclasses.in/blog/2019/08/26/the-need-for-data-science-in-healthcare-industry/>

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Data

Technically, data refers to **individual facts, statistics, or items of information**, often numeric, that are collected through observation.



Data vs Information

→ Data

- Raw facts, usually formatted in a special way.
- Based on records, observations etc.
- Unorganized.

→ Information

- A collection of facts organized in such a way that they have additional value beyond the value of the facts themselves.
- Based on analysis of data.
- Organized and always depends on data.

Ex : Data – thermometer readings of temperature
taken every hour: (16.0, 17.0, 16.0, 18.5, 17.0, 15.5....)
[on transformation]



Information – today's high: 18.5, today's low: 15.5

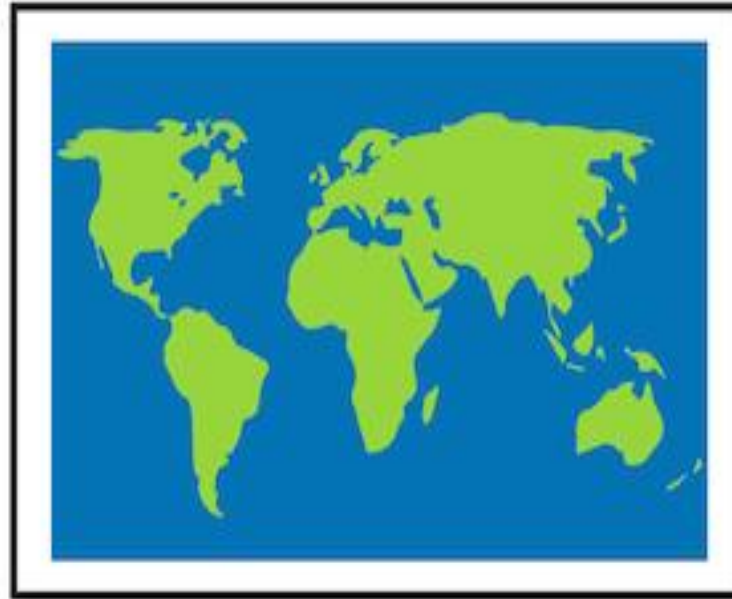
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Data vs Information

DATA



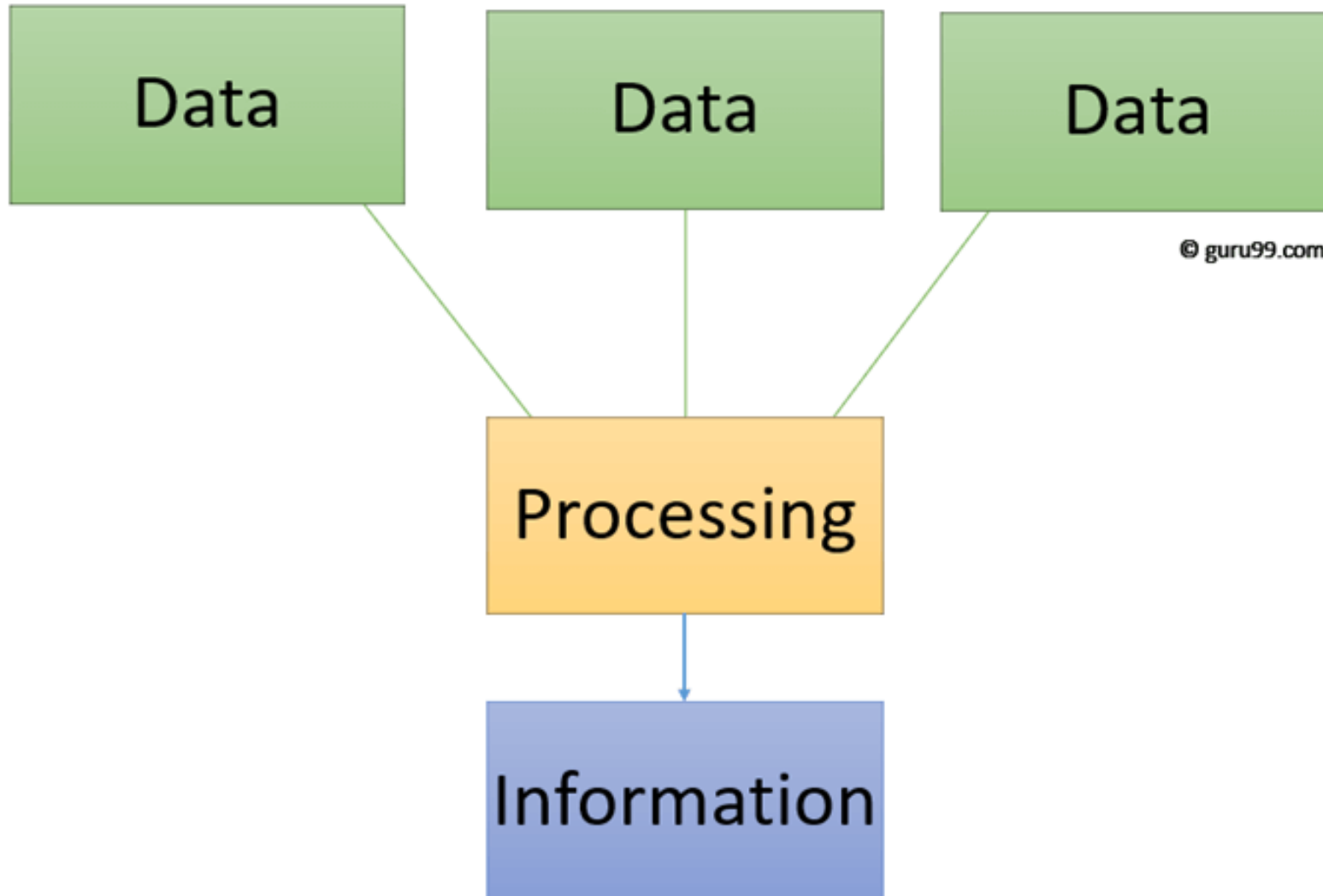
INFORMATION



Source: <https://effectualsystems.com/data-need-information/>

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Data → Information



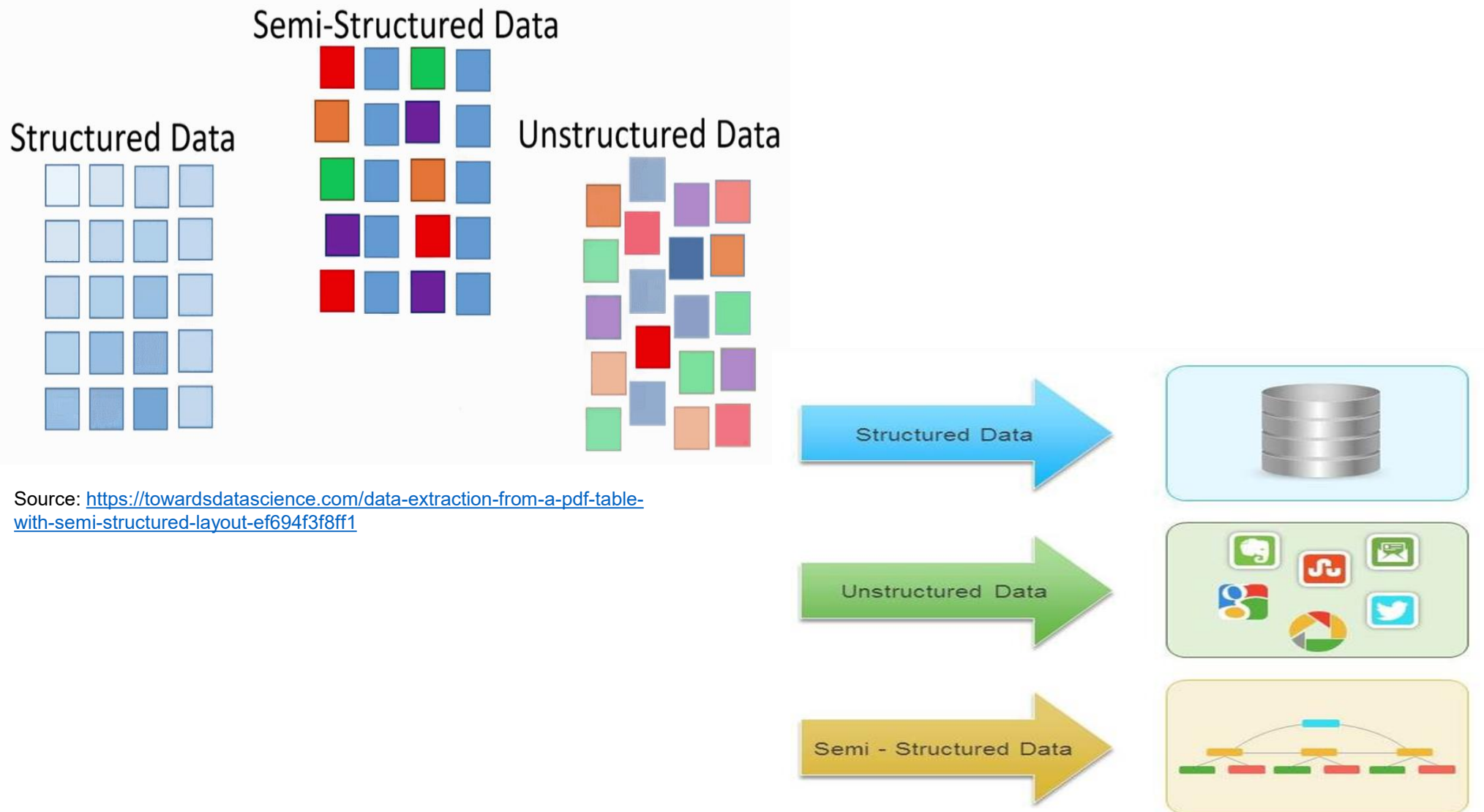
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Types of Data

Data	Represented by
Alphanumeric data	Numbers, letters, and other characters
Image data	Graphic images or pictures
Audio data	Sound, noise, tones
Video data	Moving images or pictures

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Structured, Unstructured & Semi-structured Data



Source: <https://towardsdatascience.com/data-extraction-from-a-pdf-table-with-semi-structured-layout-ef694f3f8ff1>

Structured, Unstructured & Semi-structured Data

Structured Data:

Structured data is the data whose elements are addressable for effective analysis. The data is organized into a formatted repository that is typically a database. Ex: Relational data.

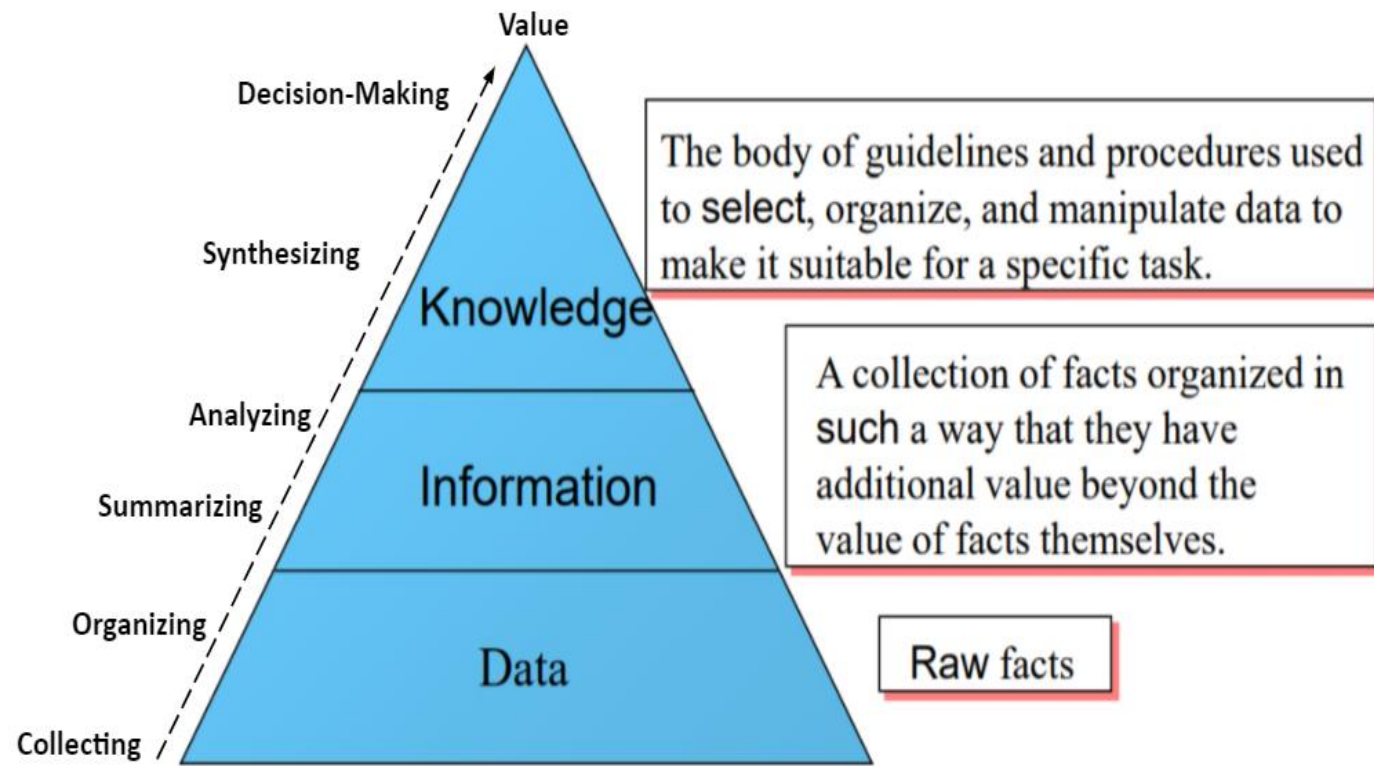
Semi-Structured Data:

It is the data that doesn't reside in relational database but has some organizational properties that make it easier to analyse. Ex: XML data.

Unstructured Data:

It is the data which is not organized in a predefined manner or doesn't have a predefined data model, thus not a good fit for a mainstream relational database. Ex: Word, pdf, text etc.

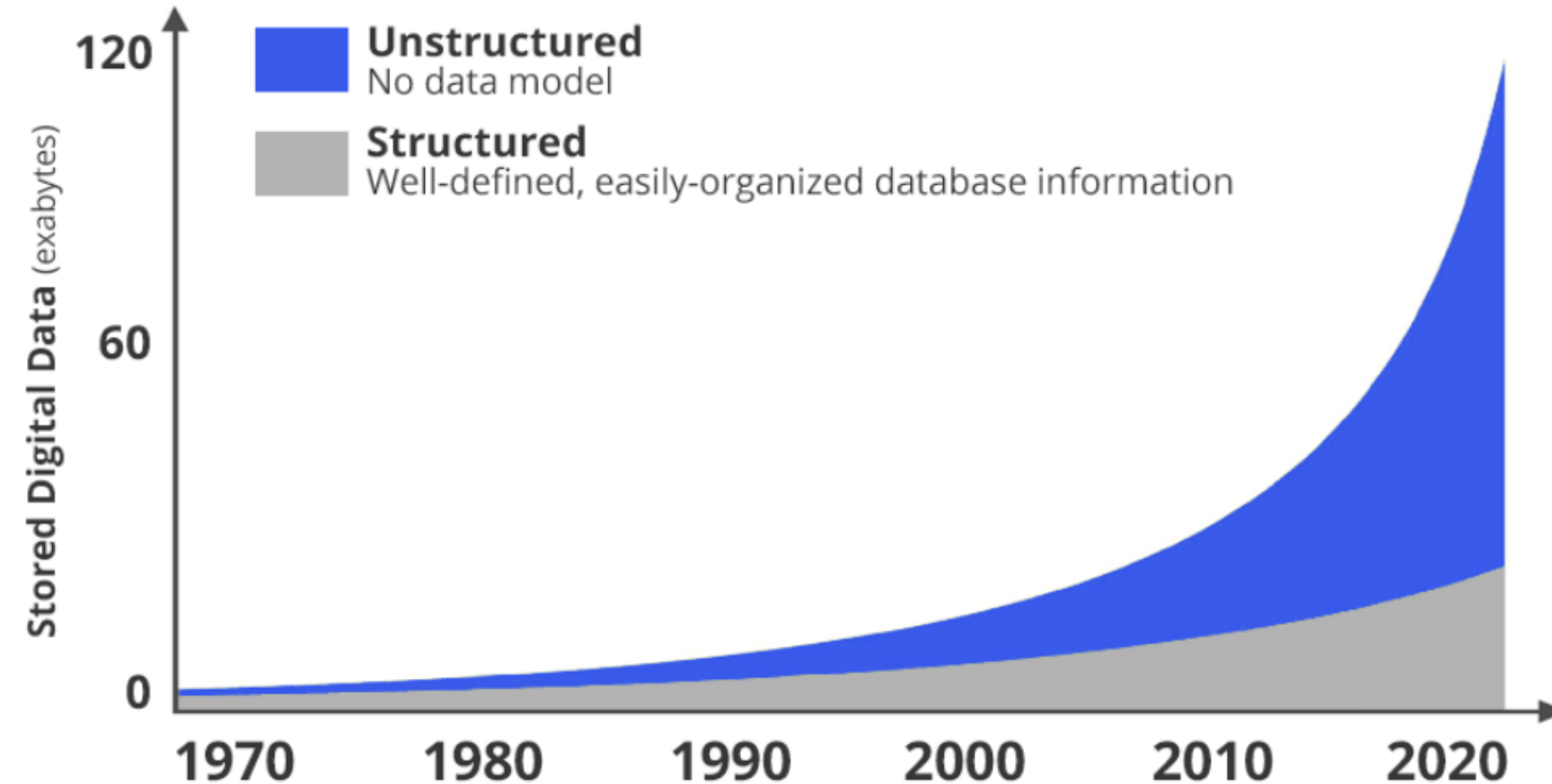
Information Concepts



As we move up the pyramid: Data becomes more refined and useful. It gains value and relevance for decisions and actions.

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Why do we need Data Science?



Source: https://static.seekingalpha.com/uploads/2020/1/14/50485001-15789998083991578_origin.png

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Why do we need Data Science?

The main reason why we need data science is the **ability to process and interpret data.** This enables users and industries to **make informed decisions** as well as helps in their **growth, optimization, and performance.**

We know that, unstructured data is generated everywhere, every second. **Unstructured data** isn't well organized or easy to access. But its **growth is enormous** and importance of analyzing and drawing inferences from this type of data is crucial.

Data Science provides a number of methods and techniques to deal with such data.

This certainly helps many businesses and industries significantly to improve their productivity.

How is Data generated?

There is tons of data getting generated each day.

Some of the major sources from which data is generated are:

web, databases, media, IoT, cloud etc.

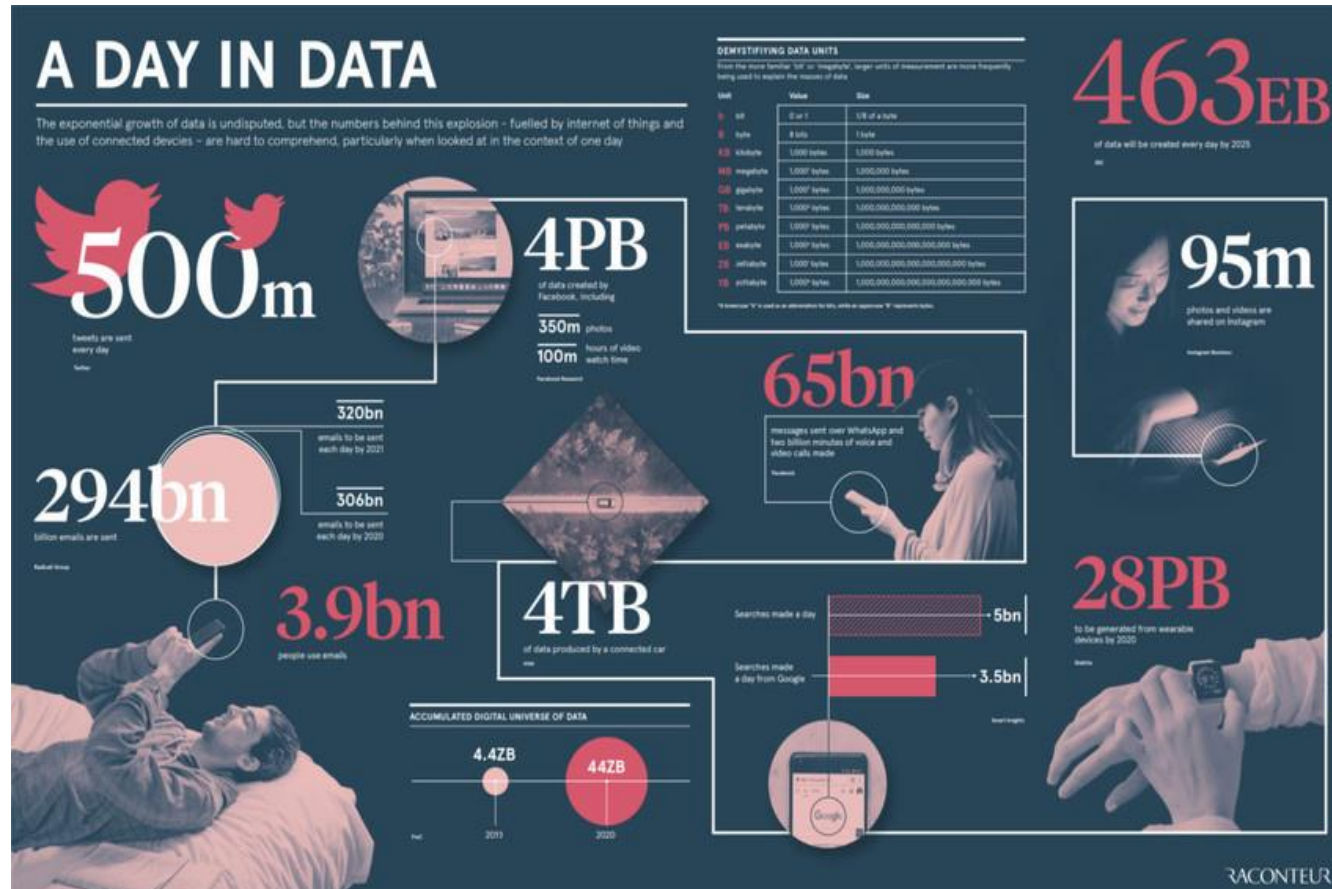
Insight into data generation in a day over the internet:

- 500 million tweets are sent
- 294 billion emails are sent
- 4 petabytes of data are created on Facebook
- 4 terabytes of data are created from each connected car
- 65 billion messages are sent on WhatsApp
- 5 billion searches are made

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How is Data generated?

By 2025, it's estimated that 463 exabytes of data will be created each day globally
– that's the equivalent of 212,765,957 DVDs per day!



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Data generation

- In 2014, Oscars-host Ellen DeGeneres' "celeb selfie" tweet that was viewed **26 million times** across the Web during a 12-hour period.
- **More than one billion hours** of TV shows and movies are streamed from Netflix per month.
- Walmart, handles **more than 1 million customer transactions** every hour, **feeding databases estimated at more than 2.5 petabytes.** (the equivalent of 167 times the books in America's Library of Congress)
- Facebook, is home to **40 billion photos.**

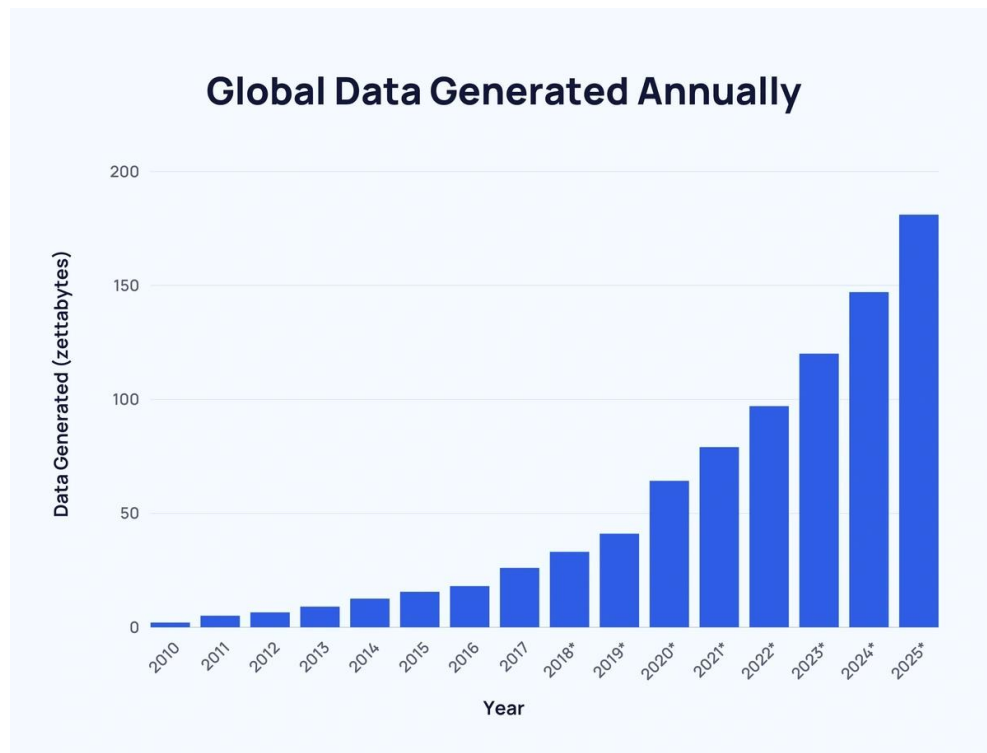
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Growth in Data generation

The total **amount of data created, captured, copied and consumed globally** has been **exponentially increasing**.

In 2020, the amount of data created & replicated was higher than expected caused by the increased demand due to the pandemic.

Up to **2025**, global data creation is projected to grow to **more than 180 zettabytes**.



Data generation: Conclusion

- The world is entering the **zettabyte era**, where data is not just massive in volume but also **increasingly complex, unstructured, and real-time.**
- This explosion of data underlines the **critical role of Data Science, AI, and Cloud Computing** in managing, analyzing, and extracting value from such vast information sources.

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How much of data is put into use?

Though there is a huge amount of data getting generated each day, it shall serve no purpose if it is left unused.

This can further **lead to information overload** where there is an overabundance of information but it is not put into work due to lack of time, resources, understanding of the information, irrelevance of the information or other reasons.

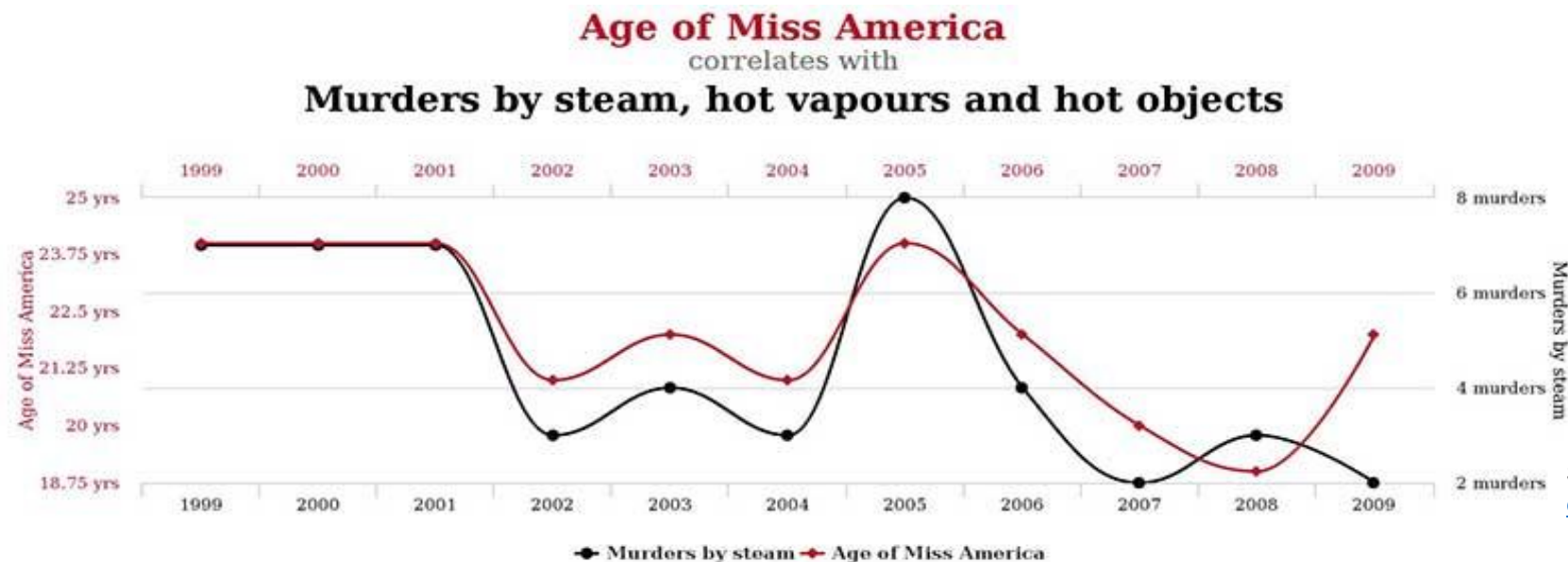
Thus, it is important to **understand the data and know how to utilize it** in the right manner.

But is data all we need?

The graph below shows a **cause & effect relationship** between 'Age of Miss America' and 'Murders by steam, hot vapour and hot objects' which **practically doesn't seem correct**.

Thus, we see that the presence of **interesting patterns need not imply their correctness**.

Blindly applying various processes and techniques on data can result in incorrect inferences.



Source: <https://i2.wp.com/boingboing.net/wp-content/uploads/2016/02/chart.jpg?fit=800%2C315&ssl=1>

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Learn how to use data



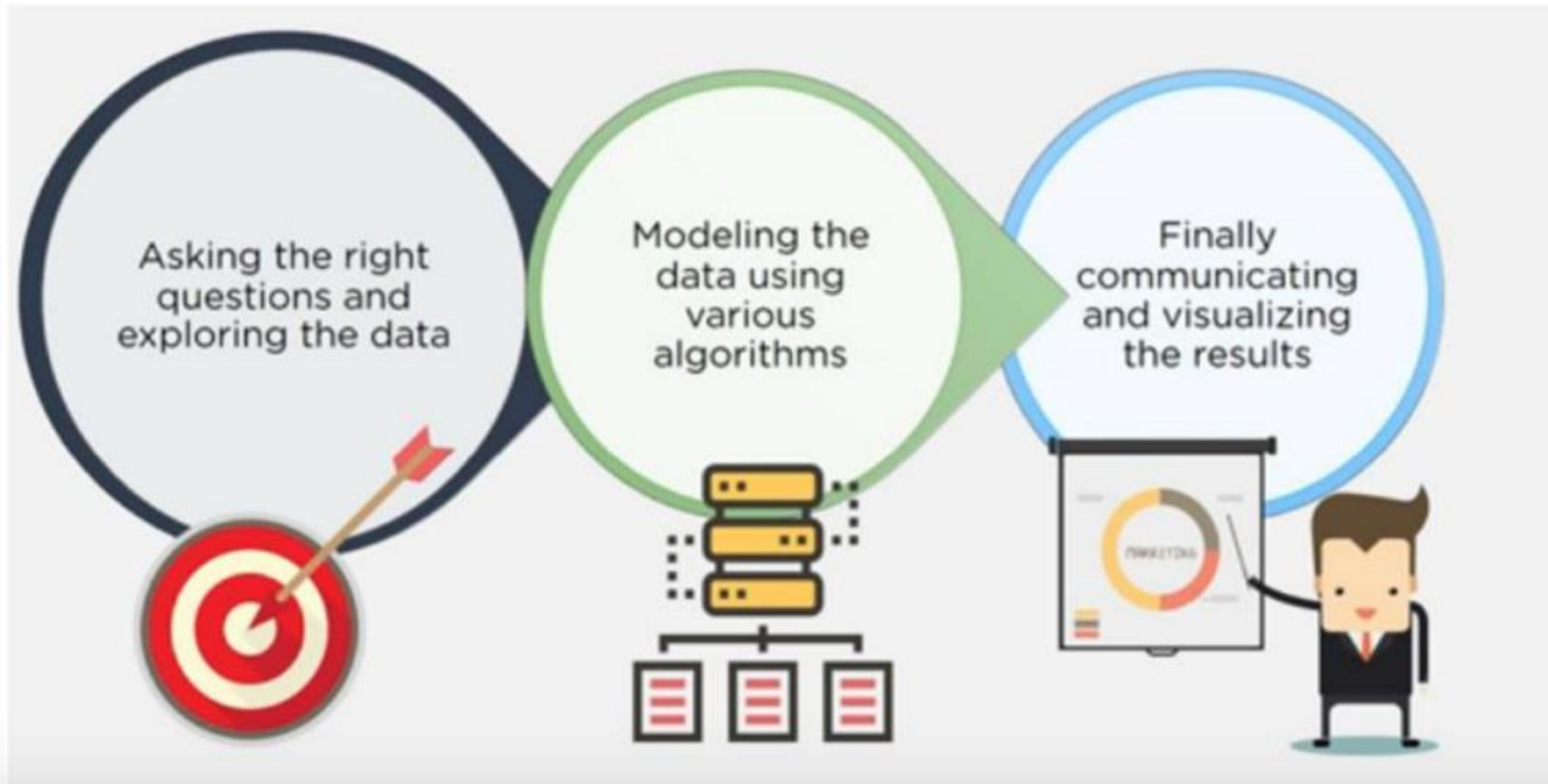
The above examples help us understand that we need to **learn how to utilize and handle** the available data in the right manner to be able to **arrive at correct results** and **draw meaningful inferences**.

- **Explore**: identify patterns
- **Predict**: make informed guesses
- **Infer**: quantify what you know

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Learn how to use data

So, Data Science or Data-driven Science is about:



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Data Scientist

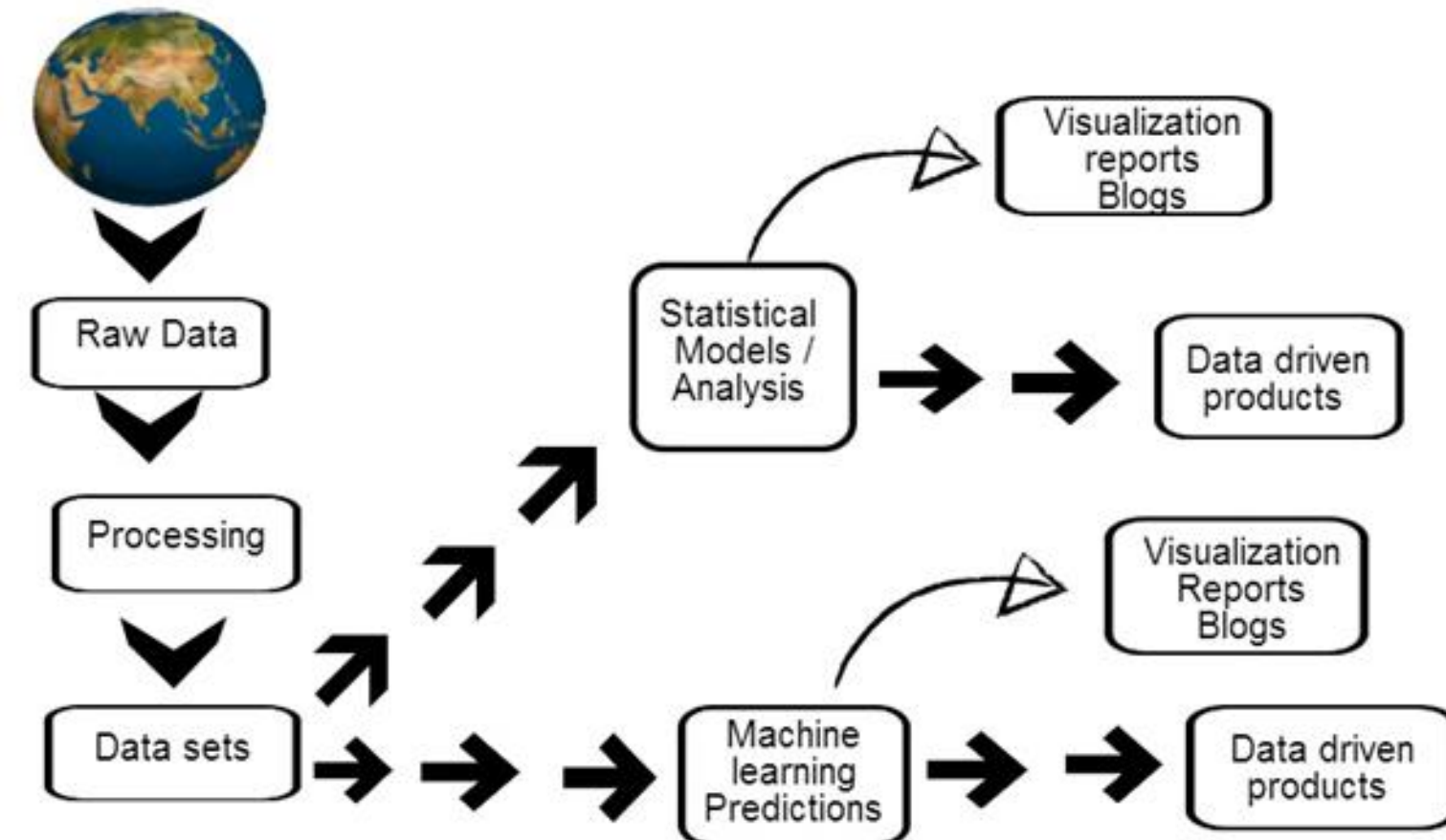
Data Scientists in simple words are those who make sense out of all the data that are available and figure out the things that can be done with it.



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What does a Data Scientist do?

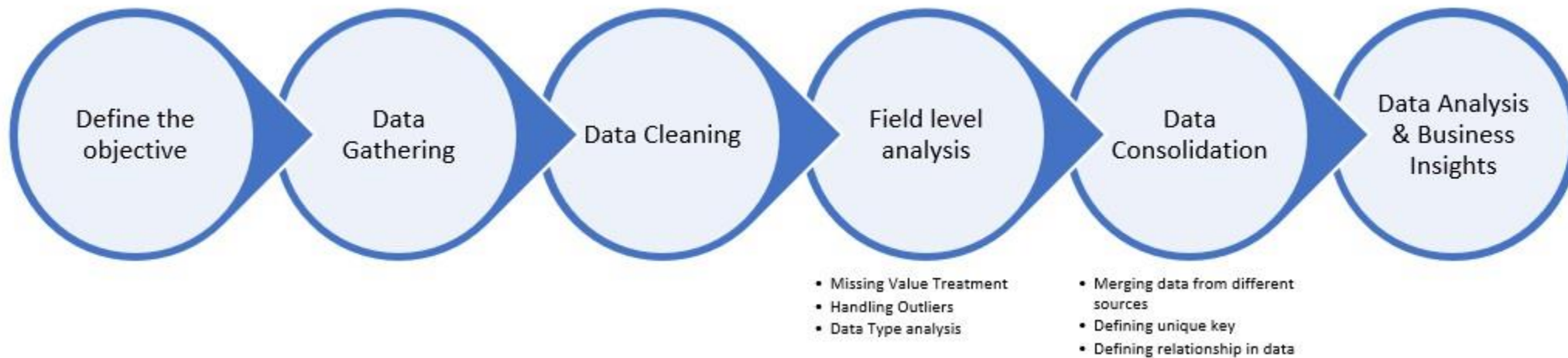
They are responsible for **collecting, analyzing, modelling and interpreting large amounts of data.** Their role combines Computer Science, Mathematics, Statistics etc.



Source: <https://edvancer.in/wp-content/uploads/2015/11/76c99311fc4be19bf4353cfc3c2e94b2.png>

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What does a Data Scientist do?



Source: medium.com

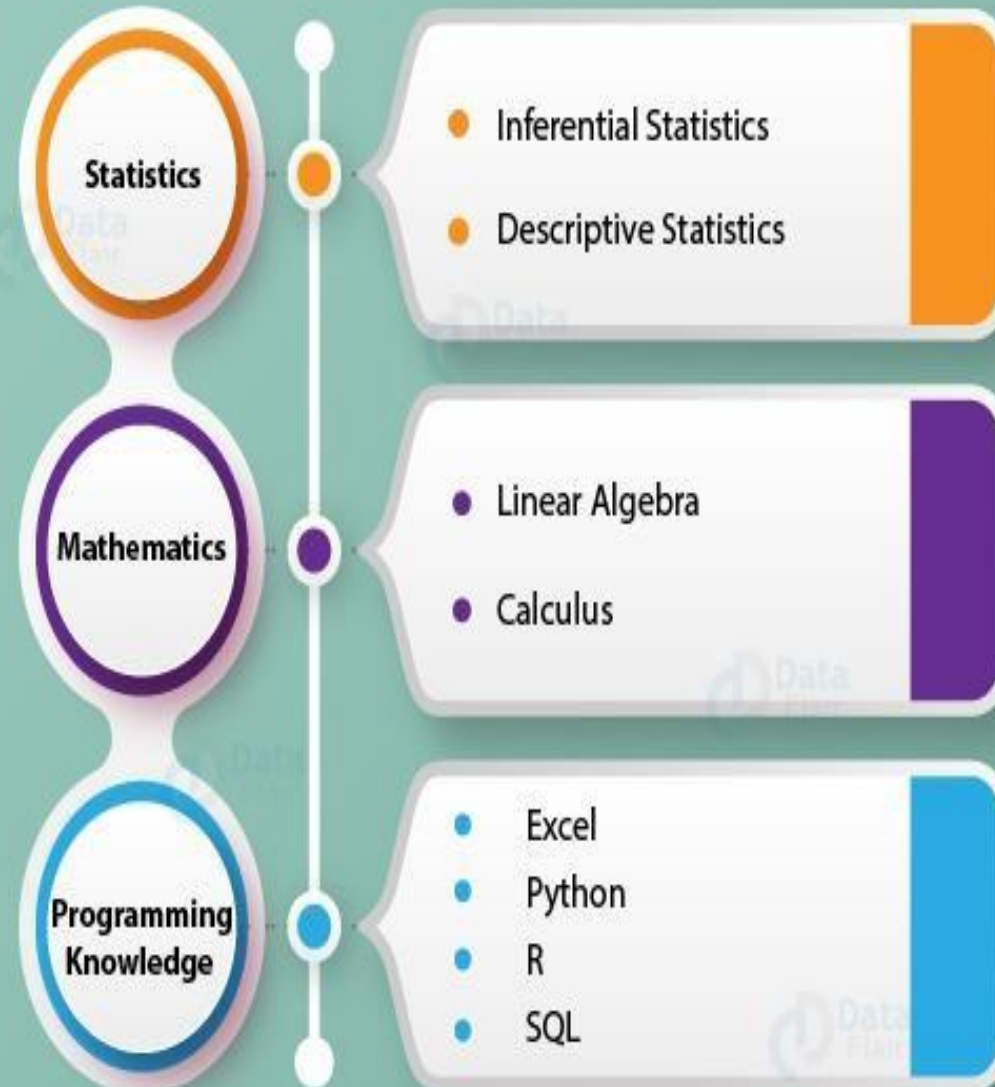
Slide courtesy: Dr. Uma

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Prerequisites for a Data Scientist



Data Science Prerequisites



Slide courtesy: Dr. Uma

Source: data-flair.training

The demand for **Data Scientists in 2025** is expected to be **exceptionally strong**, both **globally** and **in India**, due to the ever-increasing volume of data and the rise of AI-powered decision-making across sectors.

Global Demand (2025)

- According to **The U.S. Bureau of Labor Statistics (BLS)**:

Data scientist roles will grow by **35% between 2022 and 2032**, adding more than **59,000 new jobs** in the U.S. alone.

- **World Economic Forum (Future of Jobs Report)** predicts:

Data science and AI roles will be among **the top 10 fastest-growing job categories** through 2025.

- **Estimated global openings by 2025:**

11.5 million+ data-related roles, including Data Scientists, Data Engineers, ML Engineers, and AI specialists.

Demand for Data Scientist

In India-Specific Demand (2025)

- NASSCOM and AIM research (2024-25) reports:

India is projected to have **over 1.5–2 lakh** unfilled roles in Data Science & Analytics in 2025 due to a **skills gap**.

- **Top hiring sectors in India:**

- IT and Software Services
- Banking, Financial Services, and Insurance (BFSI)
- E-commerce and Retail
- Healthcare and Pharma
- Manufacturing and Logistics
- EdTech and Startups

- **Top cities for hiring:** Bengaluru, Hyderabad, Pune, Mumbai, New Delhi

How is it different from what Statisticians have been doing?

Both Statisticians and Data Scientists work closely with data.

- Statisticians **use mathematical equations and statistical models** to analyze data and arrive at conclusions.
- Data Scientists however **focus on delivering actionable results** and sometimes need to deploy the model to the production system.

Data Science vs Data Analysis

Data Science:

- Scope:**

Data science is a broader, interdisciplinary field that encompasses data analysis, data engineering, and machine learning.

- Focus:**

It focuses on extracting knowledge, building predictive models, and developing new algorithms to solve complex problems.

- Techniques:**

Data scientists utilize advanced statistical modeling, machine learning, and programming skills (like Python or R).

- Goal:**

To uncover hidden patterns, predict future trends, and create new data-driven solutions.

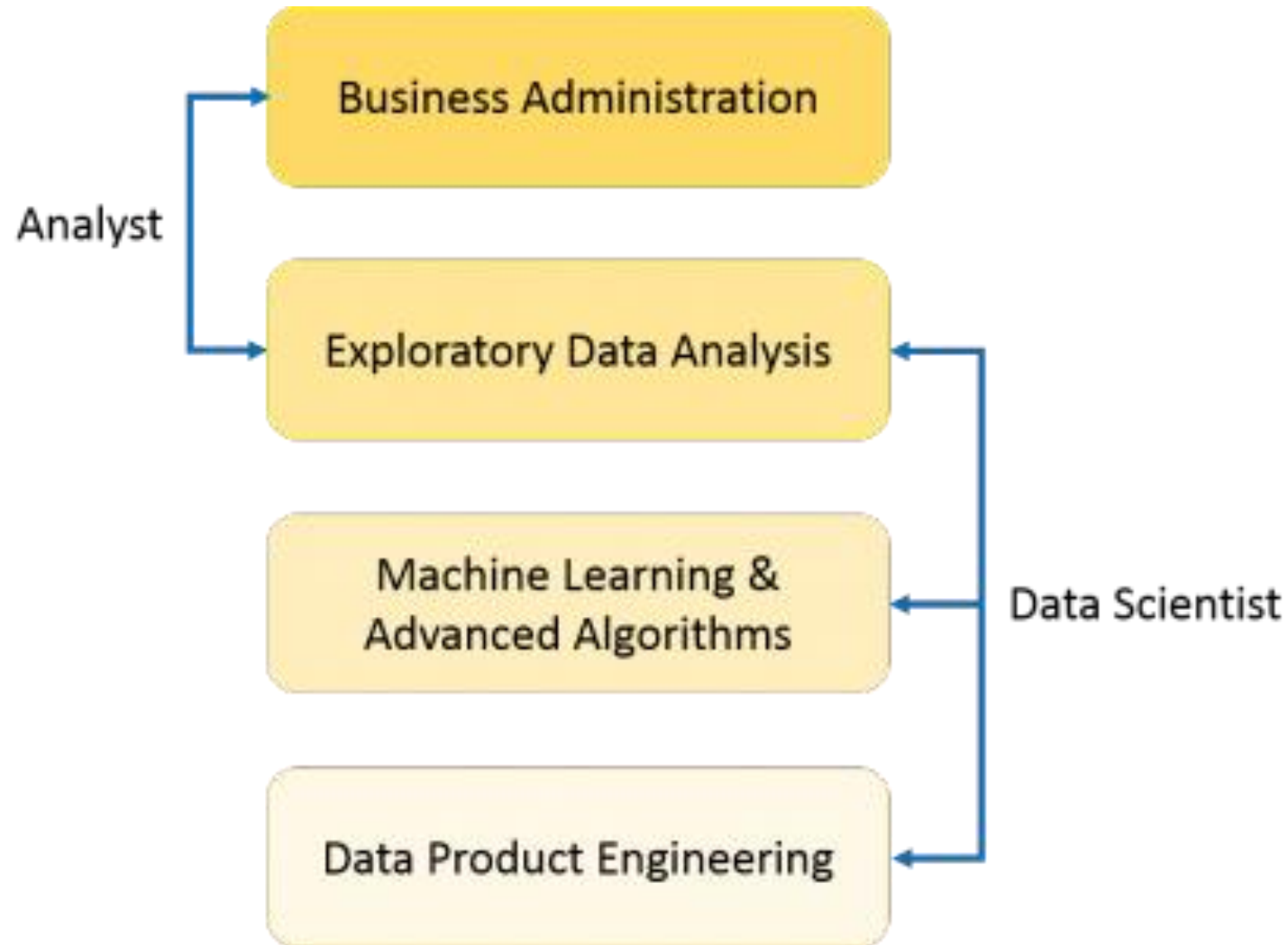
- Example:**

Building a recommendation engine for a streaming service or predicting customer churn using machine learning.

- Data Analysis includes **descriptive analytics and prediction to a certain extent.**

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Data Science vs Data Analysis



Source:

<https://d1jnx9ba8s6j9r.cloudfront.net/blog/wp-content/uploads/2017/01/Data-Analyst-vs-Data-Science-1-422x300.png>

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Common tasks in Data Science

So Data Science is mainly needed for:



Better Decision Making

Whether A or B?



Predictive Analysis

What will happen next?

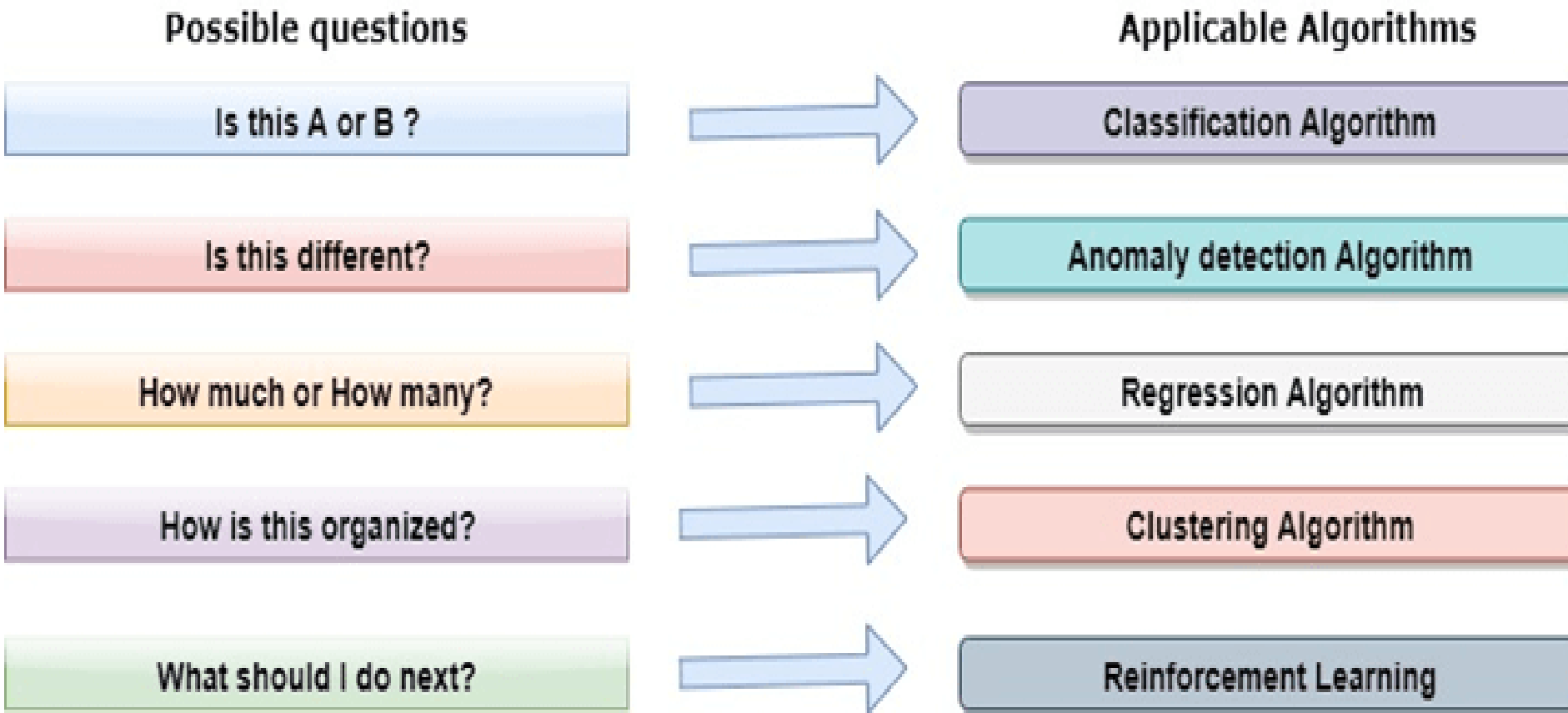


Pattern Discovery

Is there any hidden information in the data?

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Common tasks in Data Science



Source: <https://static.javatpoint.com/tutorial/data-science/images/how-to-solve-a-problem-in-data-science.png>

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References

Text Book:

Statistics for Engineers and Scientists, William Navidi, 6th Edition , McGraw Hill Education, India





THANK YOU

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Unit 1: Population & Sampling

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Topics to be covered

- ❖ Statistical Analysis
- ❖ Population
- ❖ Sample
- ❖ Sampling
- ❖ Types of Population

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Problems to be solved



Suppose, you are interested in finding

- Mean height of all male students of all the universities in India.
OR
- Average marks of all female students of PES University. OR
- Relationship between the time a student spends on studying and the grades that he gets. OR
- Impact of rise in number of student assignments on their grades.

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What is Statistical Analysis?

It's the science of **collecting, exploring and presenting large amounts of data to discover underlying patterns and trends.**

Statistics are applied every day – in research, industry and government – to become more scientific about decisions that need to be made.

The basic idea behind all statistical methods of data analysis is to **make inferences about a population by studying a relatively small sample** chosen from it.



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Population

A population is the **entire collection of objects or outcomes** about which information is sought.

As mentioned, statistical methods are based on the idea of analyzing a sample drawn from a population.

For this idea to work, **identifying the population, sample and choosing the sample** in an appropriate manner becomes important.

In research, a population doesn't always refer to people. It can mean a group containing elements of anything you want to study, such as objects, events, organizations, countries, species, organisms, etc.



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Types of population

1. **Tangible or concrete population**
2. **Conceptual population**

Tangible population

Populations where the **members are physical objects**, such as cars, bolts, apples, etc., are called **tangible or concrete populations**.

Such populations are assumed to be **always finite** and therefore **involves counting**.

After an item is sampled, the population size decreases by 1.

In principle, one could in some cases return the sampled item to the population, with a chance to sample it again, but this is rarely done in practice.



Conceptual population

Populations that **do not consist of physical or actual objects** are called **Conceptual populations**.

Conceptual populations are mostly the **result of a measurement**.

It involves **measuring something multiple times**.

Ex: length of a metal rod.

It consists of a **not well-defined group** of which all elements are not available at the time the sample is collected as **the population increases every day**.

The **size** of a conceptual population is **usually large**.

Ex: a measuring scale population can be all the possible outputs it can give. **i.e. infinite**. The measured values can be thought of as a sample from this infinite population.

Tangible & Conceptual population examples

Define the population, and state whether it is tangible or conceptual.

- A shipment of bolts is received from a vendor. To check whether the shipment is acceptable with regard to shear strength, an engineer reaches into the container and selects 10 bolts, one by one to test.

Ans: All the bolts in the shipment: **Tangible population**

- The resistance of a certain resistor is measured 5 times with the same ohmmeter.

Ans: All measurements that could be made on that resistor with that ohmmeter : **Conceptual population**

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Tangible & Conceptual population examples

Define the population, and state whether it is tangible or conceptual.

- A geologist weighs a rock several times on a sensitive scale.

Ans: All the readings that the scale could produce: **Conceptual population**

- A pollster samples 1000 registered voters in a certain state and asks them which candidate they support for governor.

Ans: All registered voters in that state : **Tangible population**

Tangible & Conceptual population examples

Define the population, and state whether it is tangible or conceptual.

- A quality engineer needs to estimate the percentage of bolts manufactured on a certain day that meet a strength specification. At 3:00 in the afternoon he samples the last 100 bolts to be manufactured.

Ans: All bolts manufactured on that day : Tangible population

- In a clinical trial to test a new drug that is designed to lower cholesterol, 100 people with high cholesterol levels are recruited to try the new drug.

Ans: All people with high cholesterol level: Tangible population

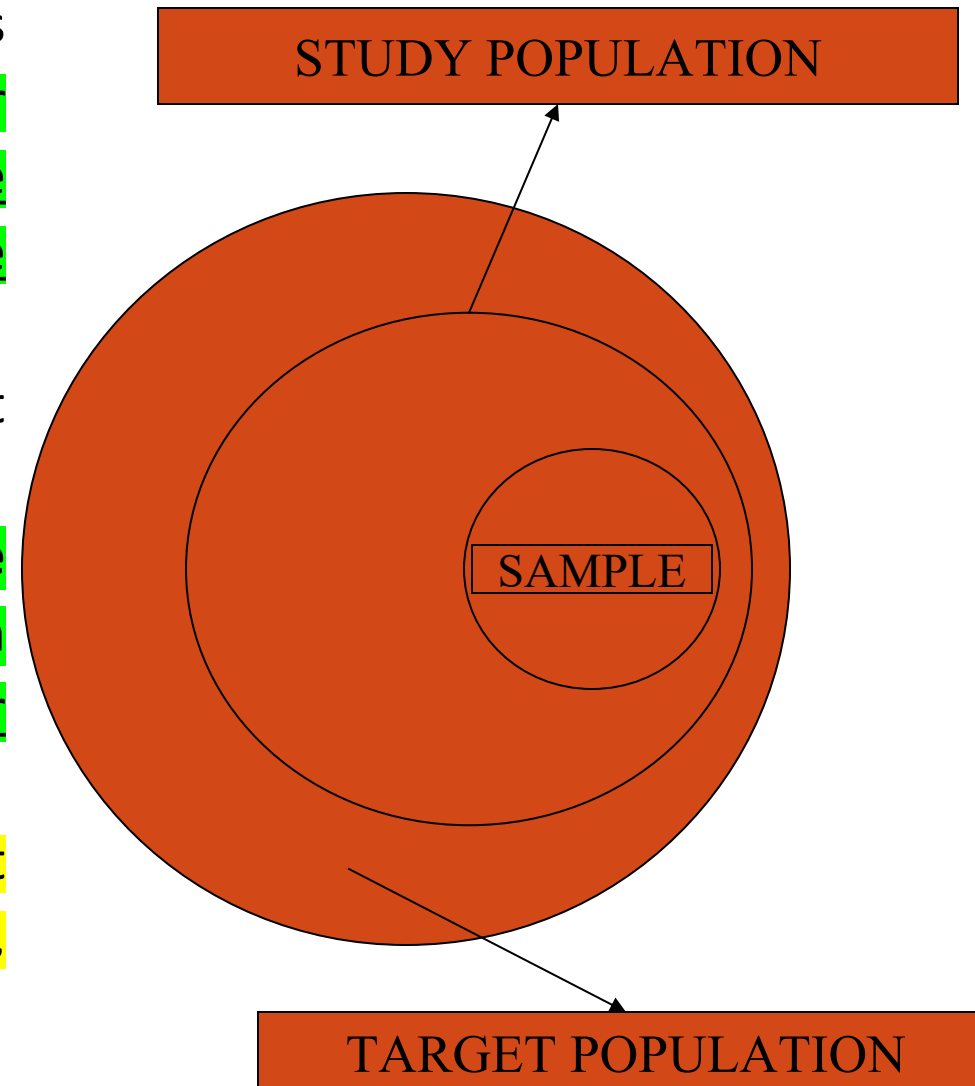
Target and Study population

- **Target or Theoretical population** refers to the entire group of individuals or objects to which researchers are interested in generalizing the conclusions.

It must meet a set of criteria of interest to the researchers.

- **Study population or accessible population** is the population to which the researches can apply their conclusions to.

It is a **subset** of the target population. It may be limited to region, state, city, county, or institution



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Target and Study population examples

Target Population	Study Population
All institutionalized elderly with Alzheimer's	All institutionalized elderly with Alzheimer's in St. Louis county nursing homes
All people with AIDS	All people with AIDS in the metropolitan St. Louis area
All low birth weight infants	All low birth weight infants admitted to the neonatal ICUs in St. Louis city & county
All school-age children with asthma	All school-age children with asthma treated in pediatric asthma clinics in university-affiliated medical centers in the Midwest

Terminologies related to Sampling

- **Target or Theoretical Population:** The population to which the investigator wants to generalize his results.
- **Sampling Frame :** The sampling frame is the list from which the potential respondents are drawn.
Ex: List of Universities, List of Students, List of Airline Companies, Telephone Directory
- **Sampling Unit :** Smallest Unit from which sample can be selected.
- **Sampling Scheme:** Method of selecting sampling units from sampling frame.
- **Sample:** All selected respondents form a sample.

Sampling Breakdown

Study : Find the mean weight of all students of all universities in India.

- Whom do you want to generalize results?
All universities in India
→ **Target or Theoretical population**
- What population can you get access to?
All universities in Karnataka
→ **Study population**
- How can you get access to them?
List of Universities in Karnataka
→ **Sampling frame**
- Who is in your study?
Two Universities from Karnataka
→ **Sample**

1. Which of the following is an example of a tangible population?

- A) All diabetic patients treated at a hospital in 2024
- B) All future patients who will develop diabetes
- C) All possible outcomes of a clinical trial simulation
- D) All hypothetical patients with a rare disease

2. Which of the following best describes a conceptual population in a vaccine study?

- A) Children vaccinated in a specific village last year
- B) All batches of vaccine produced in a factory
- C) All future children who will receive the vaccine
- D) Children currently enrolled in the study

3. Which of the following is not a tangible population?

- A) Students enrolled in Class 10 in 2025
- B) Blood pressure readings from a clinical trial
- C) All cancer patients who will be diagnosed next year
- D) Employees in a specific company today

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Answers

1. A
2. C
3. C



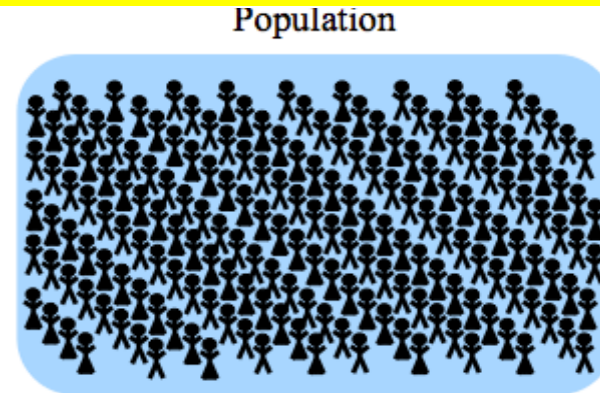
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Sample

A sample is a **subset of a population**, containing the objects or outcomes that are actually observed.

Sample size: The number of items in a sample is called a sample size. The **size of the sample is always less than the total size of the population.**

The **process of taking a predetermined number of observations from a larger population is called sampling.**



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Population vs Sample

Population	Sample
The population is a complete set .	The sample is a subset of the population
Population is hard to define and observe in real life.	A sample is much easier to contact and observe .
It is time consuming and costly to study a population	It is relatively less time consuming and low cost to study a sample.
Population contains all members of a specified group .	Sample is a subset that represents the entire population .
Reports on a population are a true representation of opinion .	Reports on a sample are have a margin of error .

Populations & Samples

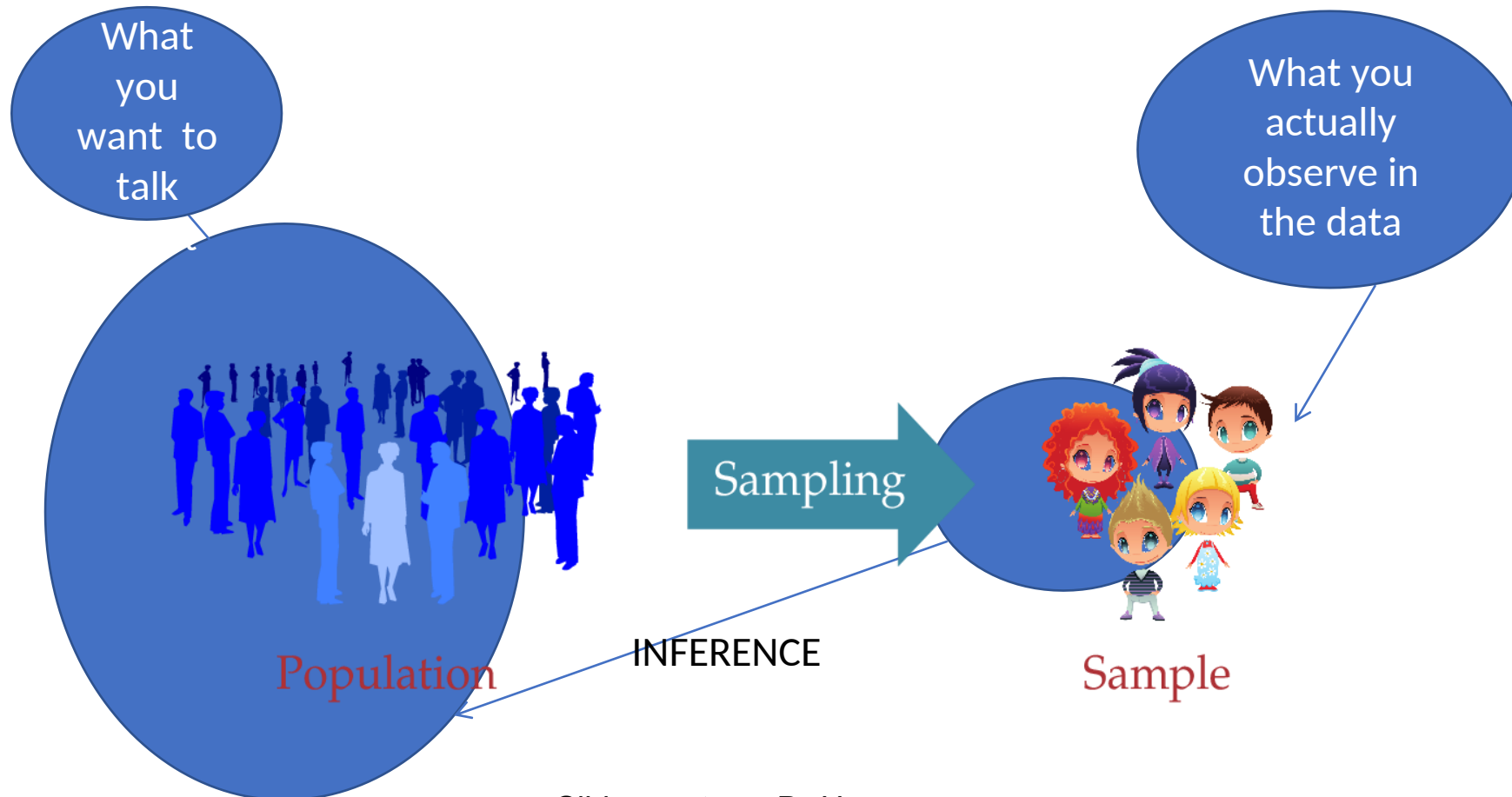
A city council member wanted to know how her constituents felt about a planned rezoning. She randomly selected 75 names from the city phone directory and conducted a phone survey. Identify the population and sample in this setting.

Answer:

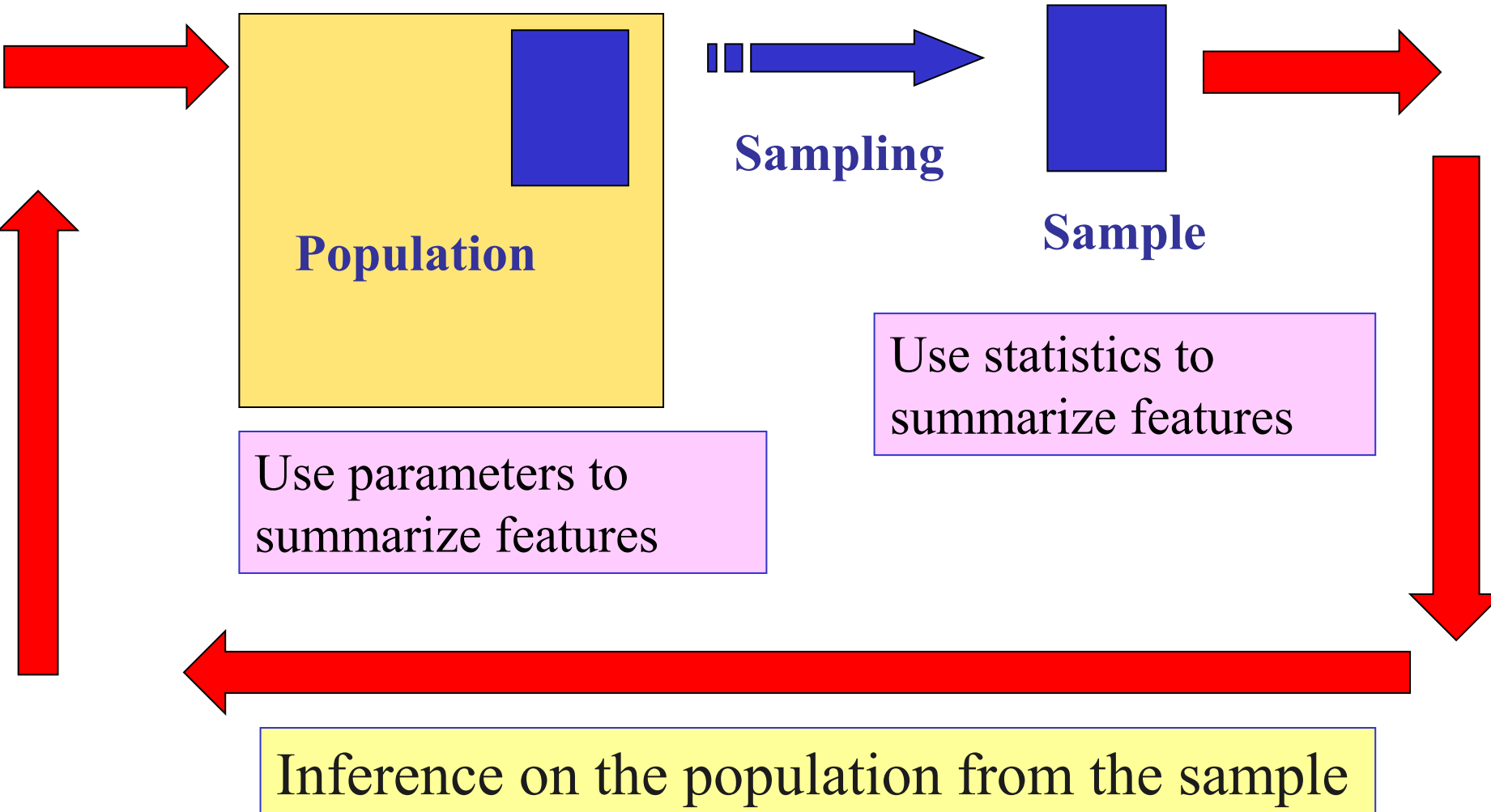
- The population is everyone listed in the city phone directory
- The sample is the 75 people selected to conduct a phone survey.

What is Sampling?

The **process of selecting observations(a sample)** in order to make an inference that can be generalized to the population.



Sampling



Why sampling?

We know that resources such as time, money and people are limited.

When the population is large in size, geographically dispersed, or difficult to contact, it's necessary to use a sample. Thus, most projects aim to gather data from a sample, rather than from the entire population. Some reasons for sampling are:

- **Necessity:** Sometimes it's simply not possible to study the whole population due to its size or inaccessibility.
- **Practicality:** It's easier and more efficient to collect data from a sample.
- **Cost-effectiveness:** There are fewer participant, laboratory, equipment, and researcher costs involved.
- **Manageability:** Storing and running statistical analyses on smaller datasets is easier and reliable.
- **Saves time:** As sample size is relatively less, it increases data-collection speed

Characteristics of a sample

- A sample must be **representative of the population**.
- It must be **appropriately sized**. i.e. it **must be sufficiently large to represent the population and provide statistical stability or reliability**.
- It must be **unbiased**. It should contain **all types of groups/units present in the population in fair proportions**.
- It must be **selected at random**. This means that any item in the group has an equal chance of being and selected and included in the sample.
- It must be **economical**. The objectives of the survey must be achieved in as minimum of cost and effort as possible.
- It must be **goal-oriented**. It must be **oriented to the research objectives and fitted to the survey conditions**.

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Is it a good sample?

Study : Survey of the job prospects of the students studying in a university.

Sample: Taking survey from the students who are in Canteen.



Is it a good sample?

This is not an example of a good sample as,

- The students in the canteen are not completely representative of the students studying in the university.
- The size of the sample (i.e. the number of students in the canteen) might not be appropriate or sufficient enough to represent the population (students studying in the university).
- The sample selection is not performed at random as each student studying in the university doesn't have an equal chance of getting selected.

Is it a good sample?

→ **Study** : A test of the effectiveness of a new high school curriculum introduced

Sample : Dividing an area by school district, then choosing a school or set number of schools at random and sampling students from each school.

This type of sampling results in a unbiased sample as it each school district in an area has its representation in the sample. Also, each school has an equal chance of getting chosen.

Is it a good sample?

→ **Study:** Conduct observations to ensure that employees are employing best practices in the company.

Sample: Each employee is assigned a random number using computer software. The same software is used periodically to choose a number of the employees and are observer. This is a good sample as each employee has an equal chance of being selected.

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THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Sampling Methods

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Sampling Methods

Mamatha H R

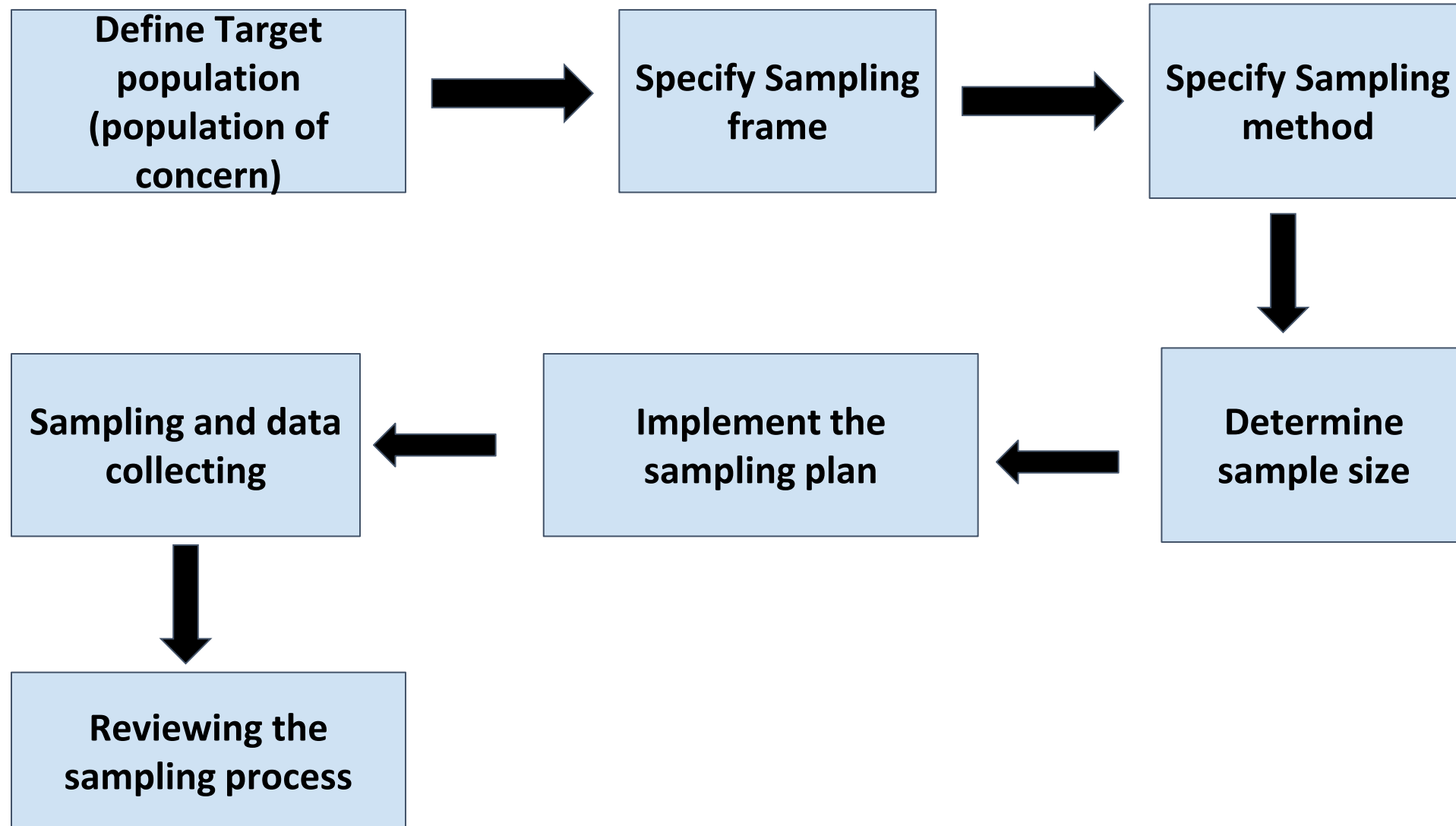
Department of Computer Science and Engineering

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Topics to be covered

- ❖ Sampling methods
- ❖ Sampling process
- ❖ Probability and Non-probability sampling
- ❖ Advantages and disadvantages of different sampling methods

Sampling process



Sampling

→ Factors that influence sample representativeness:

- Sampling procedure
- Sample size
- Participation (response)

→ When might you sample the entire population?

- When your population is very small
- When you have extensive resources
- When you don't expect a very high response

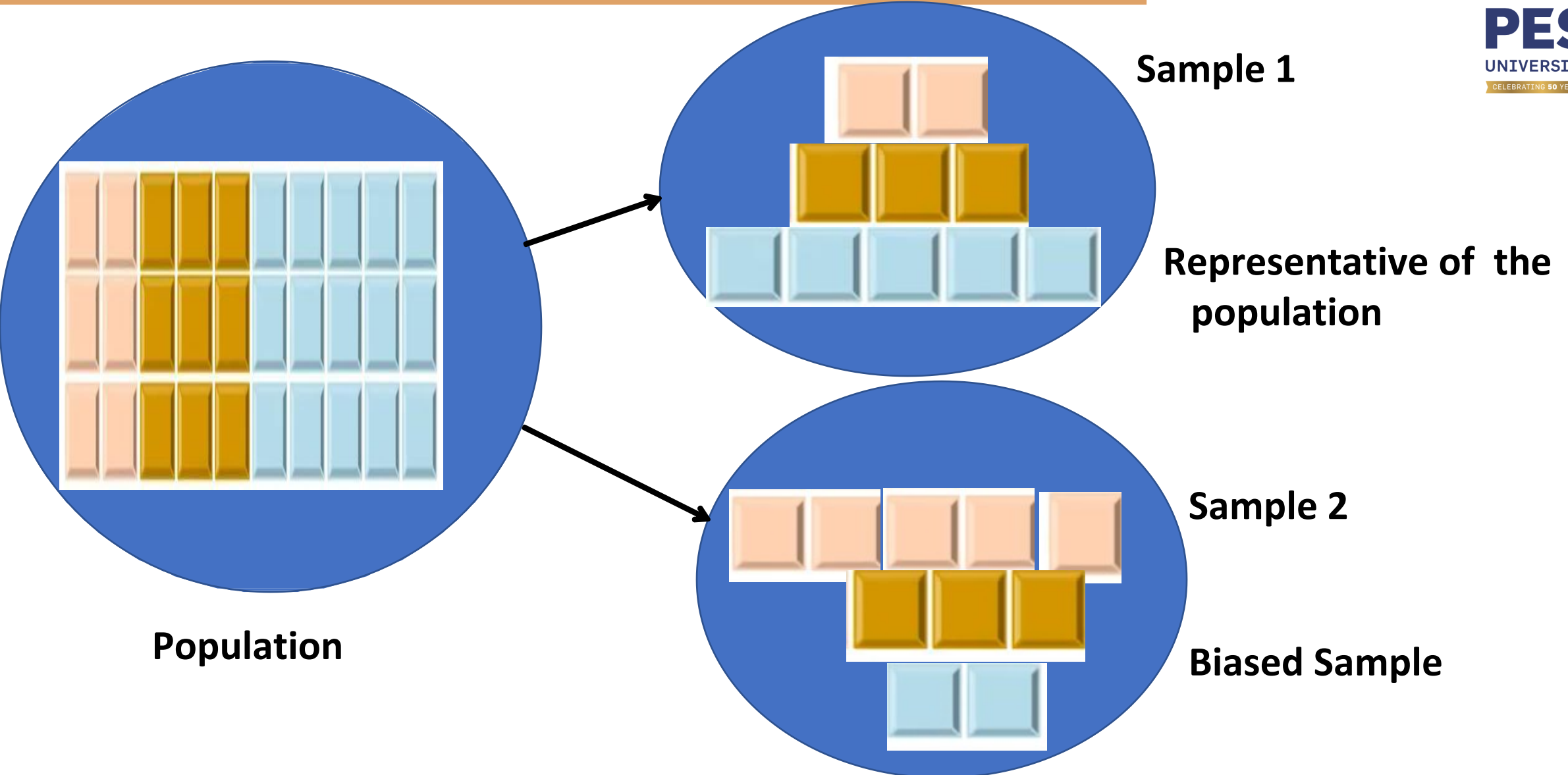


Sampling Frame

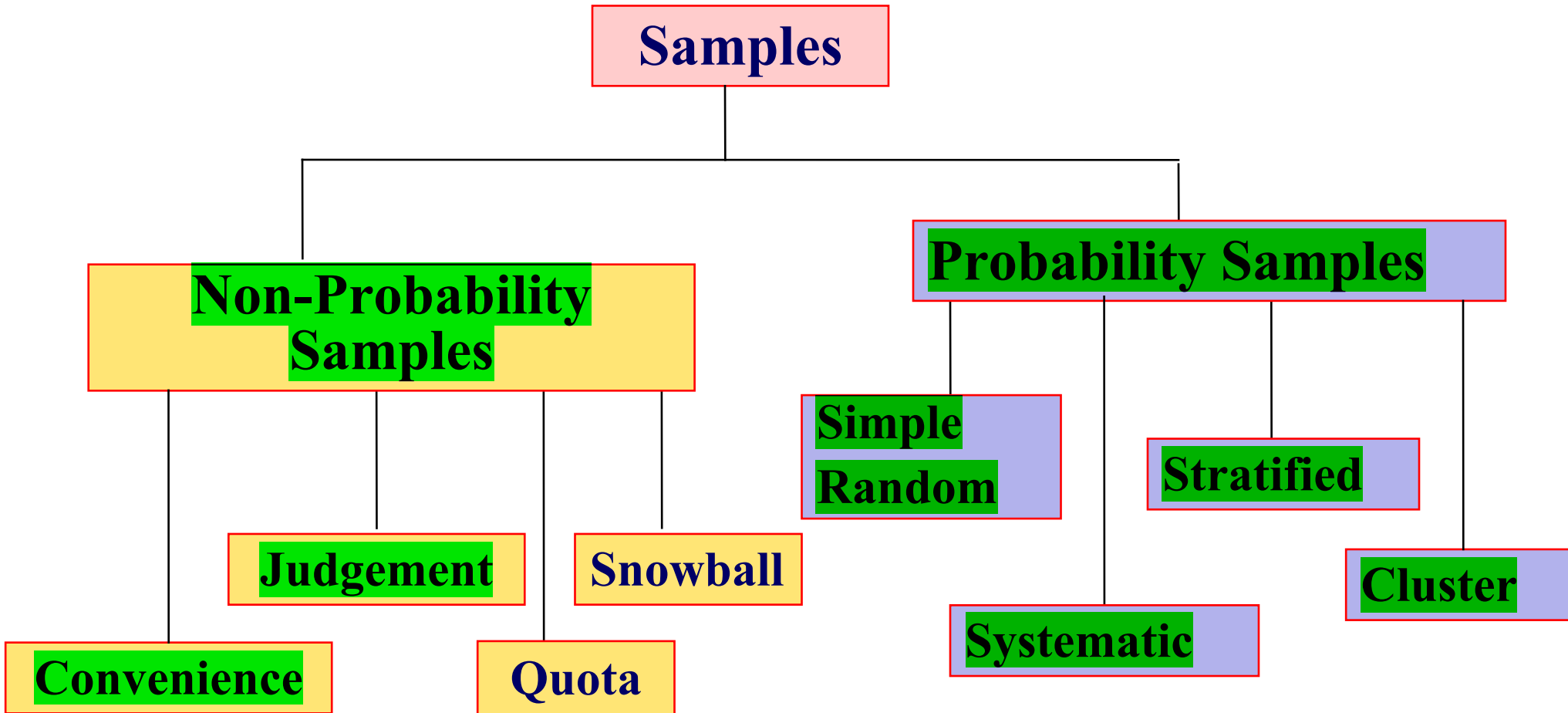
Sampling frame is the list of items or events from which the potential respondents are drawn or which are possible to measure.

- Sometimes, it is possible to identify and measure every single item in the population and to include any one of them in our sample.
- However, in the more general case this is not possible.
- There is no way to identify all rats in the set of all rats.
- As a remedy, we seek a **sampling frame** which has the property that we can identify every single element and include any of them in our sample.
- The sampling frame **must be representative of the population**.

Representative & Biased Sample

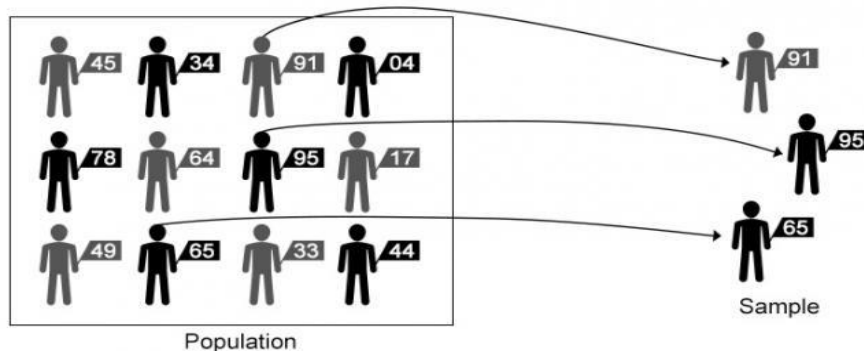


Types of Sampling methods



Probability Sampling

- **Probability sampling** is a type of sampling in which every unit in the population has a chance/probability (greater than zero) of being selected in the sample, and this probability can be accurately determined.
- This type of sampling decreases bias and sampling error in the selection process.
- When every element in the population does have the same probability of selection, this is known as an '**equal probability of selection**' (EPS) design. Such designs are also referred to as '**self-weighting**' because all sampled units are given the same weight.



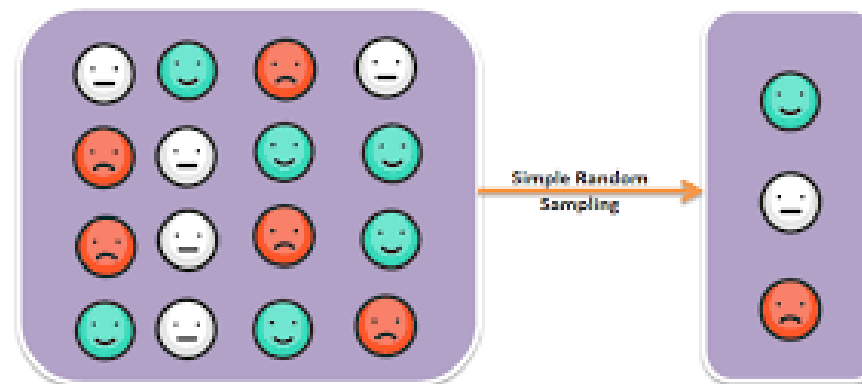
Non-Probability Sampling

- **Non-Probability sampling** is a type of sampling in which every unit in the population doesn't have a chance/probability (greater than zero) of being selected in the sample.
- Here, some elements of the population have no chance of selection (these are sometimes referred to as '**out of coverage**'/'**undercovered**'), or the probability of selection can't be accurately determined.
- It involves the selection of elements based on assumptions regarding the population of interest, which forms the criteria for selection.
- The **selection** of elements is **non random.**
- Thus, non-probability sampling does not allow the estimation of sampling errors.
- It is more likely to produce a biased sample and restricts generalization.
- It is not an appropriate data collection method for most of the statistical analysis.

Simple Random Sampling

Simple random sampling, as the name suggests, is an entirely random method of selecting the sample.

- Here, each subject or unit in the population has an equal chance of being selected.
- The sampling frame should include the whole population.
- A table of random number or lottery system is used to determine which units are to be selected.
- Simple random sampling is always an EPS design, but not all EPS designs are simple random sampling.



Simple Random Sampling

- **Purpose:** It is random and thus results in a representative-sample.
- **When to Use:** Best to use when population is small as it produces a better representative-sample.
- **Key Aspect:** Each member of the population has an equal probability of getting selected.
- **General Procedure:** Assign numbers to all members of the population & select randomly.
 - For a small population: Manual lottery method can be used for selection.
 - For a larger population : System generated numbers can be used to select elements from the population.

Simple Random Sampling Examples

In a company consisting of 10,000 employees, 25 employees are selected to survey the average number of hours a day they are present in the office.

- Population frame: List of all employees numbered from 1-10,000
- Sample : Random number table consisting of 25 random employees.
- Probability of selection of each employee :

$$N = 10,000; n = 25$$

$$\text{probability} = (25/10,000) \times 100 = 0.25\%$$

Row #	A	B	C	D	E
1	6315	9556	5014	2126	8472
2	434	854	958	2572	9220
3	1961	2395	6882	7590	5329
4	18	8580	9313	454	3511
5	7912	7194	855	753	9016

Simple Random Sampling: Advantages

→ Advantages:

- This method is simple to use.
- Estimates are easy to calculate.
- Random samples are usually fairly representative since they don't favor certain members of the population.
- Low sampling error.
- It needs only a minimum knowledge of the study group of population in advance.

Simple Random Sampling: Disadvantages

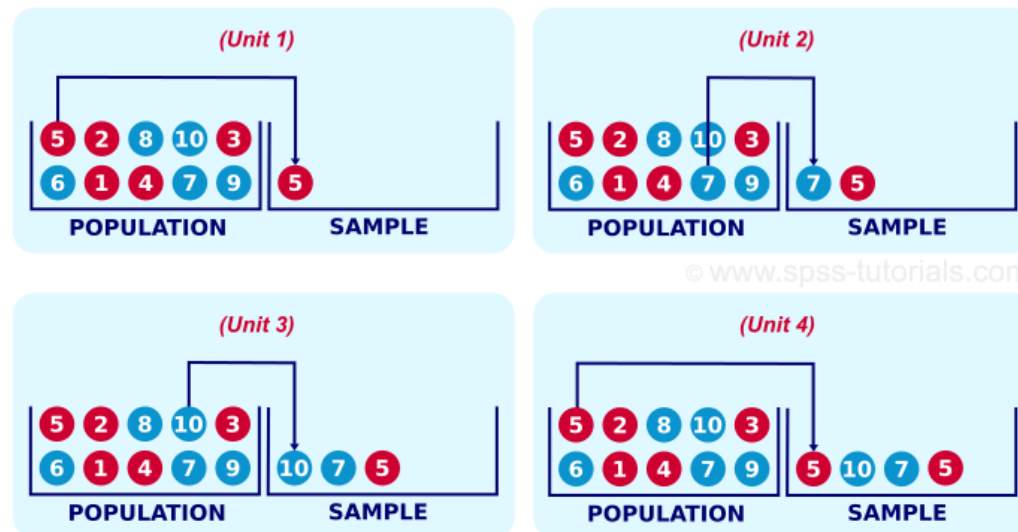
→ Disadvantages:

- If sampling frame is large, this method impracticable.
- Minority subgroups of interest in population may not be present in sample in sufficient numbers for study.
- This type of sampling can't be employed where the units of the population are heterogeneous in nature.
- Sometimes, it is difficult to have a completely cataloged universe.
- This method lacks the use of available knowledge concerning the population.

Simple Random Sampling with replacement

- This is a sampling procedure in which each sampling unit randomly selected from the population is measured or recorded and then returned to the population. Thus, a sampling unit may be sampled multiple times.
- When sampling the first marble, each marble has the same chance of 0.1 of being sampled. When sampling the second marble and all the subsequent marbles, each marble still has a 0.1 chance of being sampled.
- Each time we sample a unit, all units have similar chances of being sampled.

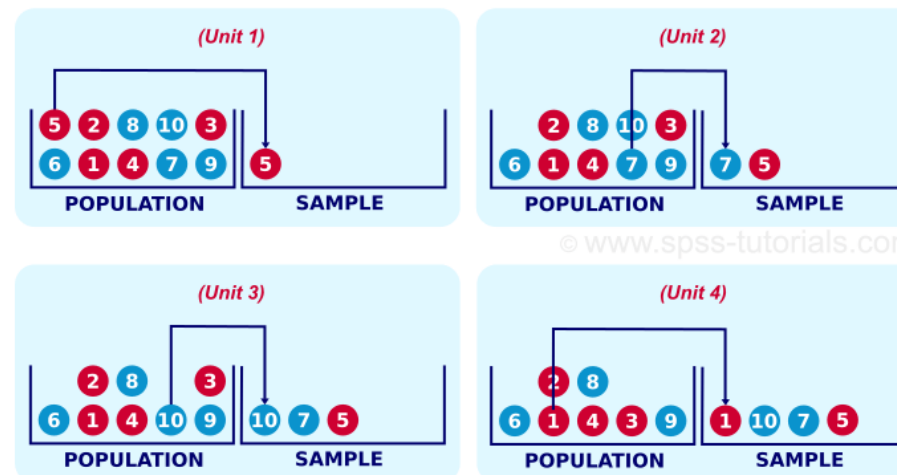
SIMPLE RANDOM SAMPLING WITH REPLACEMENT



Simple Random Sampling without replacement

- This is a sampling procedure in which sampling units are selected from a population of without replacement such that every sample unit has an equal probability of being selected.
- No element can be selected more than once in the same sample.
- For the first marble sampled, each marble has a 0.1 chance of being sampled. However, the first unit we sampled has a zero chance of being sampled again.
- Thus, the other 9 units each have a chance of $1 \text{ in } 9 = 0.11$ of being sampled as the second unit.

SIMPLE RANDOM SAMPLING WITHOUT REPLACEMENT

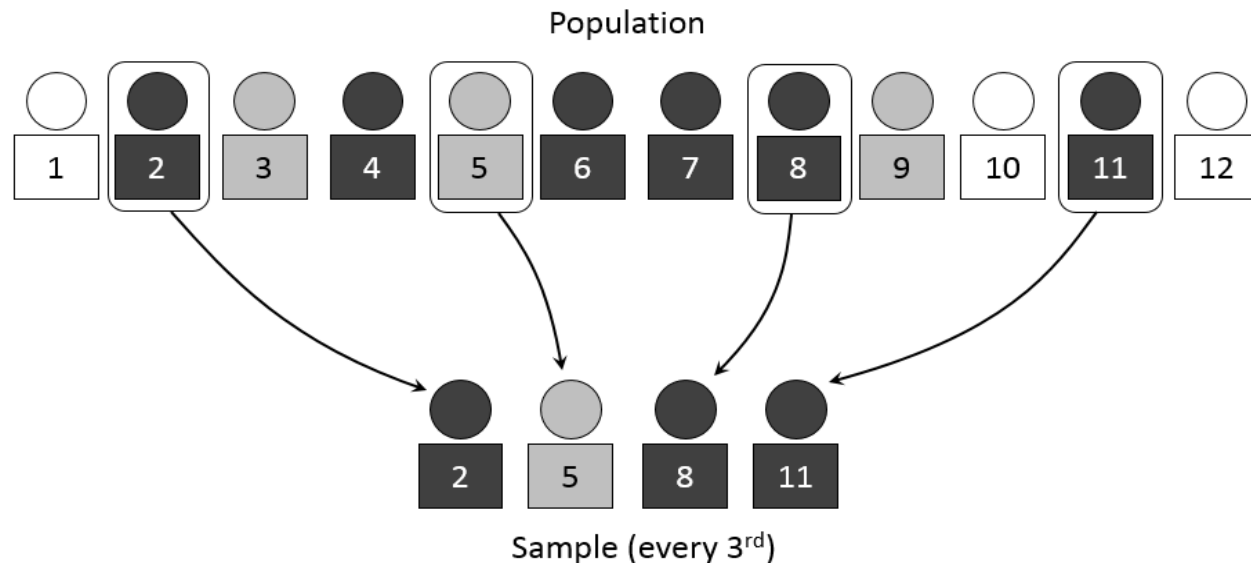


Systematic Sampling

- Systematic sampling relies on **arranging the target population according to some ordering scheme** and then **selecting elements at regular intervals** through that ordered list.
- The **first element** is selected **randomly**.
- Then it proceeds **with the selection of every kth element**. Where k is the size of the selection interval. $k = (\text{population size} / \text{sample size})$
- It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first to the kth element in the list.
- A simple example would be to select every 10th name from the telephone directory (an 'every 10th' sample, also referred to as 'sampling with a skip of 10').

Systematic Sampling

- Systematic sampling is an Equal Probability Sampling method, as all elements have the same probability of selection (in the below example given, one in twelve).
- It is not 'simple random sampling' because different subsets of the same size have different selection probabilities
- Ex: the set {2,5,8,11} has a one-in-twelve probability of selection, but the set {1,3,6,7} has zero probability of selection.



Systematic Sampling

- **When to Use:** When project budget is tight and less time to complete.
- **Key Aspect:** Find the kth value to select every kth member.
 $k = N / n$
- **General Procedure:**
 - Assign numbers to each population element.
 - Order the population elements in an ordered sequence
 - Find 'k' the size of the selection interval.
 - Select the first sample element randomly from the first k population elements.
 - Thereafter, select the sample elements at a constant interval, k, from the ordered sequence frame.

Systematic Sampling: Advantages

- Sample is easy to select.
- Suitable sampling frame can be identified easily.
- Sample evenly spreads over entire reference population.
- It is a cost effective sampling method.
- It guarantees that the entire population is evenly sampled.
- Systematic sampling also carries a low-risk factor because there is a low chance that the data can be contaminated.

Systematic Sampling: Disadvantages

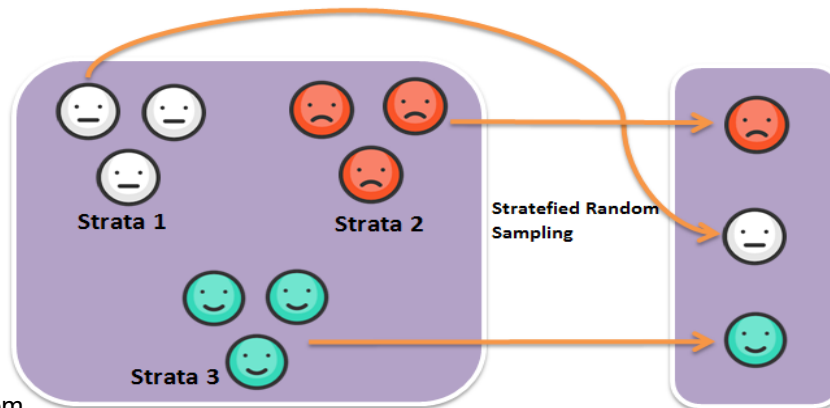
- This type of sampling **might lead to bias** if there is an underlying pattern/periodicity in the population which coincides with the selection.

Ex : If the HR database groups employees by team, and team members are listed in order of seniority, there is a risk that the interval might skip over people in junior roles, resulting in a sample that is skewed towards senior employees.

- Difficult to assess precision of estimate from one survey.
- Each element does not have an equal chance in getting selected
- Ignorance of all the elements between two k th elements.
- The size of the population is needed. Without knowing the specific number of participants in a population, systematic sampling does not work well.

Stratified Sampling

- Stratified sampling is the type of sampling in which the population is divided into 2 or more groups called strata based on a shared characteristic or trait.
- Then simple random samples are selected from each group.
- The selected 2 or more samples are combined into one.
- The strata or groups don't overlap. But, they represent the entire population.
- The shared characteristics based on which the population is divided could be gender, educational attainment, income, age etc.



Stratified Sampling

- **Each stratum** is sampled as an **independent sub-population**.
- Every unit in a stratum has same chance of being selected.
- Using same sampling fraction for all strata **ensures proportionate representation in the sample**.
- Adequate representation of minority subgroups of interest can be ensured by stratification & varying sampling fraction between strata as required.
- Since each stratum is treated as an independent population, different sampling approaches can be applied to different strata.

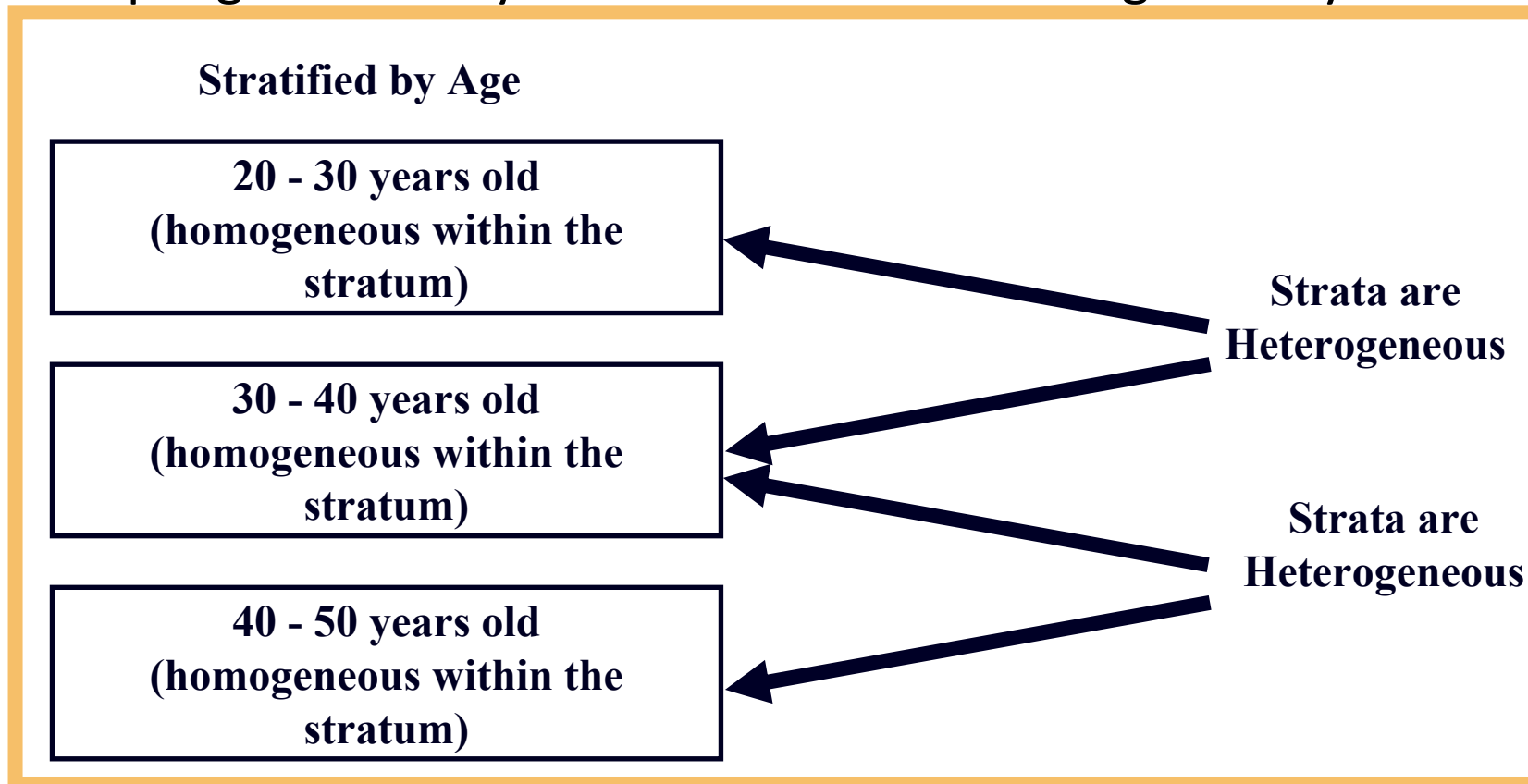
Stratified Sampling

- **Purpose:** To obtain an unbiased random sample from a larger population.
- **When to Use:** When population proportion must be reflected in sample.
- **Key Aspect:** Sample proportion is same as Population proportion, Strata is homogeneous.
- **General Procedure:**
 - Divide the population into Strata or Groups.
 - Criteria for division could be: Gender, Hair Color, Eye Color, Salary, Designation, Age etc.
 - Selection of sample: Simple Random Sampling approach is used to sample units from each strata.

Stratified Sampling examples

To find out the most popular song among the FM radio listeners.

- All listeners are stratified by age.
- Listeners from each age group are selected using simple random sampling and surveyed for their favourite song of the year.



Stratified Sampling: Advantages

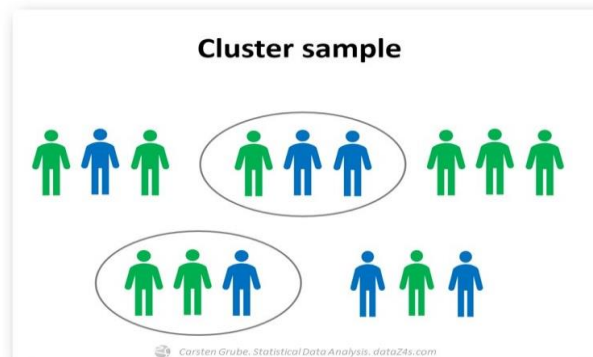
- It enhances the representativeness of the sample.
- It is easy to carry out.
- It has higher statistical efficiency.
- A stratified sample can provide a higher precision than a simple random sample of the same size.
- As it provides a greater precision, this type of sampling often requires a smaller sized sample which saves money.

Stratified Sampling: Disadvantages

- Sampling frame of the entire population has to be prepared separately for each stratum.
- When examining multiple criteria to divide the population, stratifying variables may be related to some but not to others further complicating the design and potentially reducing the utility of the strata.
- In some cases (such as designs with a large number of strata, or those with a specified minimum sample size per group), stratified sampling can potentially require a larger sample than other methods.
- It is time consuming and expensive.
- It leads to classification errors.

Cluster Sampling

- In cluster sampling, population is divided into non-overlapping clusters or areas similar to Stratified sampling.
- Each cluster is a miniature or microcosm of the population.
- Each cluster should have similar characteristics to the whole sample.
- Instead of sampling individuals from each subgroup like in stratified sampling, in cluster sampling entire clusters are randomly selected.
- A subset of the clusters is selected randomly for the sample.
- If the number of elements in the subset of clusters is larger than the desired value of n (sample size), these clusters may be subdivided to form a new set of clusters and subjected to a random selection process.



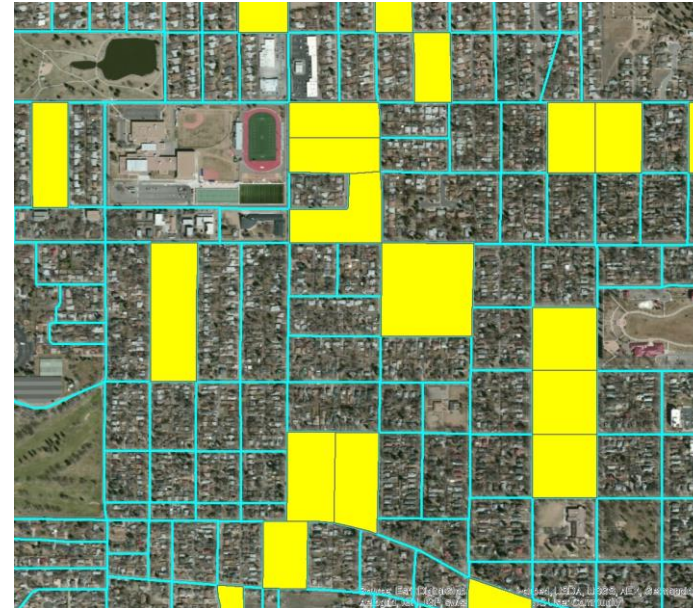
Cluster Sampling

- **When to Use:** When population is already broken up into groups(clusters).
- **Key Aspect:** Heterogeneous members in each group.
- **General Procedure:**
 - Population is divided into non-overlapping areas(clusters).
 - Each cluster is a miniature or microcosm of a population.
 - Clusters are selected randomly.
 - All elements of the selected-clusters are included in the sample or elements from the selected-clusters are chosen using simple random sampling.

Cluster Sampling examples

The municipal council of a small city wants to investigate the use of health care services by residents.

- The council first obtains electoral subdivision maps that identify and label each city block. From these maps, the council creates a list of all city blocks. This list will serve as the sampling frame.
- Every household in that city belongs to a city block, and each city block represents a cluster of households. The council randomly picks a number of city blocks.



Cluster Sampling: Advantages

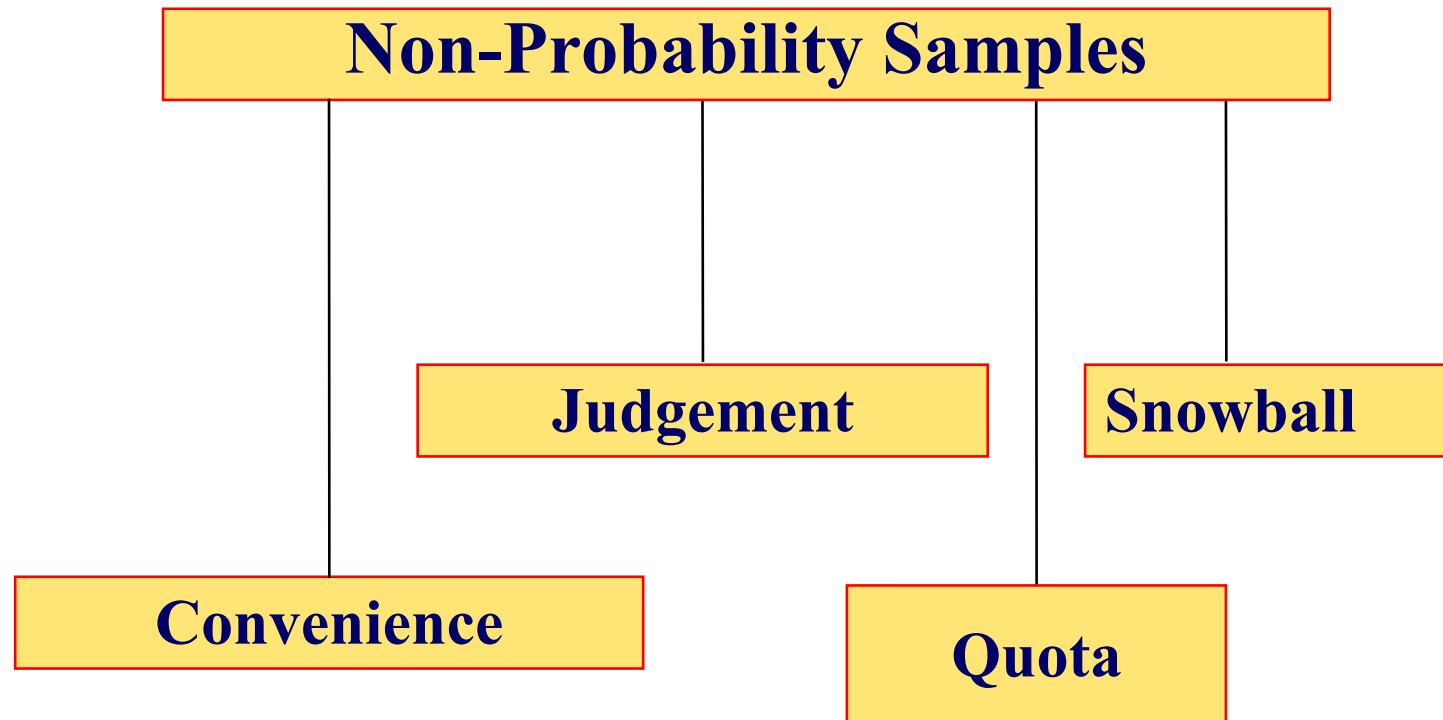
- It is more convenient for geographically dispersed populations.
- It can **reduce the travel costs** to contact sample elements.
- It **simplifies the administration of the survey**.
- It is **more feasible**. The division of the entire population into homogeneous groups increases the feasibility of the sampling.
- Since each cluster represents the entire population, **more subjects can be included in the study**.
- **Requires fewer resources**. Since cluster sampling selects only certain groups from the entire population, the method requires fewer resources for the sampling process.

Cluster Sampling: Disadvantages

- It is statistically less efficient when the cluster elements are similar.
- Costs and the number of problems occurring are greater than that of simple random sampling.
- There is higher sampling error.
- The method is **prone to biases**. If the clusters representing the entire population were formed under a biased opinion, the inferences about the entire population would be biased as well.
- It's difficult to guarantee that the sampled clusters are really representative of the whole population.

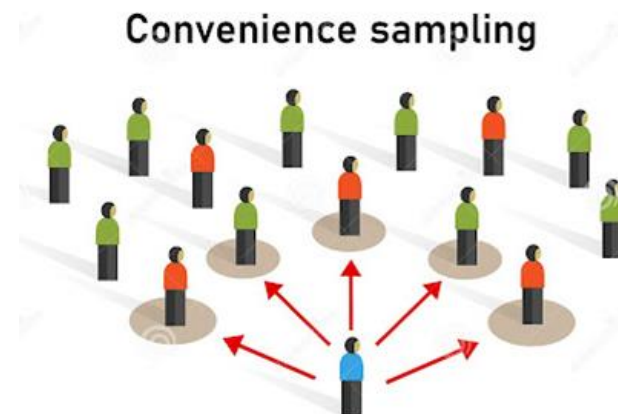
Non-probability Sampling

Non-Probability sampling is a type of sampling in which every unit in the population doesn't have a chance/probability (greater than zero) of being selected in the sample.



Convenience Sampling

- Sometimes it is also known as **grab or opportunity sampling** or **accidental or haphazard sampling**.
- This is a type of nonprobability sampling which involves the **sample being drawn from that part of the population which is close to hand.** That is, readily available and convenient.
- Here, **sample elements are selected for the convenience of the researcher.**
- The **researcher using such a sample cannot scientifically make generalizations about the total population from this sample because it would not be representative enough.**



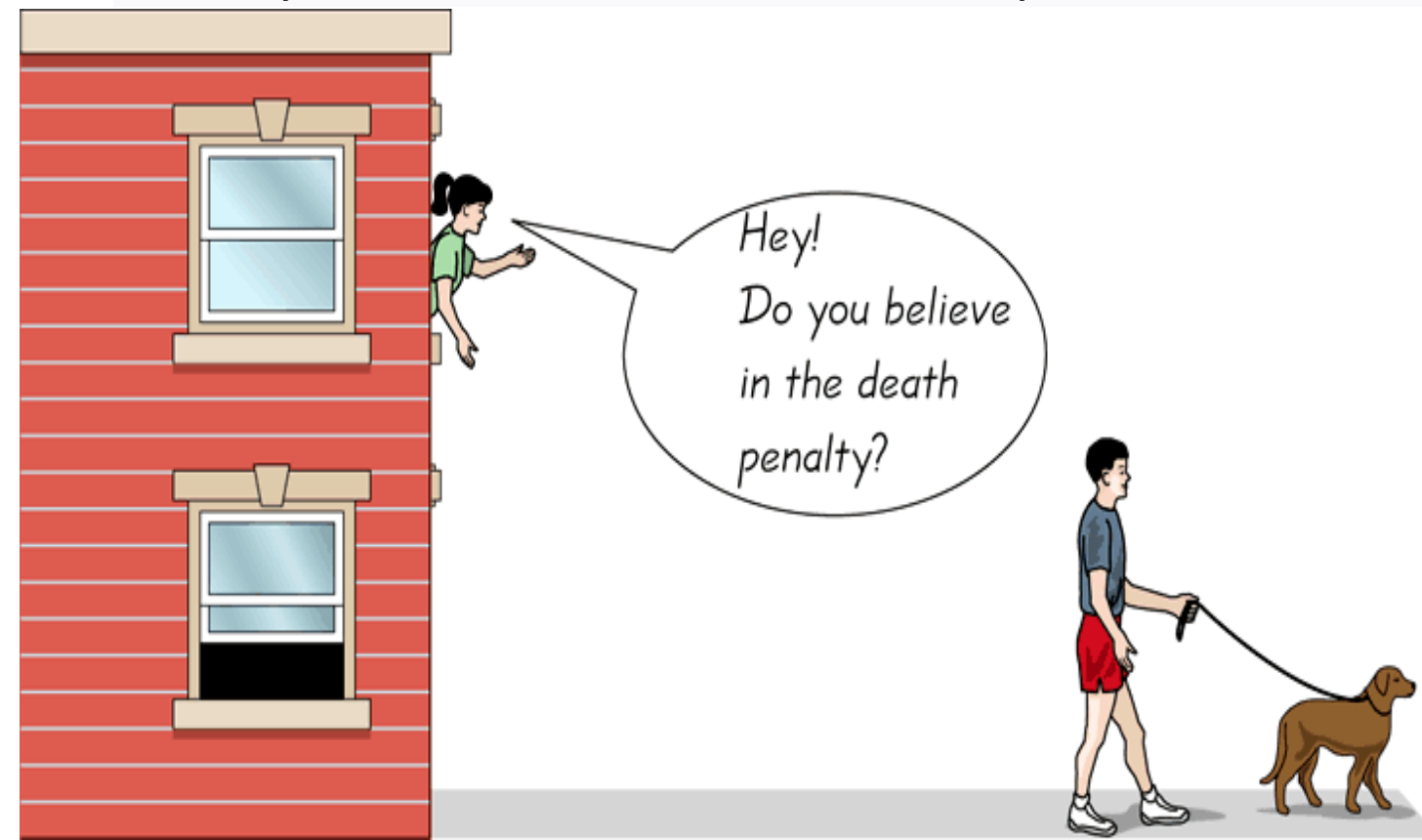
Convenience Sampling

- **When to Use:** When population is not clearly defined or sampling unit is not clear or complete source list is not available.
- **Key Aspect:** Subjects for a study are easily available within the proximity of the researcher.
- **General procedure:**
 - It is done at the “convenience” of the researcher.
 - Selection : The individuals that are convenient and easiest to reach are selected to be included in the sample.

Convenience Sampling examples

To record the popular opinions of people about the current laws of the city.

- The researcher surveys all people that pass by his house.
- Again, this is a convenient way of studying the opinions of people living in the city. But, it doesn't reflect the opinions of all the residents of the city.



Convenience Sampling: Advantages & Disadvantages

→ Advantages:

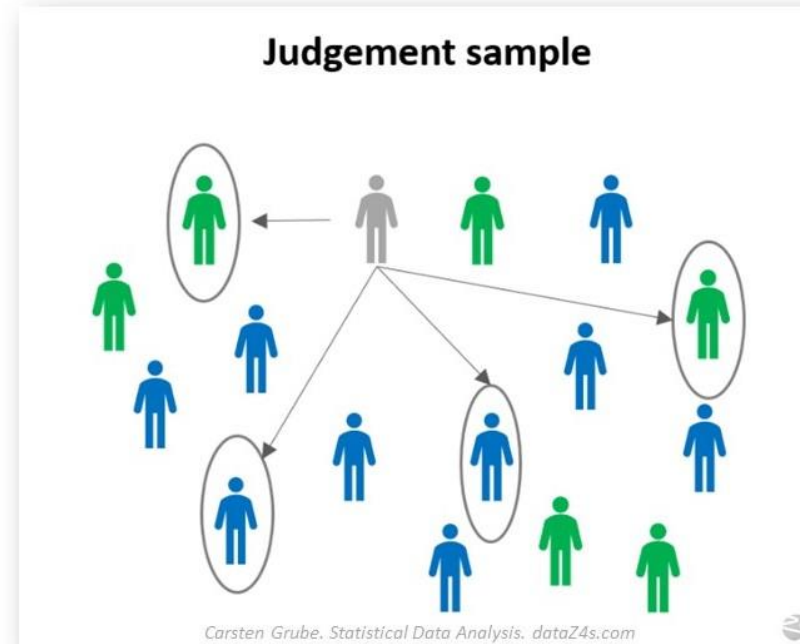
- This type of sampling is useful in pilot study.
- It costs less and is an inexpensive way to gather initial data for the research.
- It saves time.
- It is relatively easy to get a sample.
- It is simple and easy to implement.

→ Disadvantages:

- It is prone to **significant bias** as the sample may not be representative of the characteristics of the population.
- Since the same may not be representative of the population, this type of sampling can't produce generalizable results.
- It might lead to sampling errors.
- A study conducted on a convenience sample will have limited external validity.

Judgemental Sampling

- Judgemental or Purposive sampling is a type of non-probability sampling where the researcher chooses the sample based on who they think would be appropriate for the study.
- This is used primarily when there is a limited number of people that have expertise in the area being researched.
- The sample depends on the judgement of the experts conducting the study.
- It is not a scientific method of sampling.



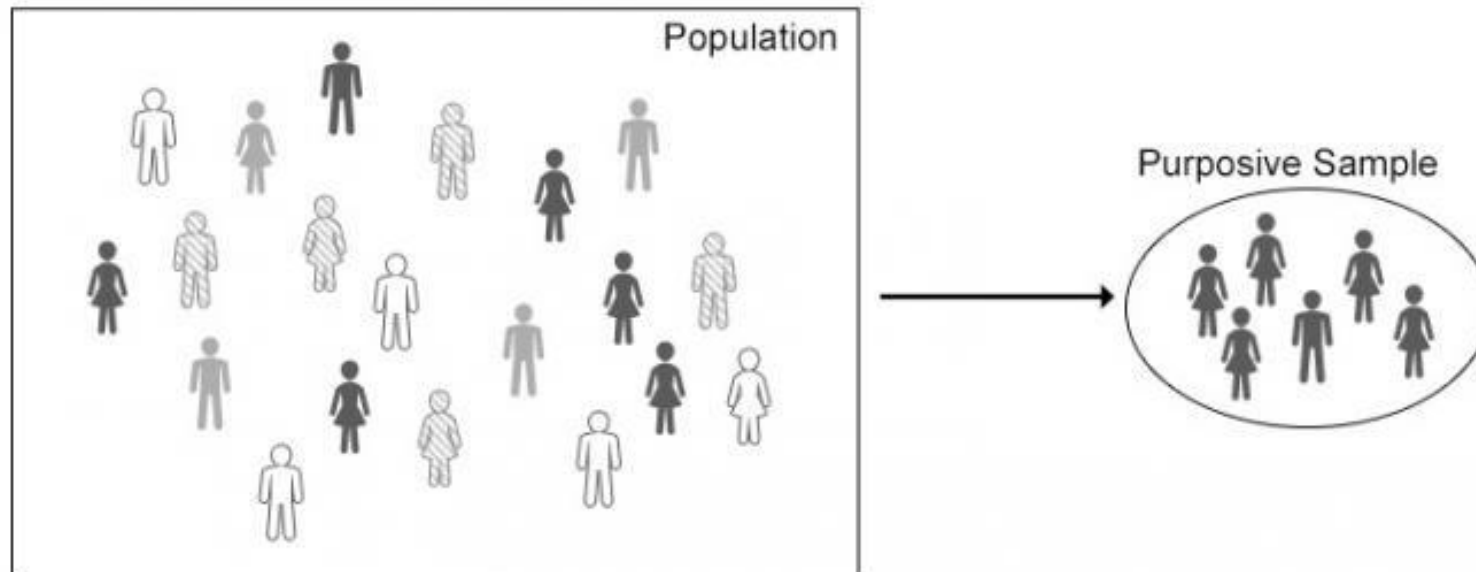
Judgemental Sampling

- **When to Use:** This is used primarily when there is a **limited number of people that have expertise in the area being researched.**
Also, the researcher must be confident that the chosen sample is truly representative of the entire population.
- **Key Aspect:** The researcher selects a sample based on experience or knowledge of the group to be sampled.
- **General Procedure:**
 - On the basis of the researcher's knowledge and judgment elements of the population are sampled.
 - Selection : Elements that own the qualities expected by the researcher.

Judgemental Sampling examples

To know more about the opinions and experiences of disabled students at your university

- You purposefully select a number of students with different support needs at your university in order to gather a varied range of data on their experiences with student services.



Purposive Sampling

Judgemental Sampling: Advantages & Disadvantages

→ Advantages:

- It consumes minimum time.
- The researcher is given an opportunity to bring his judgement and expertise to play.
- No special knowledge of statistics is needed.
- Real time results can be obtained.

→ Disadvantages:

- It is prone to errors in judgment by researcher.
- Low level of reliability and high levels of bias.
- Inability to generalize research findings to the entire population.
- It is difficult to choose the appropriate sample size.

Quota sampling

- In this type of sampling, sample elements are selected until the quota controls are satisfied.
- The population is first segmented into mutually exclusive sub-groups, just as in stratified sampling.
- Then judgment is used to select subjects or units from each segment based on a specified proportion.
- The population units are selected based on predetermined characteristics of the population.
- It is similar to Stratified sampling but it doesn't involve random selection.
- Ex: recruiting the first 50 men and first 50 women that meet inclusion criteria.

Quota sampling

- **When to Use:** If a study aims to investigate a trait or a characteristic of a certain subgroup, this type of sampling is the ideal technique.
- **Key Aspect:** Sample elements are selected until the quota controls are satisfied.
- **General Procedure:**
 - Divide the population into subgroups.
 - Identify proportions or weightage in which the subgroups are present in the population.
 - Select an appropriate sample size while maintaining the proportions of the subgroups.
 - Conduct the surveys according to the quotas defined

Quota sampling examples

A cool drinks company wants to find out what age group prefers what brand of drinks in a particular city.

- The researcher applies quotas on the age groups of 11-21, 22-31, 32-41, 42-51.
- The researcher then samples people from each quota and surveys them to gauge the trend among the population of the city.



Symbol	Age Group	No.
	11-21 Years	11
	22-31 Years	16
	32-41 Years	15
	42-51 Years	18
Total	11-51 Years	60

Quota Sampling

Quota sampling examples: Advantages & Disadvantages

→ Advantages:

- It is a cost effective method.
- There is convenience in execution of this sampling.
- It is a speedy process.
- The information can be deciphered once the sampling is done.
- It improves the representation of certain groups within the population and also ensures that they are not over-represented.

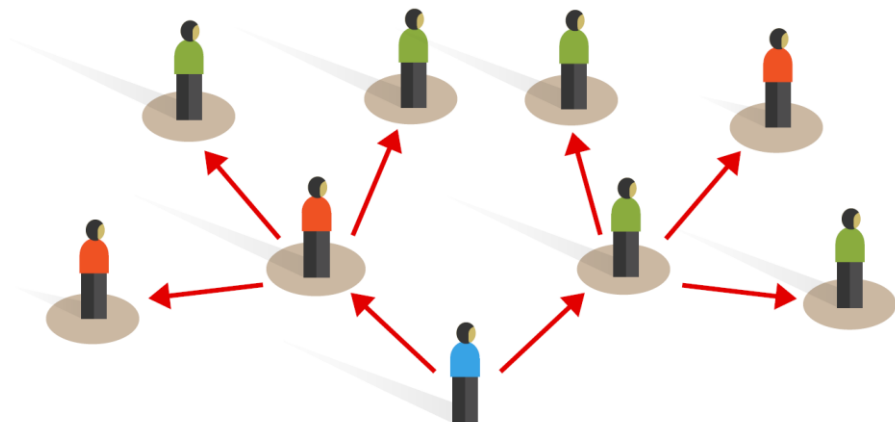
→ Disadvantages:

- Impossible to determine sampling error as the sample is not chosen using random selection.
- Can result in sampling bias if the selection of units was based on ease of access and cost considerations.
- It is not possible to make statistical inferences from the sample to the population leading to the problems of generalization.

Snowball sampling

- In this type of sampling, survey subjects are selected based on referral from other survey respondents.
- Existing subjects are asked to nominate further subjects known to them so that the sample increases in size like a rolling snowball.
- This method of sampling is effective when a sampling frame is difficult to identify.
- Usually applied when the subjects are difficult to trace. Ex: it will be extremely challenging to survey shelter less people or illegal immigrants.

Snowball sampling



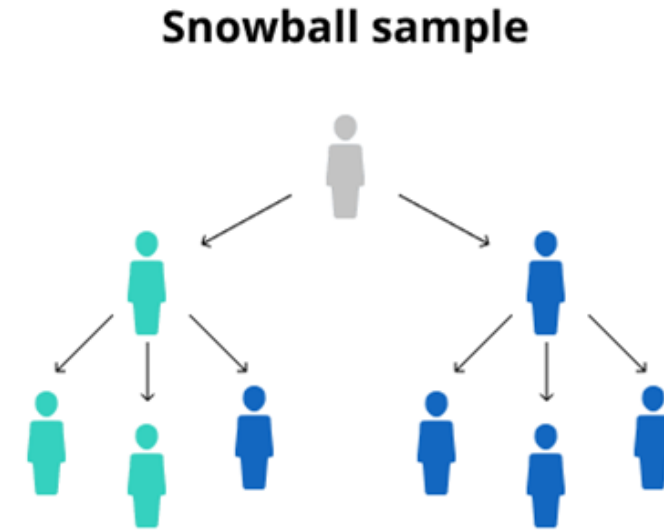
Snowball sampling

- **When to Use:** When the desired sample characteristic is rare.
- **Key Aspect:** Research starts with a key person and introduce the next one to become a chain. It may be extremely difficult or cost prohibitive to locate respondents in these situations.
- **How:**
 - Identify an initial subject and ask these people to identify others.
 - Selection : This technique relies on referrals from initial subjects to generate additional subjects.

Snowball sampling examples

To study the level of customer satisfaction among the members of an elite country club.

- It is extremely difficult to collect primary data sources unless a member of the club agrees to have a direct conversation with you and provides the contact details of the other members of the club.
- Thus the primary data source is randomly selected and it nominates other potential data sources that will be able to participate in the research studies.



Snowball sampling examples: Advantages & Disadvantages

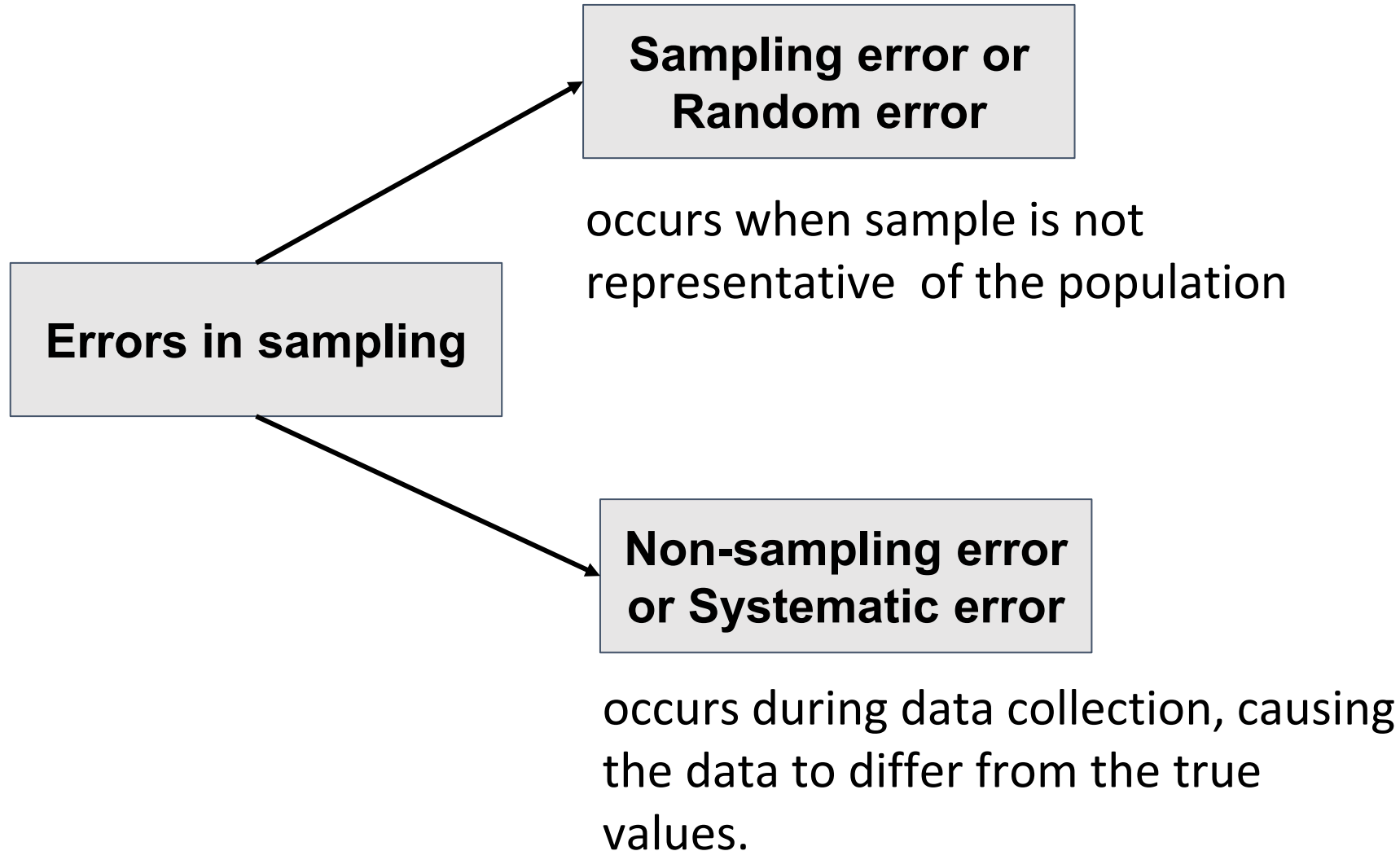
→ Advantages:

- The chain referral process allows the researcher to reach populations that are difficult to sample when using other sampling methods.
- The process is cheap, simple and cost-efficient.
- This sampling technique needs little planning and fewer workforce compared to other sampling techniques.

→ Disadvantages:

- There is a significant risk of selection bias in snowball sampling, as the referenced individuals will share common traits with the person who recommends them.
- It is usually impossible to determine the sampling error or make inferences about populations based on the obtained sample.
- The researcher has little control over the sampling method.
- Representativeness of the sample is not guaranteed.

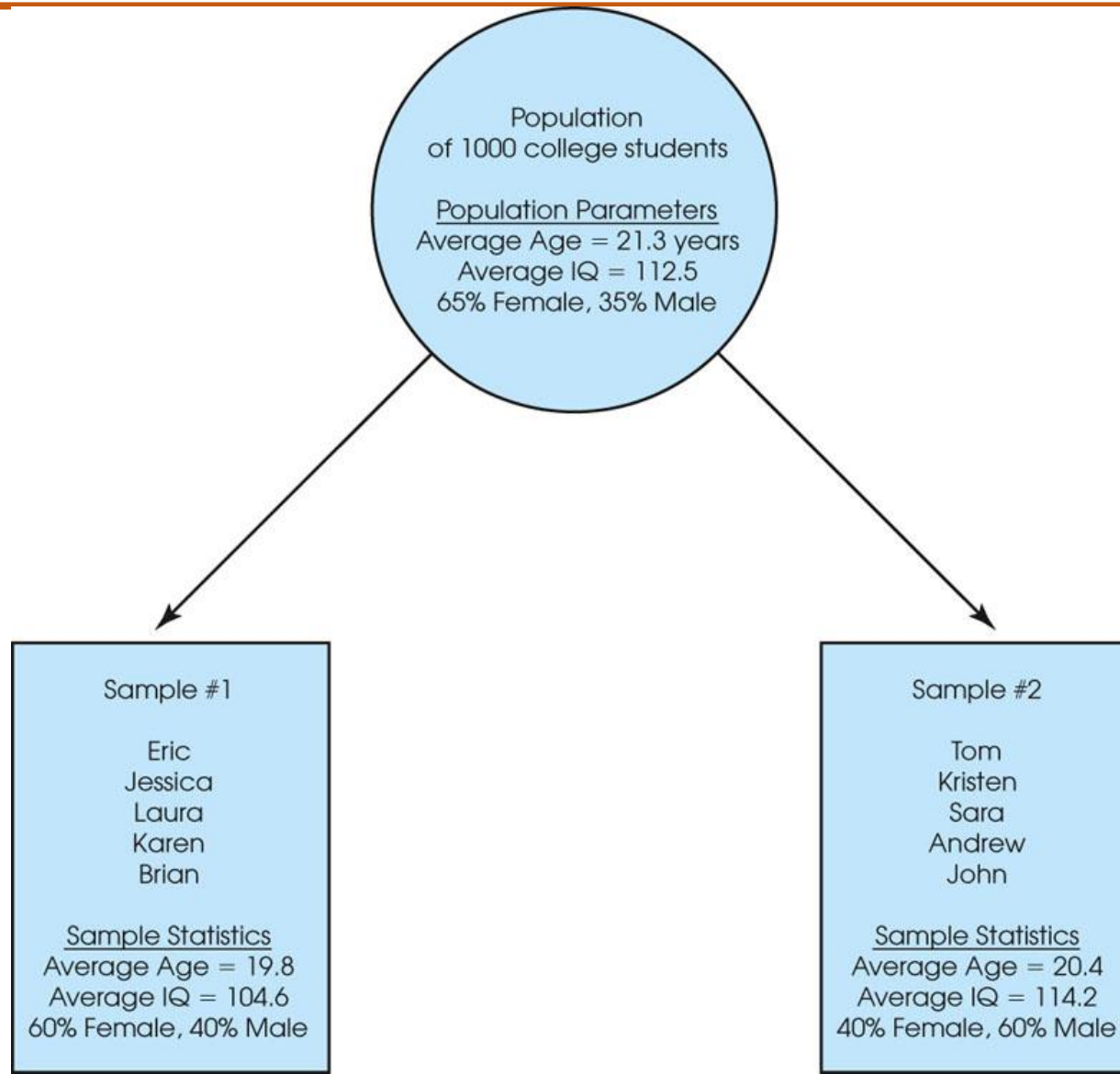
Errors in sampling



Sampling error

- The discrepancy between a sample statistic and its population parameter is called sampling error.
- Defining and measuring sampling error is a large part of inferential statistics.
- It occurs when the sample is not representative of the population.
- The sampling error for a given sample is unknown but when the sampling is random, for some estimates (for example, sample mean, sample proportion) theoretical methods may be used to measure the extent of the variation caused by sampling error.

Sampling error



As we can see there is a difference between population parameters and sample parameters. This is due to sampling error.

Two samples of same population have differing parameters. This is due to sampling variation. It is also the reason why scientific experiments produce different result under identical scenarios.

Non-sampling error

- Non-sampling errors are the results of mistakes made in implementing data collection and data processing, such as
 - failure to locate and interview the correct household
 - errors in understanding of the questions by either the interviewer or the respondent
 - data entry errors
 - missing Data
 - poorly conceived concepts, unclear definitions, and defective questionnaires
 - response errors occurring when people are unaware, refuse to answer, or overstate in their answers
- Major sources : Sampling Bias, Non-response Bias.

Sampling Bias

- Sampling bias occurs when a chosen sample is not representative of the larger population.
- It occurs due to the sampling technique/method used to perform data collection.
- It can be either **selection bias and non-response bias.**
- A sampling method has a sampling bias if all subjects in the population are not equally likely to be included in a sample.
- That is, a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.

Selection Bias & Nonresponse bias

→ Selection bias:

- It is a bias in which a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.
- It results in a biased sample of a population in which all individuals, or instances, were not equally likely to have been selected.

→ Nonresponse bias:

- Nonresponse bias is a type of sampling bias that occurs because of the absence of certain objects or subjects from a sample.
- For example, some subjects don't respond to surveys because they refuse, cannot be contacted, or have a lack of interest in the survey content.

Bias ex:

Q) A new chemical process is run 10 times each morning for five consecutive mornings. If the new process is put into production, it will be run 10 hours each day, from 7 A.M. until 5 P.M. Is it reasonable to consider the 50 yields to be a simple random sample?

Ans) Since the new process runs during both morning and afternoon, the population consists of all the yields that would ever be observed, including both morning and afternoon runs.

The sample however is drawn only from that portion of the population that consists of morning runs, and thus it is not a simple random sample. It exhibits a bias is not representative of the population intended to be studied.

Sampling variation

- Simple random samples always differ from their populations in some ways, and occasionally may be substantially different.
- Two different samples from the same population will differ from each other as well.
- This phenomenon is known as **sampling variation**.
- Sampling variation is one of the reasons that scientific experiments produce somewhat different results when repeated, even when the conditions appear to be identical.

Independence

- The items in a sample are said to be independent if knowing the values of some of them does not help to predict the values of the others.
- With a finite, tangible population, the items in a simple random sample are not strictly independent, because as each item is drawn, the population changes.
- This change can be substantial when the population is small.
- However, when the population is very large, this change is negligible and the items can be treated as if they were independent
- The sample can be considered independent if sample size is smaller than 5% of population size.
- Since conceptual population have infinite/very large size the sample obtained (ex: measuring a rock) is always independent

Questions

(Q1.) A physical education professor wants to study the physical fitness levels of students at her university. There are 20,000 students enrolled at the university, and she wants to draw a sample of size 100 to take a physical fitness test. She obtains a list of all 20,000 students, numbered from 1 to 20,000. She uses a computer random number generator to generate 100 random integers between 1 and 20,000 and then invites the 100 students corresponding to those numbers to participate in the study. Which sampling technique is used?

Answer:

The simple random sampling technique is used.

Note that it is analogous to a lottery in which each student has a ticket and 100 tickets are drawn.

Questions

(Q2) A quality engineer wants to inspect rolls of wallpaper in order to obtain information on the rate at which flaws in the printing are occurring. She decides to draw a sample of 50 rolls of wallpaper from a day's production. Each hour for 5 hours, she takes the 10 most recently produced rolls and counts the number of flaws on each. Is this a simple random sample?

Answer:

No. Not every subset of 50 rolls of wallpaper is equally likely to comprise the sample. To construct a simple random sample, the engineer would need to assign a number to each roll produced during the day and then generate random numbers to determine which rolls comprise the sample.

Questions

(Q3) A construction engineer has just received a shipment of 1000 concrete blocks, each weighing approximately 50 pounds. The blocks have been delivered in a large pile. The engineer wishes to investigate the crushing strength of the blocks by measuring the strengths in a sample of 10 blocks. Which sampling method is suitable?

Answer:

To draw a simple random sample would require removing blocks from the center and bottom of the pile, which might be quite difficult. For this reason, the engineer might construct a sample simply by taking 10 blocks off the top of the pile.

convenience sample

Questions

(Q4) A quality inspector draws a simple random sample of 40 bolts from a large shipment and measures the length of each. He finds that 34 of them, or 85%, meet a length specification. He concludes that exactly 85% of the bolts in the shipment meet the specification. The inspector's supervisor concludes that the proportion of good bolts is likely to be close to, but not exactly equal to, 85%. Which conclusion is appropriate?

Answer:

Because of sampling variation, simple random samples don't reflect the population perfectly. However, they are often fairly close. It is therefore appropriate to infer that the proportion of good bolts in the lot is likely to be close to the sample proportion, which is 85%. It is not likely that the population proportion is equal to 85%.

Questions

(Q5) Another inspector repeats the study with a different simple random sample of 40 bolts. She finds that 36 of them, or 90%, are good. The first inspector claims that she must have done something wrong, since his results showed that 85%, not 90%, of bolts are good. Is he right?

Answer:

No, he is not right. This is sampling variation at work. Two different samples from the same population will differ from each other and from the population.

Questions

(Q6) A geologist weighs a rock several times on a sensitive scale. Each time, the scale gives a slightly different reading. Under what conditions can these readings be thought of as a simple random sample? What is the population?

Answer:

If the physical characteristics of the scale remain the same for each weighing, so that the measurements are made under identical conditions, then the readings may be considered to be a simple random sample. The population is conceptual. It consists of all the readings that the scale could in principle produce.

Questions

(Q7) What sampling method can be recommended?

- Determining proportion of undernourished five year olds in a village.
- Investigating nutritional status of preschool children.
- In estimation of immunization coverage in a province, data on seven children aged 12-23 months in 30 clusters are used to determine proportion of fully immunized children in the province. Give reasons why cluster sampling is used in this survey.

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS



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THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Sampling Methods

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Sampling Methods

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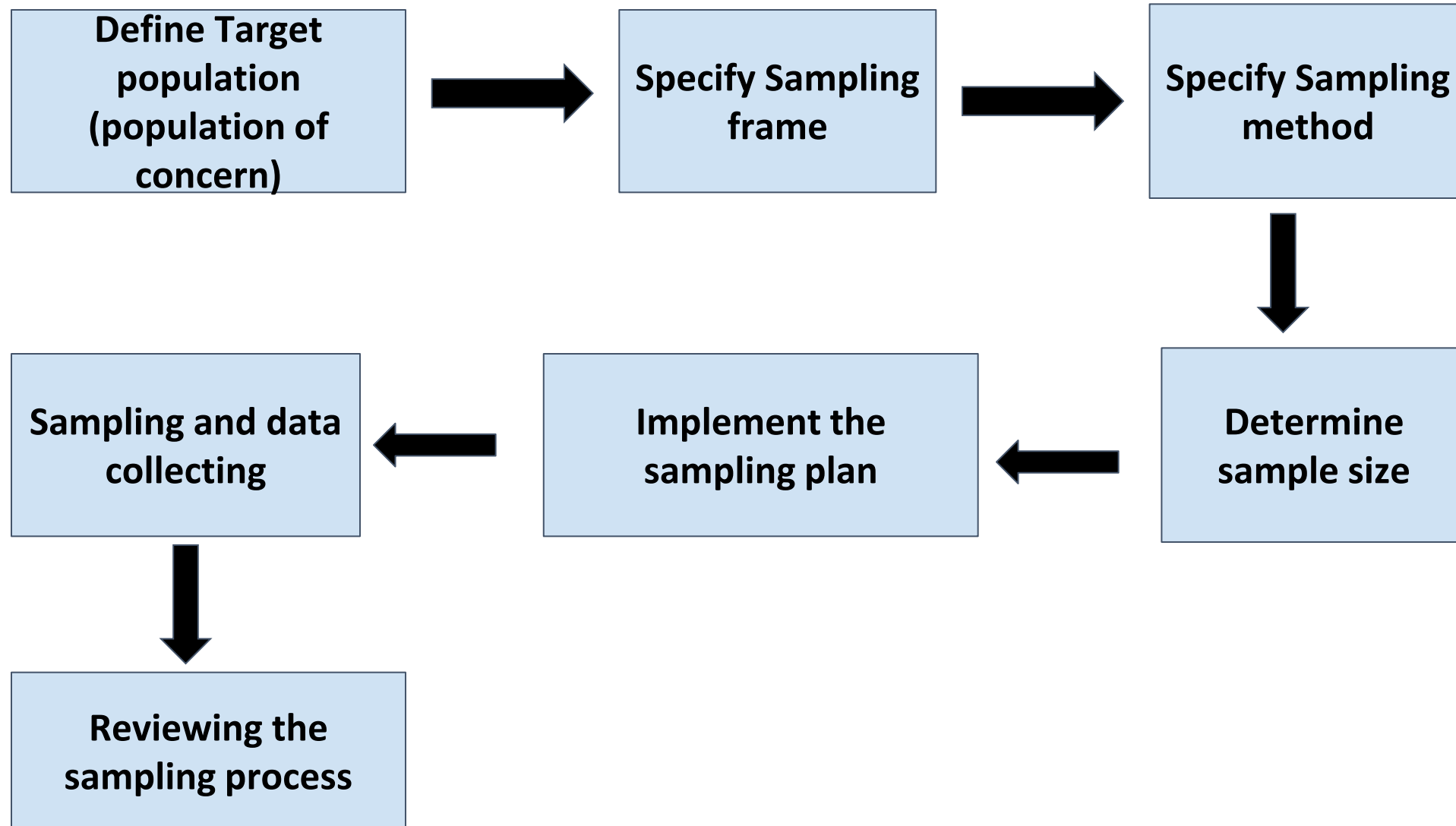
Department of Computer Science and Engineering

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Topics to be covered

- ❖ Sampling methods
- ❖ Sampling process
- ❖ Probability and Non-probability sampling
- ❖ Advantages and disadvantages of different sampling methods

Sampling process



Sampling

→ **Factors that influence sample representativeness:**

- Sampling procedure
- Sample size
- Participation (response)

→ **When might you sample the entire population?**

- When your population is very small
- When you have extensive resources
- When you don't expect a very high response

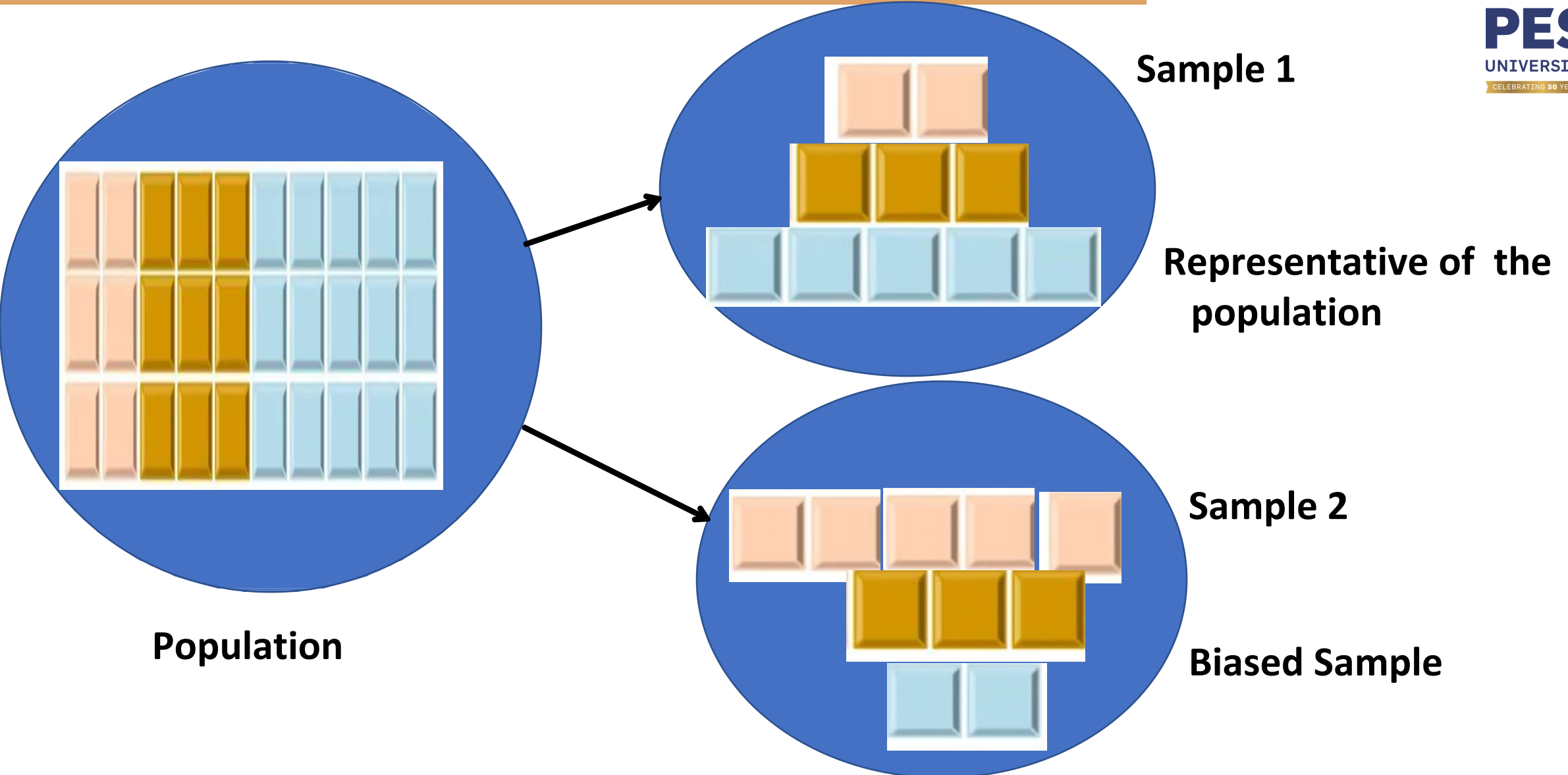


Sampling Frame

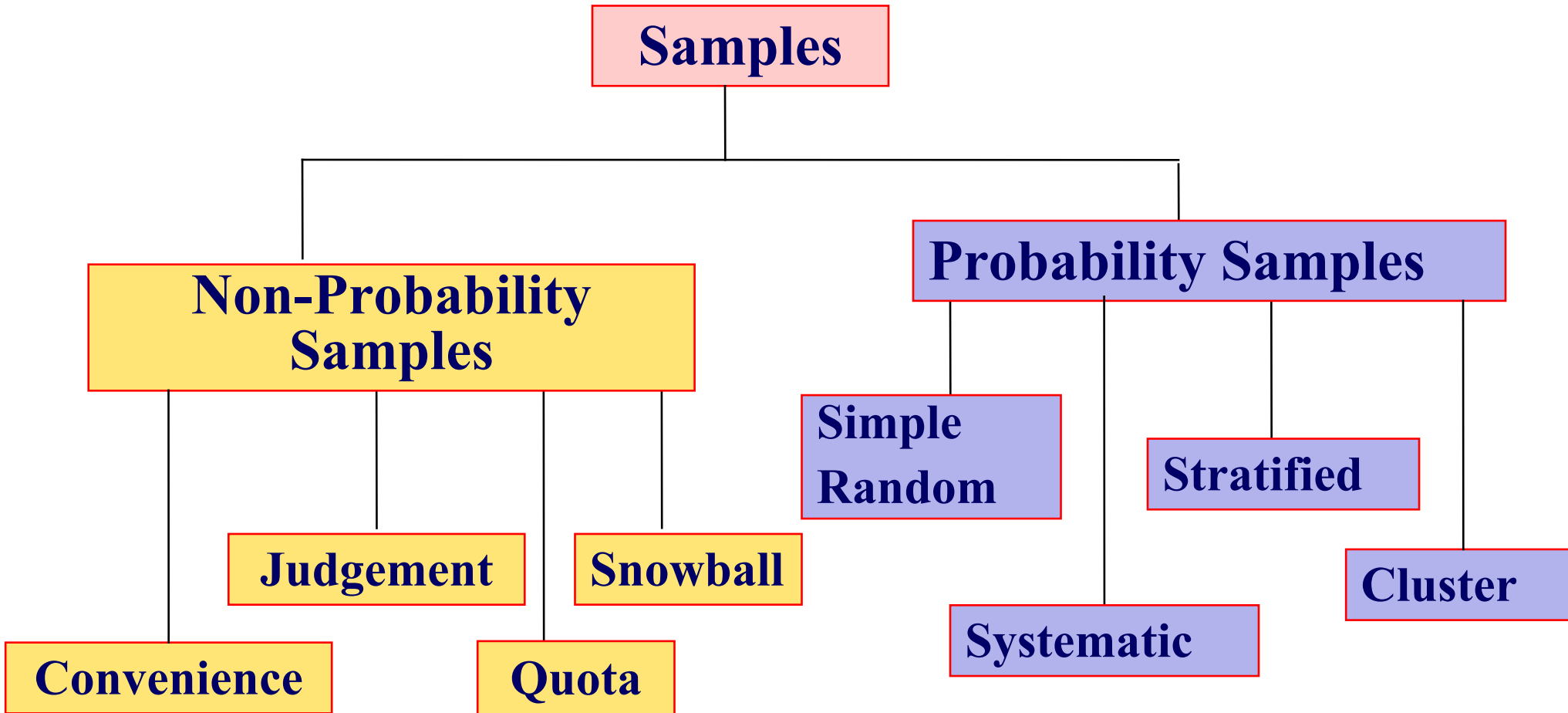
Sampling frame is the list of items or events from which the potential respondents are drawn or which are possible to measure.

- Sometimes, it is possible to identify and measure every single item in the population and to include any one of them in our sample.
- However, in the more general case this is not possible.
- There is no way to identify all rats in the set of all rats.
- As a remedy, we seek a **sampling frame** which has the property that we can identify every single element and include any of them in our sample.
- The sampling frame **must be representative of the population**.

Representative & Biased Sample

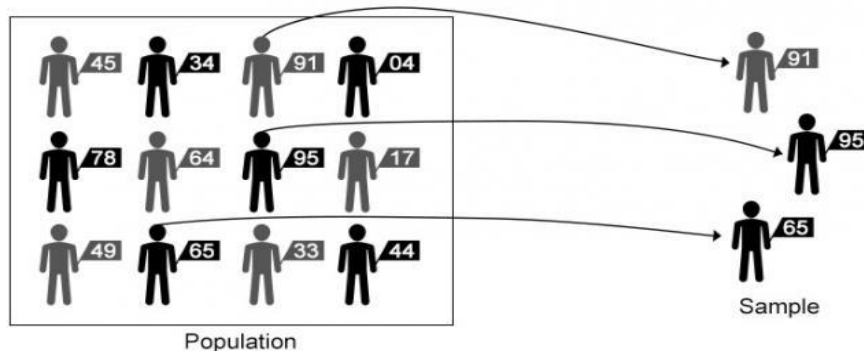


Types of Sampling methods



Probability Sampling

- **Probability sampling** is a type of sampling in which every unit in the population has a chance/probability (greater than zero) of being selected in the sample, and this probability can be accurately determined.
- This type of sampling decreases bias and sampling error in the selection process.
- When every element in the population does have the same probability of selection, this is known as an '**equal probability of selection**' (EPS) design. Such designs are also referred to as '**self-weighting**' because all sampled units are given the same weight.



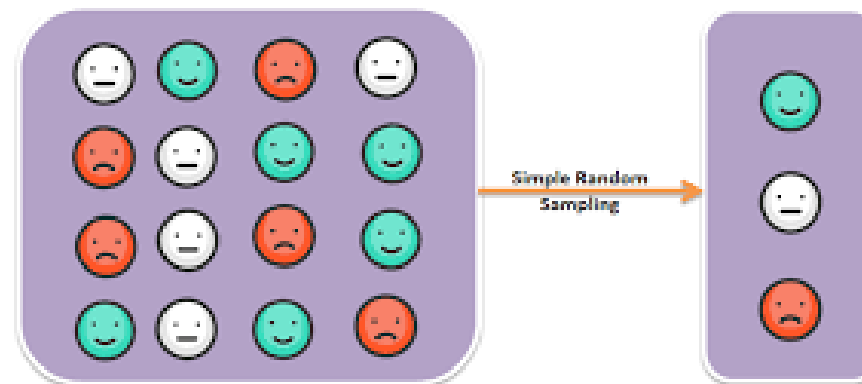
Non-Probability Sampling

- **Non-Probability sampling** is a type of sampling in which every unit in the population doesn't have a chance/probability (greater than zero) of being selected in the sample.
- Here, some elements of the population have no chance of selection (these are sometimes referred to as '**out of coverage**'/'**undercovered**'), or the probability of selection can't be accurately determined.
- It involves the **selection** of elements **based on assumptions regarding the population of interest**, which forms the criteria for selection.
- The **selection** of elements is **non random**.
- Thus, non-probability sampling does not allow the estimation of sampling errors.
- It is more likely to produce a **biased sample** and **restricts generalization**.
- It is not an appropriate data collection method for most of the statistical analysis.

Simple Random Sampling

Simple random sampling, as the name suggests, is an entirely random method of selecting the sample.

- Here, each subject or unit in the population has an equal chance of being selected.
- The sampling frame should include the whole population.
- A table of random number or lottery system is used to determine which units are to be selected.
- Simple random sampling is always an EPS design, but not all EPS designs are simple random sampling.



Simple Random Sampling

- **Purpose:** It is random and thus results in a representative-sample.
- **When to Use:** Best to use when population is small as it produces a better representative-sample.
- **Key Aspect:** Each member of the population has an equal probability of getting selected.
- **General Procedure:** Assign numbers to all members of the population & select randomly.
 - For a small population: Manual lottery method can be used for selection.
 - For a larger population : System generated numbers can be used to select elements from the population.

Simple Random Sampling Examples

In a company consisting of 10,000 employees, 25 employees are selected to survey the average number of hours a day they are present in the office.

- Population frame: List of all employees numbered from 1-10,000
- Sample : Random number table consisting of 25 random employees.
- Probability of selection of each employee :

$$N = 10,000; n = 25$$

$$\text{probability} = (25/10,000) \times 100 = 0.25\%$$

Row #	A	B	C	D	E
1	6315	9556	5014	2126	8472
2	434	854	958	2572	9220
3	1961	2395	6882	7590	5329
4	18	8580	9313	454	3511
5	7912	7194	855	753	9016

Simple Random Sampling: Advantages

→ Advantages:

- This method is simple to use.
- Estimates are easy to calculate.
- Random samples are usually fairly representative since they don't favor certain members of the population.
- Low sampling error.
- It needs only a minimum knowledge of the study group of population in advance.

Simple Random Sampling: Disadvantages

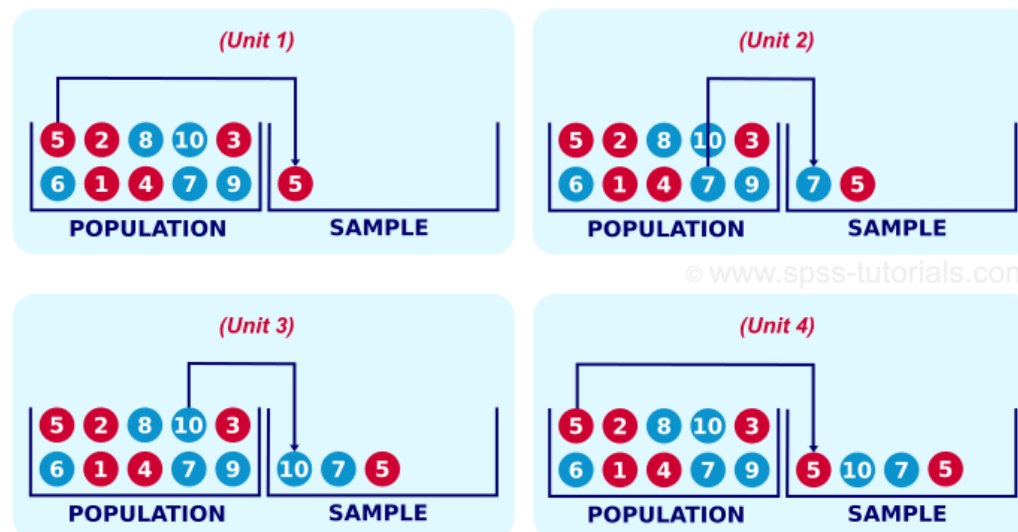
→ Disadvantages:

- If sampling frame is large, this method impracticable.
- Minority subgroups of interest in population may not be present in sample in sufficient numbers for study.
- This type of sampling can't be employed where the units of the population are heterogeneous in nature.
- Sometimes, it is difficult to have a completely cataloged universe.
- This method lacks the use of available knowledge concerning the population.

Simple Random Sampling with replacement

- This is a sampling procedure in which each sampling unit randomly selected from the population is measured or recorded and then returned to the population. Thus, a sampling unit may be sampled multiple times.
- When sampling the first marble, each marble has the same chance of 0.1 of being sampled. When sampling the second marble and all the subsequent marbles, each marble still has a 0.1 chance of being sampled.
- **Each time we sample a unit, all units have similar chances of being sampled.**

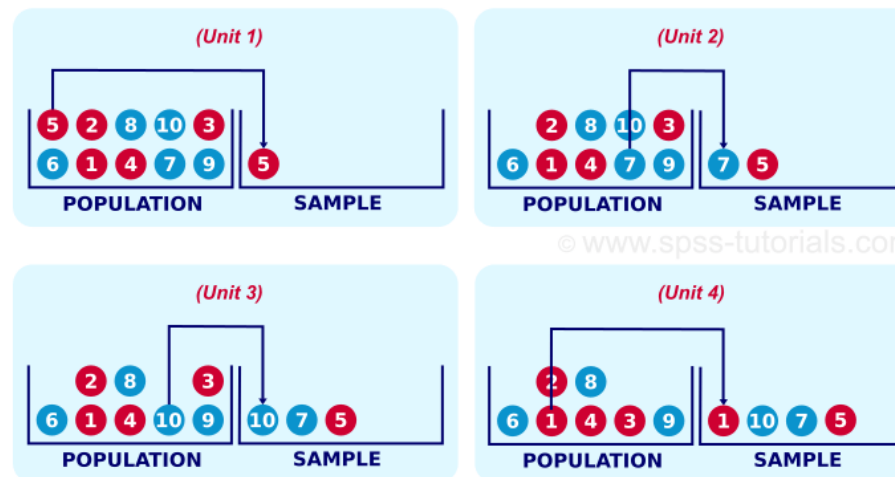
SIMPLE RANDOM SAMPLING WITH REPLACEMENT



Simple Random Sampling without replacement

- This is a sampling procedure in which sampling units are selected from a population of without replacement such that every sample unit has an equal probability of being selected.
- No element can be selected more than once in the same sample.
- For the first marble sampled, each marble has a 0.1 chance of being sampled. However, the first unit we sampled has a zero chance of being sampled again.
- Thus, the other 9 units each have a chance of $1 \text{ in } 9 = 0.11$ of being sampled as the second unit.

SIMPLE RANDOM SAMPLING WITHOUT REPLACEMENT

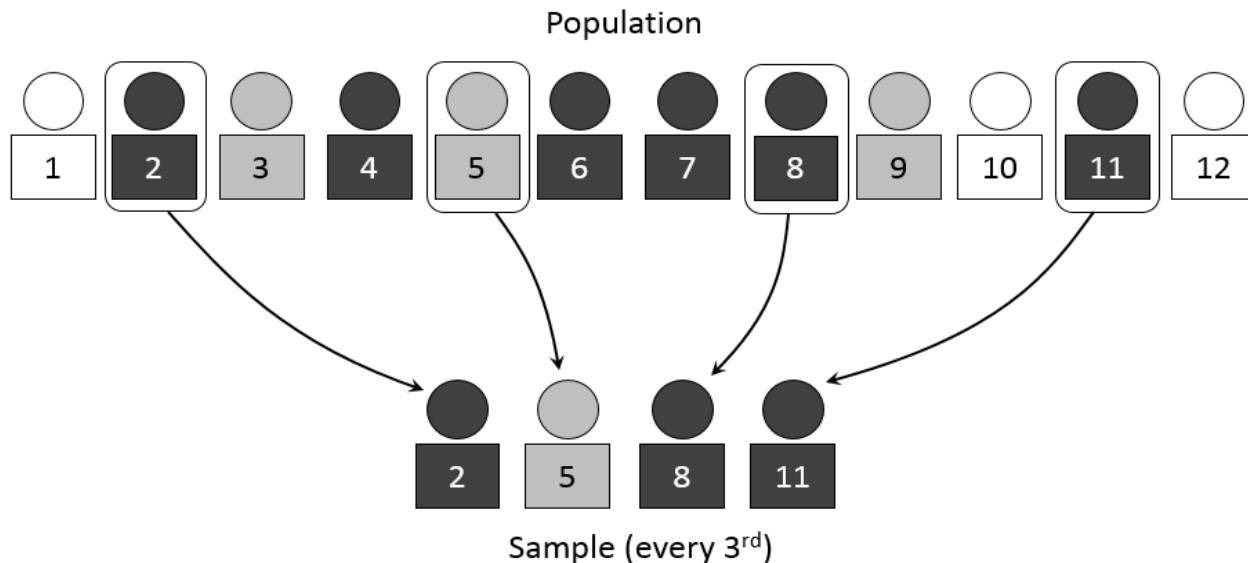


Systematic Sampling

- Systematic sampling relies on **arranging the target population according to some ordering scheme** and then **selecting elements at regular intervals** through that ordered list.
- The **first element** is selected **randomly**.
- Then it proceeds with the **selection of every kth element**. Where k is the size of the selection interval. $k = (\text{population size} / \text{sample size})$
- It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first to the kth element in the list.
- A simple example would be to select every 10th name from the telephone directory (an 'every 10th' sample, also referred to as 'sampling with a skip of 10').

Systematic Sampling

- Systematic sampling is an Equal Probability Sampling method, as all elements have the same probability of selection (in the below example given, one in twelve).
- It is not 'simple random sampling' because different subsets of the same size have different selection probabilities
- Ex: the set {2,5,8,11} has a one-in-twelve probability of selection, but the set {1,3,6,7} has zero probability of selection.



Systematic Sampling

- **When to Use:** When project budget is tight and less time to complete.
- **Key Aspect:** Find the kth value to select every kth member.
 $k = N / n$
- **General Procedure:**
 - Assign numbers to each population element.
 - Order the population elements in an ordered sequence
 - Find 'k' the size of the selection interval.
 - Select the first sample element randomly from the first k population elements.
 - Thereafter, select the sample elements at a constant interval, k, from the ordered sequence frame.

Systematic Sampling: Advantages

- Sample is easy to select.
- Suitable sampling frame can be identified easily.
- Sample evenly spreads over entire reference population.
- It is a cost effective sampling method.
- It guarantees that the entire population is evenly sampled.
- Systematic sampling also carries a low-risk factor because there is a low chance that the data can be contaminated.

Systematic Sampling: Disadvantages

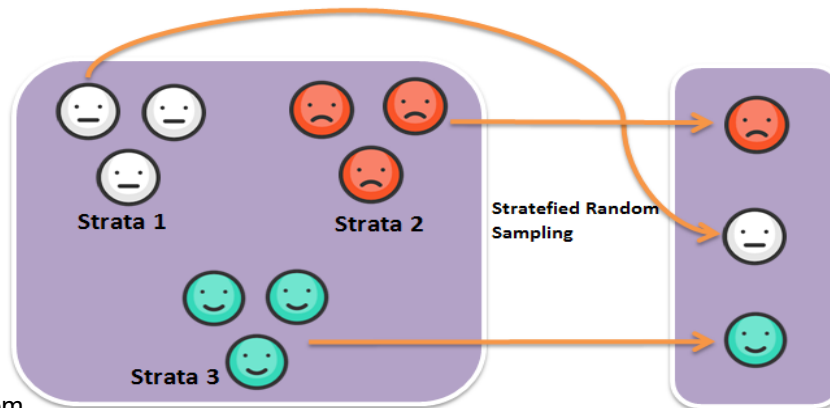
- This type of sampling **might lead to bias** if there is an underlying pattern/periodicity in the population which coincides with the selection.

Ex : If the HR database groups employees by team, and team members are listed in order of seniority, there is a risk that the interval might skip over people in junior roles, resulting in a sample that is skewed towards senior employees.

- Difficult to assess precision of estimate from one survey.
- Each element does not have an equal chance in getting selected
- Ignorance of all the elements between two k th elements.
- The size of the population is needed. Without knowing the specific number of participants in a population, systematic sampling does not work well.

Stratified Sampling

- Stratified sampling is the type of sampling in which the population is divided into 2 or more groups called strata based on a shared characteristic or trait.
- Then simple random samples are selected from each group.
- The selected 2 or more samples are combined into one.
- The strata or groups don't overlap. But, they represent the entire population.
- The shared characteristics based on which the population is divided could be gender, educational attainment, income, age etc.



Stratified Sampling

- **Each stratum** is sampled as an **independent sub-population**.
- Every unit in a stratum has same chance of being selected.
- Using same sampling fraction for all strata **ensures proportionate representation in the sample**.
- Adequate representation of minority subgroups of interest can be ensured by stratification & varying sampling fraction between strata as required.
- Since each stratum is treated as an independent population, different sampling approaches can be applied to different strata.

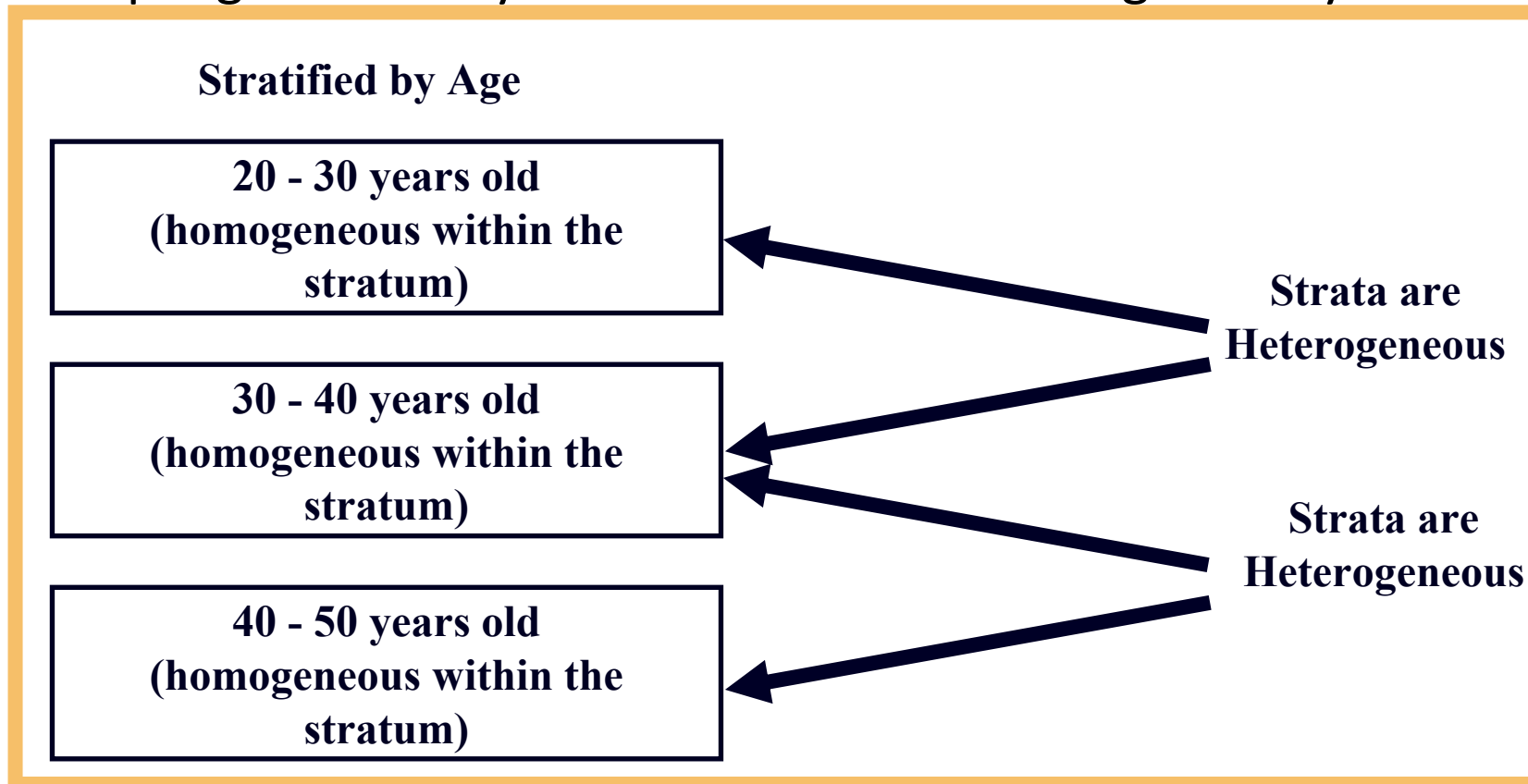
Stratified Sampling

- **Purpose:** To obtain an unbiased random sample from a larger population.
- **When to Use:** When population proportion must be reflected in sample.
- **Key Aspect:** Sample proportion is same as Population proportion, Strata is homogeneous.
- **General Procedure:**
 - Divide the population into Strata or Groups.
 - Criteria for division could be: Gender, Hair Color, Eye Color, Salary, Designation, Age etc.
 - Selection of sample: Simple Random Sampling approach is used to sample units from each strata.

Stratified Sampling examples

To find out the most popular song among the FM radio listeners.

- All listeners are stratified by age.
- Listeners from each age group are selected using simple random sampling and surveyed for their favourite song of the year.



Stratified Sampling: Advantages

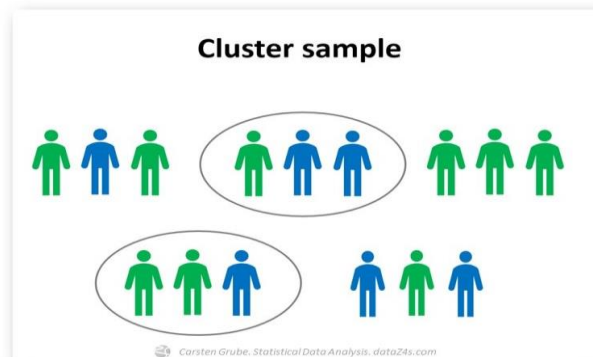
- It enhances the representativeness of the sample.
- It is easy to carry out.
- It has higher statistical efficiency.
- A stratified sample can provide a higher precision than a simple random sample of the same size.
- As it provides a greater precision, this type of sampling often requires a smaller sized sample which saves money.

Stratified Sampling: Disadvantages

- Sampling frame of the entire population has to be prepared separately for each stratum.
- When examining multiple criteria to divide the population, stratifying variables may be related to some but not to others further complicating the design and potentially reducing the utility of the strata.
- In some cases (such as designs with a large number of strata, or those with a specified minimum sample size per group), stratified sampling can potentially require a larger sample than other methods.
- It is time consuming and expensive.
- It leads to classification errors.

Cluster Sampling

- In cluster sampling, population is divided into non-overlapping clusters or areas similar to Stratified sampling.
- Each cluster is a miniature or microcosm of the population.
- Each cluster should have similar characteristics to the whole sample.
- Instead of sampling individuals from each subgroup like in stratified sampling, in cluster sampling entire clusters are randomly selected.
- A subset of the clusters is selected randomly for the sample.
- If the number of elements in the subset of clusters is larger than the desired value of n (sample size), these clusters may be subdivided to form a new set of clusters and subjected to a random selection process.



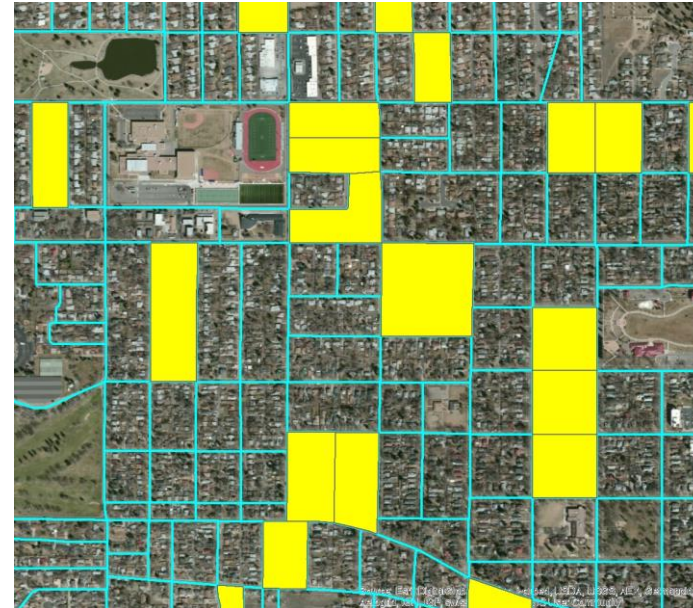
Cluster Sampling

- **When to Use:** When population is already broken up into groups(clusters).
- **Key Aspect:** Heterogeneous members in each group.
- **General Procedure:**
 - Population is divided into non-overlapping areas(clusters).
 - Each cluster is a miniature or microcosm of a population.
 - Clusters are selected randomly.
 - All elements of the selected-clusters are included in the sample or elements from the selected-clusters are chosen using simple random sampling.

Cluster Sampling examples

The municipal council of a small city wants to investigate the use of health care services by residents.

- The council first obtains electoral subdivision maps that identify and label each city block. From these maps, the council creates a list of all city blocks. This list will serve as the sampling frame.
- Every household in that city belongs to a city block, and each city block represents a cluster of households. The council randomly picks a number of city blocks.



Cluster Sampling: Advantages

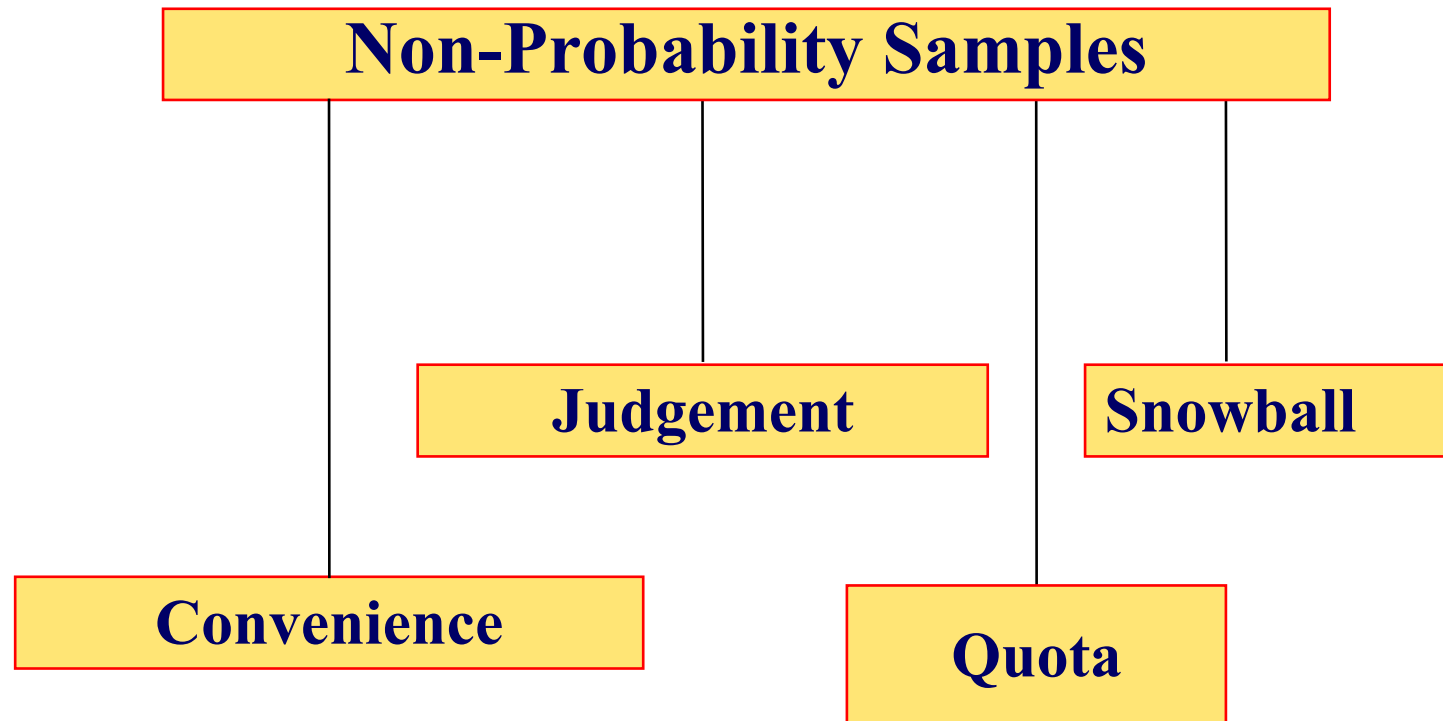
- It is more convenient for geographically dispersed populations.
- It can **reduce the travel costs** to contact sample elements.
- It **simplifies the administration of the survey**.
- It is **more feasible**. The division of the entire population into homogeneous groups increases the feasibility of the sampling.
- Since each cluster represents the entire population, **more subjects can be included in the study**.
- **Requires fewer resources**. Since cluster sampling selects only certain groups from the entire population, the method requires fewer resources for the sampling process.

Cluster Sampling: Disadvantages

- It is statistically less efficient when the cluster elements are similar.
- Costs and the number of problems occurring are greater than that of simple random sampling.
- There is higher sampling error.
- The method is **prone to biases**. If the clusters representing the entire population were formed under a biased opinion, the inferences about the entire population would be biased as well.
- It's difficult to guarantee that the sampled clusters are really representative of the whole population.

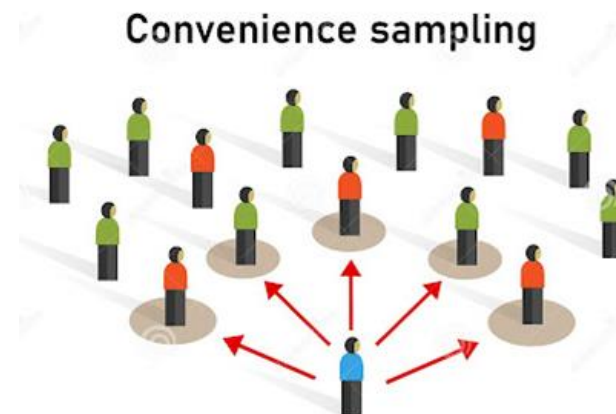
Non-probability Sampling

Non-Probability sampling is a type of sampling in which every unit in the population doesn't have a chance/probability (greater than zero) of being selected in the sample.



Convenience Sampling

- Sometimes it is also known as **grab or opportunity sampling** or **accidental or haphazard sampling**.
- This is a type of nonprobability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.
- Here, sample elements are selected for the convenience of the researcher.
- The researcher using such a sample cannot scientifically make generalizations about the total population from this sample because it would not be representative enough.



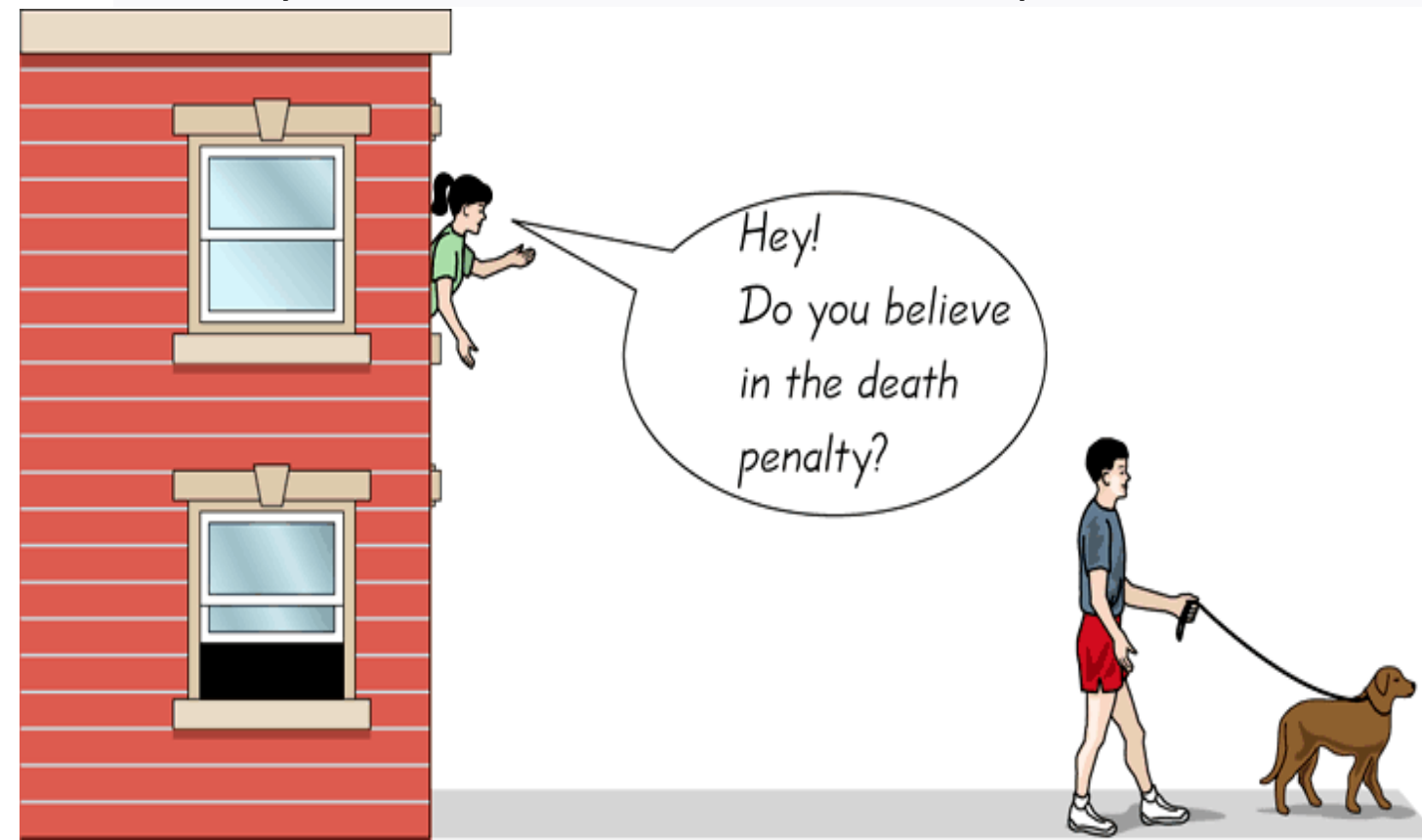
Convenience Sampling

- **When to Use:** When population is not clearly defined or sampling unit is not clear or complete source list is not available.
- **Key Aspect:** Subjects for a study are easily available within the proximity of the researcher.
- **General procedure:**
 - It is done at the “convenience” of the researcher.
 - Selection : The individuals that are convenient and easiest to reach are selected to be included in the sample.

Convenience Sampling examples

To record the popular opinions of people about the current laws of the city.

- The researcher surveys all people that pass by his house.
- Again, this is a convenient way of studying the opinions of people living in the city. But, it doesn't reflect the opinions of all the residents of the city.



Convenience Sampling: Advantages & Disadvantages

→ Advantages:

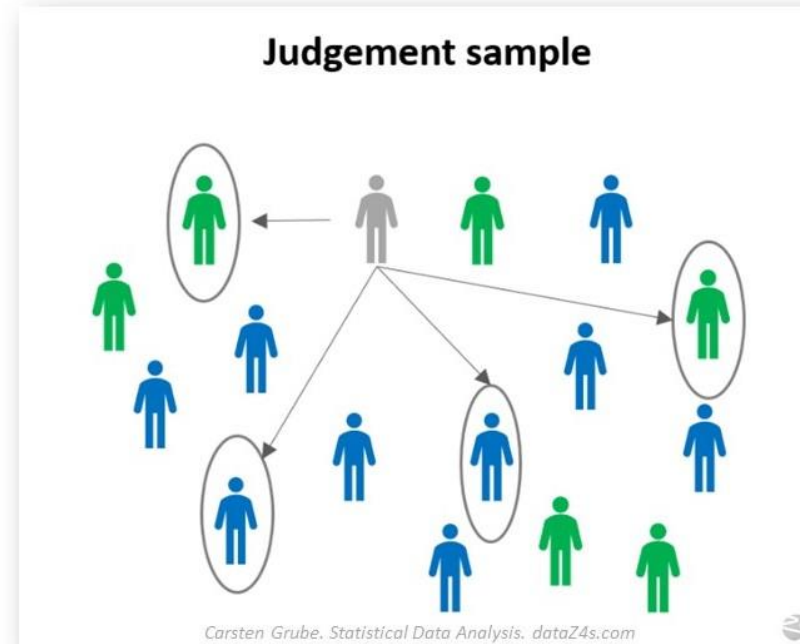
- This type of sampling is useful in pilot study.
- It costs less and is an inexpensive way to gather initial data for the research.
- It saves time.
- It is relatively easy to get a sample.
- It is simple and easy to implement.

→ Disadvantages:

- It is prone to significant bias as the sample may not be representative of the characteristics of the population.
- Since the same may not be representative of the population, this type of sampling can't produce generalizable results.
- It might lead to sampling errors.
- A study conducted on a convenience sample will have limited external validity.

Judgemental Sampling

- Judgemental or Purposive sampling is a type of non-probability sampling where the **researcher chooses the sample** based on who they think would be appropriate for the study.
- This is used primarily when there is a limited number of people that have expertise in the area being researched.
- The sample depends on the judgement of the experts conducting the study.
- It is not a scientific method of sampling.



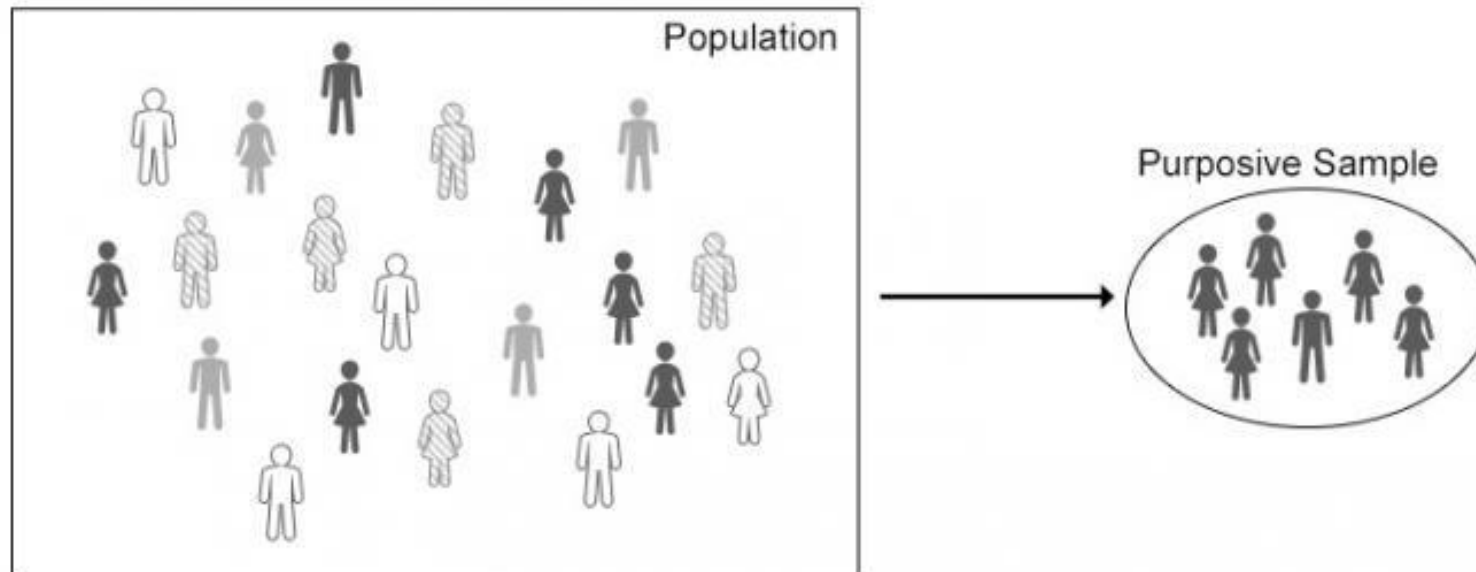
Judgemental Sampling

- **When to Use:** This is used primarily when there is a limited number of people that have expertise in the area being researched. Also, the researcher must be confident that the chosen sample is truly representative of the entire population.
- **Key Aspect:** The researcher selects a sample based on experience or knowledge of the group to be sampled.
- **General Procedure:**
 - On the basis of the researcher's knowledge and judgment elements of the population are sampled.
 - Selection : Elements that own the qualities expected by the researcher.

Judgemental Sampling examples

To know more about the opinions and experiences of disabled students at your university

- You purposefully select a number of students with different support needs at your university in order to gather a varied range of data on their experiences with student services.



Purposive Sampling

Judgemental Sampling: Advantages & Disadvantages

→ Advantages:

- It consumes minimum time.
- The researcher is given an opportunity to bring his judgement and expertise to play.
- No special knowledge of statistics is needed.
- Real time results can be obtained.

→ Disadvantages:

- It is prone to errors in judgment by researcher.
- Low level of reliability and high levels of bias.
- Inability to generalize research findings to the entire population.
- It is difficult to choose the appropriate sample size.

Quota sampling

- In this type of sampling, sample elements are selected until the quota controls are satisfied.
- The population is first segmented into mutually exclusive sub-groups, just as in stratified sampling.
- Then judgment is used to select subjects or units from each segment based on a specified proportion.
- The population units are selected based on predetermined characteristics of the population.
- It is similar to Stratified sampling but it doesn't involve random selection.
- Ex: recruiting the first 50 men and first 50 women that meet inclusion criteria.

Quota sampling

- **When to Use:** If a study aims to investigate a trait or a characteristic of a certain subgroup, this type of sampling is the ideal technique.
- **Key Aspect:** Sample elements are selected until the quota controls are satisfied.
- **General Procedure:**
 - Divide the population into subgroups.
 - Identify proportions or weightage in which the subgroups are present in the population.
 - Select an appropriate sample size while maintaining the proportions of the subgroups.
 - Conduct the surveys according to the quotas defined

Quota sampling examples

A cool drinks company wants to find out what age group prefers what brand of drinks in a particular city.

- The researcher applies quotas on the age groups of 11-21, 22-31, 32-41, 42-51.
- The researcher then samples people from each quota and surveys them to gauge the trend among the population of the city.



Symbol	Age Group	No.
	11-21 Years	11
	22-31 Years	16
	32-41 Years	15
	42-51 Years	18
Total	11-51 Years	60

Quota Sampling

Quota sampling examples: Advantages & Disadvantages

→ Advantages:

- It is a cost effective method.
- There is convenience in execution of this sampling.
- It is a speedy process.
- The information can be deciphered once the sampling is done.
- It improves the representation of certain groups within the population and also ensures that they are not over-represented.

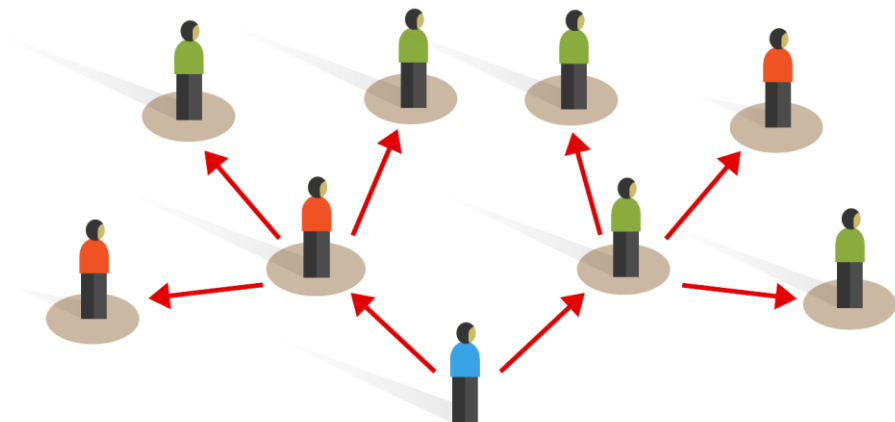
→ Disadvantages:

- Impossible to determine sampling error as the sample is not chosen using random selection.
- Can result in sampling bias if the selection of units was based on ease of access and cost considerations.
- It is not possible to make statistical inferences from the sample to the population leading to the problems of generalization.

Snowball sampling

- In this type of sampling, survey subjects are selected based on referral from other survey respondents.
- Existing subjects are asked to nominate further subjects known to them so that the sample increases in size like a rolling snowball.
- This method of sampling is effective when a sampling frame is difficult to identify.
- Usually applied when the subjects are difficult to trace. Ex: it will be extremely challenging to survey shelter less people or illegal immigrants.

Snowball sampling



Source: cuttingedgepr.com

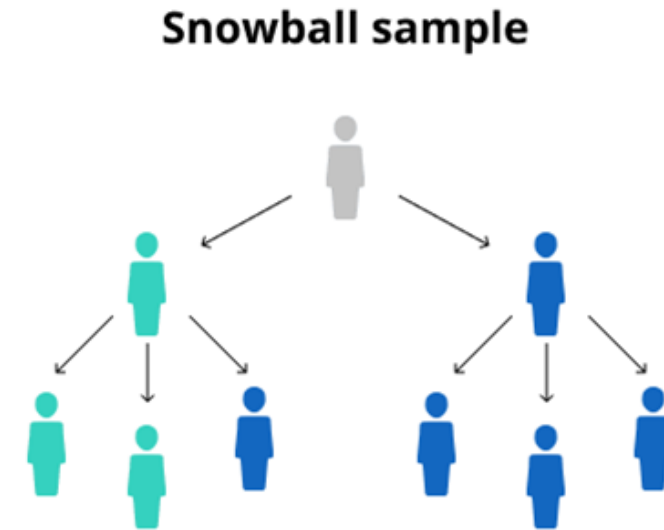
Snowball sampling

- **When to Use:** When the desired sample characteristic is rare.
- **Key Aspect:** Research starts with a key person and introduce the next one to become a chain. It may be extremely difficult or cost prohibitive to locate respondents in these situations.
- **How:**
 - Identify an initial subject and ask these people to identify others.
 - Selection : This technique relies on referrals from initial subjects to generate additional subjects.

Snowball sampling examples

To study the level of customer satisfaction among the members of an elite country club.

- It is extremely difficult to collect primary data sources unless a member of the club agrees to have a direct conversation with you and provides the contact details of the other members of the club.
- Thus the primary data source is randomly selected and it nominates other potential data sources that will be able to participate in the research studies.



Snowball sampling examples: Advantages & Disadvantages

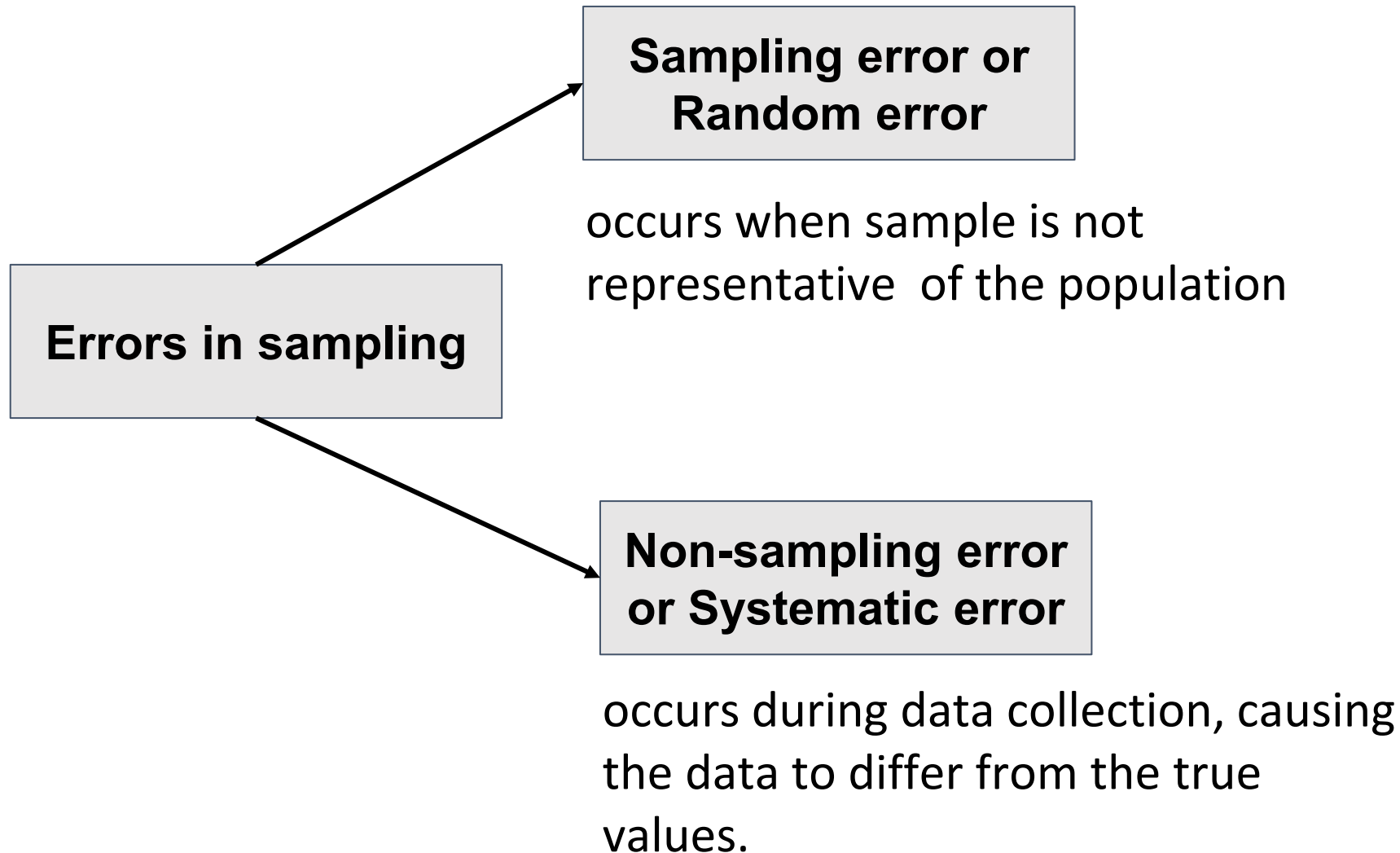
→ Advantages:

- The chain referral process allows the researcher to reach populations that are difficult to sample when using other sampling methods.
- The process is cheap, simple and cost-efficient.
- This sampling technique needs little planning and fewer workforce compared to other sampling techniques.

→ Disadvantages:

- There is a significant risk of selection bias in snowball sampling, as the referenced individuals will share common traits with the person who recommends them.
- It is usually impossible to determine the sampling error or make inferences about populations based on the obtained sample.
- The researcher has little control over the sampling method.
- Representativeness of the sample is not guaranteed.

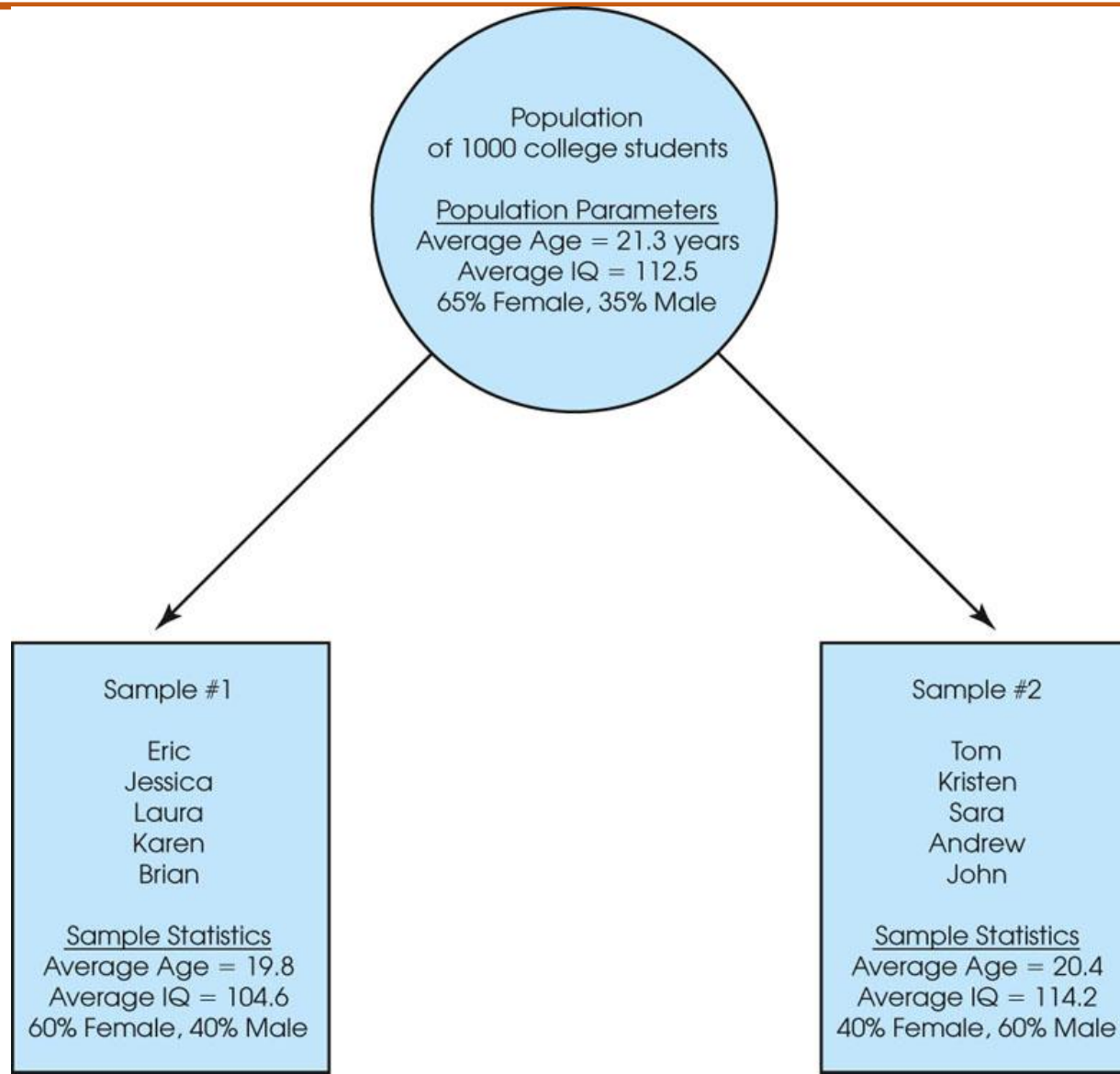
Errors in sampling



Sampling error

- The discrepancy between a sample statistic and its population parameter is called sampling error.
- Defining and measuring sampling error is a large part of inferential statistics.
- It occurs when the sample is not representative of the population.
- The sampling error for a given sample is unknown but when the sampling is random, for some estimates (for example, sample mean, sample proportion) theoretical methods may be used to measure the extent of the variation caused by sampling error.

Sampling error



As we can see there is a difference between population parameters and sample parameters. This is due to sampling error.

Two samples of same population have differing parameters. This is due to sampling variation. It is also the reason why scientific experiments produce different result under identical scenarios.

Non-sampling error

- Non-sampling errors are the results of mistakes made in implementing data collection and data processing, such as
 - failure to locate and interview the correct household
 - errors in understanding of the questions by either the interviewer or the respondent
 - data entry errors
 - missing Data
 - poorly conceived concepts, unclear definitions, and defective questionnaires
 - response errors occurring when people are unaware, refuse to answer, or overstate in their answers
- Major sources : Sampling Bias, Non-response Bias.

Sampling Bias

- Sampling bias occurs when a chosen sample is not representative of the larger population.
- It occurs due to the sampling technique/method used to perform data collection.
- It can be either **selection bias and non-response bias**.
- A sampling method has a sampling bias if all subjects in the population are not equally likely to be included in a sample.
- That is, a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.

Selection Bias & Nonresponse bias

→ Selection bias:

- It is a bias in which a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.
- It results in a biased sample of a population in which all individuals, or instances, were not equally likely to have been selected.

→ Nonresponse bias:

- Nonresponse bias is a type of sampling bias that occurs because of the absence of certain objects or subjects from a sample.
- For example, some subjects don't respond to surveys because they refuse, cannot be contacted, or have a lack of interest in the survey content.

Bias ex:

Q) A new chemical process is run 10 times each morning for five consecutive mornings. If the new process is put into production, it will be run 10 hours each day, from 7 A.M. until 5 P.M. Is it reasonable to consider the 50 yields to be a simple random sample?

Ans) Since the new process runs during both morning and afternoon, the population consists of all the yields that would ever be observed, including both morning and afternoon runs.

The sample however is drawn only from that portion of the population that consists of morning runs, and thus it is not a simple random sample. It exhibits a bias is not representative of the population intended to be studied.

Sampling variation

- Simple random samples always differ from their populations in some ways, and occasionally may be substantially different.
- Two different samples from the same population will differ from each other as well.
- This phenomenon is known as **sampling variation**.
- Sampling variation is one of the reasons that scientific experiments produce somewhat different results when repeated, even when the conditions appear to be identical.

Independence

- The items in a sample are said to be independent if knowing the values of some of them does not help to predict the values of the others.
- With a finite, tangible population, the items in a simple random sample are not strictly independent, because as each item is drawn, the population changes.
- This change can be substantial when the population is small.
- However, when the population is very large, this change is negligible and the items can be treated as if they were independent
- The sample can be considered independent if sample size is smaller than 5% of population size.
- Since conceptual population have infinite/very large size the sample obtained (ex: measuring a rock) is always independent

Questions

(Q1.) A physical education professor wants to study the physical fitness levels of students at her university. There are 20,000 students enrolled at the university, and she wants to draw a sample of size 100 to take a physical fitness test. She obtains a list of all 20,000 students, numbered from 1 to 20,000. She uses a computer random number generator to generate 100 random integers between 1 and 20,000 and then invites the 100 students corresponding to those numbers to participate in the study. Which sampling technique is used?

Answer:

The simple random sampling technique is used.

Note that it is analogous to a lottery in which each student has a ticket and 100 tickets are drawn.

Questions

(Q2) A quality engineer wants to inspect rolls of wallpaper in order to obtain information on the rate at which flaws in the printing are occurring. She decides to draw a sample of 50 rolls of wallpaper from a day's production. Each hour for 5 hours, she takes the 10 most recently produced rolls and counts the number of flaws on each. Is this a simple random sample?

Answer:

No. Not every subset of 50 rolls of wallpaper is equally likely to comprise the sample. To construct a simple random sample, the engineer would need to assign a number to each roll produced during the day and then generate random numbers to determine which rolls comprise the sample.

Questions

(Q3) A construction engineer has just received a shipment of 1000 concrete blocks, each weighing approximately 50 pounds. The blocks have been delivered in a large pile. The engineer wishes to investigate the crushing strength of the blocks by measuring the strengths in a sample of 10 blocks. Which sampling method is suitable?

Answer:

To draw a simple random sample would require removing blocks from the center and bottom of the pile, which might be quite difficult. For this reason, the engineer might construct a sample simply by taking 10 blocks off the top of the pile.

convenience sample

Questions

(Q4) A quality inspector draws a simple random sample of 40 bolts from a large shipment and measures the length of each. He finds that 34 of them, or 85%, meet a length specification. He concludes that exactly 85% of the bolts in the shipment meet the specification. The inspector's supervisor concludes that the proportion of good bolts is likely to be close to, but not exactly equal to, 85%. Which conclusion is appropriate?

Answer:

Because of sampling variation, simple random samples don't reflect the population perfectly. However, they are often fairly close. It is therefore appropriate to infer that the proportion of good bolts in the lot is likely to be close to the sample proportion, which is 85%. It is not likely that the population proportion is equal to 85%.

Questions

(Q5) Another inspector repeats the study with a different simple random sample of 40 bolts. She finds that 36 of them, or 90%, are good. The first inspector claims that she must have done something wrong, since his results showed that 85%, not 90%, of bolts are good. Is he right?

Answer:

No, he is not right. This is sampling variation at work. Two different samples from the same population will differ from each other and from the population.

Questions

(Q6) A geologist weighs a rock several times on a sensitive scale. Each time, the scale gives a slightly different reading. Under what conditions can these readings be thought of as a simple random sample? What is the population?

Answer:

If the physical characteristics of the scale remain the same for each weighing, so that the measurements are made under identical conditions, then the readings may be considered to be a simple random sample. The population is conceptual. It consists of all the readings that the scale could in principle produce.

Questions

(Q7) What sampling method can be recommended?

- Determining proportion of undernourished five year olds in a village.
- Investigating nutritional status of preschool children.
- In estimation of immunization coverage in a province, data on seven children aged 12-23 months in 30 clusters are used to determine proportion of fully immunized children in the province. Give reasons why cluster sampling is used in this survey.

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<https://www.analyticsvidhya.com/blog/2019/09/data-scientists-guide-8-types-of-sampling-techniques/>

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Statistics for Engineers and Scientists, William Navidi.



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Types of Data & Experiments

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Types of Data & Experiments

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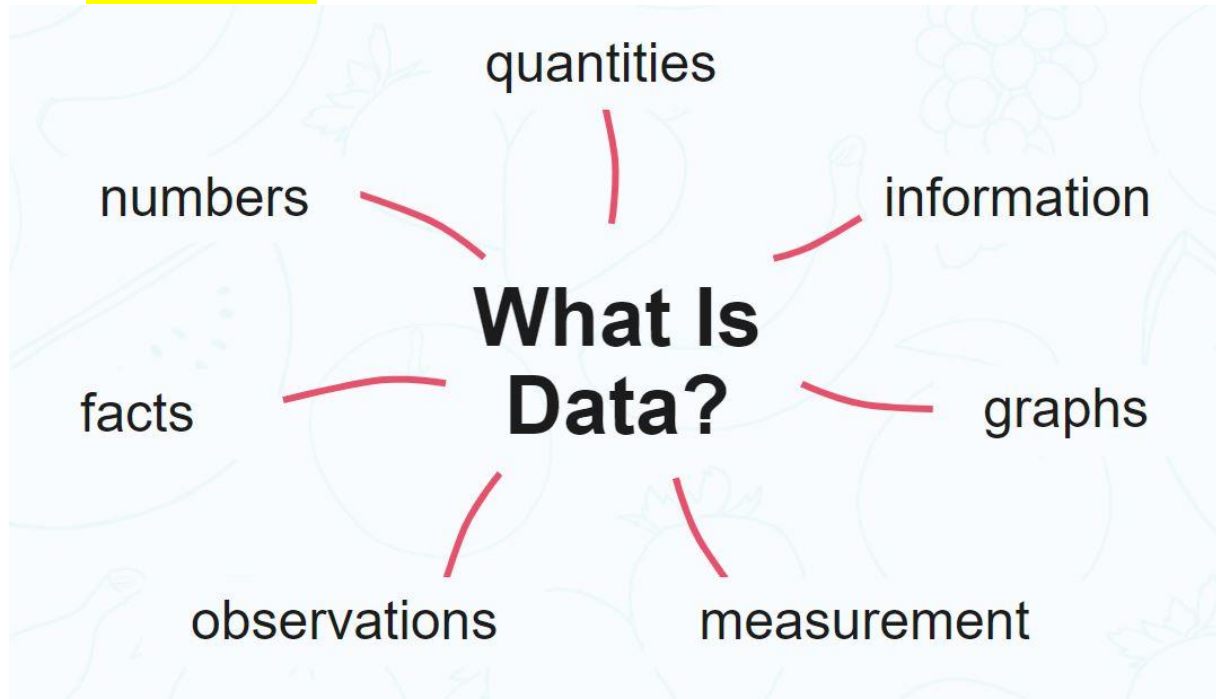
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Topics to be covered

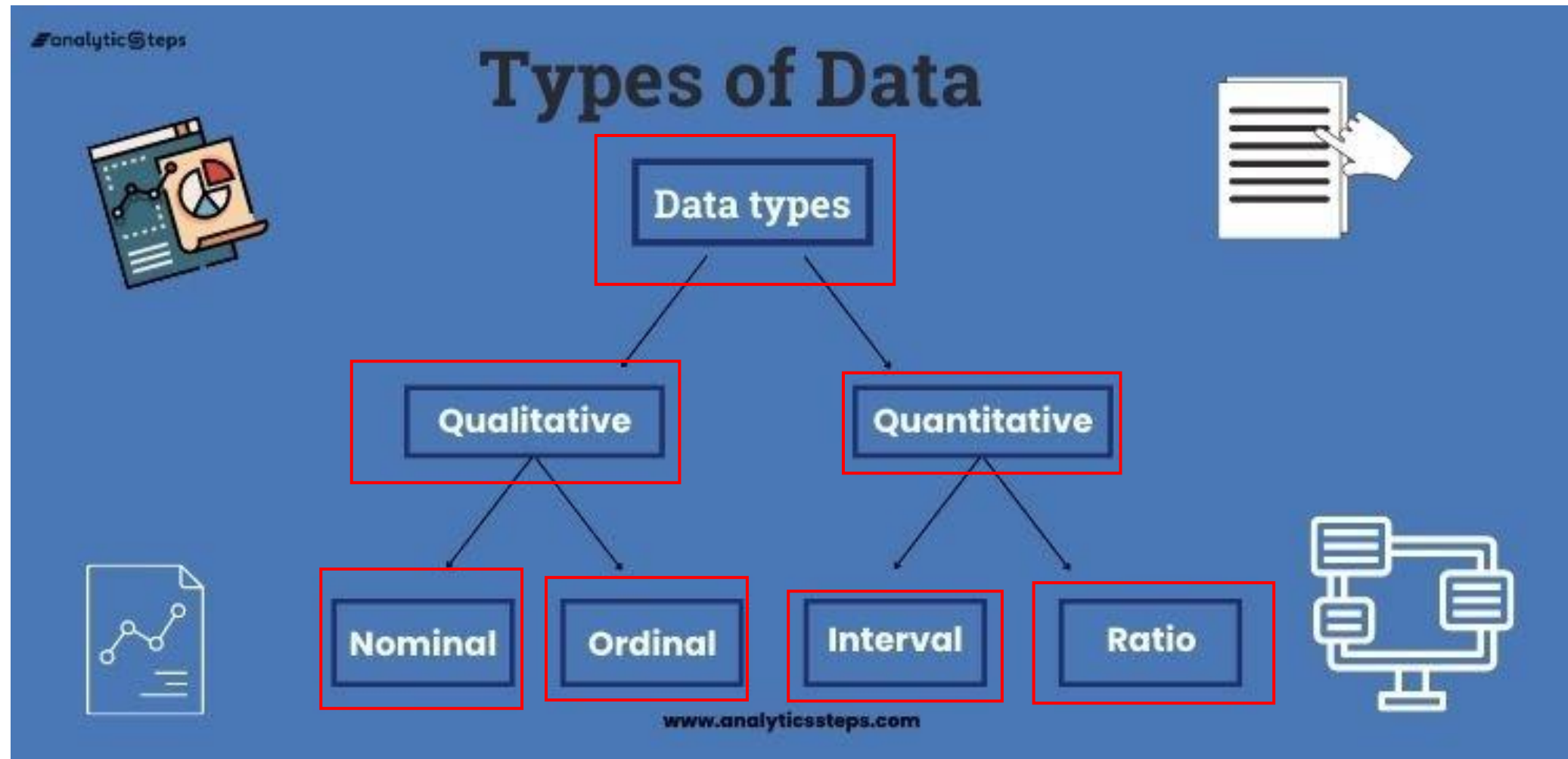
- ❖ Types of data
- ❖ Variables or Attributes
- ❖ Types of studies

Data

- Data refers to **individual facts, statistics, or items of information** that are collected through observation.
- It can also be defined as the **facts and figures** collected, summarized, analyzed and interpreted.
- The **data collected** in a **particular study** are referred to as the **data set**.



Types of data



Quantitative data

- Quantitative Data are measurements that are recorded on a naturally occurring numerical scale.
- These are easily open for statistical manipulation and can be represented by a wide variety of statistical types of graphs and charts like line charts, bar graphs, scatter plots, etc.
- These types of data tries to find the answers to questions such as
 - “how many,
 - “how much” and
 - “how often”
- Example: Age, GPA, Salary, Cost of books this semester, Scores of tests and exams, weight of a person, temperature in a room etc.
- There are 2 general types of quantitative data:
 - Discrete data
 - Continuous data

Qualitative data

- Qualitative Data are measurements that cannot be recorded on a natural numerical scale, but are recorded in categories.
- It is also known as Categorical Data as the information can be sorted by category, not by number.
- Example: Year in school, Live on/off campus, Major, Gender, colors etc.
- These can answer the questions like:
 - “how this has happened”, or
 - “why this has happened”.
- In general, there are 2 types of qualitative data:
 - Nominal data
 - Ordinal data

Qualitative data: Nominal data

- This data type is used just for labeling variables, without having any quantitative value.
- Here, the term 'nominal' comes from the Latin word "nomen" which means 'name'.
- They are categories without any particular order or direction.
- The nominal data sometimes is referred to as "labels".
- Their use is restricted to keeping track of people, objects and events.
- They are least powerful in measurement with no arithmetic origin or order.
- Hence, nominal data is of restricted or limited use.
- Examples: Gender (Women, Men), Hair color (Blonde, Brown, Brunette, Red, etc.), Marital status (Married, Single, Widowed) etc.

Qualitative data: Nominal data

- It can't be manipulated using mathematical operators.
- But, it can be visualized using pie chart.
- Nominal data can be both quantitative and qualitative.
- Quantitative labels lack a relationship.

NOMINAL DATA

Nominal data divides variables into mutually exclusive, labeled categories.

Examples

Eye color



Smartphone



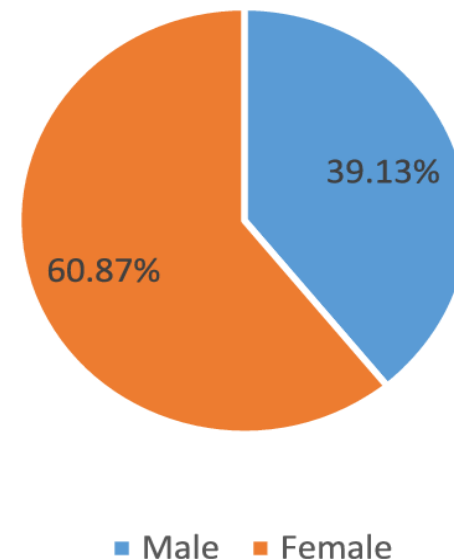
Transport



How is nominal data analyzed?

Descriptive statistics:
Frequency distribution and mode

Non-parametric
statistical tests



Qualitative data: Nominal data

→ How to analyze Nominal Data?

- Using grouping method.
- Group them into categories.
- For each category, frequency or percentage can be calculated.
- Hypothesis testing is carried out using nonparametric tests such as Chi-Square test.
- To determine whether there is a significant difference between the expected frequency and the observed frequency.

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Qualitative data: Nominal data examples

Gender, marital status or any alphabetic / numeric code without intrinsic order or ranking.

Sl. No.	Subject	Code
1	Physics	P
2	Chemistry	C
3	Mathematics	M
4	Biology	B

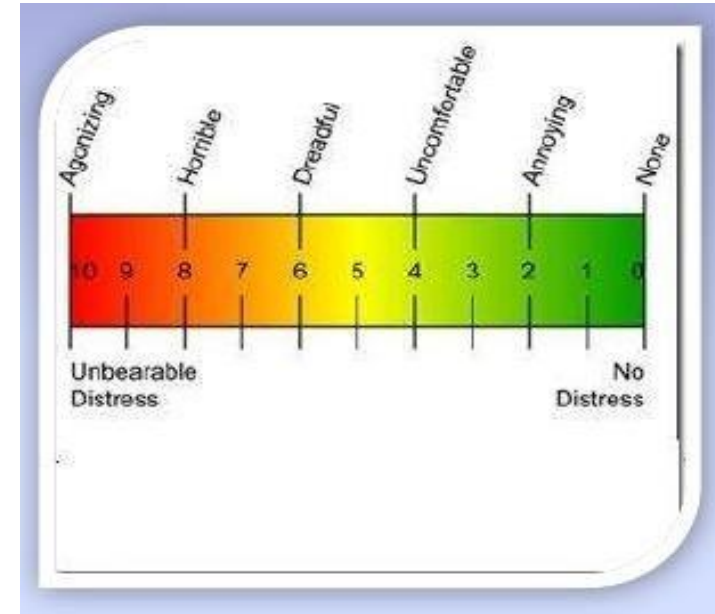
Source:
www.slideshare.net

Qualitative data: Ordinal data

- In simple words, we can understand the ordinal data as qualitative data for which the values are ordered.
- Ordinal data may indicate superiority.
- But, we cannot do arithmetic operations with ordinal data because they only show the sequence.
- Based on the relative position, we can also assign numbers to ordinal data. For example, “first, second, third...etc.”
- Ordinal data allows for setting up inequalities, but it has no absolute value.
- More precise comparisons are not possible.
- Examples:
 - Ranking of users in a competition: The first, second, and third, etc.
 - Rating of a product taken by the company on a scale of 1-10.
 - Economic status: low, medium, and high.

Qualitative data: Ordinal data

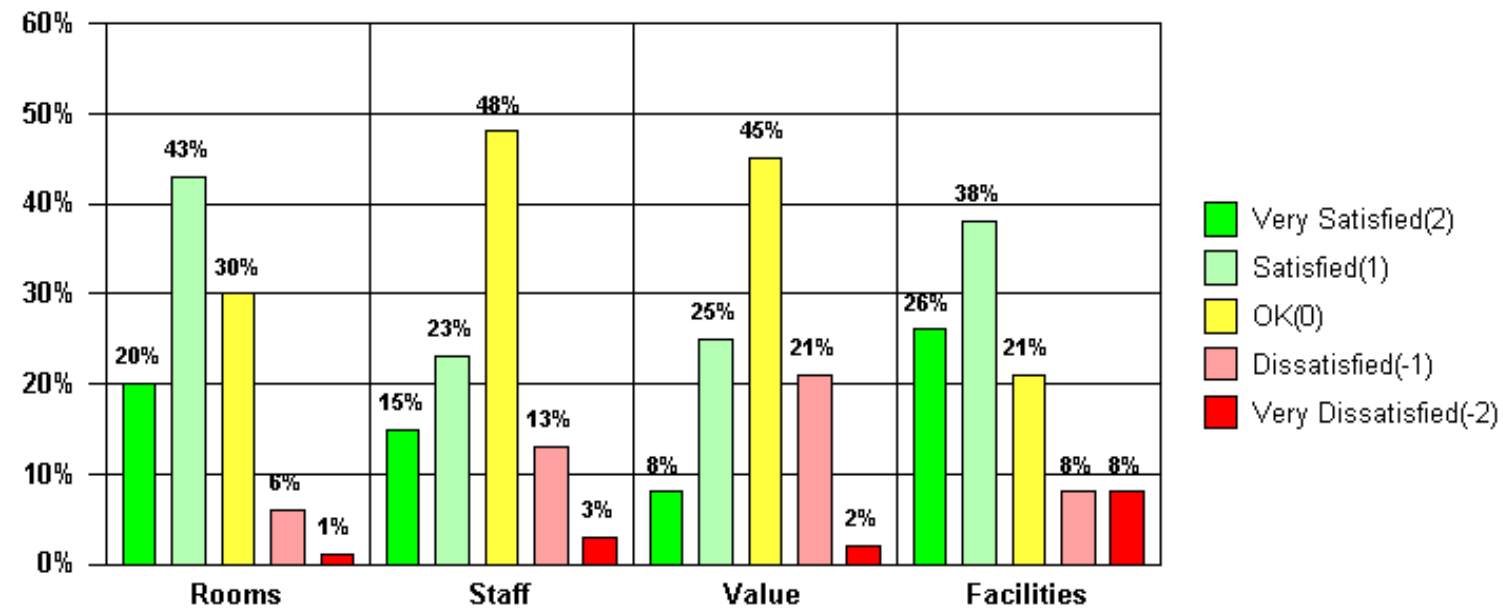
- Here, the order matters but not the difference between values.
- Example: Pain Scales
 - Patients are asked to express the amount of pain they are feeling on a scale of 1 to 10.
 - A score of 7 means more pain than a score of 5, and that is more pain than a score of 3.
 - But the difference between the 7 and the 5 may not be the same as that between 5 and 3.
 - The values simply express an order.



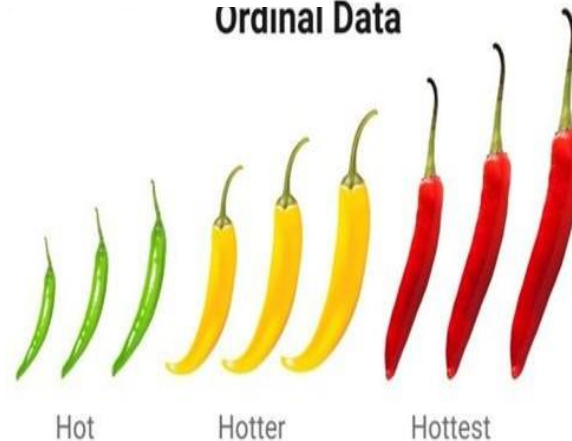
Source: Questionpro, slideshare.net

Qualitative data: Ordinal data examples

Chart of Ordinal data



Ordinal Data

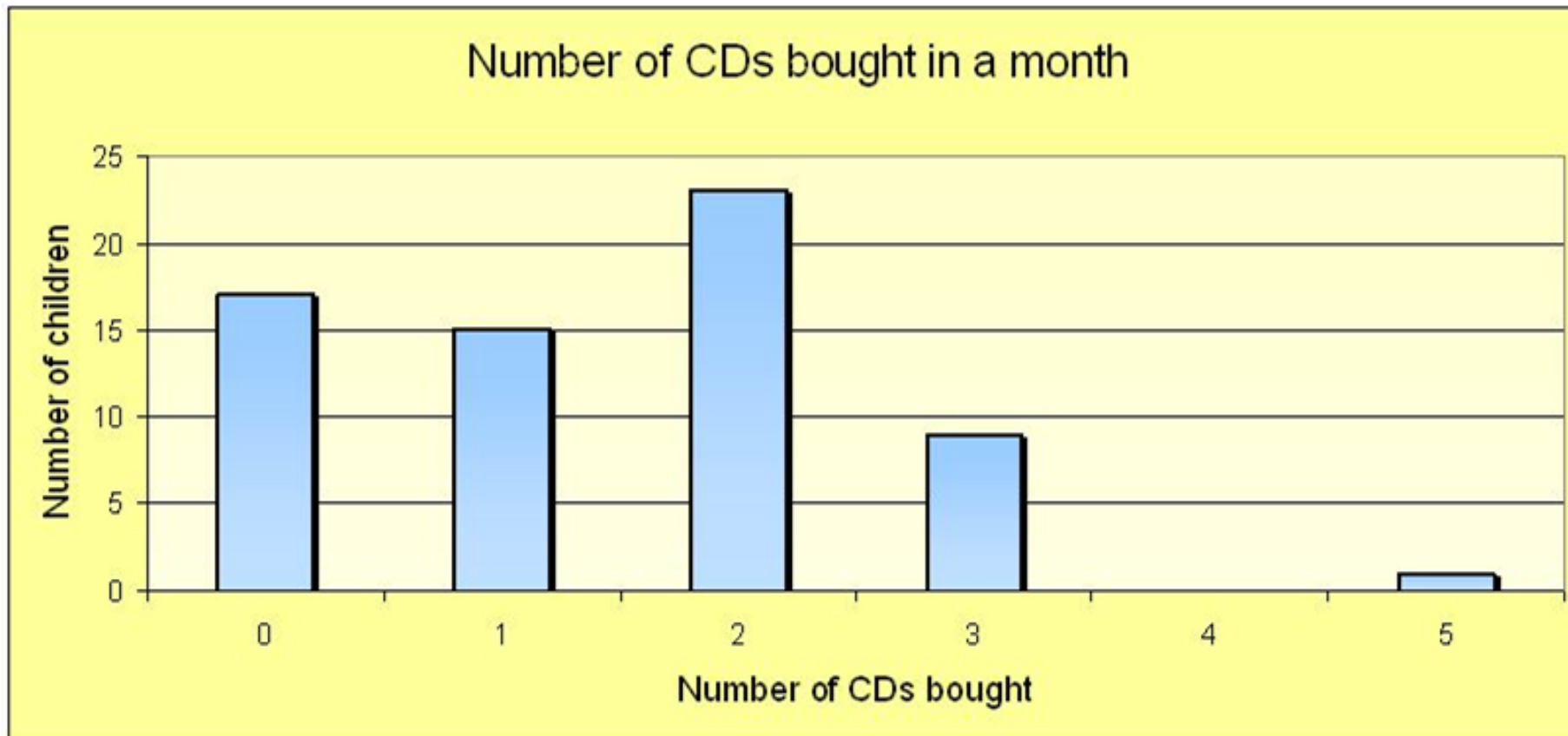


Quantitative data: Discrete data

- A set of data is said to be discrete if the values belonging to the set are discrete and separate.
- The data values cannot be divided into smaller parts. For Example, the number of students in a class is an example of discrete data since we can count whole individuals but can't count in fractions like 2.5, 3.75, kids.
- It has a limited number of possible values e.g. days of the month.
- Discrete data can take only certain values by a finite 'jumps' i.e. It 'jumps' from one value to another but does not take any intermediate value between them.
- Examples of discrete data:
 - The number of students in a class.
 - The number of workers in a company.
 - The number of test questions you answered correctly.

Quantitative data: Discrete data

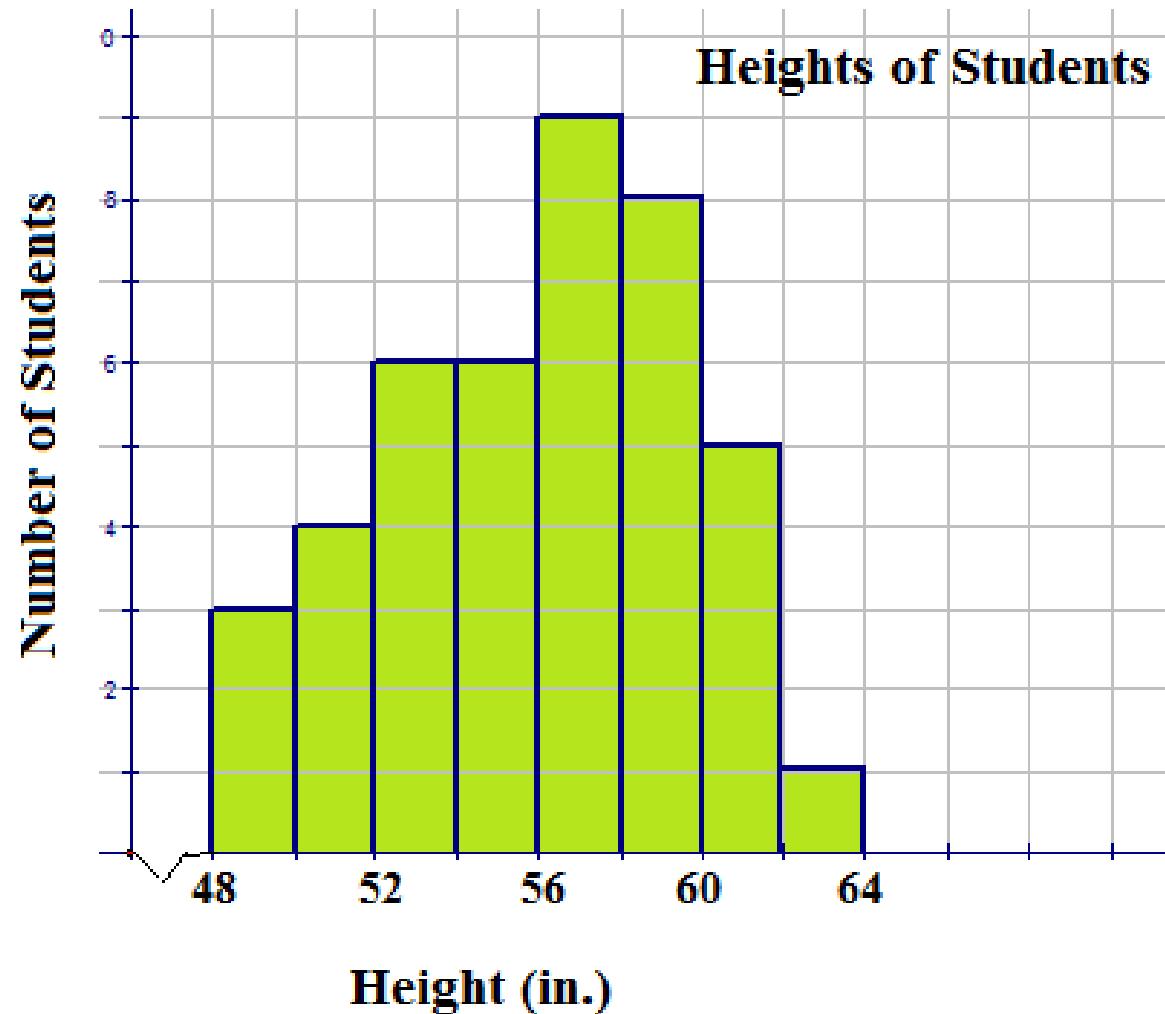
- Bar charts can be used to display discrete numerical data.
- For example, the bar chart below shows the number of CDs bought by a group of children in a given month.



Quantitative data: Continuous data

- A set of data is said to be continuous if the values belonging to the set can take on any value within a finite or infinite interval.
- It represents the information that could be meaningfully divided into its finer levels.
- It can be measured on a scale or continuum and can have almost any numeric value. For Example, we can measure our height at very precise scales in different units such as meters, centimeters, millimeters, etc.
- Examples of continuous data:
 - The amount of time required to complete a project.
 - The height of children.
 - The speed of cars.

Quantitative data: Continuous data



Quantitative data: Interval data

- It is a data type which is measured along a scale, in which each point is placed at equal distance from one another.
- These data types are measurable and ordered with the nearest items but have no meaningful zero.
- Interval scales not only educate us about the order of the items but in addition, give information about the value between every item.
- There are some descriptive statistics that we can calculate for interval data such as :
 - Central measures of tendency (mean, median, mode)
 - Range (minimum, maximum)
 - Spread (percentiles, interquartile range, and standard deviation).
- Examples: Temperature ($^{\circ}\text{C}$ or F, but not Kelvin),
Dates (1055, 1297, 1976 etc), Time Gap on a 12-hour clock (6 am, 6pm)

Quantitative data: Interval data

INTERVAL DATA

Interval data is measured along a numerical scale that has equal intervals between adjacent values.

Examples

Temperature



IQ score



Income ranges



Quantitative data: Ratio data

- Ratio data **classifies and ranks data**, and uses measured, continuous intervals, just like interval data.
- However, unlike interval data, ratio data **has a true zero**.
- This **basically means that zero is an absolute, below which there are no meaningful values**.
- Speed, age, or weight are all excellent examples since none can have a negative value (you cannot be -10 years old or weigh -160 pounds)
- **These data are also in the ordered units that have the same difference.**
- Ratio data allow for forming quotients in addition to setting up inequalities and forming differences.
- All mathematical operations (manipulations with real numbers) are possible on ratio data.
- **It is the most precise data and allow for application of all statistical techniques.**

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Quantitative data: Ratio data



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Quantitative data: Ratio data

- Example:

Roll No.	Name	Gender	Rank	Height	Weight In Kgs
1	Amar	M	9	4' 8"	51
2	Asha	F	1	3' 10"	39
3	Bhaskar	M	5	4' 5"	48
4	Chandru	M	3	4' 3"	41

Identify the type of data

- Number of cartons of milk manufactured each day.
Quantitative data, Discrete data
- Temperatures of airplane interiors at a given airport in Celsius.
Quantitative data, Continuous data, Interval data.
- College major of each student in a class.
Qualitative data, Nominal data
- Method of payment
Qualitative data, Nominal data
- Incomes of college students on work study programs.
Quantitative data, Discrete data
- Weights of newborn calves.
Quantitative data, Continuous data, Ratio data.

Identify the type of data

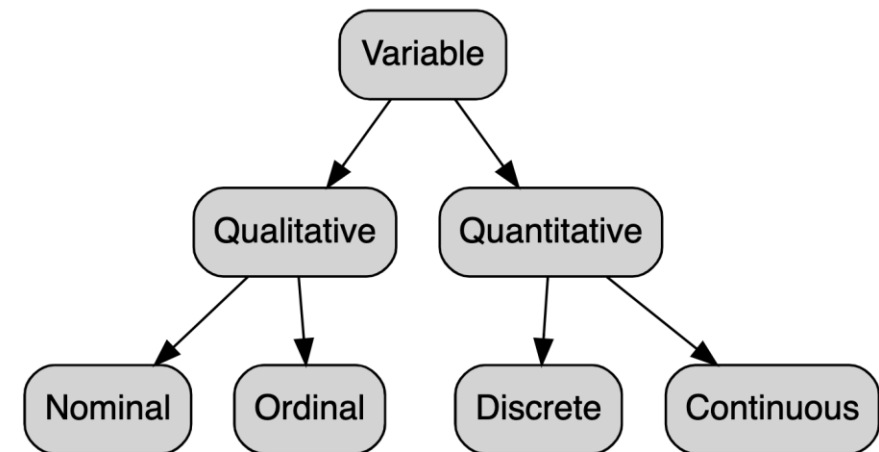
- Gender of each employee at a company.
Qualitative data, Nominal data
- Number of tomatoes on each plant in a field.
Quantitative data, Discrete data
- Number of defective items in a lot.
Quantitative data, Discrete data
- Salaries of CEOs of oil companies.
Quantitative data, Discrete data

Attribute or Variable

- **Attribute**(or variable, feature, dimension) is a data field, representing a characteristic or feature of a data object.
- It is a property of a data object which is measured for each observation or record.
- It can vary from one observation to another.
- Example : name, age, Student-ID, address, marks, gender etc.
- There are different types of attributes or variables such as:
 - Nominal
 - Ordinal
 - Discrete
 - Continuous
 - Binary
 - Interval
 - Ratio

Quantitative Variable

- A variable that can be measured numerically is called a quantitative variable.
- The data collected on a quantitative variable are called quantitative data.
- Thus, a quantitative variable represents a measure and is numeric.
Its values can be recorded on a numeric scale.
- Example: a country's population, a book's price, height, weight, number of items sold to a shopper, time in 100 yard dash etc.



Quantitative Variable: Discrete

- A variable whose values are countable is called a discrete variable.
- In other words, a discrete variable can assume only certain values with no intermediate values.
- Its number of values is finite or limited.
- Example: number of oranges in a bag, number of students in a classroom, shoe size etc.



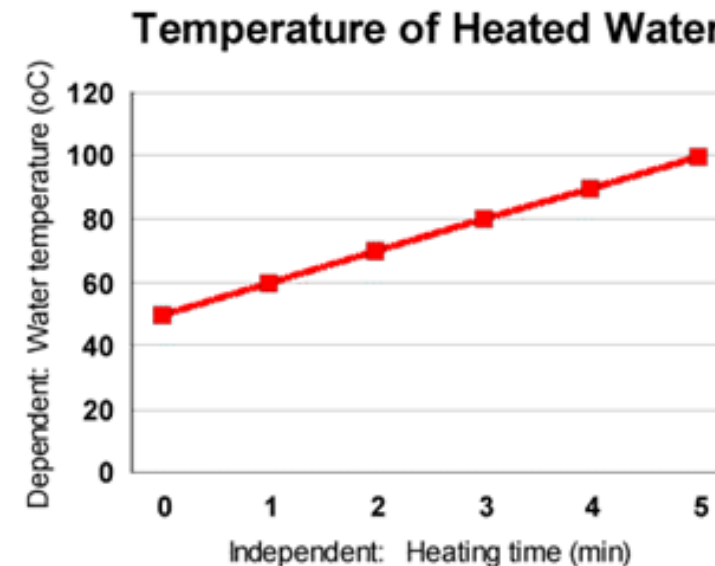
Days	Number of students present
Monday	24
Tuesday	26
Wednesday	28
Thursday	30
Friday	29
Saturday	22

Quantitative Variable: Continuous

- A variable that can assume any numerical value over a certain interval or intervals is called a continuous variable.
- It represents the numerical data as accurately as possible.
- It can take unlimited number of values between the lowest and highest points of measurements.
- Continuous attributes are typically represented as floating-point variables.
- Example: height, weight, temperature

Continuous Variables	Quantile				
	Zeroth	First	Second	Third	Fourth
Apparent Temperature	28.11	67.25	75.09	82.495	107.23
Cloud Cover	0.0	0.03	0.1	0.22	1.0
Dew Point	16.55	58.16	66.0	73.08	82.14
Relative Humidity	0.16	0.62	0.79	0.89	1.0
Temperature	35.61	67.25	75.09	80.37	94.99
Wind Speed	0.0	3.87	5.66	7.82	26.55

Sources: [slideplayer.com](https://www.slideplayer.com), [researchgate.com](https://www.researchgate.com)



Qualitative Variable

- A variable that can not assume a numerical value but can be classified into two or more non-numeric categories is called a qualitative
- It is also known as a **categorical variable**.
- The **data collected** on such a variable are called **qualitative data**.
- The values of this variable are not numeric as they do not result from counting or measuring.
- Thus, arithmetic operations can't be applied on these variables.
- Example: hair colour, favorite books, religion, political party in power, profession, name etc.

Qualitative Variable: Nominal

- A nominal variable is a type of variable that is used to name, label or categorize particular attributes that are being measured.
- Nominal means “relating to names” .
- The utilities of a nominal attribute are sign or title of objects .
- Each value represents some kind of category, code or state.
- It takes qualitative values representing different categories.
- There is no intrinsic ordering of these categories.
- Example: Gender- male, female;
Marital status- married, unmarried;
Skin colour- dark, white, brown;

Qualitative Variable: Ordinal

- An ordinal variable is a type of categorical variable that **takes values with an order or rank.**
- Qualitative variables **have natural, ordered categories and the distances between the categories is not known.**
- Example: Likert scale, or the survey question **"Is your general health poor, reasonable, good, or excellent?"** may have those answers coded respectively as **1, 2, 3, and 4.**
- Educational status- **undergraduate, postgraduate, matriculate**

Likert Scale

5 Point Likert Scale					
Numbering	1	2	3	4	5
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I like Stock Market	☐	☐	☐	☐	☐
I like Stocks	☐	☐	☐	☐	☐
I Like Money	☐	☐	☐	☐	☐
I Like Return	☐	☐	☐	☐	☐
Total	4	8	12	16	20

Source: cdn.wallstreetmojo.com

Quantitative Variable: Interval

- It is interval measured on a scale of equal-sized units
- Values of inter variables have order
- It has no true zero-point.
- Interval variables allow to rank the items measured in order.
- They also allow to quantify and compare the magnitude of differences between them.
- Example: temperature in ° C or ° F, calendar dates etc.

Quantitative Variable: Ratio

- Ratio variables represent the highest level of measurement.
- A ratio variable has an inherent or true zero-point.
- The numerical relationship between the values of a ratio variable is meaningful.
- We can speak of values as being an order of magnitude larger than the unit of measurement (10 K is twice as high as 5 K).
- Example: temperature in Kelvin, length, counts, monetary quantities etc.

Properties of Attributes

The type of an attribute depends on which of the following properties it possesses:

- Distinctness: $=, \neq$
- Order: $<, >$
- Addition: $+, -$
- Multiplication: $*, /$

- **Nominal attribute:** distinctness
- **Ordinal attribute:** distinctness & order
- **Interval attribute:** distinctness, order & addition
- **Ratio attribute:** all 4 properties

Properties of Attributes

THE FOUR LEVELS OF MEASUREMENT:

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

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Examples



In the table below identify which columns represent qualitative variables and which columns represent quantitative variables.

Name	Height	River	State	Completed
Oroville	754	Feather	CA	1968
Hoover	725	Colorado	NV	1936
Dworshak	718	N Fork Clearwater	ID	1873
Glen Canyon	708	Colorado	AZ	1966
New Bullards Bar	636	North Yuba	CA	1970
New Melones	626	Stanislaus	CA	1979
Swift	610	Lewis	WA	1958
Mossyrock	607	Cowlitz	WA	1968
Shasta	600	Sacramento	CA	1945
Hungry Horse	564	S Fork Flathead	MT	1953
Grand Coulee	551	Columbia	WA	1942
Ross	541	Skagit	WA	1949

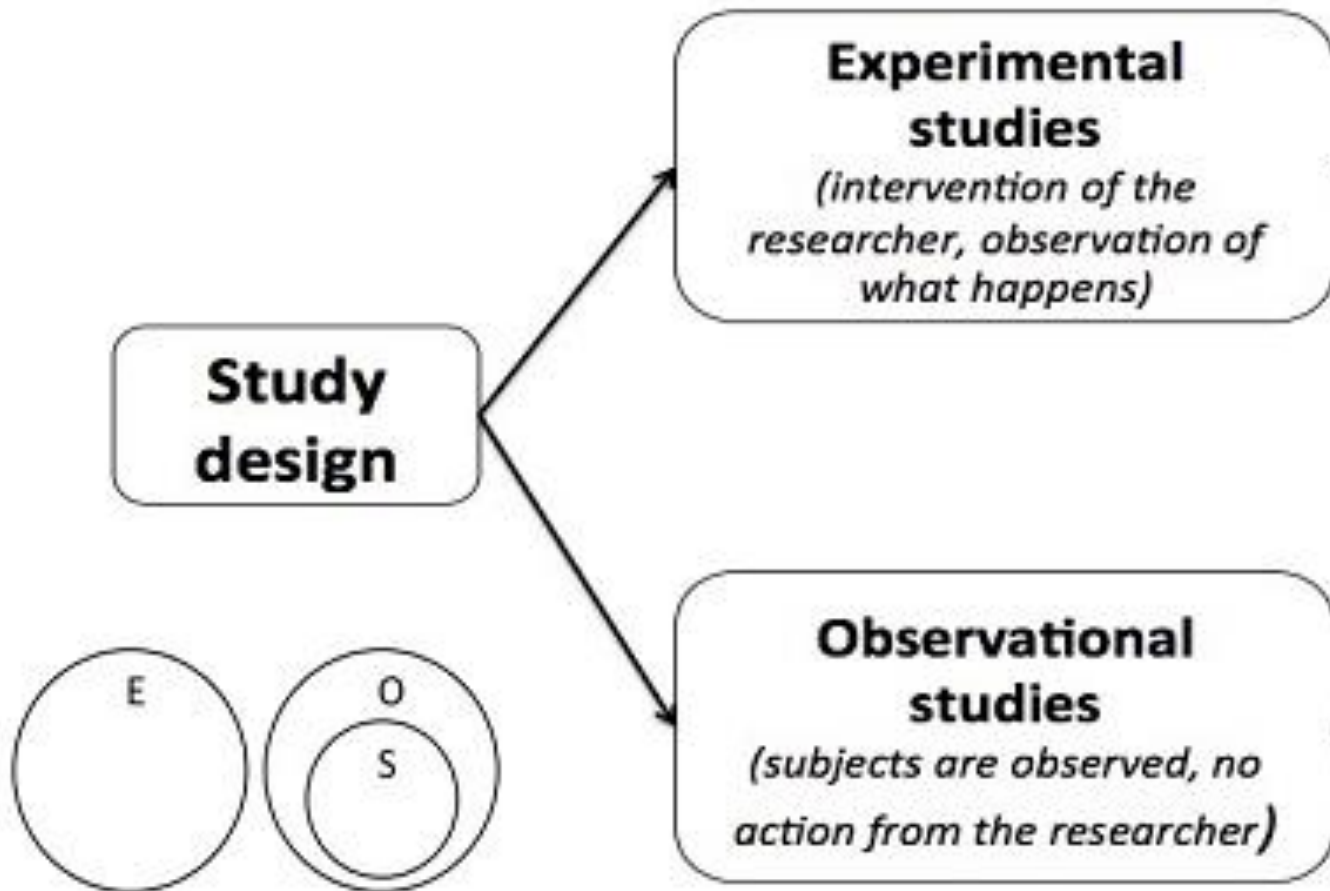
Answer: Qualitative variables: Name, River, State
Quantitative variable: height, Completed

Types of studies

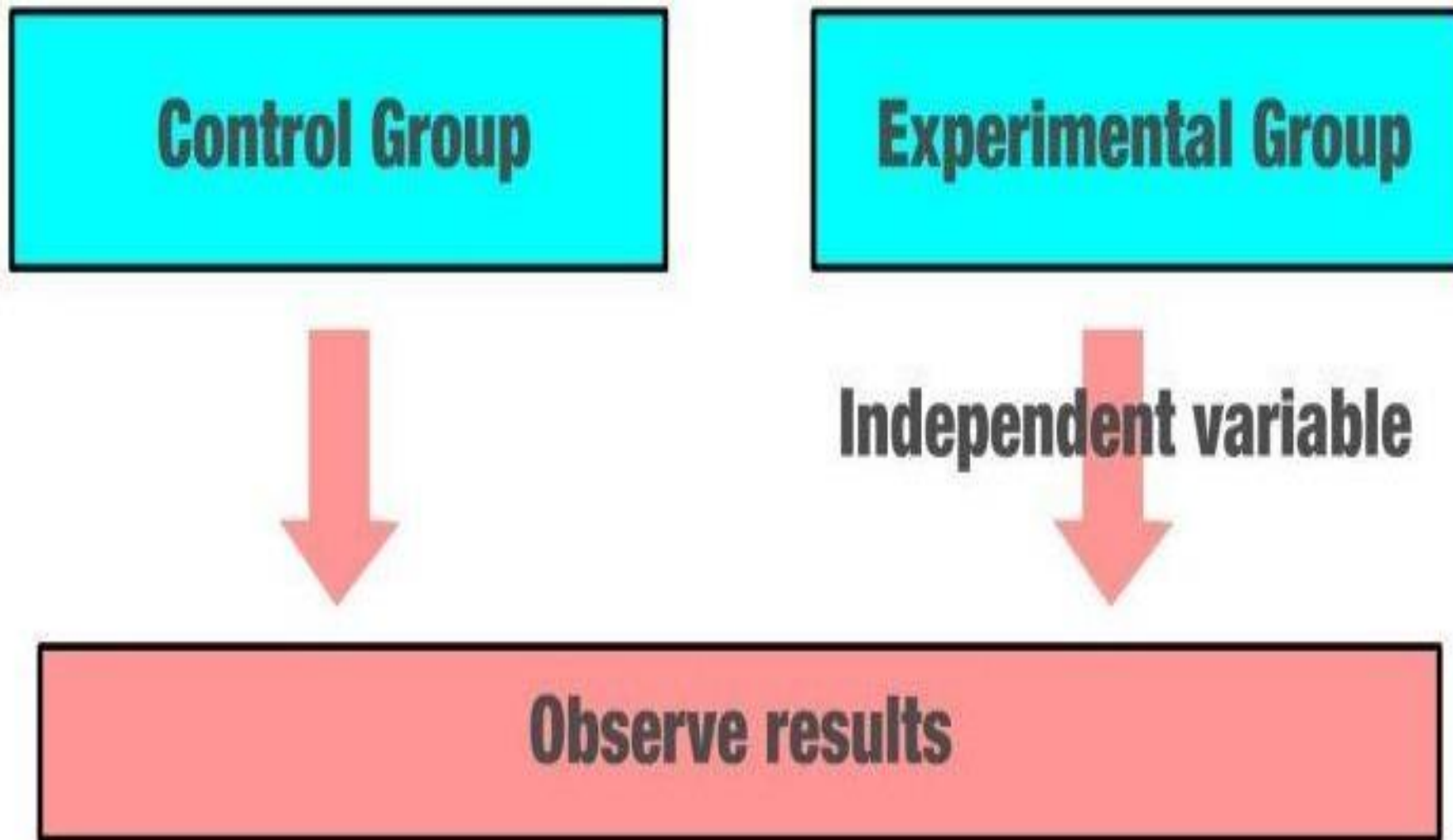
We do studies to gather information and draw conclusions. The type of conclusion we draw depends on the study method used:

- I. **Observational study:** In an observational study, we measure or survey members of a sample without trying to affect them.
- II. **Controlled study:** In a controlled experiment, we assign people or things to groups and apply some treatment to one of the groups, while the other group does not receive the treatment.

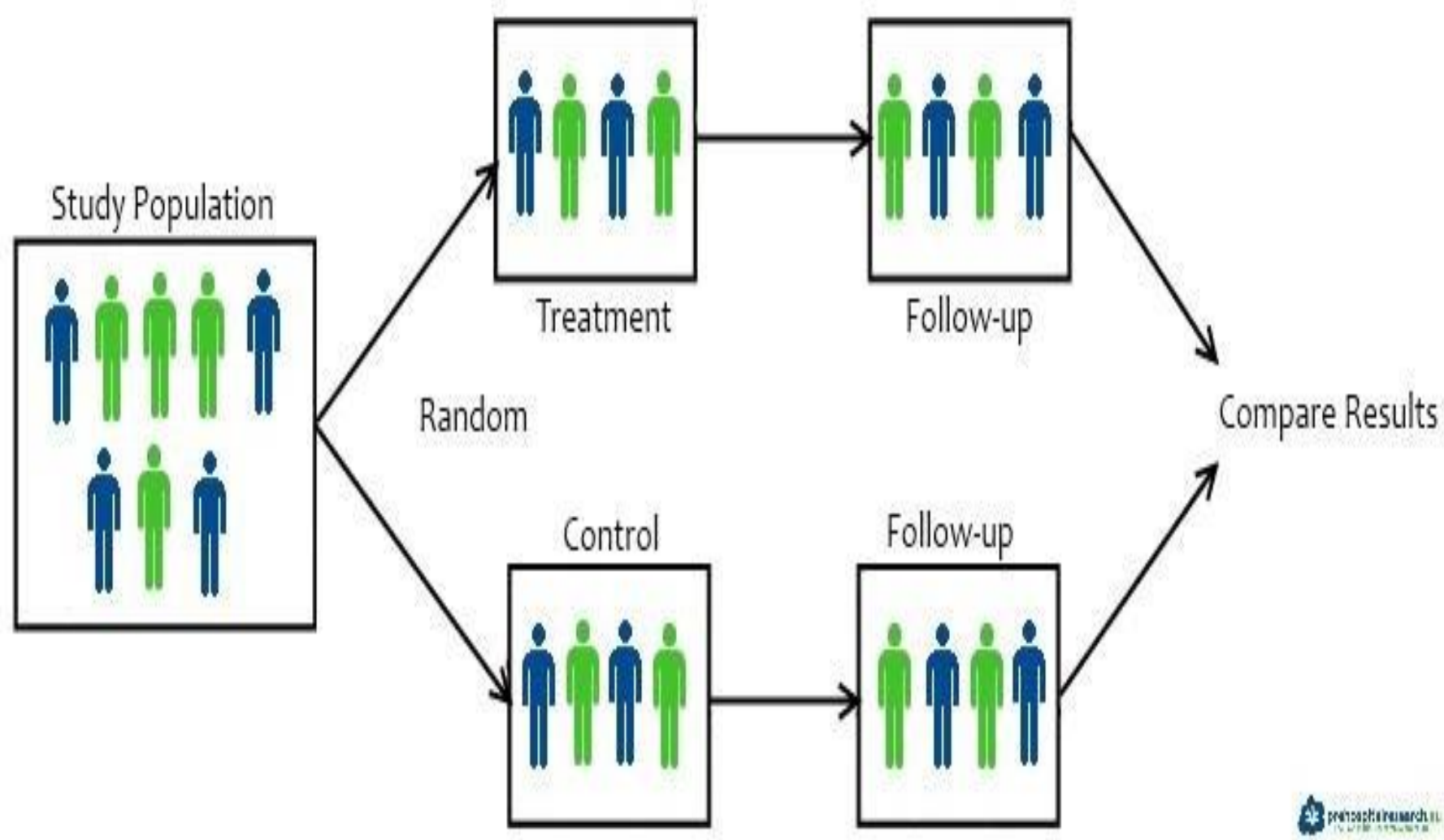
Types of Statistical Studies



Controlled Study

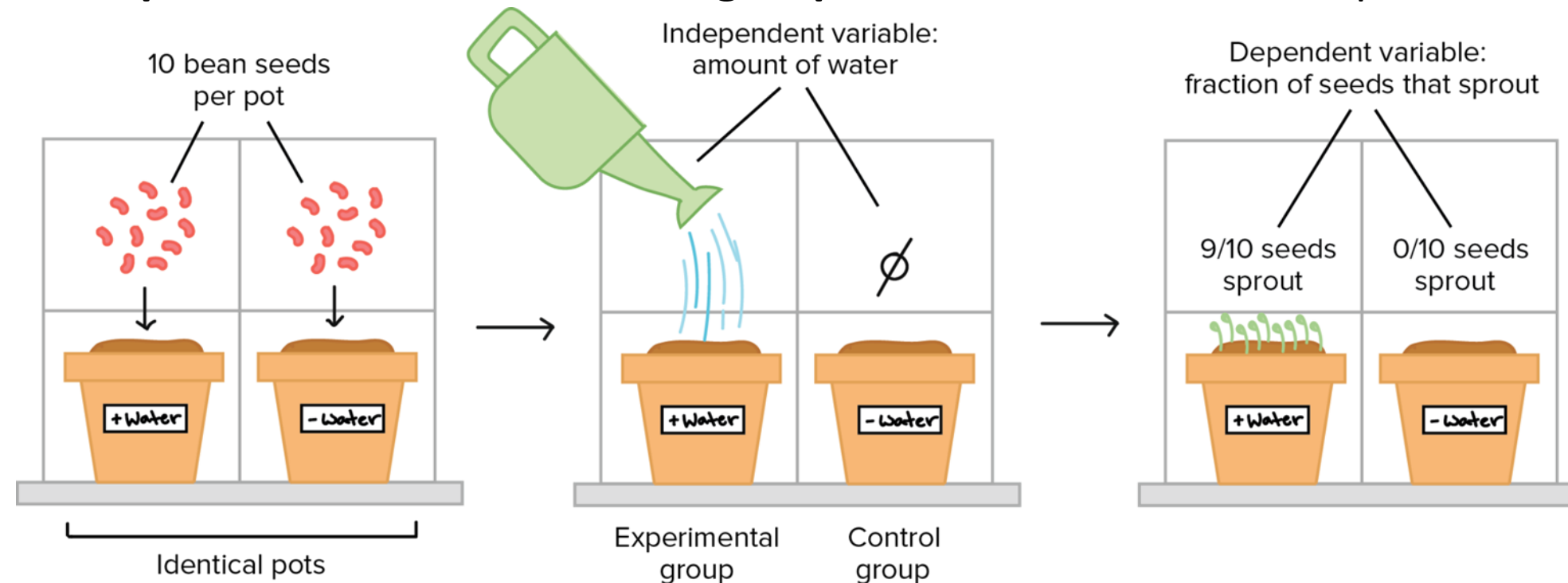


Controlled Study



Controlled Experiment

- While all experiments have an experimental group, not all experiments require a control group.
- Controls are extremely useful where the experimental conditions are complex and difficult to isolate.
- Experiments that use control groups are called controlled experiments.



Observational study vs. Experimental study

Observational Study	Experimental Study
Observe only, no “treatment” assigned.	“Treatment” assigned.
Generally a control group is not needed.	Uses control group for comparison.
Reports an association.	Report a cause and effect.
May (or not) use random sample sets.	Randomization of sample group.
May (or not) generalize to population.	Generalize to population.

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Identify the types of study

Q1.A study took random sample of adults and asked them about their bedtime habits. The data showed that people who drank a cup of tea before bedtime were more likely to go to sleep earlier than those who didn't drink tea.

Answer : Observation Study

Q2.A study took a group of adults and randomly divided them into two groups. One group was told to drink tea every night for a week, while the other group was told not to drink tea that week. Researchers then compared when each group fell asleep.

Answer : Experimental Study

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Identify the types of study

Q3.A study randomly assigned volunteers to one of two groups:

One group was directed to use social media sites as they usually do.

One group was blocked from social media sites.

Answer : Experimental Study

Q4.A study took a random sample of people and examined their social media habits. Each person was classified as either a light, moderate, or heavy social media user. The researchers looked at which groups tended to be happier.

Answer : Observation Study

References

<https://www.analyticsvidhya.com/blog/2021/06/complete-guide-to-data-types-in-statistics-for-data-science/>

Text Book:

Statistics for Engineers and Scientists, William Navidi.



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Types of Statistics & Summary Statistics

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Department of Computer Science and Engineering

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 1: Types of Statistics & Summary Statistics

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Topics to be covered

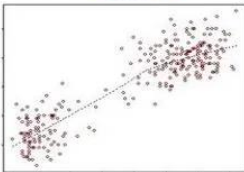
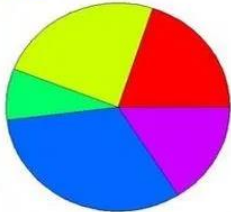
- ❖ Statistics
- ❖ Types of Statistics
- ❖ Summary Statistics

Statistics

- Statistics is the science of data. It involves collecting, classifying, summarizing, organizing, analyzing, and interpreting numerical information.
- It involves study and manipulation of data, including ways to gather, review, analyze, and draw conclusions from data.

Statistics

"Statistics is the science of collecting, organizing, presenting, analyzing and interpreting numerical data to assist in making more effective decisions."



Source: i0.wp.com

Slide Courtesy: Dr. Uma

Statistics

Statistics involves:

- Collecting Data
 - Ex: Survey
- Presenting Data
 - Ex: Charts & Tables
- Characterizing Data
 - Ex: Average

Sample Statistic and Population Parameter

→ Sample statistic:

- It is a numerical measurement describing some **characteristic** of a **sample**.
- Example: sample average, median, sample standard deviation, and percentiles.

→ Population parameter:

- It is a numerical measurement describing some **characteristic** of a **population**.
- Example: mean and variance of a population are population parameters.

Sample Statistic and Population Parameter

Statistic vs Parameter

Sample

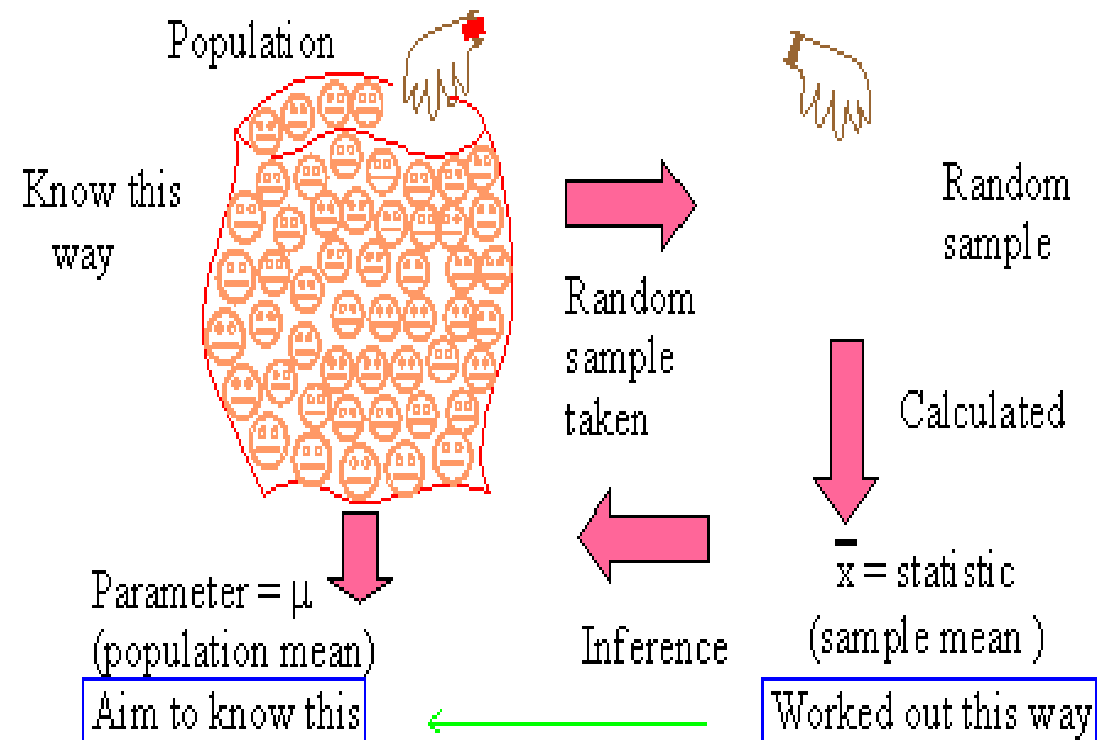
Population

$\bar{x}$ $\leftarrow$ mean $\rightarrow$ μ

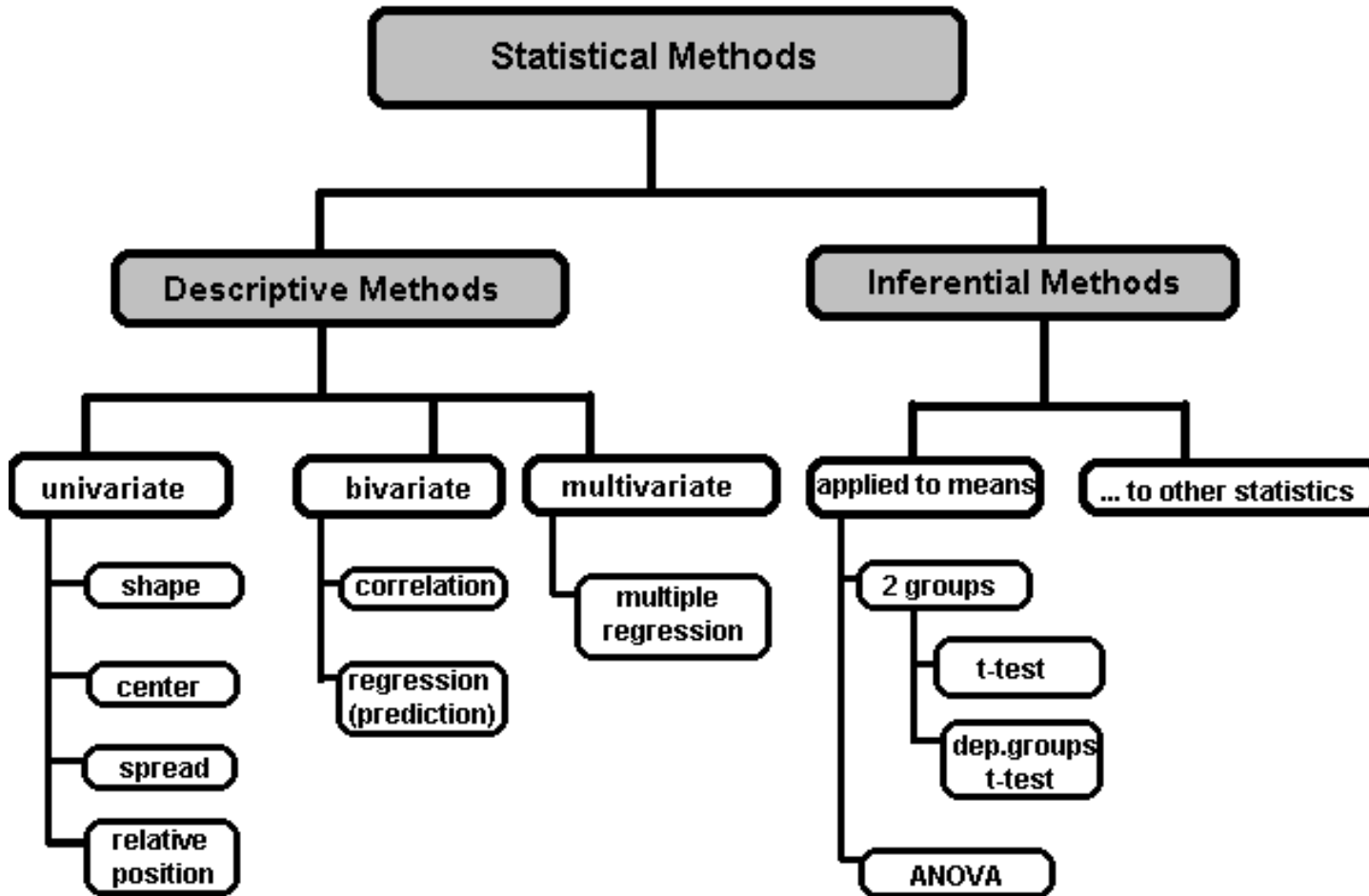
s $\leftarrow$ st. dev. $\rightarrow$ σ

$\hat{p}$ $\leftarrow$ proportion $\rightarrow$ p

n $\leftarrow$ size $\rightarrow$ N



Taxonomy of Statistics



Descriptive Statistics

- Descriptive statistics are methods for organizing and summarizing data.
- Descriptive statistics utilizes numerical and graphical methods to look for patterns in a data set, to summarize the information revealed in a data set and to present that information in a convenient form.
- A descriptive value for a population is called a parameter and a descriptive value for a sample is called a statistic.
- For example, tables or graphs are used to organize data, and descriptive values such as the average score are used to summarize data.

Descriptive Statistics

- Purpose: To describe data

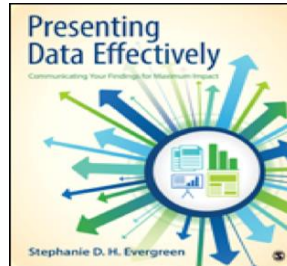
- Collect Data

 - e.g. Survey



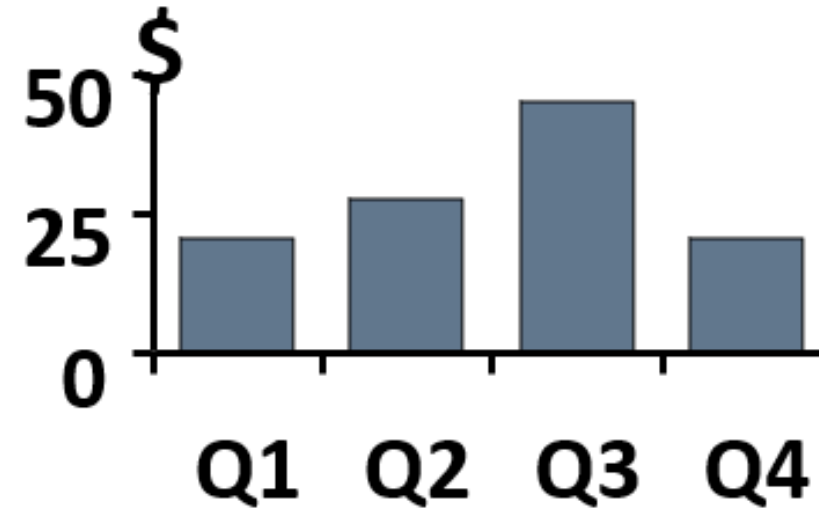
- Present Data

 - e.g. Tables and graphs



- Characterize Data

 - e.g. Sample mean $\bar{X} = \frac{\sum x}{n}$



$$\bar{X} = 30.5 \quad S^2 = 113$$

Descriptive Statistics

Types of descriptive statistics:

■ Organize Data

■ Tables

- Frequency Distributions
- Relative Frequency Distributions

■ Graphs

- Bar Chart or Histogram
- Stem and Leaf Plot
- Frequency Polygon

Descriptive Statistics

→ Summarizing Data:

- **Central Tendency** (or Groups' "Middle Values")

- Mean
- Median
- Mode

- **Variation** (or Summary of Differences Within Groups)

- Range
- Interquartile Range
- Variance
- Standard Deviation

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Why is Descriptive Statistics used?

An Illustration : Which Group is Smarter?

Class A--IQs of 13 Students

102	115
128	109
131	89
98	106
140	119
93	97
110	

Class B--IQs of 13 Students

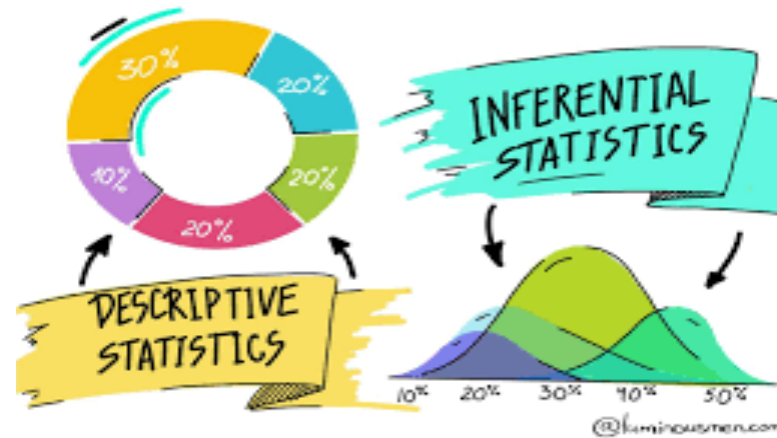
127	162
131	103
96	111
80	109
93	87
120	105
109	

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Why is Descriptive Statistics used?

An Illustration : Which Group is Smarter?

Class A--IQs of 13 Students		Class B--IQs of 13 Students	
102	115	127	162
128	109	131	103
131	89	96	111
98	106	80	109
140	119	93	87
93	97	120	105
110		109	



Which group is smarter now?

Class A--Average IQ

110.54

Class B--Average IQ

110.23

They're roughly the same!

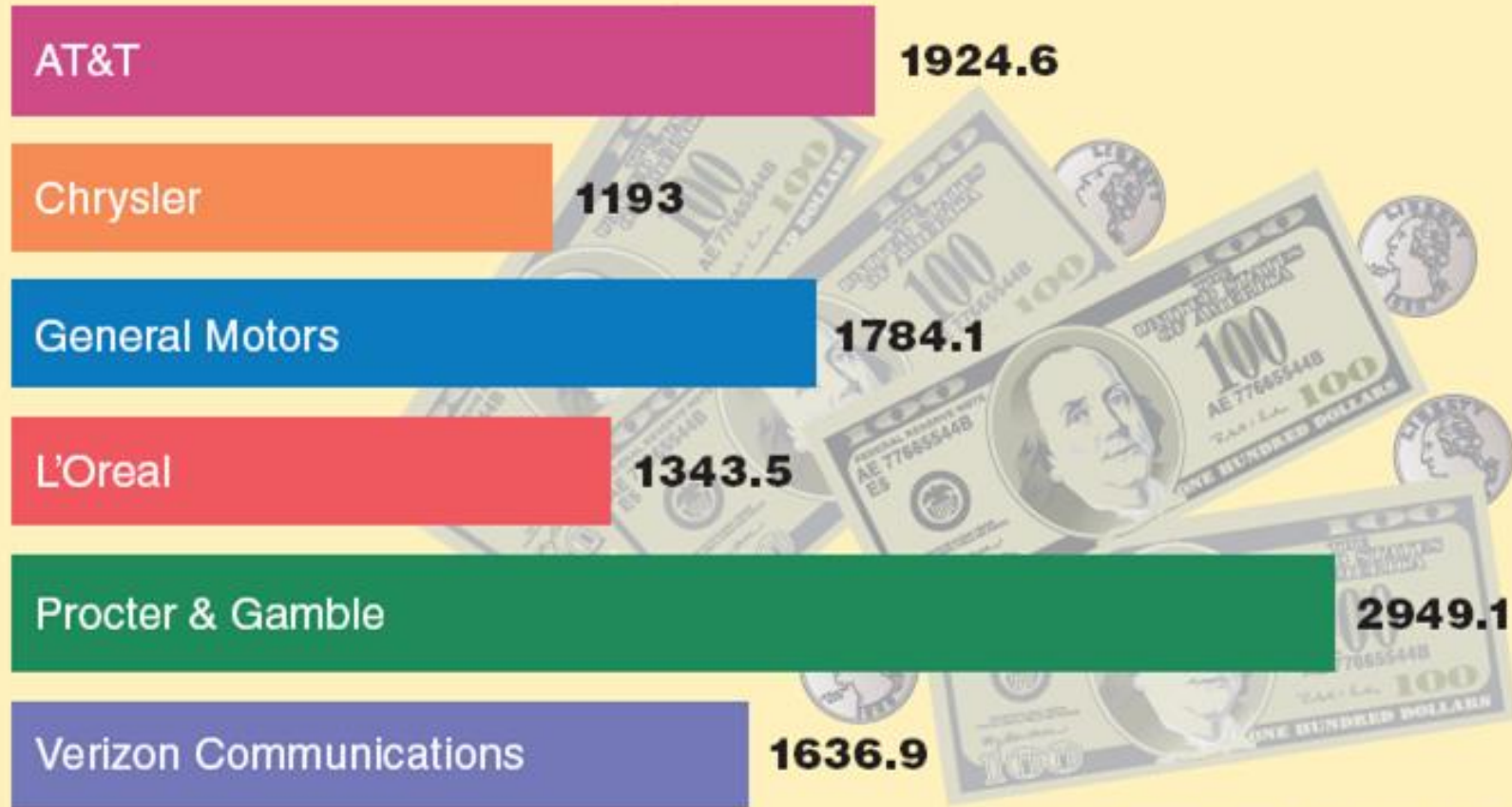
With a summary descriptive statistic, it is much easier to answer our question.

Figure speaks it all !!!

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Descriptive Statistics Examples

HOW MUCH DID COMPANIES SPEND ON ADS IN 2011?



Note: Amounts are in millions of dollars.

Inferential Statistics

- Inferential statistics utilizes sample data to make estimates, decisions, predictions or other generalizations about a larger set of data.
- There are two main areas of inferential statistics:
 1. **Estimating parameters:** This means taking a statistic from the sample data (for example the sample mean) and using it to say something about a population parameter (for example the population mean).
 2. **Hypothesis tests:** This is where sample data can be used to answer research questions.

For example,

- a. one might be interested in knowing if a new cancer drug is effective;
- b. if breakfast helps children perform better in schools.

Why is Inferential Statistics used?

Suppose you want to know the mean income of the subscribers of Netflix.

- Mean (μ) — a parameter of a population.
- You draw a random sample of 100 subscribers and determine that their mean income is \$27,500.
- Mean ($\bar{x}$) = \$27,500 (a summary statistic).
- Conclusion : You conclude that the population mean income μ is likely to be close to \$27,500 as well.
- This is an example of statistical inference.

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Descriptive Statistics vs Inferential Statistics

S. No	Descriptive Statistics	Inferential Statistics
1	Concerned with the describing the target population	Make inferences from the sample and generalize them to the population.
2	Organize, analyze and present the data in a meaningful manner	Compares, test and predicts future outcomes.
3	Final results are shown in form of charts, tables and Graphs	Final result is the probability scores.
4	Describes the data which is already known	Tries to make conclusions about the population that is beyond the data available.
5	Tools- Measures of central tendency (mean/median/ mode), Spread of data (range, standard deviation etc.)	Tools- hypothesis tests, Analysis of variance etc.

Descriptive Statistics vs Inferential Statistics

Descriptive Statistics

- Organize
- Summarize
- Simplify
- Presentation of data



Describing data

Inferential Statistics

- Generalize from samples to population
- Hypothesis testing
- Relationships among variables



Make predictions

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Questions

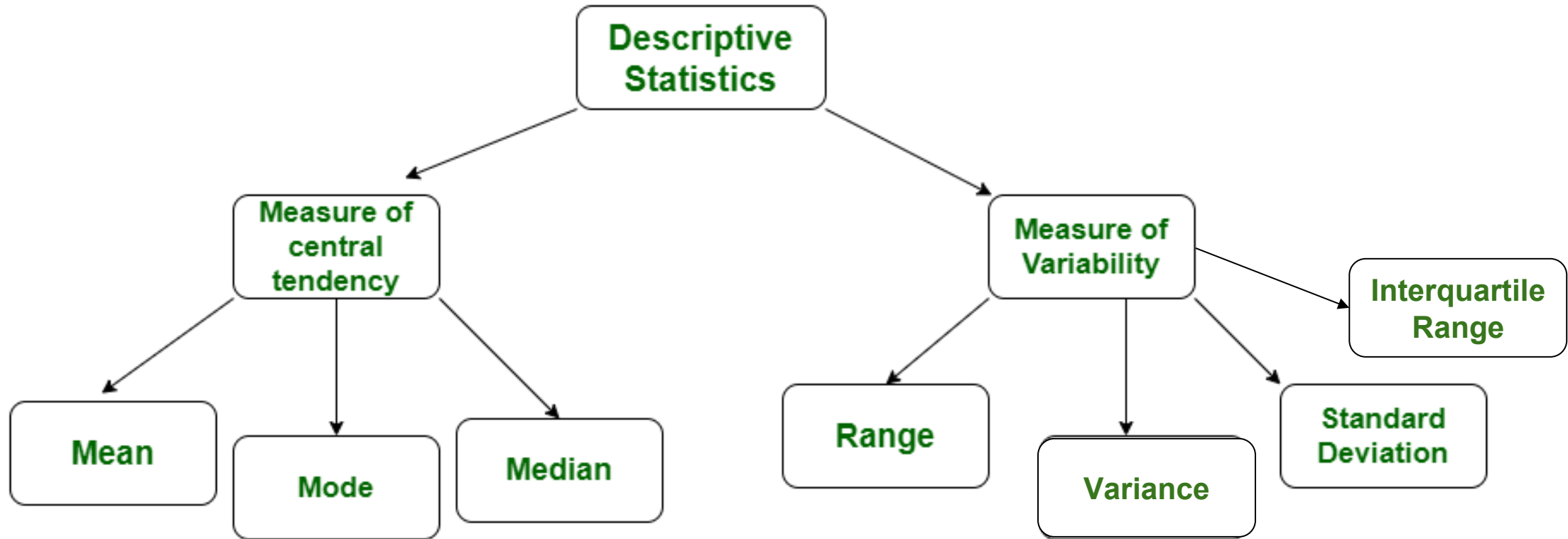
Q2. A burger outlet wanted to perform market research to determine what type of burgers their customers liked. The outlet is researching to figure out the favourite tastes of their customers to provide better services and dishes to the customers. The outlet gathered a customer sample size of a 100 customers in different age groups and regular nearby customers at the outlet. The outlet was able to determine that 80% of the customers liked their burgers to be spicy and crispy while the rest liked it non-crunchy and non-spicy.

What type of statistics was applied to arrive that conclusion?

Ans: Inferential statistics



Types of Descriptive Statistics



Measures of central tendency

- There are three different types of 'average'.
- These are the mean, the median and the mode.
- They are used by statisticians as a way of summarizing where the 'centre' of the data is.

$$\text{Mean} = \frac{\text{sum of all values}}{\text{total number of values}}$$

$$\text{Median} = \text{middle value (when the data are arranged in order)}$$

$$\text{Mode} = \text{most common value}$$

Measures of central tendency: Mean

- Mean is the **arithmetic average** computed by summing all the values in the dataset and dividing the sum by the number of data values.
- The population mean is represented by **Greek letter μ** .
- For a **finite set of dataset with measurement values $X_1, X_2, \dots, X_n$** (a set of n numbers), it is defined by the formula:

a) Population mean:

$$\mu_x = \sum_{i=1}^N \frac{x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

where N is the population size

a) Sample mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where n is the sample size

Measures of central tendency: Weighted mean

- Weighted mean is an average where certain values of the data set contribute more to the mean value.
- For a finite set of dataset with measurement values $X_1, X_2, \dots, X_n$

(a set of n numbers), and the corresponding weights $w_1, w_2, \dots, w_n$

it is defined by the formula:

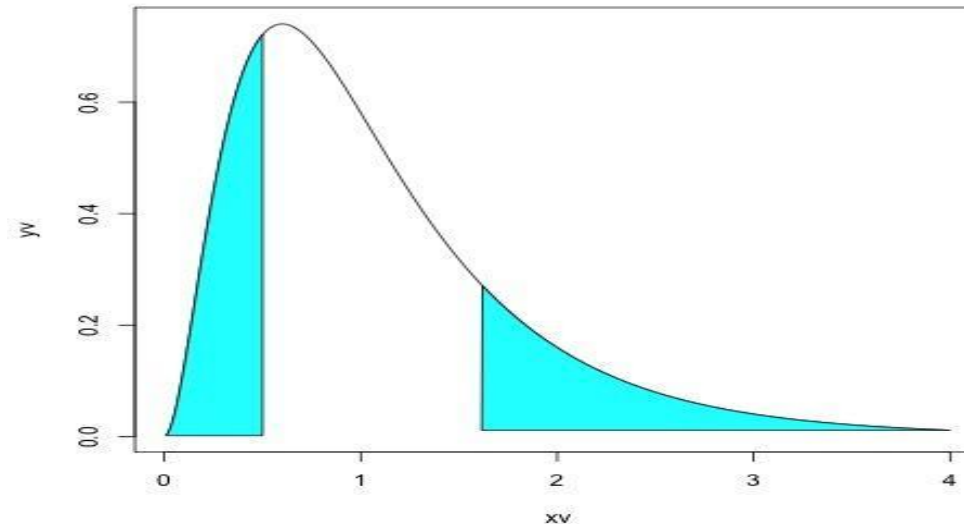
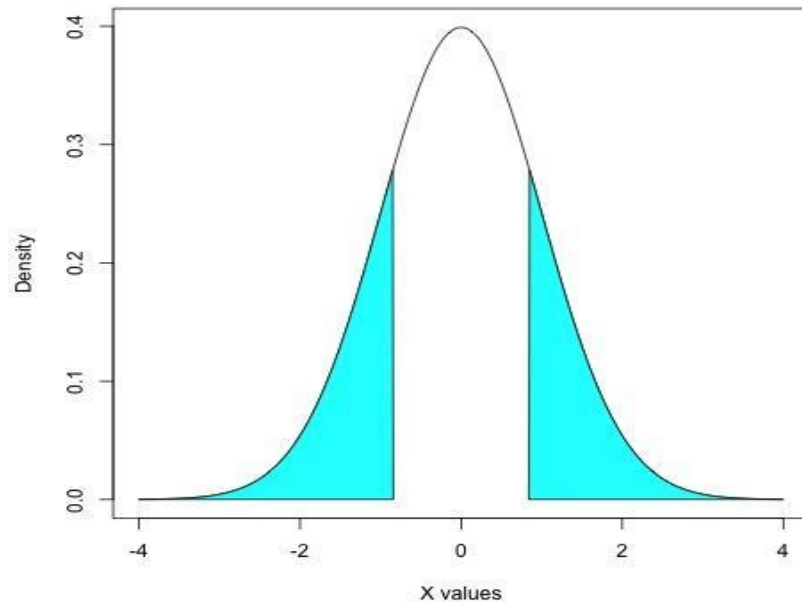
$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Measures of central tendency: Trimmed mean

- The trimmed mean is computed by arranging the sample values in order, “trimming” an equal number of them from each end, and computing the mean of those remaining.
- If $p\%$ of the data are trimmed from each end, the resulting trimmed mean is called the “ **$p\%$ trimmed mean**”.
- There are no hard-and-fast rules on how many values to trim.
- The most commonly used trimmed means are the 5%, 10%, and 20% trimmed means.

Measures of central tendency: Trimmed mean

- If the sample size is denoted by n , and a $p\%$ trimmed mean is desired, the number of data points to be trimmed is $np/100$
- It is used to reduce the effects of outliers on the calculated average.
- This method is best suited for data with large, erratic deviations or extremely skewed distributions.



Measures of central tendency: Mean

→ Advantages:

- It takes into account all the available information.
- It can be combined with means of other groups to give the overall mean.
- Easy and quick way to represent the entire data values by a single or unique number due to its straightforward method of calculation.
- Each data set has a unique mean value.

→ Disadvantages:

- It is a very sensitive measure.
- Thus, its value is easily affected by extreme values known as the outliers.
- It can only be used on interval or ratio data.

Measures of central tendency: Question

For the following data

30 75 79 80 80 105 126 138 149 179 179 191
223 232 232 236 240 242 245 247 254 274 384 470

Compute the mean, median, and the 5%, 10%, and 20% trimmed means.

Solution:

- The mean is found by averaging together all 24 numbers, which produces a value of 195.42.
- The median is the average of the 12th and 13th numbers, which is $(191 + 223)/2 = 207.00$.
- To compute the 5% trimmed mean, we must drop 5% of the data from each end.
- This comes to $(0.05)(24) = 1.2$ observations.
- We round 1.2 to 1, and trim one observation off each end.

Measures of central tendency: Question

- The 5% trimmed mean is the average of the remaining 22 numbers: $75 + 79 + \dots + 274 + 384 / 22 = 190.45$
- To compute the 10% trimmed mean, round off $(0.1)(24) = 2.4$ to 2.
Drop 2 observations from each end, and then average the remaining 20:
 $79 + 80 + \dots + 254 + 274 / 20 = 186.55$
- To compute the 20% trimmed mean, round off $(0.2)(24) = 4.8$ to 5.
- Drop 5 observations from each end, and then average the remaining 14:
 $105 + 126 + \dots + 242 + 245 / 14 = 194.07$

Measures of central tendency: Median

- **Median** is the value separating the higher half from the lower half of a data sample, a population, or a probability distribution.
- It is the middle number in a sorted, ascending or descending, list of numbers and can be more descriptive of that data set than the average.
- For a data set, it may be thought of as "the middle" value.
- The basic feature of the median in describing data compared to the mean (often simply described as the "average") is that it is not affected by a small proportion of extremely large or small values, and therefore provides a better representation of a "typical" value.

Measures of central tendency: Median

→ Process of calculating median:

1. Arrange all the values of the data set in ascending order.

$X_1, X_2, X_3, \dots, X_n$

1. Find the **middle position**.

3. The element corresponding to middle position is **considered as median if odd number of elements are present.**

i.e. if **n is odd, median = $(n+1/2)$ th element's value**

4. If there are even number of elements present then the average of the elements present in the middle positions is considered as median. i.e. if n is even,

median = $((n/2)$ th element's value + $(n/2 + 1)$ th element's value) / 2

Measures of central tendency: Median

Odd no. of elements

5, 13, 9, 7, 1, 9, 2, 9, and 11

put in
ascending order

1, 2, 5, 7, 9, 9, 9, 11, 13

Median
(middle value)

Even no. of

9.25, 12.31, 35.12, 56.13, 10.01, and 22.15

arrange in
ascending order

9.25, 10.01, 12.31, 22.15, 35.12, 56.13

Median = average of the two middle values

Measures of central tendency: Median

→ Advantages:

- Not affected by the outliers in the data set.
- An outlier is a data point that is radically “distant” or “away” from common trends of values in a given set.
- It does not represent a typical number in the set.
- The concept of the median is intuitive and thus can easily be explained as the center value.
- Each set has a unique median value.

→ Disadvantages:

- Its value is perceived as it is.
- It cannot be utilized for further algebraic treatment.

Measures of central tendency: Mode

- The **mode** is the value that appears most often in a set of data values
- Like the statistical mean and median, the mode is a way of expressing, in a (usually) single number.
- To calculate the mode, we need to look at which **value appears the most often.**
- Example, the mode of the sample [1, 3, 6, 6, 6, 6, 7, 7, 12, 12, 17] is 6.
- Given the list of data [1, 1, 2, 4, 4] its mode is not unique.
It has 2 modes: 1 and 4
- A dataset, in such a case, is said to be **bimodal**, while a set with more than two modes may be described as **multimodal**.
- Empirical formula:

$$mean - mode = 3 \times (mean - median)$$

Measures of central tendency: Mode

→ Advantages:

- Quick and easy to compute.
- Unaffected by extreme values.
- Can be used at any level of measurement.
- Useful to find the most “popular” or common item. This includes data sets that do not involve numbers.

→ Disadvantages:

- It is a terminal statistic.
- A given subgroup could make this measure unrepresentative of the population's centre.
- If the set contains **no repeating values**, the **mode is irrelevant**.
- In contrast, if there are many values that have the same count, then mode can be meaningless.

Measures of central tendency: Questions

Alex did a survey of how many games each of his 20 friends owned, and got this:

9, 15, 11, 12, 3, 5, 10, 20, 14, 6, 8, 8, 12, 12, 18, 15, 6, 9, 18, 11

Find the mean, median and mode.

Ans:

Sorting in ascending order:

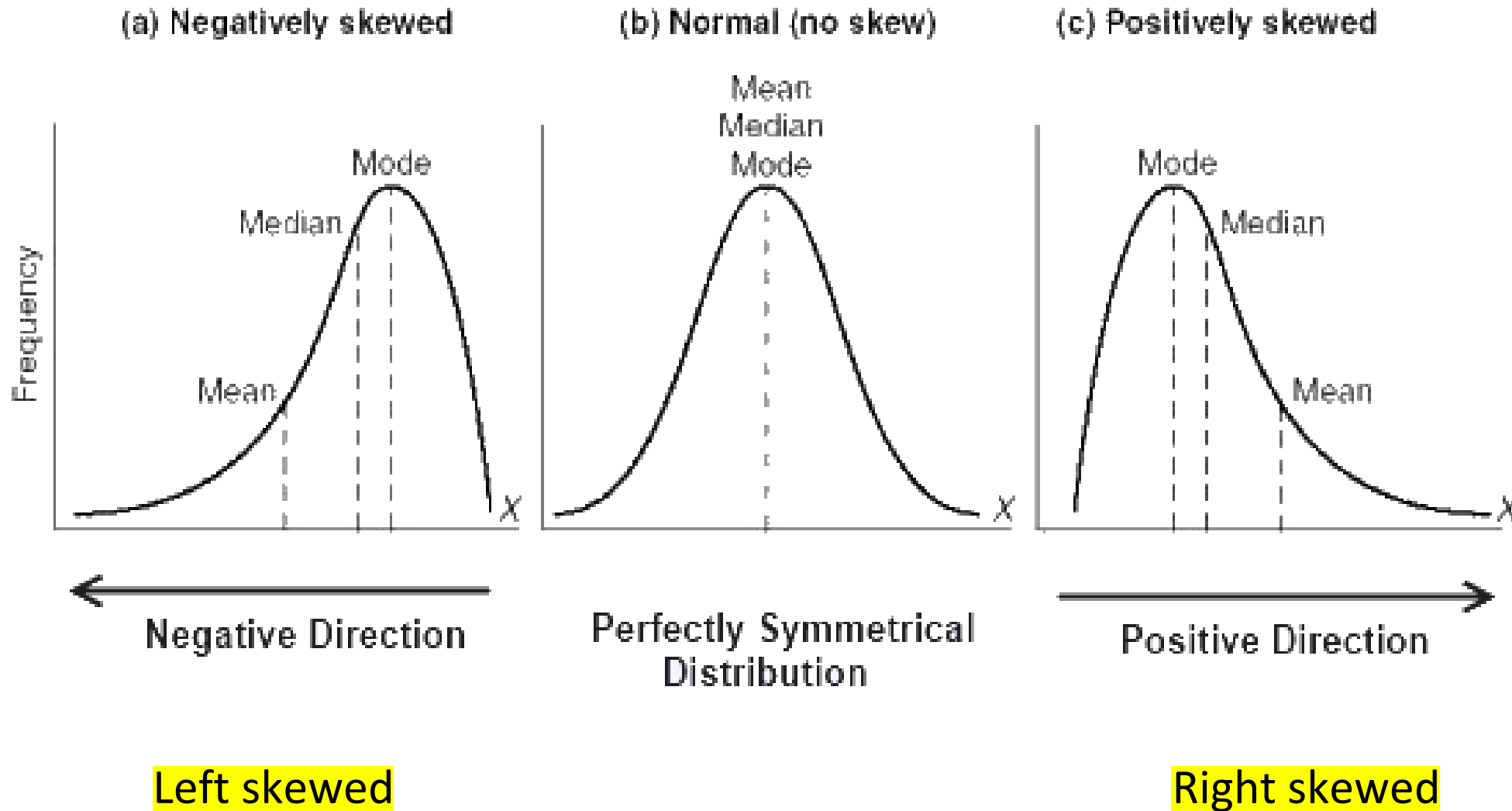
3, 5, 6, 6, 8, 8, 9, 9, 10, 11, 11, 12, 12, 12, , 14, 15, 15, 18, 18, 20

- Mean = $222/20 = 11.1$
- Median = $(11+11)/2 = 11$
- Mode = 12

Skewed and Symmetric distributions

- **Skewness** is a measure of the asymmetry of the distribution of about its mean.
- The skewness value can be positive, zero, negative, or undefined.
- **Symmetric Distribution:** A symmetric distribution is one where the left and right hand sides of the distribution are roughly equally balanced around the mean.
- In symmetric distributions, the mean, median, and mode are the same.
- **Skewed Distribution:** A skewed distribution is one where the left and right hand sides of the distribution are not balanced around the mean.
- In skewed data, the mean and median lie further toward the skew than the mode.
- The greater the distance of mean and median, the greater is the skewness of the distribution.

Skewed and Symmetric distributions



Skewed and Symmetric distributions

- **Distribution of a variable:** tells us what values the variable takes and how often it takes these values.
- **Shape:** It is the “shape” of the distribution of the data.
- If **mean = median = mode**, the shape of the distribution is **symmetric**.
- If **mode < median < mean**, the shape of the distribution trails to the right, is **positively skewed**.
- If **mean < median < mode**, the shape of the distribution trails to the left, is **negatively skewed**.
- Distributions of various “shapes” have different properties and names such as the **“normal” distribution**, which is also known as the **“bell curve”** (among mathematicians it is called the **Gaussian distribution**)

Measures of central tendency

Various central tendency measures can be applied on different types of data.

- **Quantitative data:**
 - Mode – the most frequently occurring observation
 - Median – the middle value in the data
 - Mean – arithmetic average
- **Qualitative data:**
 - Mode – always appropriate
Ex : Maximum Type of Color
 - Mean – never appropriate
Ex : Average value of Yellow color

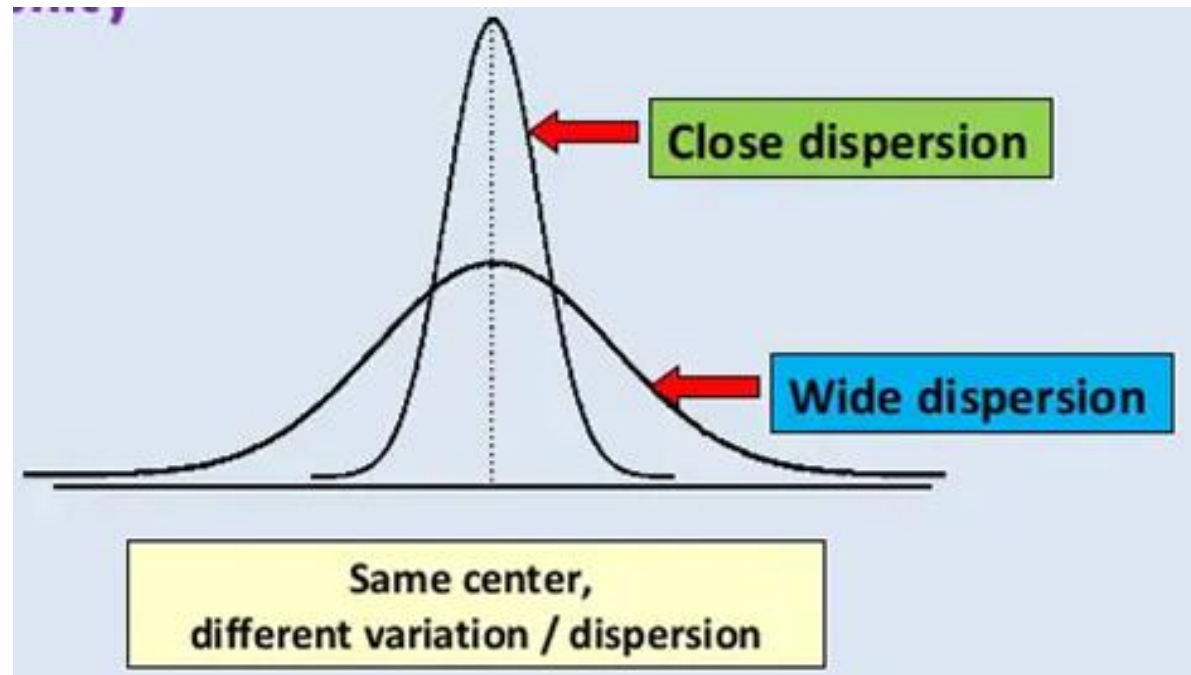
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When to use mean, median and mode?

TYPE OF VARIABLE	BEST MEASURE OF CENTRAL TENDENCY
Nominal	Mode
Ordinal	Median
Interval / Ratio (not skewed)	Mean
Interval / Ratio (skewed)	Median

Measures of Spread/Dispersion

- In statistics, the measures of dispersion help to interpret the variability of data
- It helps to know how much homogeneous or heterogeneous the data is.
- In simple terms, it shows how squeezed or scattered the variable is
- There are two main types of dispersion methods in statistics which are:
 - (i) Absolute Measure of Dispersion
 - (ii) Relative Measure of Dispersion



Measures of Spread/Dispersion

- **Absolute Measure of Dispersion:**

It contains the same unit as the original data set. Absolute dispersion method expresses the variations in terms of the average of deviations of observations like standard or mean deviations. It includes range, standard deviation, quartile deviation, etc.

- **Relative Measure of Dispersion:**

The relative measures of dispersion are used to compare the distribution of two or more data sets. This measure compares values without units. Common relative dispersion methods include: Coefficient of Range, Coefficient of Variation, Coefficient of Standard Deviation, Coefficient of Quartile Deviation, Coefficient of Mean Deviation.

Measures of Spread: Range

- Range is the most common and easily understandable measure of dispersion.
- It is the difference between two extreme observations of the data set.
- If $X_{\max}$ and $X_{\min}$ are the two extreme observations then

$$\text{Range} = X_{\max} - X_{\min}$$

5, 13, 9, 7, 1, 9, 2, 9, and 11

put in
ascending order

1, 2, 5, 7, 9, 9, 9, 11, 13



lowest

highest

Source: Chilimath.com

Here, range = $13 - 1$

= 12

Measures of Spread: Range

Class A

AGES OF STUDENTS

13,13,14,14,14,15,15,15,15,16,16,16

$$\begin{aligned}\text{Range} &= \text{highest} - \text{lowest} \\ &= 16 - 13 \\ \text{Range} &= 3\end{aligned}$$

Class B

AGES OF STUDENTS

11,13,13,14,14,15,15,15,15,16,16,18

$$\begin{aligned}\text{Range} &= \text{highest} - \text{lowest} \\ &= 18 - 11 \\ \text{Range} &= 7\end{aligned}$$

Observations:

Since the **range of** Class A is **smaller** than in Class B, can we claim that the age distribution in Class A is more clustered (closely related) than in Class B? In other words, are the ages listed in Class A more uniform than in Class B?

Measures of Spread: Range

Range Can Be Misleading:

- The range can sometimes be misleading when there are extremely high or low values.
- Example: {8, 11, 5, 9, 7, 6, 3616}
lowest value : 5
highest 3616,
- So the range is $3616 - 5 = 3611$.
- The single value of 3616 makes the range large, but most values are around 10.

Measures of Spread: Range

→ Advantages:

- It is the simplest of the measure of dispersion
- Easy to calculate
- Easy to understand
- Independent of change of origin

→ Disadvantages:

- It is based on two extreme observations. Hence, get affected by fluctuations
- A range is not a reliable measure of dispersion
- Dependent on change of scale
- It can drastically be affected by outliers (values that are not typical as compared to the rest of the elements in the set).

Measures of Spread: Quantiles

When presenting or analysing measurements of a continuous variable it is sometimes helpful to group subjects into several equal groups.

For example, to create four equal groups we need the values that split the data such that 25% of the observations are in each group.

The cut off points are called quartiles, and there are three of them (the middle one also being called the median).

Likewise, we use two tertiles to split data into three groups, four quintiles to split them into five groups, and so on.

Measures of Spread

The general term for such cut off points is quantiles; other values likely to be encountered are deciles, which split data into 10 parts, and centiles, which split the data into 100 parts (also called percentiles).

Values such as quartiles can also be expressed as centiles; for example, the lowest quartile is also the 25th centile and the median is the 50th centile.

Measures of Spread:

- A quintile is a statistical value of a data set that represents 20% of a given population, so the first quintile represents the lowest fifth of the data (1% to 20%); the second quintile represents the second fifth (21% to 40%) and so on.

Example:

- Quintiles are used to create cut-off points for a given population;
- A government-sponsored socio-economic study may use quintiles to determine the maximum wealth a family could possess in order to belong to the lowest quintile of society.
- This cut-off point can then be used as a prerequisite for a family to receive a special government subsidy aimed to help society's less fortunate.

Measure of Spread:Percentile

- A **percentile** is a comparison measure between a particular value and the values of the rest of the data set.
- It shows the percentage of values that a particular element has surpassed.

Example

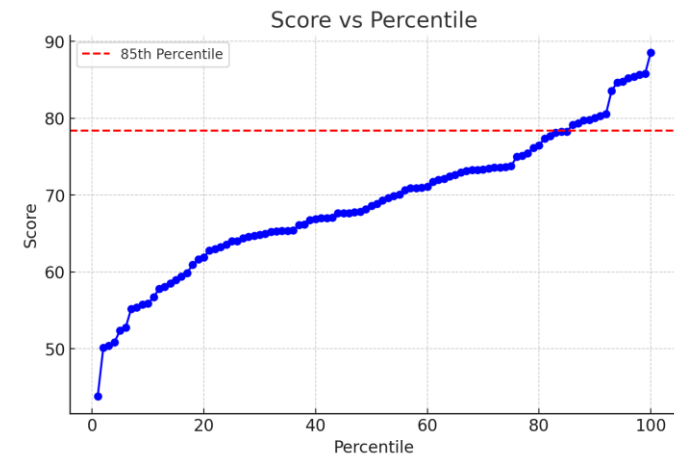
- If your score = 75 and your percentile rank = 85th, then:
- Interpretation: You performed better than 85% of the people who took the same test.
- Percentile tells you your position relative to others, not the raw score itself.
- The higher the percentile, the better your performance compared to the group.

Measure of Spread:Percentile

- The percentile rank is calculated using the formula

$R = (P/100) * (N+1)$ where P is the desired percentile and N is the number of data points.

- The pth percentile of a sample, for a number p between 0 and 100, divides the sample such that,
 - p% of the sample values are less than the pth percentile
 - (100-p%) are greater than the pth percentile



Percentile

Steps to calculate the percentile rank:

1. Order the n samples values from smallest to largest.
2. Compute the quantity $(P/100)(n+1)$, where n is the sample size.
3. If the above quantity is an integer, the sample value in this position is the percentile.
4. Otherwise, average the two sample values at the preceding and succeeding integer positions with respect to the quantity obtained in step 3.

Percentile example

If the scores of a set of students in a math test are 25, 7, 9, 13, 2 and 8 what is the 15th percentile and 75th percentile?

Ans: Arrange the numbers in ascending order and give the rank ranging from 1(the lowest number) to 5 (the highest number)

Score	2	7	8	9	13	25
-------	---	---	---	---	----	----

$$\begin{aligned} R &= (P/100)(N+1) \\ &= (15/100) (6+1) \\ &= 1.05 \text{ (is it not an integer)} \end{aligned}$$

$$\begin{aligned} \text{Percentile value} &= (1\text{st element} + 2\text{nd element value})/2 \\ &= (2+7)/2 \\ &= 4.5 \text{ it is 15th percentile} \end{aligned}$$

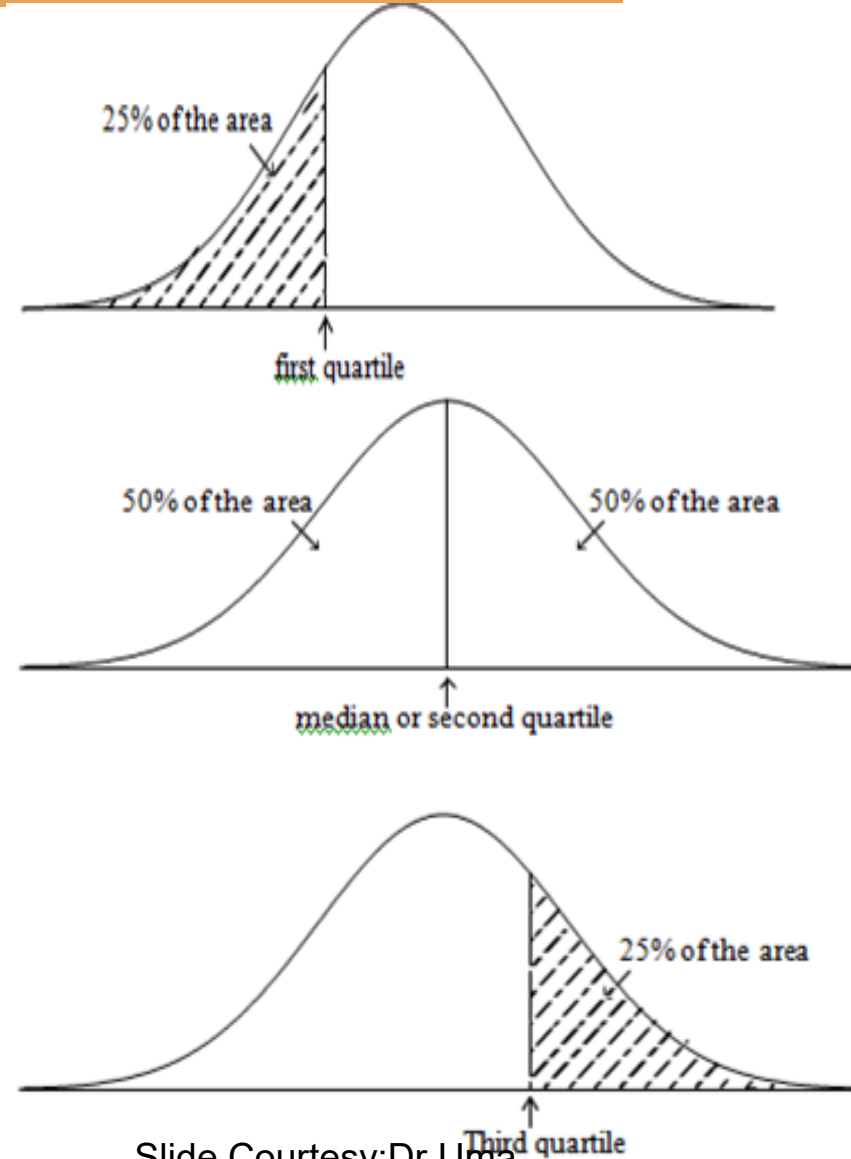
Thus, score 19 is the 75th percentile

Measures of Spread: Quartiles

- Quartiles are the values that **divide a list of numbers into quarters**.
- Quartiles are obtained by first putting the list of numbers in order and then cutting the list into four equal parts.
- The Quartiles are at the "cuts" in the data.
- The first quartile, (Q1) is the middle number between the smallest number and the median of the data.
- The second quartile, (Q2) is the median of the data set.
- The third quartile, (Q3) is the middle number between the median and the largest number.

Measures of Spread: Quartiles

- The first quartile is the 25th percentile
- The median is the 50th percentile
- The third quartile is the 75th percentile

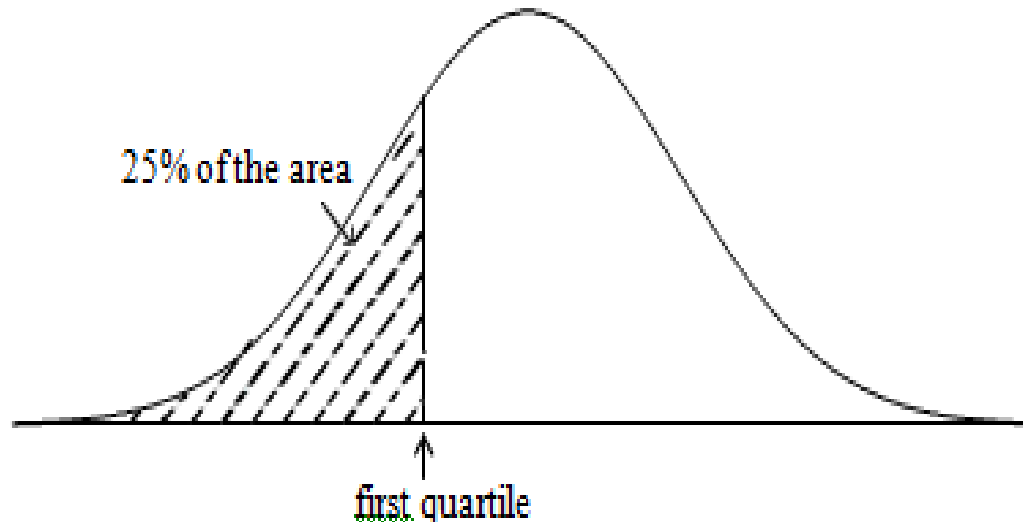


Slide Courtesy: Dr. Uma

Measures of Spread: Quartiles

→ The First Quartile:

- The first quartile is the point which gives us 25% of the area to the left of it and 75% to the right of it.
- This means that 25% of the observations are less than or equal to the first quartile and 75% of the observations greater than or equal to the first quartile.
- The first quartile is also called the 25th percentile.



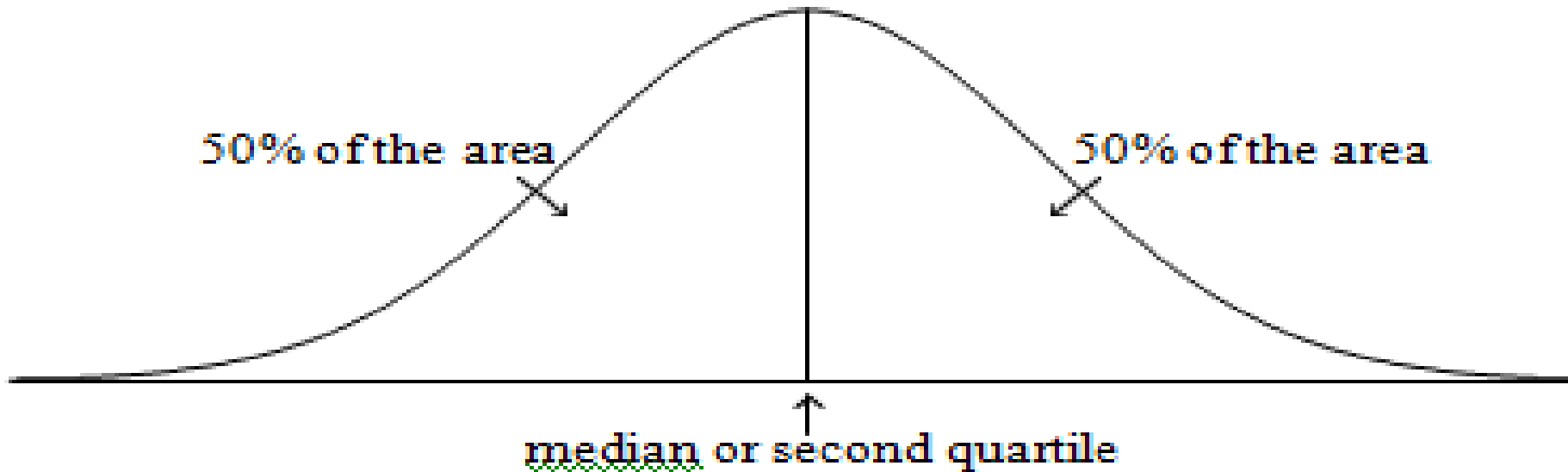
Measures of Spread: Quartiles

- To find the first quartile, compute the value $0.25(n + 1)$.
- If this is an integer, then the sample value in that position is the first quartile.
- If not, then take the average of the sample values on either side of this value.

Measures of Spread: Quartiles

→ The Second Quartile or median:

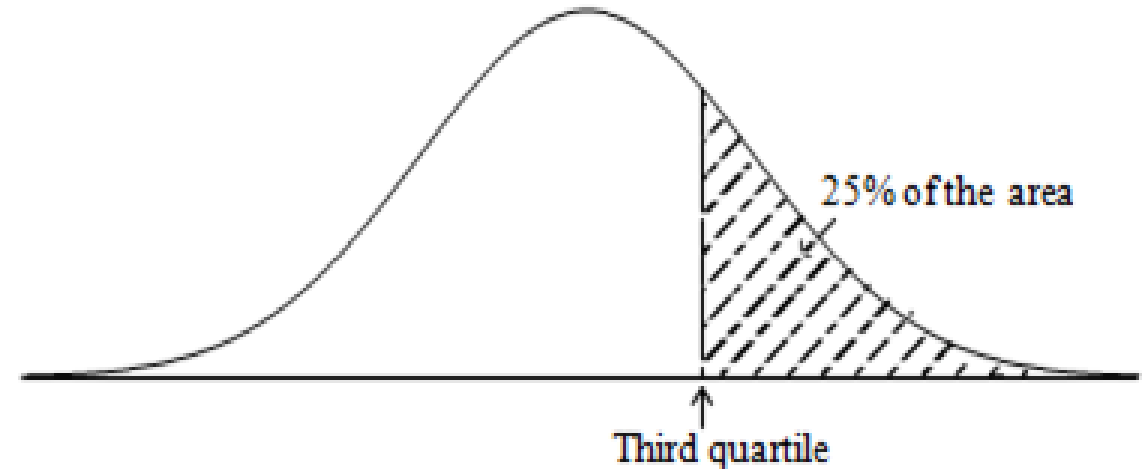
- It is easy to see how to divide the area in Figure into two equal parts, since the graph is symmetric.
- The point which gives us 50% of the area to the left of it and 50% to the right of it is called the second quartile or median
- Second quartile is calculated using the value $0.5(n+1)$



Measures of Spread: Quartiles

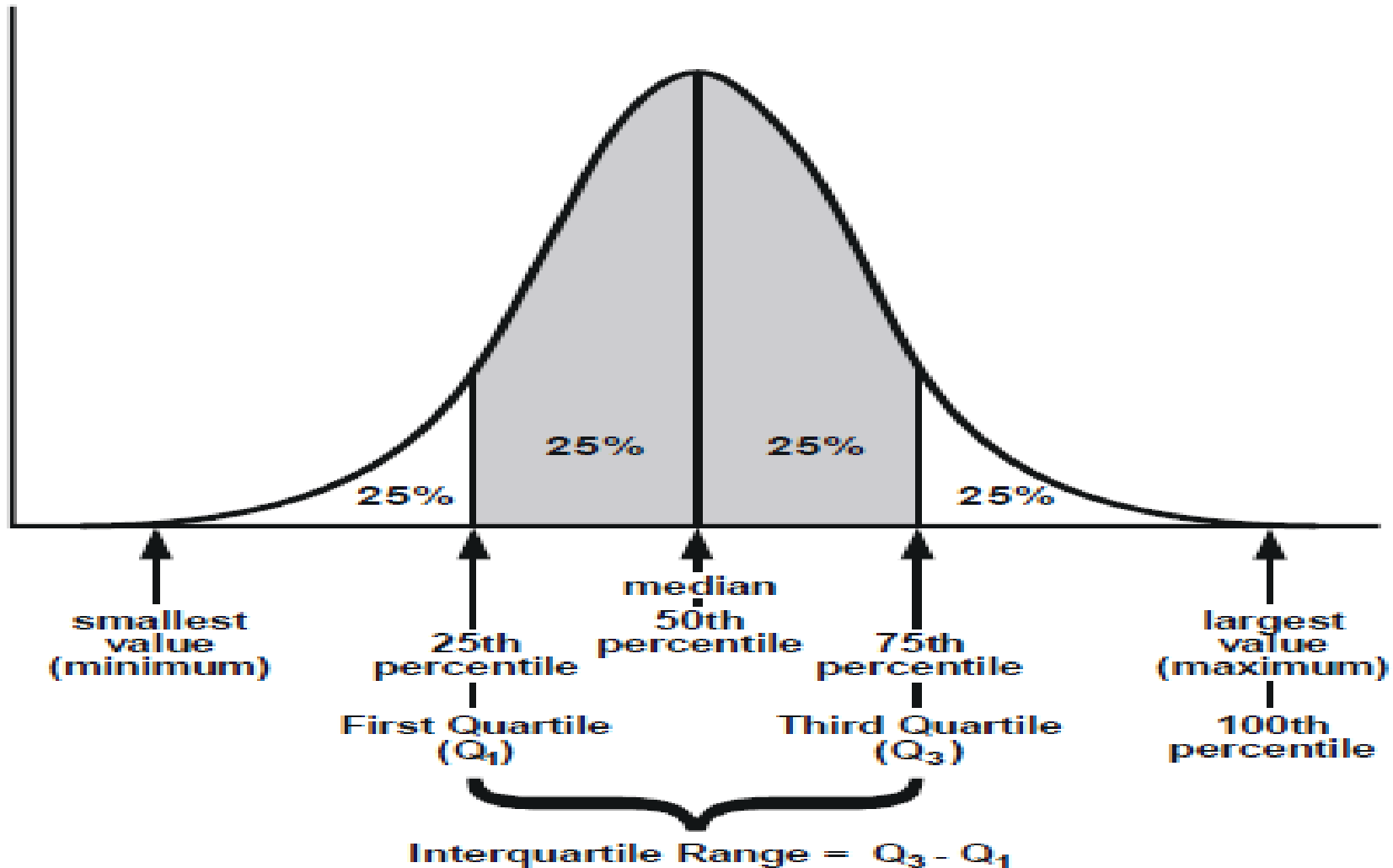
→ The Third Quartile:

- The third quartile is the point which gives us 75% of the area to the left of it and 25% of the area to the right of it.
- This means that 75% of the observations are less than or equal to the third quartile and 25% of the observation are greater than or equal to the third quartile.
- The third quartile is also called the 75th percentile.
- The third quartile is computed in the same way, except that the value $0.75(n+1)$ is used.



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Measures of Spread: Quartile Summary



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Measures of Spread: Quartile example

Example: 5, 8, 4, 4, 6, 3, 8

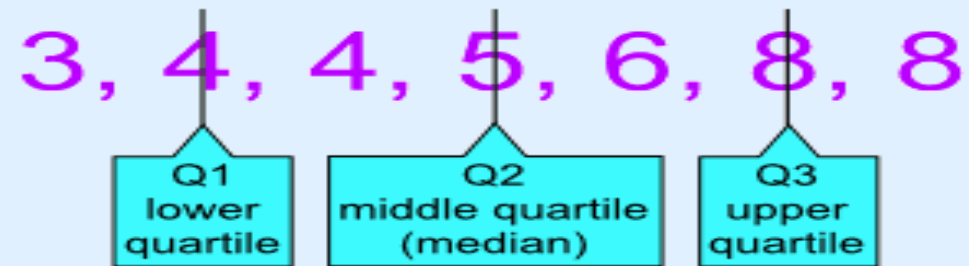
Calculate Q_1 , Q_2 , Q_3 and IQR

Measures of Spread: Quartile example

Example: 5, 8, 4, 4, 6, 3, 8

Put them in order: 3, 4, 4, 5, 6, 8, 8

Cut the list into quarters:



And the result is:

- Quartile 1 (Q1) = 4
- Quartile 2 (Q2), which is also the Median, = 5
- Quartile 3 (Q3) = 8

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Measures of Spread: Quartile example

Example: 1, 3, 3, 4, 5, 6, 6, 7, 8, 8

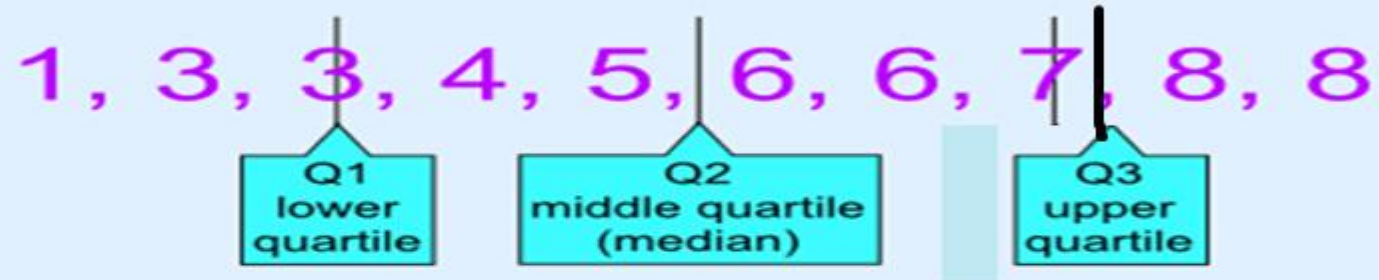
Calculate Q_1 , Q_2 , Q_3

Measures of Spread: Quartile example

Example: 1, 3, 3, 4, 5, 6, 6, 7, 8, 8

The numbers are already in order

Cut the list into quarters:



In this case Quartile 2 is half way between 5 and 6:

$$Q2 = (5+6)/2 = 5.5$$

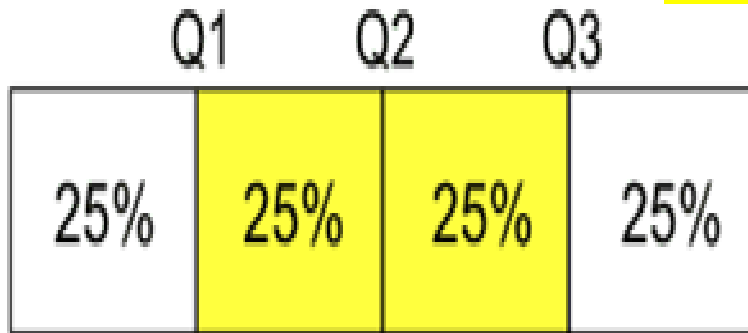
And the result is:

- Quartile 1 (Q1) = 3
- Quartile 2 (Q2) = 5.5
- Quartile 3 (Q3) = 7.5

Measures of Spread: Inter-quartile Range

- Interquartile range is the distance or range between the 25th percentile and the 75th percentile.
- That is, quantifies the difference between the third and first quartiles.
- **Interquartile Range = Upper Quartile(Q3) – Lower Quartile(Q1)**

$$\text{IOR} = Q3 - Q1$$



Interquartile Range
 $= Q3 - Q1$

Measures of Spread: Inter-quartile Range Median approach)

→ Steps to find IQR :

1. Arrange the data scores in ascending order.
2. Find the median of the data set(the number in the middle).
3. Find the median of the lower half of the scores (Q1).
4. Find the median of the upper half of the scores (Q3).

Note: If the number of scores is even, the median is the average of the two middle scores.

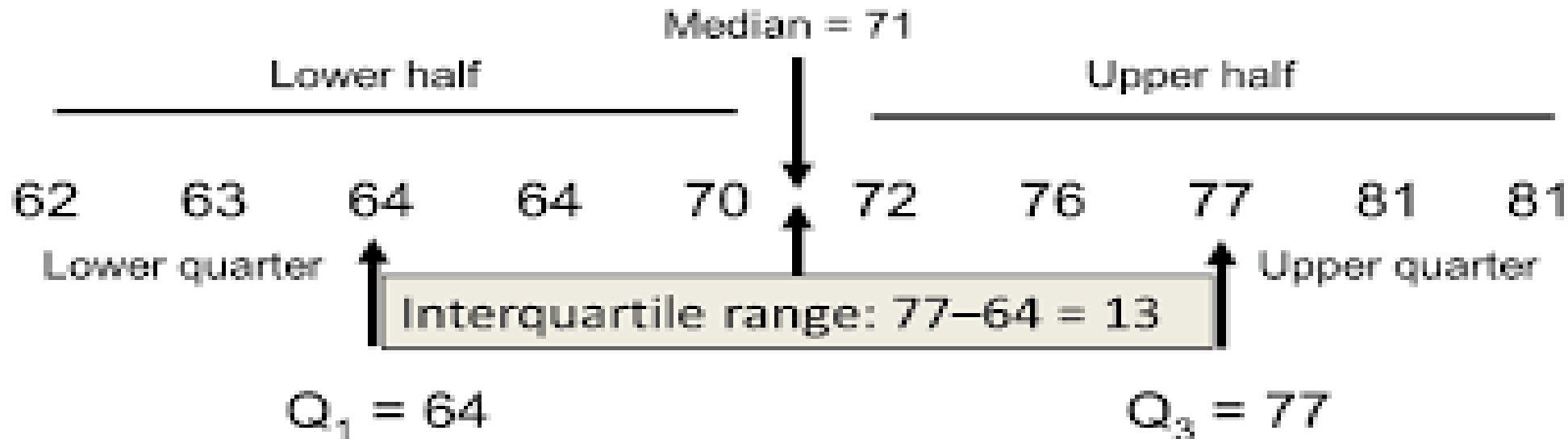
Measures of Spread: Inter-quartile Range (Median approach)

The marks obtained by 10 students in a test are:

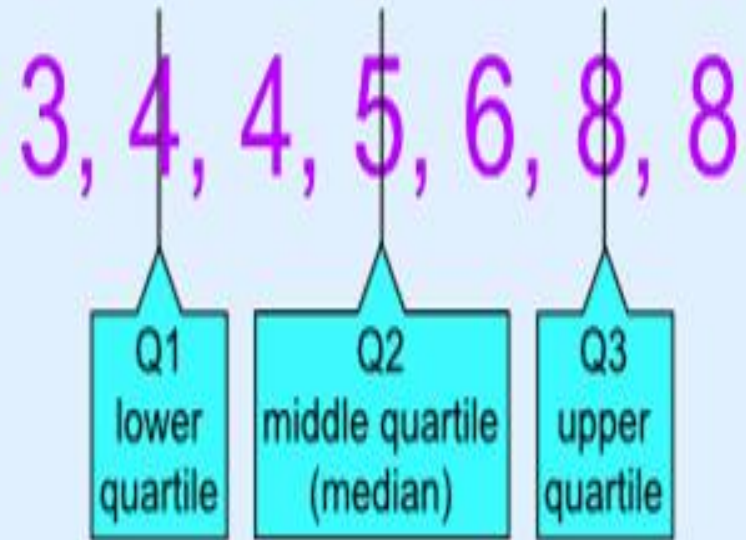
62, 63, 64, 64, 70, 72, 76, 77, 81, 81.

Using the data, find:

1. The first quartile (Q_1)
2. The third quartile (Q_3)
3. The interquartile range (IQR)



Measures of Spread: Inter-quartile Range example



The Interquartile Range is:

$$Q3 - Q1 = 8 - 4 = 4$$

Measures of Spread: Inter-quartile Range question

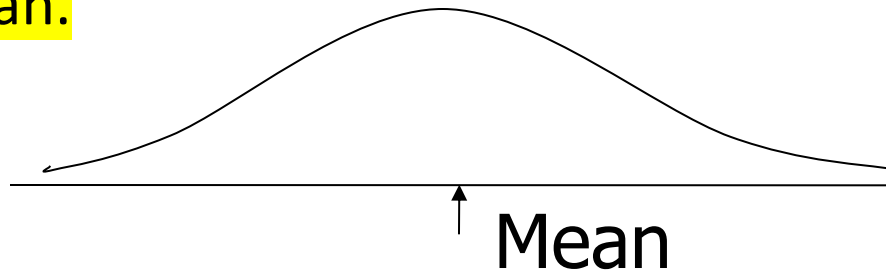
For the following data sets, calculate the quartiles and find the interquartile range.

The following numbers represent the time in minutes that twelve employees took to get to work on a particular day.

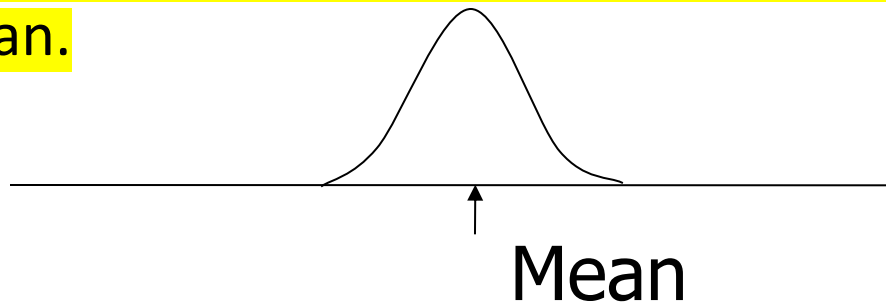
18 34 68 22 10 92 46 52 38 29 45 37

Measures of Spread: Variance

- Variance is a measure of the spread of the recorded values on a variable.
- It is a measure of dispersion, meaning it is a measure of how far a set of numbers is spread out from their average value.
- The larger the variance, the further the individual cases are from the mean.



- The smaller the variance, the closer the individual scores are to the mean.



Measures of Spread: Variance

- It is the average of the distance that each score is from the mean
(Squared deviation from the mean)
- Steps to calculate variance:
 1. Find the mean value of the given data values.
 2. Subtract mean from each data value.
 3. Square each value that is obtained from step2.
 4. Find the sum of all values that is obtained from step 3.
 5. Divide the result that is obtained from step 4 by N(for population) and n-1(for sample).

Variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} \quad \text{Sample Variance}$$

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{Population Variance}$$

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Measures of Spread: Variance - example

Calculate the variance of the following dataset: 46, 69, 32, 60, 52, 41

Solution:

- Find the mean ($\bar{x}$): $\bar{x} = (46 + 69 + 32 + 60 + 52 + 41) \div 6 = 50$
- **Find each score's deviation from the mean.** Subtract the mean from each score to get the deviations from the mean. Since $\bar{x} = 50$, take away 50 from each score.

Score	Deviation from the mean
46	$46 - 50 = -4$
69	$69 - 50 = 19$
32	$32 - 50 = -18$
60	$60 - 50 = 10$
52	$52 - 50 = 2$
41	$41 - 50 = -9$

Measures of Spread: Variance - example

- Square each deviation from the mean
- Add up all of the squared deviations. This is called the sum of squares.
 $16 + 361 + 324 + 100 + 4 + 81 = 886$
- Divide the sum of the squares by $n - 1$ (for a sample) or N (for a population).

Since we're working with a sample, we'll use $n - 1$, where $n = 6$.

$$886 \div (6 - 1) = 886 \div 5 = 177.2$$

- Variance = 177.2

Squared deviations from the mean

$$(-4)^2 = 4 \times 4 = 16$$

$$19^2 = 19 \times 19 = 361$$

$$(-18)^2 = -18 \times -18 = 324$$

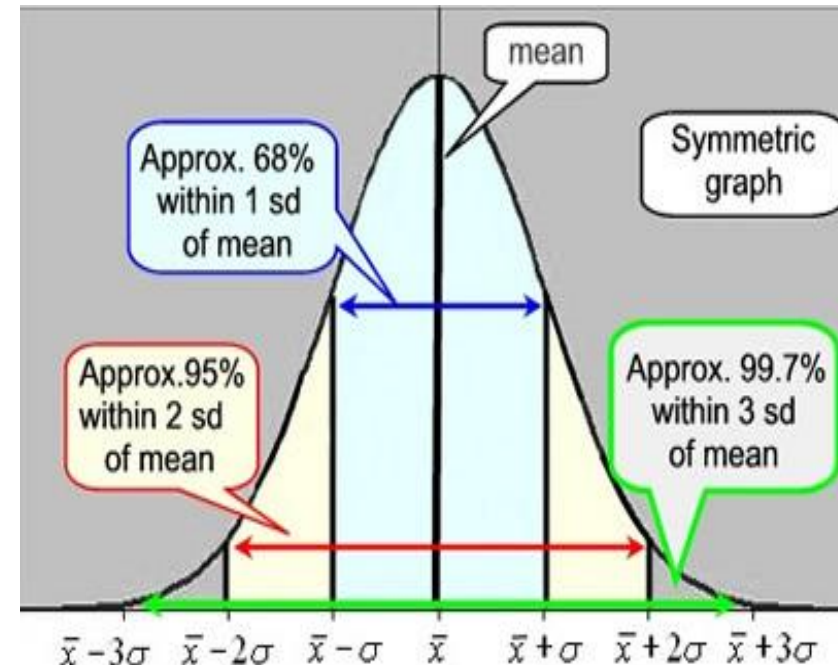
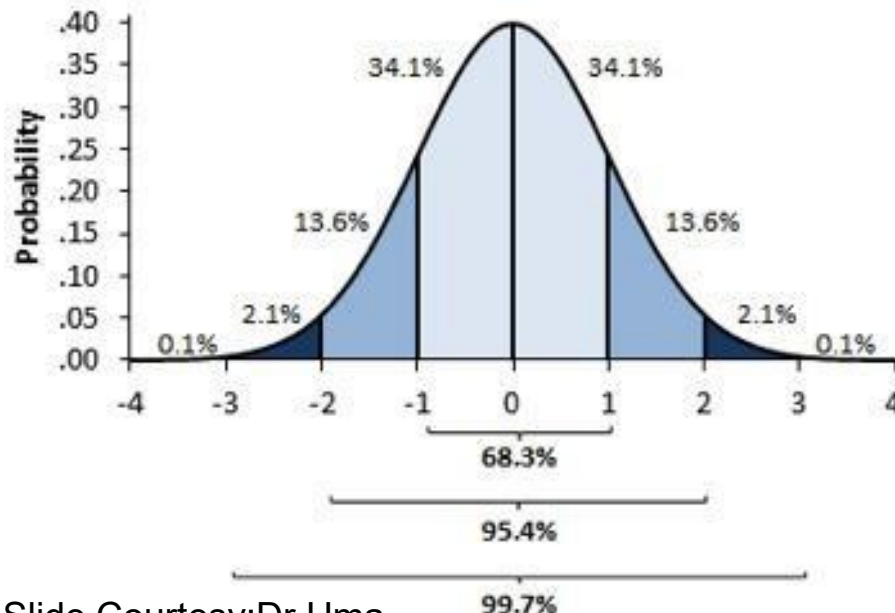
$$10^2 = 10 \times 10 = 100$$

$$2^2 = 2 \times 2 = 4$$

$$(-9)^2 = -9 \times -9 = 81$$

Measures of Spread: Standard Deviation

- Standard deviation signifies the deviation of the elements of the data set from the mean value of the distribution.
- It quantifies the amount of variation of a set of data values.
- It is a measure of the variability of a single item.
- The standard deviation does not decline as the sample size increases.
- The estimate of the standard deviation becomes more stable as the sample size increases.



Measures of Spread: Standard Deviation

- Larger the standard deviation, greater amounts of variation around the mean.
- Std deviation = 0 only when all values are the same (only when you have a constant and not a “variable”)
- If you were to “rescale” a variable, the s.d. would change by the same magnitude.
- Like the mean, the standard deviation will be inflated by an outlier case value.

Measures of Spread: Standard Deviation

Standard Deviation = Square root of Variance

Example

Find the standard deviation and variance

x	$x - \bar{x}$	$(x - \bar{x})^2$
30	4	16
26	0	0
<u>22</u>	-4	16
78		32

Mean = 26

Sum = 0

The variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = 32 / 2 = 16$$

The standard deviation

$$s = \sqrt{16} = 4$$

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Practical Application for Understanding Variance and Standard Deviation

Even though we live in a world where we pay real dollars for goods and services (not percentages of income), most American employers issue raises based on percent of salary. Why do supervisors think the most fair raise is a percentage raise?

Answer:

- 1) Because higher paid persons win the most money.
- 2) The easiest thing to do is raise everyone's salary by a fixed percent.

If your budget went up by 5%, salaries can go up by 5%.

The problem is that the flat percent raise gives unequal increased rewards.

Questions

1. Which of the following is a **measure of the spread (dispersion) of data?**

A) Mean B) Mode C) Median **D) Range**

2. The range of the dataset [3, 5, 7, 9, 11] is:

A) 5 **B) 8** C) 6 D) 11

3. Which measure of spread is most affected by outliers?

A) Interquartile range **B) Range** C) Standard deviation D) Variance

4. If all values in a dataset are the same, then the range is:

A) 1 B) Undefined **C) 0** D) Cannot be determined

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Answers

1. **Correct Answer: D**
2. **Correct Answer: B**
3. **Correct Answer: B**
4. **Correct Answer: C**



Questions

1. The type of data represented by "blood group (A, B, AB, O)" is:

- A) Nominal
- B) Ordinal
- C) Interval
- D) Ratio

2. Which of the following is an example of ordinal data?

- A) Body temperature
- B) Educational qualification (Primary, Secondary, Graduate)
- C) Weight
- D) Blood glucose

3. Temperature measured in Celsius is considered:

- A) Nominal
- B) Ordinal
- C) Interval
- D) Ratio

Questions

4. Which of these is a discrete quantitative variable?

- A) Systolic blood pressure
- B) Number of children in a family
- C) Body mass index
- D) Income in rupees

5. In a controlled study, the control group is the group that:

- A) Receives a double dose of treatment
- B) Is observed but not given any treatment
- C) Is randomly eliminated from the study
- D) Receives the same treatment as the experimental group

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Questions



1. **Correct Answer: A**
2. **Correct Answer: B**
3. **Correct Answer: C**
4. Correct Answer: B
5. Correct Answer: B

References

Text Book:

Statistics for Engineers and Scientists, William Navidi.



THANK YOU

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Chebyshev's Inequality

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Department of Computer Science and Engineering

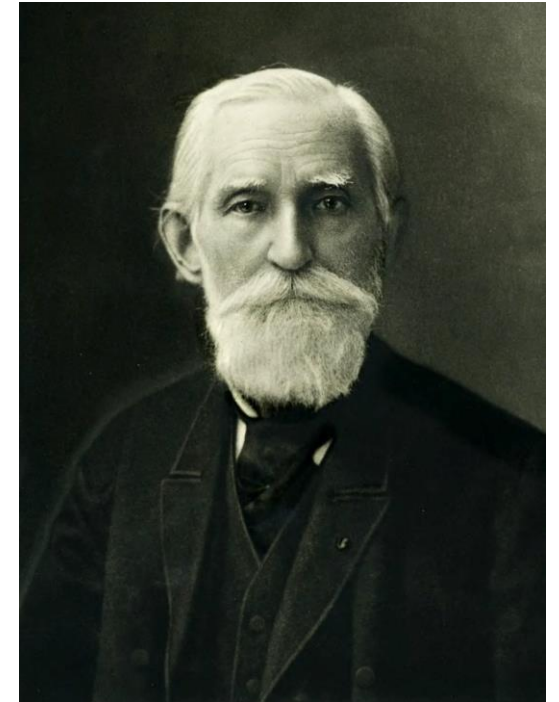
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Chebyshev's Inequality



Chebyshev's inequality

- The mean of a random variable is a measure of the centre of its distribution, and the standard deviation is a measure of the spread.
- *Chebyshev's inequality* relates the mean and the standard deviation by providing a bound on the probability that a random variable takes on a value that differs from its mean by more than a given multiple of its standard deviation.
- Specifically, the probability that a random variable differs from its mean by k standard deviations or more is never greater than $1/k^2$.



Theorem by Russian mathematician Pafnuty Chebyshev

Statement of Chebyshev's Inequality

Chebyshev's inequality states that at least $1 - 1/K^2$ of data from a sample must fall within K standard deviations from the mean, where K is any positive real number greater than one.

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Chebyshev's Inequality is used to describe the percentage of values in a distribution within an interval centered at the mean.

Formula

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

X = random variable

μ = expected value

σ = standard deviation

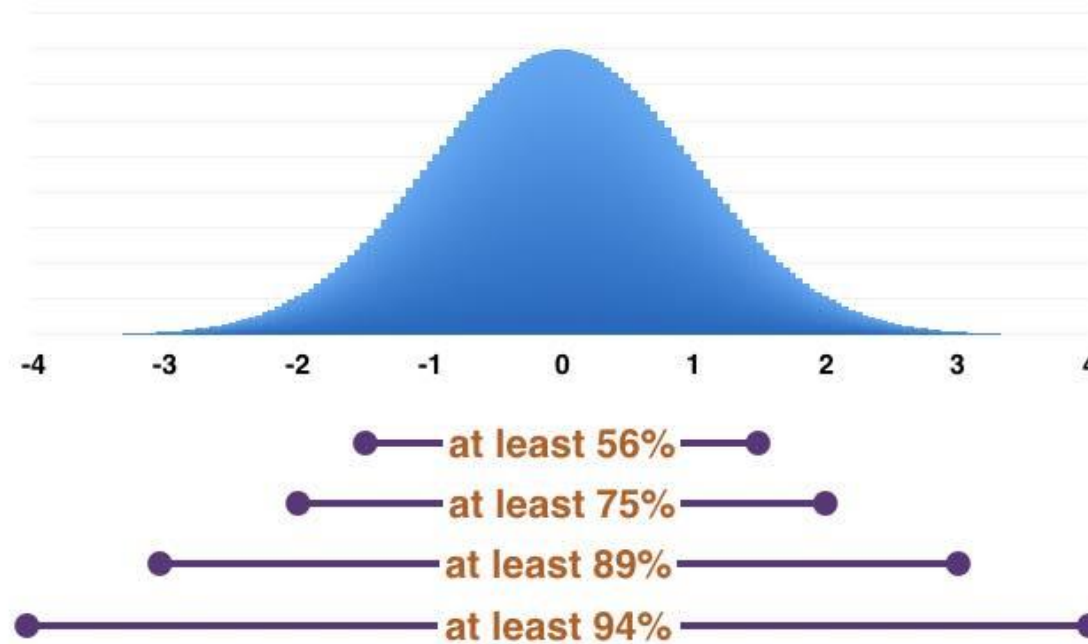
k = number of standard deviations from the mean

Only the case $k > 1$ is useful. When $k \leq 1$ the right hand $1/k^2 \geq 1$ and the inequality is trivial as all probabilities are ≤ 1 .

Chebyshev's Inequality

The proportion of observations within k standard deviations is at least $1 - 1/k^2$

No. Std Devs	$1 - 1/k^2$
1.5	0.56
2.0	0.75
3.0	0.89
4.0	0.94



Problem 1: The length of a rivet manufactured by a certain process has mean $\mu_X = 50$ mm and standard deviation $\sigma_X = 0.45$ mm. What is the largest possible value for the probability that the length of the rivet is outside the interval 49.1–50.9 mm?

Solution

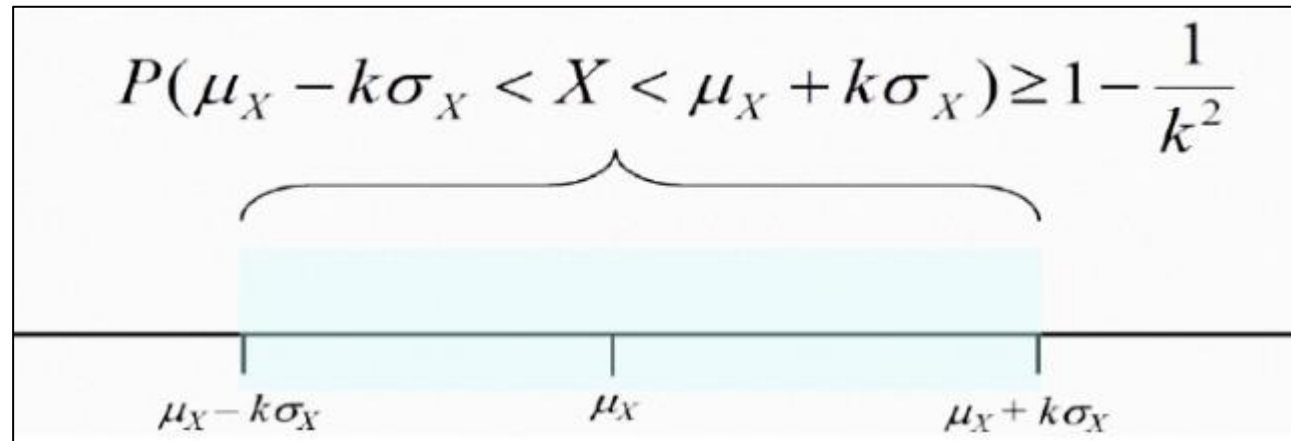
Let X denote the length of a randomly sampled rivet. We must find $P(X \leq 49.1 \text{ or } X \geq 50.9)$. Now

$$P(X \leq 49.1 \text{ or } X \geq 50.9) = P(|X - 50| \geq 0.9) = P(|X - \mu_X| \geq 2\sigma_X)$$

Applying Chebyshev's inequality with $k = 2$, we conclude that

$$P(X \leq 49.1 \text{ or } X \geq 50.9) \leq \frac{1}{4}$$

Chebyshev's inequality states that at least $1-1/K^2$ of data from a sample must fall within K standard deviations from the mean, where K is any positive real number greater than one.



Problem-2:

Assume that the probability density function for X , the length of a rivet in Example 2.46, is

$$f_X(x) = \begin{cases} [477 - 471(x - 50)^2]/640 & 49 \leq x \leq 51 \\ 0 & \text{otherwise} \end{cases}$$

It can be verified that $\mu_X = 50$ and $\sigma_X = 0.45$. Compute the probability that the length of the rivet is outside the interval 49.1–50.9 mm. How close is this probability to the Chebyshev bound of $1/4$?

Solution

$$\begin{aligned}P(X \leq 49.1 \text{ or } X \geq 50.9) &= 1 - P(49.1 < X < 50.9) \\&= 1 - \int_{49.1}^{50.9} \frac{477 - 471(x - 50)^2}{640} dx \\&= 1 - \left. \frac{477x - 157(x - 50)^3}{640} \right|_{49.1}^{50.9} \\&= 0.01610\end{aligned}$$

The actual probability is much smaller than the Chebyshev bound of $1/4$.

- Because the Chebyshev bound is generally much larger than the actual probability, it should only be used **when the distribution of the random variable is unknown.**
- When the distribution is known, then the probability density function or probability mass function should be used to compute probabilities.

- To illustrate the inequality, we will look at it for a few values of K:
 - For $K = 2$ we have $1 - 1/K^2 = 1 - 1/4 = 3/4 = 75\%$.
Chebyshev's inequality says that at least 75% of the data values of any distribution must be within two standard deviations of the mean.
 - For $K = 3$ we have $1 - 1/K^2 = 1 - 1/9 = 8/9 = 89\%$.
So Chebyshev's inequality says that at least 89% of the data values of any distribution must be within three standard deviations of the mean.

Problem 1

Computers from a particular company are found to last on average for three years without any hardware malfunction, with standard deviation of two months. At least what percent of the computers last between 31 months and 41 months?

Use Chebyshev's inequality.

- Mean $\mu = 36$ months, standard deviation $\sigma = 2$ months.
- Interval: 31 to 41 months $\rightarrow$ distance from mean = 5 months.
- Hence $k = \frac{5}{2} = 2.5$.

Chebyshev:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} = 1 - \frac{1}{(2.5)^2} = 1 - \frac{1}{6.25} = 0.84.$$

At least 84% of the computers last between 31 and 41 months.

Problem 2

What is the smallest number of standard deviations from the mean that we must go if we want to ensure that we have at least 50% of the data of a distribution?

Use Chebyshev's inequality:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Require at least 50% inside:

$$1 - \frac{1}{k^2} \geq 0.5 \Rightarrow \frac{1}{k^2} \leq 0.5 \Rightarrow k^2 \geq 2 \Rightarrow k \geq \sqrt{2} \approx 1.414.$$

Smallest number: $\boxed{\sqrt{2} \text{ standard deviations } (\approx 1.414\sigma)}$.

MCQ

1. According to Chebyshev's inequality, the proportion of data within **k standard deviations** of the mean is at least:
 - A) $1/k$
 - B) $1 - 1/k^2$
 - C) $1 - k$
 - D) $1 + 1/k^2$

2. Using Chebyshev's inequality, what proportion of data lies within 3 standard deviations of the mean?
 - A) At least 66.7%
 - B) At least 75%
 - C) At least 88.9%
 - D) At least 95%

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MCQ

1. Option B
2. Option C



Do It Yourself !!!

The length of a metal pin manufactured by a certain process has mean 50 mm and standard deviation 0.45mm.

What is the largest possible value for the probability that the length of the metal pin is outside the interval $[49.1, 50.9]$ mm?



THANK YOU

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Mathematics For Computer Science Engineers

Linear Functions on Random Variables

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Linear Functions on Random Variables

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- We often construct new random variables by performing arithmetic operations on other random variables.
- For example, we might
 - Add a constant to a random variable
 - Multiply a random variable by a constant
 - Add two or more random variables together.

(Combine multiple random variables)



Linear Functions of Random variables – Different Transformations

Addition – Adding a constant to each value of X .

Subtraction – Subtracting a constant from each value of X .

Multiplication – Multiplying each value of X by a constant.

Division – Dividing each value of X by a constant.

where, X represents a Random Variable.

- When a constant is added to a random variable
 - the mean is increased by the value of the constant
 - the variance and standard deviation remains unchanged

If X is a random variable and b is a constant, then

$$\mu_{X+b} = \mu_X + b$$

$$\sigma_{X+b}^2 = \sigma_X^2$$

Multiplying by a Constant

- Often we need to multiply a random variable by a constant
- Multiplication by a constant affects the mean, variance, and standard deviation of a random variable:

- Mean gets multiplied by the constant

If X is a random variable and a is a constant, then

$$\mu_{aX} = a\mu_X$$

- Variance is multiplied by square of the constant
- Standard Deviation gets multiplied by modulus value of the constant

If X is a random variable and a is a constant, then

$$\sigma_{aX}^2 = a^2\sigma_X^2$$

$$\sigma_{aX} = |a|\sigma_X$$

Multiplying and Adding by a Constant

- Multiplication and Addition by a constant affects the mean, variance, and standard deviation of a random variable (Say constant 'a' is multiplied and 'b' is added):
 - Mean gets multiplied by the constant 'a' and the product value is increased by constant 'b'
 - Variance is multiplied by square of the constant 'a'
 - Standard Deviation gets multiplied by modulus value of the constant 'a'

Mean: $\mu(aX + b) = a\mu_X + b$ or $E(aX + b) = aE(X) + b$

Variance: $\sigma^2(aX + b) = a^2\sigma_X^2$ or $V(aX + b) = a^2V(X)$

Standard Deviation: $\sigma(aX + b) = |a|\sigma(X)$

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Applying Transformations to Random Variable X

Transformation	Effect on mean	Effect on Variance	Effect on shape of probability histogram
+ or - a Constant	✓ Changes	X Doesn't change	X Doesn't change
* or / by a Constant	✓ Changes	✓ Changes	X Doesn't change

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Applying Transformations to Random Variable X

Tranformation	Effect on mean	Effect on Variance	Effect on SD	Effect on shape of probability histogram
Add Ex : $Y = X + 2$	$E(Y) = E(X) + 2$	$Var(Y) = Var(X)$	$SD(Y) = SD(X)$	Doesn't change
Subtract Ex : $Y = X - 2$	$E(Y) = E(X) - 2$	$Var(Y) = Var(X)$	$SD(Y) = SD(X)$	Doesn't change
Multiplying by a constant Ex: $Y = X * 2$	$E(Y) = E(X) * 2$	$Var(Y) = Var(X) * 2^2$	$SD(Y) = SD(X) * 2$	Doesn't change
Dividing by a constant Ex: $Y = X/2$	$E(Y) = E(X) / 2$	$Var(Y) = Var(X) / 2^2$	$SD(Y) = SD(X) / 2$	Doesn't change

Example

- Marie has a part-time job walking dogs to earn money on weekends. The following probability distribution represents the probability of having a particular number of clients on any given day. If she earns \$2.75 per client, how much could she expect to earn each day, on average, and what is the standard deviation of her expected earnings?

# clients	20	25	30	35	40
probability	.15	.35	.30	.15	.05

Example

- Start by finding the mean:
- $\mu_x = 20 \times .15 + 25 \times .35 + 30 \times .3 + 35 \times .15 + 40 \times .05 = 28$
- Use the mean to find the variance:
- $\sigma^2_x = (20-28)^2 \times .15 + (25-28)^2 \times .35 + (30-28)^2 \times .30 + (35-28)^2 \times .15 + (40-28)^2 \times .05$
- $= 28.5$
- Use the variance to find the standard deviation:
 $\sigma_x = \sqrt{28.5} = 5.3$
- Now we can find her average income by multiplying the mean, 28 by Marie's rate, \$2.75, to get her **average daily income of \$77**
- Finally, we can multiply the calculated standard deviation, 5.3, by the rate, \$2.75, to get the standard deviation of her income: $5.3 \times \$2.75 = \14.58
- What all this means is that Marie can expect to average \$77 per day, on average, give or take about \$14.58

Combining Random Variables

- There are many instances where it might require us to examine more than a single random variable at once
- We can form new distributions by combining random variables.

If $X_1, \dots, X_n$ are random variables and $c_1, \dots, c_n$ are constants, then the random variable

$$c_1 X_1 + \dots + c_n X_n$$

is called a **linear combination** of $X_1, \dots, X_n$.

- Consider the below scenario where X and Y are two random variables and let T be a combination of both the random variables X and Y then

Mean

$$\text{Adding: } T = X + Y \quad \mu_T = \mu_X + \mu_Y$$

$$\text{Subtracting: } D = X - Y \quad \mu_D = \mu_X - \mu_Y$$

- More generally,

If $X_1, X_2, \dots, X_n$ are random variables, then the mean of the sum $X_1 + X_2 + \dots + X_n$ is given by

$$\mu_{X_1+X_2+\dots+X_n} = \mu_{X_1} + \mu_{X_2} + \dots + \mu_{X_n} \quad (2.47)$$

If X and Y are random variables, and a and b are constants, then

$$\mu_{aX+bY} = \mu_{aX} + \mu_{bY} = a\mu_X + b\mu_Y \quad (2.48)$$

More generally, if $X_1, X_2, \dots, X_n$ are random variables and $c_1, c_2, \dots, c_n$ are constants, then the mean of the linear combination $c_1 X_1 + c_2 X_2 + \dots + c_n X_n$ is given by

$$\mu_{c_1 X_1 + c_2 X_2 + \dots + c_n X_n} = c_1 \mu_{X_1} + c_2 \mu_{X_2} + \dots + c_n \mu_{X_n} \quad (2.49)$$

What are Independent random Variables?

- Two random variables are independent if **knowledge concerning one of them does not affect the probabilities of the other.**
- The notion of independence for random variables is very much like the notion of independence for events.
- When **two events are independent**, the probability that both occur is found by multiplying the probabilities for each event i.e.
$$P(X=a \text{ and } Y=b) = P(X=a) * P(Y=b)$$

If X and Y are **independent** random variables, and S and T are sets of numbers, then

$$P(X \in S \text{ and } Y \in T) = P(X \in S)P(Y \in T) \quad (2.50)$$

More generally, if $X_1, \dots, X_n$ are independent random variables, and $S_1, \dots, S_n$ are sets, then

$$\begin{aligned} P(X_1 \in S_1 \text{ and } X_2 \in S_2 \text{ and } \dots \text{ and } X_n \in S_n) = \\ P(X_1 \in S_1)P(X_2 \in S_2) \dots P(X_n \in S_n) \end{aligned} \quad (2.51)$$

Example

Rectangular plastic covers for a compact disc (CD) tray have specifications regarding length and width. Let X be the length and Y be the width, each measured to the nearest millimeter, of a randomly sampled cover. The probability mass function of X is given by $P(X = 129) = 0.2$, $P(X = 130) = 0.7$, and $P(X = 131) = 0.1$. The probability mass function of Y is given by $P(Y = 15) = 0.6$ and $P(Y = 16) = 0.4$. The area of a cover is given by $A = XY$. Assume X and Y are independent. Find the probability that the area is 1935 mm^2 .

Example

Solution

The area will be equal to 1935 if $X = 129$ and $Y = 15$. Therefore

$$\begin{aligned}P(A = 1935) &= P(X = 129 \text{ and } Y = 15) \\&= P(X = 129)P(Y = 15) \quad \text{since } X \text{ and } Y \text{ are independent} \\&= (0.2)(0.6) \\&= 0.12\end{aligned}$$

If $X_1, X_2, \dots, X_n$ are independent, then the variance of $Y = c_1X_1 + c_2X_2 + \dots + c_nX_n$ is

$$V(Y) = c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \dots + c_n^2\sigma_n^2,$$

When all Constants $c_1, c_2, \dots, c_n = 1$ then

$$\sigma_{X_1+X_2+\dots+X_n}^2 = \sigma_{X_1}^2 + \sigma_{X_2}^2 + \dots + \sigma_{X_n}^2$$

If X and Y are *independent* random variables with variances σ_X^2 and σ_Y^2 , then the variance of the sum $X + Y$ is

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 \quad (2.54)$$

The variance of the difference $X - Y$ is

$$\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2 \quad (2.55)$$

The fact that the variance of the difference is the *sum* of the variances may seem counterintuitive. However, it follows from Equation (2.53) by setting $c_1 = 1$ and $c_2 = -1$.

Example

A piston is placed inside a cylinder. The clearance is the distance between the edge of the piston and the wall of the cylinder and is equal to one-half the difference between the cylinder diameter and the piston diameter. Assume the piston diameter has a mean of 80.85 cm with a standard deviation of 0.02 cm. Assume the cylinder diameter has a mean of 80.95 cm with a standard deviation of 0.03 cm. Find the mean clearance. Assuming that the piston and cylinder are chosen independently, find the standard deviation of the clearance.

Example

Solution

Let X_1 represent the diameter of the cylinder and let X_2 the diameter of the piston. The clearance is given by $C = 0.5X_1 - 0.5X_2$. Using Equation (2.49), the mean clearance is

$$\begin{aligned}\mu_C &= \mu_{0.5X_1 - 0.5X_2} \\ &= 0.5\mu_{X_1} - 0.5\mu_{X_2} \\ &= 0.5(80.95) - 0.5(80.85) \\ &= 0.050\end{aligned}$$

Since X_1 and X_2 are independent, we can use Equation (2.53) to find the standard deviation σ_C :

$$\begin{aligned}\sigma_C &= \sqrt{\sigma_{0.5X_1 - 0.5X_2}^2} \\ &= \sqrt{(0.5)^2\sigma_{X_1}^2 + (-0.5)^2\sigma_{X_2}^2} \\ &= \sqrt{0.25(0.02)^2 + 0.25(0.03)^2} \\ &= 0.018\end{aligned}$$

- Here's a few important points about combining random variables:
 - Make sure that the **random variables are independent** or that it's reasonable to assume independence, before combining variances.
 - **Even when we subtract random variables, we still add their variances**; subtracting two variables increases the overall variability in the outcomes.
 - We can find the **standard deviation of the combined distributions** by taking the square root of the combined variances.

Independence and Simple Random Samples

- When a simple random sample of numerical values is drawn from a population, each item in the sample can be considered as random variables.
- Also the items in a simple random sample may be treated as **independent** when the sample is a **small proportion** ($\leq 5\%$) of a finite population.

If $X_1, X_2, \dots, X_n$ is a simple random sample then $X_1, X_2, \dots, X_n$ can be treated as independent random variables all with the same distribution

- When $X_1, X_2, \dots, X_n$ are independent random variables, all with the same distribution, it is sometimes said that $X_1, X_2, \dots, X_n$ are ***independent and identically distributed (i.i.d.)***.
- **Independent** : outcome of one observation does not affect the outcome of other observation.
- **Identically Distributed**: They have same mean and variance.

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Example

If X and Y are independent random variables such that $E(X) = 9.5$ and $E(Y) = 6.8$, $SD(X) = 0.4$ and $SD(Y) = 0.1$

Find Means and SD of the following:

- 1) $3X$
- 2) $Y - X$
- 3) $X + 4Y$

Example

Solution:

	Mean	SD
X	9.5	0.4
Y	6.8	0.1
3X	$3 * 9.5 = 28.5$	$3 * 0.4 = 1.2$
$Y - X$	$6.8 - 9.5 = -2.7$	$\text{Sqrt}(0.1^2 + 0.4^2) = 0.4123$
$X + 4Y$	$9.5 + 4 * 6.8 = 36.7$	$\text{Sqrt}(0.4^2 + (4^2 * 0.1^2)) = 0.5656$

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Combinations of Two Random Variables

- Means combine very easily with addition or subtraction.
- We can't add standard deviations.
- Variances can be added,
- We can't add standard deviations. square root of variance is the standard deviation.
- Add variances even if subtracting the random variable

Problems for Practice

Example 1 (Discrete Random Variable)

Find $E(Y)$ If $Y = X^2 - 2X + 6$ given that X is a random variable with the following probability distribution:

x	0	1	2	3
$f(x)$	$\frac{1}{6}$	$\frac{1}{2}$	0	$\frac{1}{3}$

Example 1: Solution

$$E(Y) = E(X^2 - 2X + 6) \rightarrow E(Y) = E(X^2 - 2X + 6) \rightarrow E(Y) = E(X^2) - 2E(X) + 6$$

$$E(X^2) = 0^2 \left(\frac{1}{6} \right) + 1^2 \left(\frac{1}{2} \right) + 2^2 (0) + 3^2 \left(\frac{1}{3} \right) = \frac{7}{2}$$

$$E(X) = 0 \left(\frac{1}{6} \right) + 1 \left(\frac{1}{2} \right) + 2(0) + 3 \left(\frac{1}{3} \right) = \frac{3}{2}$$

$$E(Y) = E(X^2 - 2X + 6) \rightarrow E(Y) = \frac{7}{2} - 2 \left(\frac{3}{2} \right) + 6 = \frac{13}{2}$$

$$E(Y) = \frac{13}{2}$$

Example 2 (Continuous Random Variable)

If $f(x) = \frac{1}{3}x^2, -1 \leq x \leq 2$ and $g(X) = 4X + 5$, find $E(g(X))$:

Solution:

$$E(g(X)) = E(4X + 5) = 4E(X) + 5 \text{ where } E(X) = \int_a^b xf(x)dx$$

$$E(g(X)) = 4 \int_{-1}^2 \left(\frac{1}{3}x^3 \right) + 5 = 4 \left[\frac{1}{12}x^4 \right]_{-1}^2 + 5 = 4 \left(\frac{5}{4} \right) + 5$$

$$E(g(X)) = E(4X + 5) = 10$$

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Example 3

let X represent the number that occurs when a blue die is tossed and Y , the number that happens when an orange die is tossed. Calculate the expected value of the linear combination $3X - Y$ of the random variables X and Y



Example 3: Solution

$$X = \text{Blue} = \{1, 2, 3, 4, 5, 6\}$$

X	1	2	3	4	5	6
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$Y = \text{Orange} = \{1, 2, 3, 4, 5, 6\}$$

Y	1	2	3	4	5	6
$f(y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \sum xf(x) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

$$E(Y) = \sum yf(y) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

$$E(3X - Y) = 3E(X) - E(Y)$$

$$E(3X - Y) = 3\left(\frac{7}{2}\right) - \left(\frac{7}{2}\right) = 7$$

Example 4

A certain bowling league has weekly matches where two teams compete against each other. Each team has four players whose individual scores are combined to form their team's overall score that week. Based on data from previous matches, the table below shows summary statistics for the weekly scores of two teams.

Team	Mean	Standard deviation
Bowling Buds	$\mu_B = 610$	$\sigma_B = 60$
Alleycats	$\mu_A = 630$	$\sigma_A = 80$

Let A represent the Alleycats' score on a randomly chosen week, B represent the Bowling Buds' score on a randomly chosen week, and D represent the difference between their scores ($D = A - B$). Assume that the team's scores are independent from each other.

Find the standard deviation of D .

Example 4: Solution

The variance of the difference of two **independent** random variables X and Y is equal to the sum of their variances.

That is, if X and Y are **independent** random variables, $\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2$.

Since we can assume that their scores are independent, the variance of the difference between their scores is

$$\sigma_D^2 = \sigma_A^2 + \sigma_B^2$$

$$= 80^2 + 60^2$$

$$= 10,000$$

and the standard deviation is

$$\sigma_D = \sqrt{\sigma_A^2 + \sigma_B^2} = \sqrt{10,000} = 100$$

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Example 5

A factory process requires 3 steps to finish a product. The steps each have a mean completion time of $\mu = 10$ seconds and a standard deviation of $\sigma = 5$ seconds. The time it takes to complete each step is independent from the other steps. Let T be the total completion time for the 3 steps.

Find the standard deviation of T .

Example 5: Solution

The variance of the sum of **independent** random variables X , Y , and Z is equal to the sum of their variances.

That is, if X , Y , and Z are **independent** random variables, $\sigma_{X+Y+Z}^2 = \sigma_X^2 + \sigma_Y^2 + \sigma_Z^2$.

Since we were told that the times are independent, the variance of the total time is

$$\sigma_T^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2$$

$$= 5^2 + 5^2 + 5^2$$

$$= 75$$

and the standard deviation is

$$\sigma_T = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2} = \sqrt{75}$$

Example 6

SRS travels offers a half-day trip in a tourist area. There must be at least 2 passengers for the visit to run. The vehicle provided by SRS travels can hold up to 6 passengers.

SRS travels charges Rs. 150 per passenger. The amount spent on petrol and permit by SRS travels per trip is Rs. 100. Number of passengers that turn up on a randomly selected day(X) and the corresponding probabilities are given below.

X	2	3	4	5	6
$p(x)$	0.15	0.25	0.35	0.20	0.05

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Example 6: Solution

X	2	3	4	5	6
p(x)	0.15	0.25	0.35	0.20	0.05

Define new random variables for the following:

1. The amount SRS travels collects on a randomly selected day and write the probability distribution function
(Let the r.v. be Y).
2. Profit made by SRS travels on a randomly selected day.
(Let the r.v. be Z).

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Example 6: Solution

Probability Distribution	X	p(x)	Y (= 150 * X)	p(y)	Z (= Y - 100)	p(z)
	2	0.15	300	0.15	200	0.15
	3	0.25	450	0.25	350	0.25
	4	0.35	600	0.35	500	0.35
	5	0.20	750	0.20	650	0.20
	6	0.05	900	0.05	800	0.05
Mean	$E(X) = 3.75$		$E(Y) = 150 * E(X)$ $= 562.5$		$E(Z) = E(Y) - 100$ $= 462.5$	
SD	$SD(X) = 1.09$		$SD(Y) = 150 * SD(X)$ $= 163.5$		$SD(Z) = SD(Y) = 163.5$	

The expectation of a linear function $E[aX+b]$ is equal to

A) $aE(X)+b$

B) $E(a)E(X)+b$

C) $E(X+a+b)$

D) $aE(X+b)a$

If X is a random variable and $Y=3X+5$ then $\text{Var}(Y)=$

A) $3 \cdot \text{Var}(X)$

B) $9 \cdot \text{Var}(X)$

C) $\text{Var}(X)+5$

D) $3+\text{Var}(X)+5$

Example

Do It Yourself !!!

X – represents the number of passengers on a randomly selected trip with SRS travels.

Y – represents the number of passengers on a randomly selected trip with VRL Logistics.

The probability distributions of X and Y are given below:

X	2	3	4	5	6
$p(x)$	0.15	0.25	0.35	0.20	0.05

Y	2	3	4	5
$p(y)$	0.3	0.4	0.2	0.1

Define a new random variable T that represents number of passengers that SRS and VRL can expect on a randomly selected day.

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Example

Do It Yourself !!!

1. Define a new random variable T that represents number of passengers that SRS and VRL can expect on a randomly selected day.
2. Find mean and variance of T .



THANK YOU

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PROBABILITY PLOTS

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Normal Probability Plot

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Topics to be covered...

- ✓The Normal Probability Plot.
- ✓Understanding Q-Q Plot.
- ✓Interpreting the Probability Plots.

Are my data “normal”?

- Not all continuous random variables are normally distributed!!
- It is important to evaluate how well the data are approximated by a normal distribution

Are my data normally distributed?

1. Look at the histogram! Does it appear bell shaped?
2. Compute descriptive summary measures—are mean, median, and mode similar?
3. Do 2/3 of observations lie within 1 std dev of the mean? Do 95% of observations lie within 2 std dev of the mean?
4. Look at a normal probability plot—is it approximately linear?

What are Probability Plots basically mean?

- So far, we have always worked with randomly selected samples from some population.
- We have used an appropriate probability distribution to fit in the data accordingly.
- The probability plot is one way of accessing it through graphical representation.
- By visualizing the data, we can achieve tremendous amount of information.
- For instance our data may be skewed, or be bi-modal, and typically determines the distribution from which population it has come from.

- The data that is been plotted in the theoretical normal distribution should form a straight line. This denotes the normality of the data.
- A straight diagonal line depicts that the data is normally distributed.
- Identifies whether the data is skewed to left or right which does not fit the normal distribution.

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How can I claim that my data is normally distributed?



For larger samples,

- Histogram will have a bell shaped curve which we call as symmetric and there will not be any outliers.
- The mean, median and mode will be similar and lie at the same point.
- In the similar way, 68% of observations lies within one standard deviation of the mean. 95% within two and 99.7% with three standard deviations.

For small samples,

- Histogram does not provides good visual presence, hence to conform its normality we can use Probability Plots.

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Different ways of dividing data equally between 0 and 1.

Methods	Plotting Position Method
Blom	$(i - 0.375)/(n + 0.25)$
Benard	$(i - 0.3)/(n + 0.4)$
Hazen	$(i - 0.5)/n$
Van der Waerden	$i/(n + 1)$
Kaplan-Meier	i/n

Construction of a Probability Plot

- 1) Sort the data.
- 2) Assign evenly spaced values to the data between 0 and 1.
- 3) For each x_i in the data set,

$$\frac{(i - 0.5)}{n}$$

Where,

i is the position of the data item

n is the size of the data set.

- 4) Find theoretical quantiles - Q_i .
- 5) Plot every point (x_i, Q_i) .
- 6) Plot (x_i, x_i)
- 7) Look into the observation whether it forms approximately straight line. This helps us to understand the type of distribution.

Problem – Normal Probability Plot

Problem:

Construct a normal probability plot for the following data. Do these data appear to come from an approximately normal distribution?

3.01, 3.35, 4.79, 5.96, 7.89.

Solution:

Sort the values

3.01, 3.35, 4.79, 5.96, 7.89 / $n = 5$

i	X_i	$\frac{(i - 0.5)}{5}$
1	3.01	0.1
2	3.35	0.3
3	4.79	0.5
4	5.96	0.7
5	7.89	0.9

The value $(i - 0.5)/n$ is chosen to reflect the position of X_i in the ordered sample.

There are $i - 1$ values less than X_i , and i values less than or equal to X_i .

The quantity is a compromise between $(i - 0.5)/n$ proportions and i/n .

The distribution that the sample come from is

$$N(5, 2^2)$$

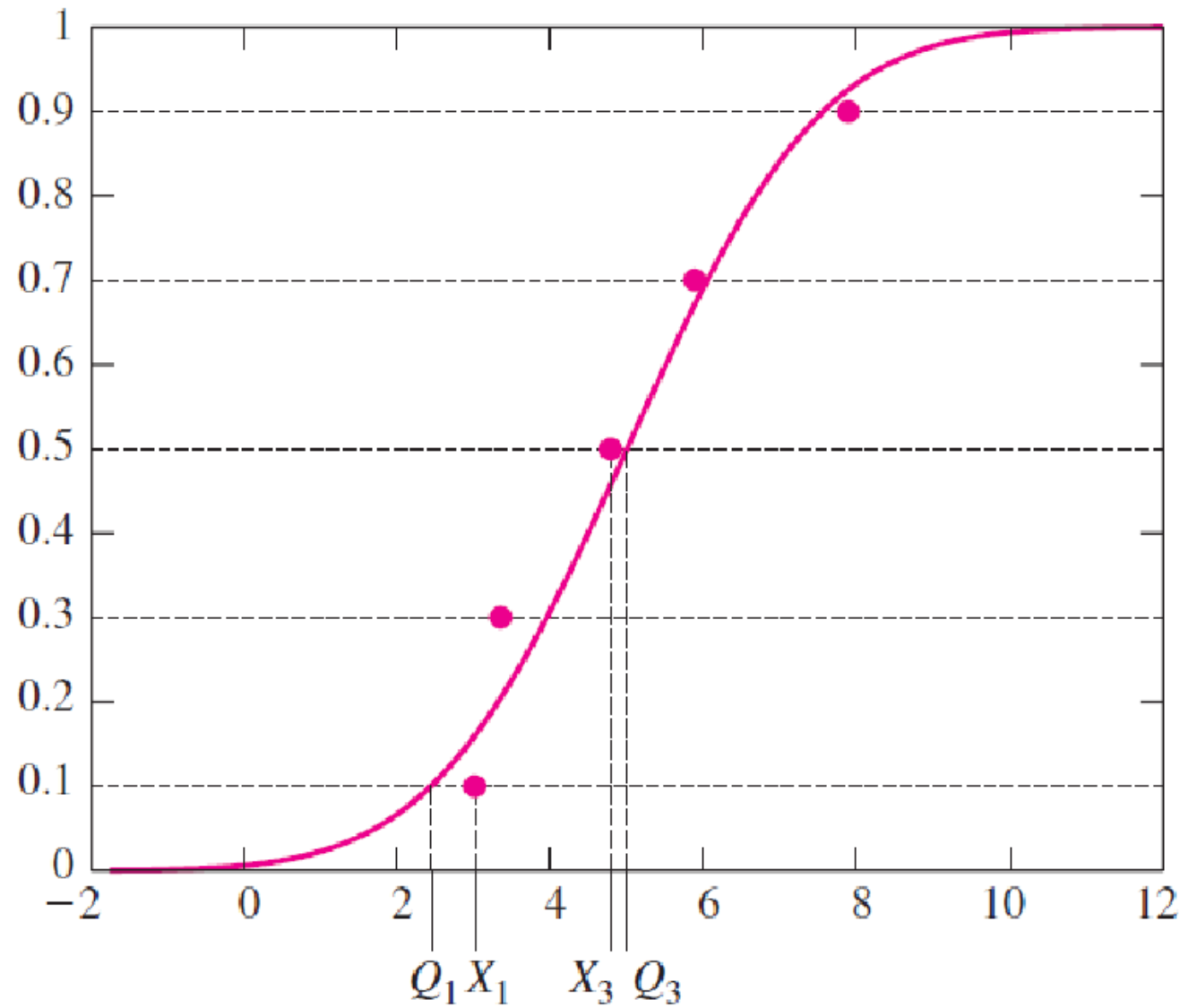


FIGURE 4.21 The curve is the cdf of $N(5, 2^2)$. If the sample points $X_1, \dots, X_5$ came from this distribution, they are likely to lie close to the curve.

Solution - Understanding behind Normal Probability Plot

- From the plot we can infer that $(X_1, 0.1)$ intersects at the point $(Q_1, 0.1)$. We understand that Q_1 is at the 10th percentile of the $N(5, 2^2)$ distribution.
- Applying similar reasoning to the remaining points, we would expect each Q_i to be close to its corresponding X_i by 20th, 30th, 40th and so on.
- The **probability plot** consists of the points (X_i, Q_i) .
- **Since the distribution that** generated the Q_i was a normal distribution, this is called a **normal probability plot**.

Solution - Understanding behind Normal Probability Plot

- If $X_1, \dots, X_n$ do in fact come from the distribution that generated the Q_i , the points should lie close to a straight line.
- To construct the plot, we must compute the Q_i .
- These are the $100(i - 0.5)/n$ percentiles of the distribution that is suspected of generating the sample.
- In this example the Q_i are the 10th, 30th, 50th, 70th, and 90th percentiles of the $N(5, 2^2)$ distribution.
- We could approximate these values by looking up the z-scores corresponding to these percentiles, and then converting to raw scores.

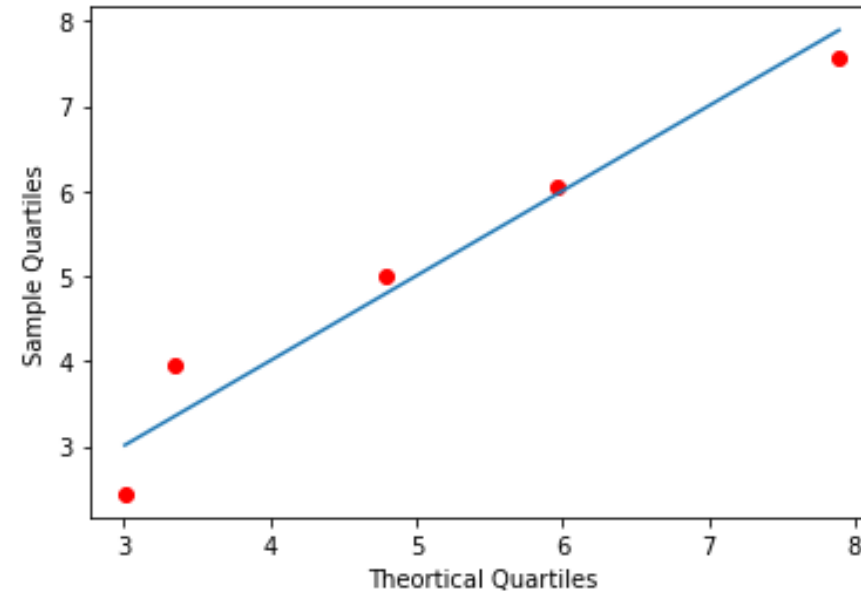
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Solution - Find the Theoretical Quartiles Q_i

i	X_i	$\frac{(i - 0.5)}{5}$	Closest Area in z - Table	Z-score	(Q_i) $X = z * \sigma + \mu$
1	3.01	0.1	0.1003	-1.28	$-1.28 * 2 + 5 = 2.44$
2	3.35	0.3	0.3015	-0.52	$-0.52 * 2 + 5 = 3.95$
3	4.79	0.5	0.5000	0.00	$0.00 * 2 + 5 = 5.00$
4	5.96	0.7	0.6985	0.52	$0.52 * 2 + 5 = 6.05$
5	7.89	0.9	0.8997	1.28	$1.28 * 2 + 5 = 7.56$

Solution - Normal Probability Plot

- The figure shows a normal probability plot for the sample $X_1, \dots, X_5$.
- A straight line is superimposed on the plot, to make it easier to judge whether the points lie close to a straight line or not.
- The sample points are close to the line, so it is quite plausible that the sample came from a normal distribution.
- The sample points $X_1, \dots, X_n$ are called empirical quantiles.



Q-Q Plot

- The points $Q_1, \dots, Q_n$ are called **quantiles** (**divides distribution into equal sized areas**) of the distribution.
- These are the points in the data below which a certain proportion of the data falls.
- The probability plot is sometimes called a **quantile–quantile plot**, or **QQ plot**.
- We can use this Q-Q plot to check the **assumption of Normality** of the data.
- Determines whether if two set of quantiles come from the populations of same distribution. If, yes roughly forms a straight line.

- Probability plots work better with larger samples.
- A good rule of thumb is to require at least 30 points before relying on a probability plot.
- Probability plots can still be used for smaller samples, but they will detect only fairly large departures from normality.

Interpreting Probability Plots

- It's best not to use hard-and-fast rules when interpreting a probability plot. Judge the straightness of the plot by eye.
- When deciding whether the points on a probability plot lie close to a straight line or not, do not pay too much attention to the points at the very ends (high or low) of the sample, unless they are quite far from the line.
- It is common for a few points at either end to stray from the line somewhat.
- However, a point that is very far from the line when most other points are close is an outlier, and deserves attention.

Questions

1. In a normal probability plot, if the data are normally distributed, the points will:

- A) Form a parabola
- B) Scatter randomly
- C) Lie approximately on a straight line
- D) Form a bell-shaped curve

2. What does it indicate if a normal probability plot shows a distinct S-shaped curve?

- A) The data is uniformly distributed
- B) The data is normally distributed
- C) The data has heavy tails (kurtosis)
- D) The data is log-normally distributed

3. Which of the following tools is most similar in purpose to a normal probability plot?

- A) Histogram
- B) Boxplot
- C) Q-Q plot
- D) Pie chart

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Answers

1. C
2. C
3. C





THANK YOU

D. Uma

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Sampling Distribution

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Sampling Distribution

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Sampling Distribution

- In inferential statistics, we want to use characteristics of the **sample** (i.e. a **statistic**) to estimate the characteristics of the **population** (i.e. a **parameter**).
- If we obtain a random sample and calculate a sample statistic from that sample, the sample statistic is a random variable .
- The population parameters, however, are fixed. If the statistic is a random variable, can we find the distribution? The mean? The standard deviation?

- The answer is yes! This is why we need to study the sampling distribution of statistics. So what is a sampling distribution?

Sampling Distributions

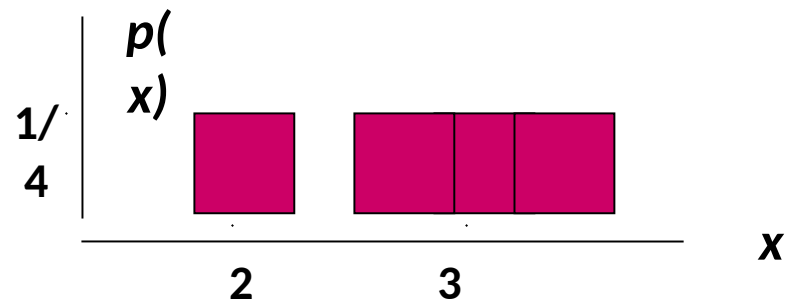
- The sampling distribution of a statistic is a probability distribution based on a large number of samples of size n from a given population.

Population: 3, 5, 2, 1

Draw samples of size $n = 3$ without replacement

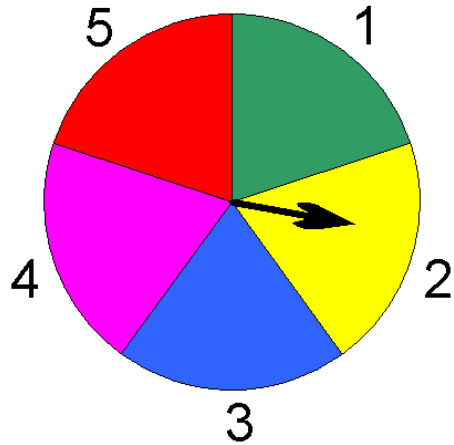
<u>Possible samples</u> $\bar{x}$	
3, 5, 2	$10/3 = 3.33$
3, 5, 1	$9/3 = 3$
3, 2, 1	$6/3 = 2$
5, 2, 1	$8/3 = 2.67$

Each value of $\bar{x}$ is
equally likely, with
probability $1/4$



Sampling Distributions

Consider a population that consists of the numbers 1, 2, 3, 4 and 5 generated in a manner that the probability of each of those values is 0.2 no matter what the previous selections were. This population could be described as the outcome associated with a spinner such as given below with the distribution next to it.



x	$p(x)$
1	0.2
2	0.2
3	0.2
4	0.2
5	0.2

Population Mean

$$\mu = E[X] = \sum k \cdot P(X=k) = (1+2+3+4+5) \cdot 0.2 = 15 \cdot 0.2 = 3$$

If the sampling distribution for the means of samples of size two is analyzed, it looks like

Sample	
1, 1	1
1, 2	1.5
1, 3	2
1, 4	2.5
1, 5	3
2, 1	1.5
2, 2	2
2, 3	2.5
2, 4	3
2, 5	3.5
3, 1	2
3, 2	2.5
3, 3	3

Sample	
3, 4	3.5
3, 5	4
4, 1	2.5
4, 2	3
4, 3	3.5
4, 4	4
4, 5	4.5
5, 1	3
5, 2	3.5
5, 3	4
5, 4	4.5
5, 5	5

	frequency	p(x)
1	1	0.04
1.5	2	0.08
2	3	0.12
2.5	4	0.16
3	5	0.20
3.5	4	0.16
4	3	0.12
4.5	2	0.08
5	1	0.04
25		

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

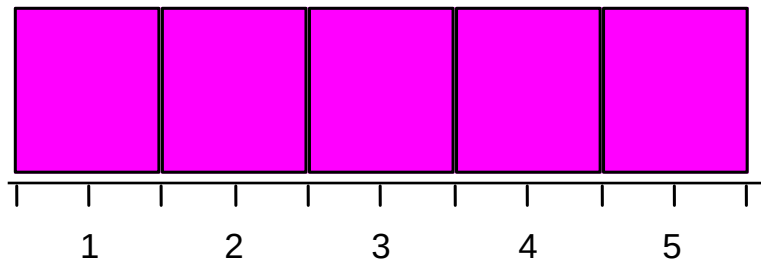
Sampling Distributions

Find the statistic means

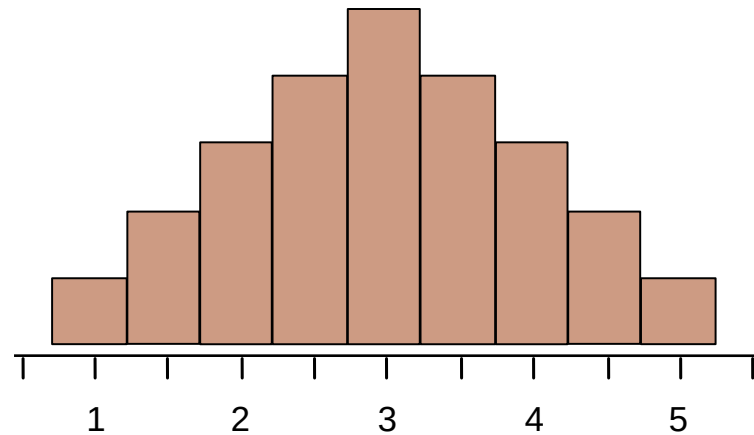
$$\bar{X} = E[X] = \sum$$

$$x \cdot P(x) = 1 \times 0.4 + 1.5 \times 0.08 + 2 \times 0.12 + 2.5 \times 0.16 + 3 \times 0.2 + 3.5 \times 0.16 + 4 \times 0.12 + 4.5 \times 0.08 + 5 \times 0.04 = 3$$

The original distribution and the sampling distribution of means of samples with $n=2$ are given below.



Original distribution

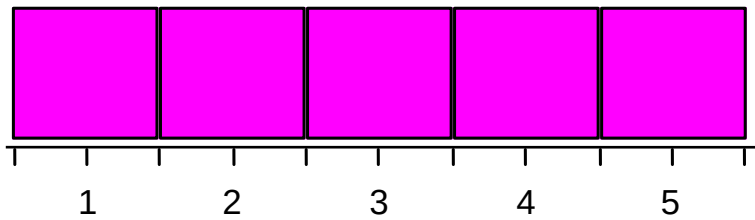


Sampling distribution

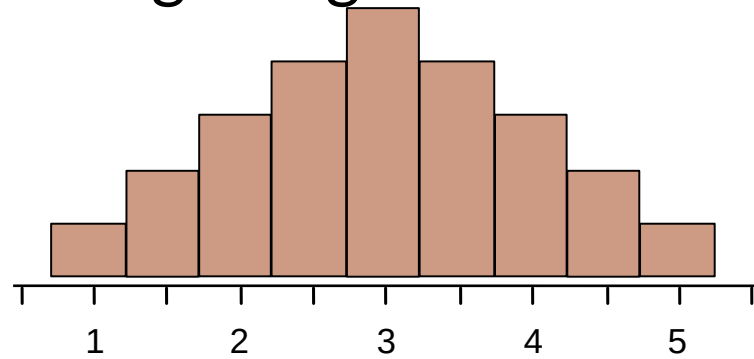
$n = 2$

Sampling Distributions

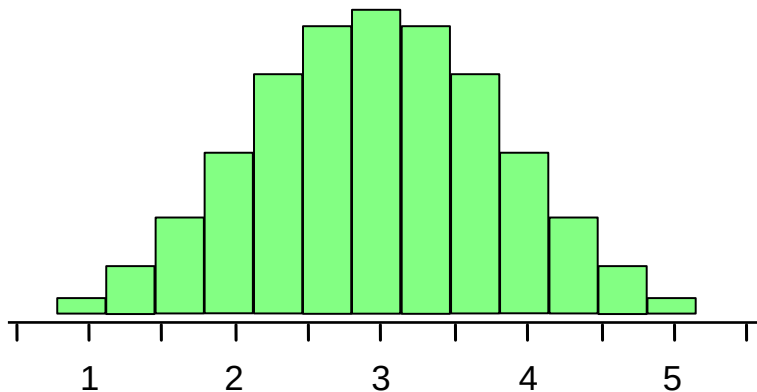
Sampling distributions for $n=3$ and $n=4$ were calculated and are illustrated below. The shape is getting closer and closer to the normal distribution.



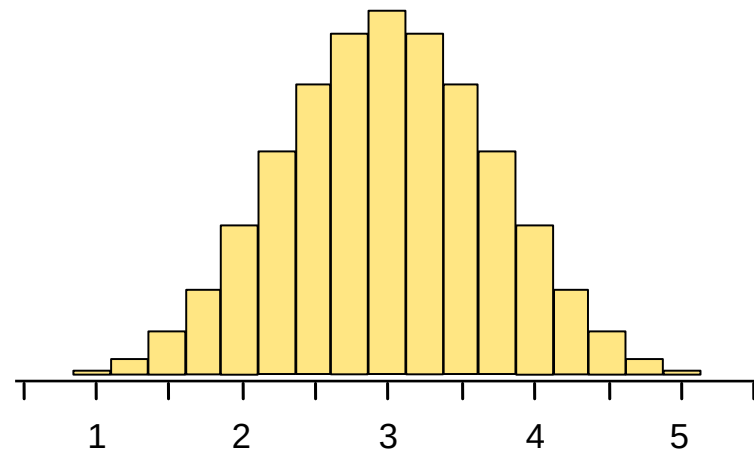
Original distribution



Sampling distribution $n = 2$



Sampling distribution $n = 3$



Sampling distribution $n = 4$

The Mean and Variance of a Sample Mean

- The most frequently encountered linear combination is the sample mean.
- Specifically, if $X_1, \dots, X_n$ is a simple random sample from a population with mean μ and variance σ^2 , then the sample mean $\bar{X}$ is the linear combination. $\bar{X}$ is a weighted sum of the random variables X_i :

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\bar{X} = \frac{1}{n} X_1 + \dots + \frac{1}{n} X_n$$

From this fact we can compute the mean and variance of $\bar{X}$.

$$\begin{aligned} \mu_{\bar{X}} &= \mu_{\frac{1}{n} X_1 + \dots + \frac{1}{n} X_n} \\ &= \frac{1}{n} \mu_{X_1} + \dots + \frac{1}{n} \mu_{X_n} \quad (\text{using Equation 2.49}) \\ &= \frac{1}{n} \mu + \dots + \frac{1}{n} \mu \\ &= (n) \left(\frac{1}{n} \right) \mu \\ &= \mu \end{aligned}$$

X_i are identically distributed, which means they have the same mean μ

The Variance of a Sample Mean

- the items in a simple random sample may be treated as independent random variables.

Therefore

$$\begin{aligned}
 \sigma_{\bar{X}}^2 &= \sigma_{\frac{1}{n}X_1 + \dots + \frac{1}{n}X_n}^2 \\
 &= \frac{1}{n^2}\sigma_{X_1}^2 + \dots + \frac{1}{n^2}\sigma_{X_n}^2 && \text{(using Equation 2.53)} \\
 &= \frac{1}{n^2}\sigma^2 + \dots + \frac{1}{n^2}\sigma^2 \\
 &= (n) \left(\frac{1}{n^2} \right) \sigma^2 \\
 &= \frac{\sigma^2}{n}
 \end{aligned}$$

The Mean and Variance of a Sample Mean

If $X_1, \dots, X_n$ is a simple random sample from a population with mean μ and variance σ^2 , then the sample mean $\bar{X}$ is a random variable with

$$\mu_{\bar{X}} = \mu \quad (2.56)$$

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} \quad (2.57)$$

The standard deviation of $\bar{X}$ is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad (2.58)$$

A process that fill the plastic bottles with a beverage has a mean fill volume of 2.013L and a standard deviation of 0.005L. A case contains 24 bottles. Assuming that the bottles in a case are a simple random sample of bottles filled by this method. Find the mean and standard deviation of the average volume per bottle in a case.

Solution

Let $V_1, \dots, V_{24}$ represent the volumes in 24 bottles in a case. This is a simple random sample from a population with mean $\mu = 2.013$ and standard deviation $\sigma = 0.005$. The average volume is $\bar{V} = (V_1 + \dots + V_{24})/24$. Using Equation (2.56),

$$\mu_{\bar{V}} = \mu = 2.013$$

Using Equation (2.58),

$$\sigma_{\bar{V}} = \frac{\sigma}{\sqrt{24}} = 0.001$$

The Central Limit Theorem

The Central Limit Theorem

Let $X_1, \dots, X_n$ be a simple random sample from a population with mean μ and variance σ^2 .

Let $\bar{X} = \frac{X_1 + \dots + X_n}{n}$ be the sample mean.

Let $S_n = X_1 + \dots + X_n$ be the sum of the sample observations.

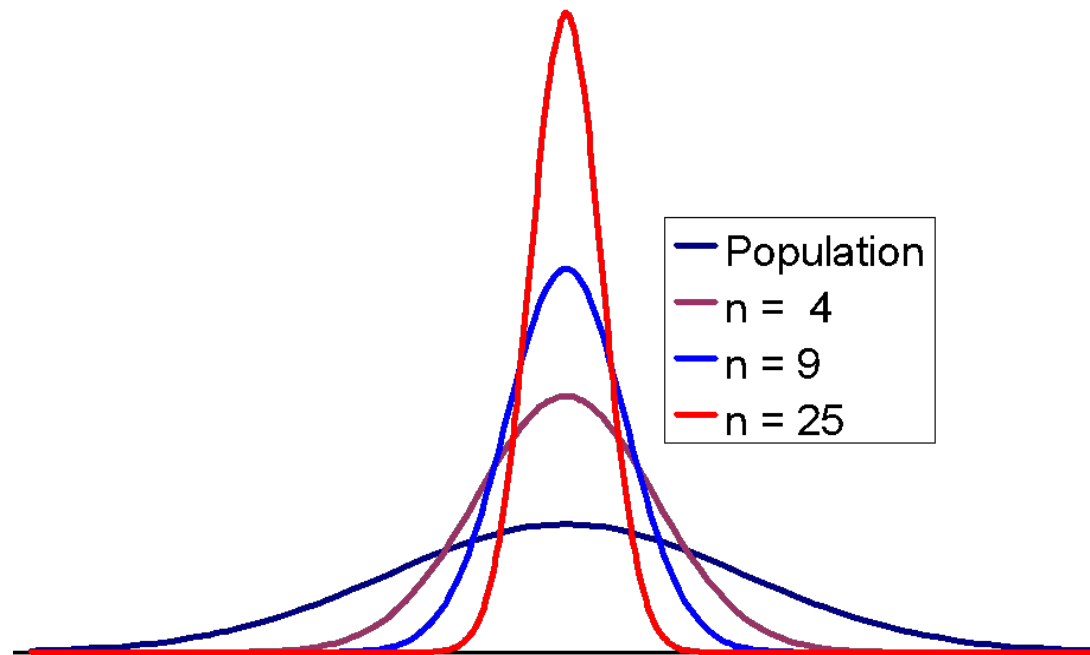
Then if n is sufficiently large,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{approximately} \quad (4.55)$$

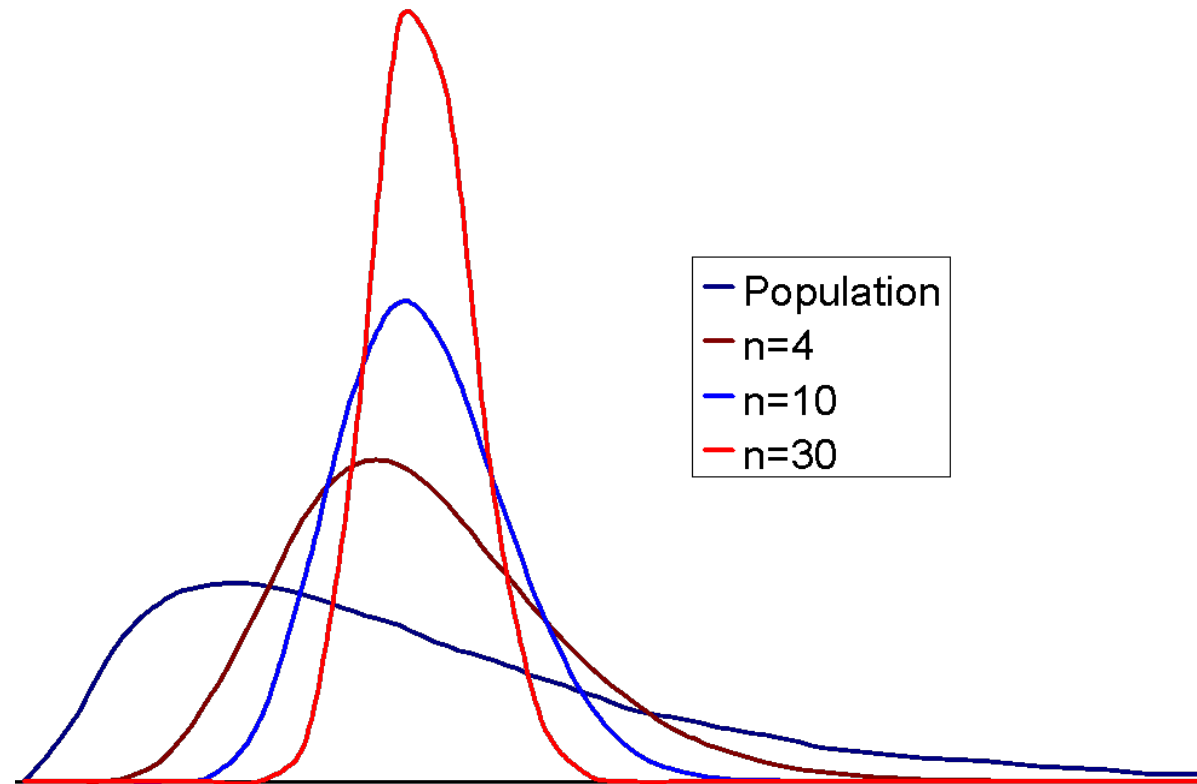
and

$$S_n \sim N(n\mu, n\sigma^2) \quad \text{approximately} \quad (4.56)$$

For most populations, if the sample size is greater than 30, the Central Limit Theorem approximation is good.

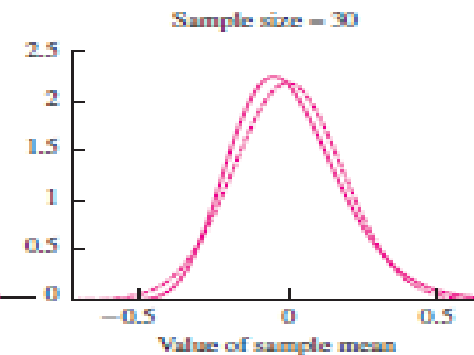
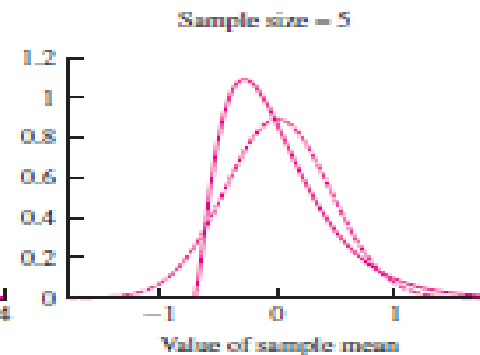
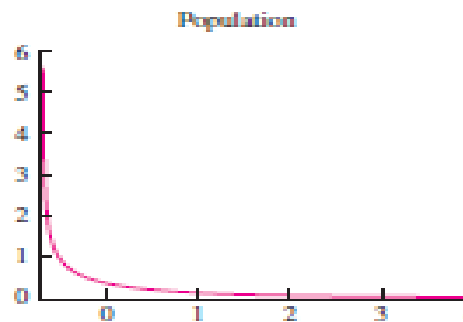
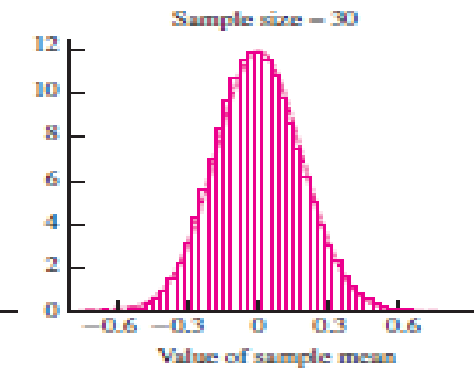
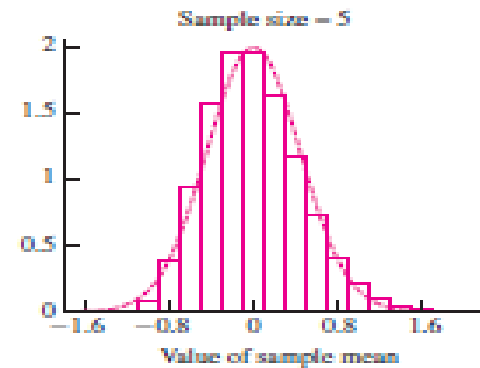
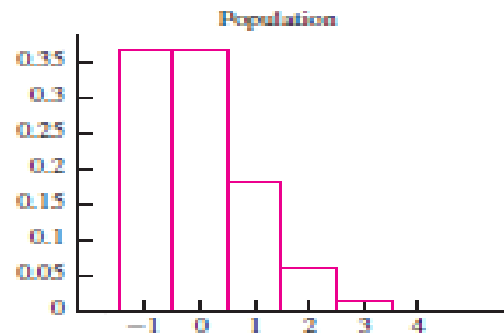
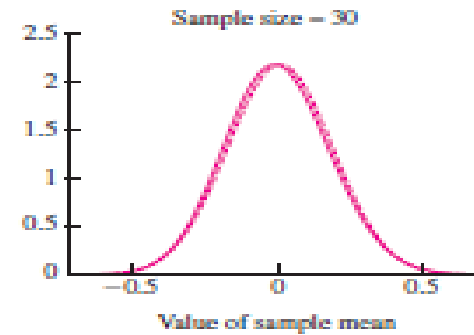
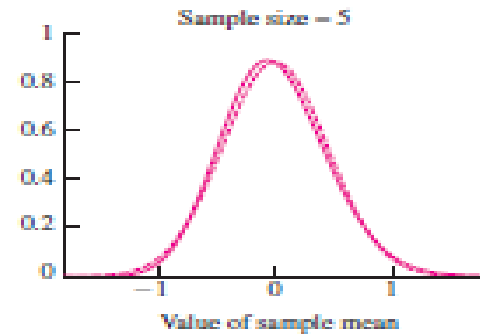
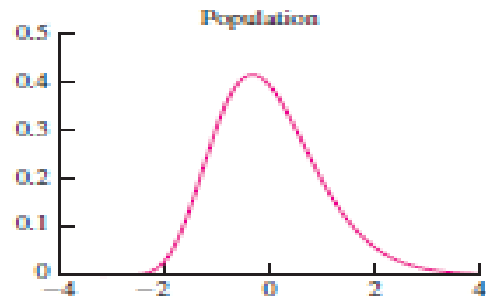


Symmetric normal like population



Skewed population

Illustrations of Sampling Distributions



✓ If the sampling distribution of $\bar{x}$ is normal or approximately normal, *standardize or rescale* the interval of interest in terms of

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

✓ Find the appropriate area using Table.

A random sample of size $n = 16$ from a normal distribution with $\mu = 10$ and $\sigma = 8$. Find the probability of $\bar{X} > 12$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Substitute $\bar{x} = 12$:

$$Z = \frac{12 - 10}{2} = \frac{2}{2} = 1$$

$$P(Z > 1) = 1 - P(Z \leq 1)$$

The cumulative probability $P(Z \leq 1)$ is approximately 0.8413. Thus:

$$P(Z > 1) = 1 - 0.8413 = 0.1587$$

∴, the probability that the sample mean $\bar{x}$ is greater than 12 is approximately 0.1587 or 15.87%.

Example

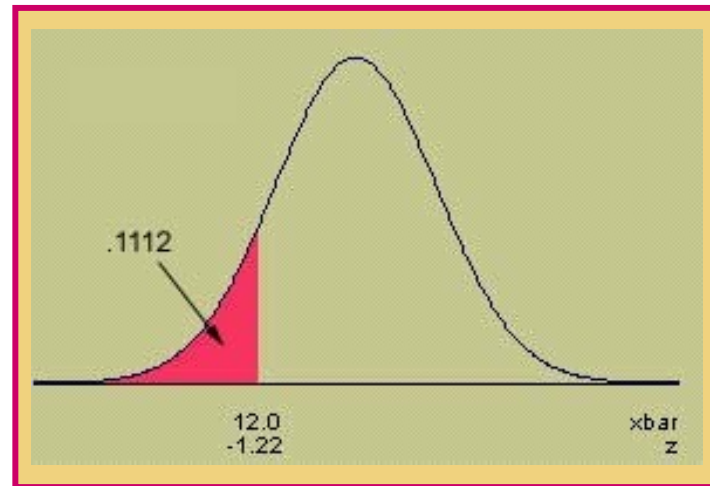
A soda filling machine is supposed to fill cans of soda with 12 fluid ounces. Suppose that the fills are actually normally distributed with a mean of 12.1 oz and a standard deviation of .2 oz.

What is the probability that the average fill for a 6-pack of soda is less than 12 oz?

$$P(\bar{x} < 12) =$$

$$P\left(\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} < \frac{12 - 12.1}{.2 / \sqrt{6}}\right) =$$

$$P(z < -1.22) = .1112$$

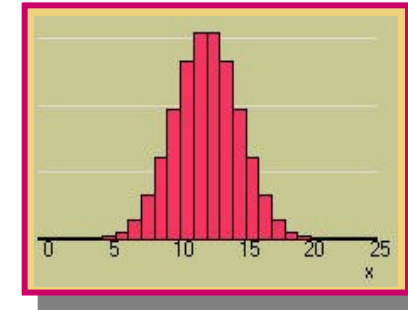


The Sampling Distribution of the Sample Proportion

✓ The **Central Limit Theorem** can be used to conclude that the binomial random variable x is approximately normal when n is large, with mean np and variance npq .

✓ The sample proportion, $\hat{p} = \frac{x}{n}$ is simply a *rescaling* of the binomial random variable x , dividing it by n .

✓ From the Central Limit Theorem, the sampling distribution of $\hat{p}$ will also be **approximately normal**, with a *rescaled* mean and standard deviation.



The Sampling Distribution of the Sample Proportion

✓ A random sample of size n is selected from a binomial population with parameter p .

✓ The sampling distribution of the sample proportion,

$$\hat{p} = \frac{x}{n}$$

$$\sqrt{\frac{pq}{n}}$$

will have mean p and standard deviation

✓ If n is large, and p is not too close to zero or one, the sampling distribution of $\hat{p}$ will be

The standard deviation of p -hat is sometimes called the STANDARD ERROR (SE) of p -hat.

The Sampling Distribution of the Sample Proportion

We define the sample proportion as:

$$\hat{p} = x / n \text{ where}$$

$x \sim \text{Bin}(n, p)$ (i.e., x follows a Binomial distribution with parameters n and p)

We know: $\mu_x = E[x] = np$

$$\sigma_x^2 = \text{Var}(x) = np(1 - p)$$

Now, derive the expected value of the sample proportion:

$$\begin{aligned} E[\hat{p}] &= E[x / n] \\ &= (1 / n) * E[x] \\ &= (1 / n) * np \\ &= p. \end{aligned}$$

Therefore, $E[\hat{p}] = p \Rightarrow \mu_{\hat{p}} = p$

The Sampling Distribution of the Sample Proportion

Derivation of the Variance of the Sample Proportion

We know:

$$\hat{p} = x / n$$

Now, compute the variance:

$$\sigma^2_{\hat{p}} = \text{Var}[\hat{p}] = \text{Var}[x / n]$$

$$= (1 / n^2) * \text{Var}[x]$$

$$= (1 / n^2) * np(1 - p)$$

$$= p(1 - p) / n$$

Therefore,

$$\sigma^2_{\hat{p}} = p(1 - p) / n$$

$$\sigma_{\hat{p}} = \text{sqrt}[p(1 - p) / n]$$

✓ If the sampling distribution of $\hat{p}$ is normal or approximately normal, *standardize or rescale* the interval of interest in terms of

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

✓ Find the appropriate area using Z Table .

If $X \sim \text{Bin}(n, p)$, and if $np > 10$ and $n(1 - p) > 10$, then

$$X \sim N(np, np(1 - p)) \quad \text{approximately}$$

$$\hat{p} \sim N\left(p, \frac{p(1 - p)}{n}\right) \quad \text{approximately}$$

Example

Suppose 3% of the people contacted by phone are receptive to a certain sales pitch and buy your product. If your sales staff contacts 2000 people, what is the probability that more than 100 of the people contacted will purchase your product?

$$n=2000, \quad p=0.03, \quad np=60, \quad nq=1940,$$

OK to use the normal approximation

$$P(\hat{p} > 100 / 2000) = P\left(z > \frac{.05 - .03}{\sqrt{\frac{.03(.97)}{2000}}}\right) = P(z > 5.24) \approx 5.6 \times 10^{-8}.$$

This is **practically zero**.

$$P(X > 100) \approx 0 \quad (\text{virtually impossible})$$

Example

Let X denote the number of flaws in a 1 in. length of copper wire. The probability mass function of X is presented in the following table.

x	$P(X = x)$
0	0.48
1	0.39
2	0.12
3	0.01

One hundred wires are sampled from this population. What is the probability that the average number of flaws per wire in this sample is less than 0.5?

Example

The population mean number of flaws is $\mu = 0.66$, and the population variance is $\sigma^2 = 0.5244$

$$\mu_X = \sum_x x P(X = x)$$

We need to find $P(\bar{x} < 0.5)$.

sample size is $n = 100$, which is a large sample. It follows from the Central Limit Theorem

$$\sigma_X^2 = \sum_x (x - \mu_X)^2 P(X = x)$$

that $\bar{x} \sim N(0.66, 0.005244)$.

The z-score of 0.5 is therefore
 $z = (0.5 - 0.66) / \sqrt{0.005244} = -2.21$

From the z table, the area to the left of -2.21 is 0.0136.

Therefore $P(\bar{x} < 0.5) = 0.0136$, so only 1.36% of samples of size 100 will have fewer than 0.5 flaws per wire.

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Example

At a large university, the mean age of the students is 22.3 years, and the standard deviation is 4 years. A random sample of 64 students is drawn. What is the probability that the average age of these students is greater than 23 years?



Example

Let $X_1, \dots, X_{64}$ be the ages of the 64 students in the sample.

Find $P(\bar{x} > 23)$.

Now the population from which the sample was drawn has mean $\mu = 22.3$ and variance $\sigma^2 = 16$.

The sample size is $n = 64$.

It follows from the Central Limit Theorem that $\bar{x} \sim N(22.3, 0.25)$.

The z-score for 23 is

$$z = 23 - 22.3 / \sqrt{0.25} = 1.40$$

From the z table, the area to the right of 1.40 is 0.0808.

Therefore $P(\bar{x} > 23) = 0.0808$.

Example

The manufacture of a certain part requires two different machine operations. The time on machine 1 has mean 0.4 hours and standard deviation 0.1 hours. The time on machine 2 has mean 0.45 hours and standard deviation 0.15 hours. The times needed on the machines are independent. Suppose that 65 parts are manufactured. What is the distribution of the total time on machine 1? On machine 2? What is the probability that the total time used by both machines together is between 50 and 55 hours?

Example

The Central Limit Theorem

Let $X_1, \dots, X_n$ be a simple random sample from a population with mean μ and variance σ^2 .

Let $\bar{X} = \frac{X_1 + \dots + X_n}{n}$ be the sample mean.

Let $S_n = X_1 + \dots + X_n$ be the sum of the sample observations.

Then if n is sufficiently large,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{approximately} \quad (4.55)$$

and

$$S_n \sim N(n\mu, n\sigma^2) \quad \text{approximately} \quad (4.56)$$

For most populations, if the sample size is greater than 30, the Central Limit Theorem approximation is good.

Example

Let $X_1, \dots, X_{65}$ represent the times of the 65 parts on machine 1.

The population from which this sample was drawn has mean $\mu_x = 0.4$ and standard deviation $\sigma_x = 0.1$.

Let $S_x = X_1 + \dots + X_{65}$ be the total time on machine 1.

It follows from the Central Limit Theorem that

$$S_x \sim N(65\mu_x, 65\sigma_x^2) = N(26, 0.65)$$

Example

Let $Y_1, \dots, Y_{65}$ represent the times of the 65 parts on machine 2.

The population from which this sample was drawn has mean $\mu_Y = 0.45$ and standard deviation $\sigma_Y = 0.15$.

Let $S_Y = Y_1 + \dots + Y_{65}$ be the total time on machine 2.

It follows from the Central Limit Theorem that

$$S_Y \sim N(65\mu_Y, 65\sigma_Y^2) = N(29.25, 1.4625)$$

Example

let $T = S_X + S_Y$ represent the total time on both machines.

Since

$S_X \sim N(26, 0.65)$, $S_Y \sim N(29.25, 1.4625)$, and S_X and S_Y are independent,
it follows that

$\mu_T = 26 + 29.25 = 55.25$, $\sigma_T^2 = 0.65 + 1.4625 = 2.1125$, and

$$T \sim N(55.25, 2.1125)$$

To find $P(50 < T < 55)$ we compute the z-scores of 50 and of 55.

$$z = (50 - 55.25) / \sqrt{2.1125} = -3.61$$

$$z = (55 - 55.25) / \sqrt{2.1125} = -0.17$$

Example

The area to the left of $z = -3.61$ is 0.0002.

The area to the left of $z = -0.17$ is 0.4325.

The area between $z = -3.61$ and $z = -0.17$ is $0.4325 - 0.0002 = 0.4323$.

The probability that the total time used by both machines together is between 50 and 55 hours is 0.4323

Questions

1. According to the Central Limit Theorem, as sample size increases, the sampling distribution of the sample mean:

- A) Becomes skewed
- B) Approaches a normal distribution
- C) Becomes uniform
- D) Becomes bimodal

2. Which of the following conditions is necessary for applying the Central Limit Theorem?

- A) The population must be normally distributed
- B) The population must be finite
- C) The sample size must be large
- D) The population mean must be zero

3. If the sample size is increased, the standard error of the mean will:

- A) Increase
- B) Remain the same
- C) Decrease
- D) Become zero

Answers

1. C
2. C
3. C



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Generation of Random Variates

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Generation of Random Variates

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Topics to be covered

- ❖ Random Numbers
- ❖ Random Variate Generator
- ❖ Random Variates
- ❖ Techniques for generating Random Variates

Random Numbers

- It is a random sequence of numbers obtained from a stochastic process.
- A random number is a number chosen as if by chance from some specified distribution such that selection of a large set of these numbers reproduces the underlying distribution.
- A random number is chosen using methods which give equal probability to all numbers occurring in the specified distribution.
- In real world, random numbers may be generated using a dice or a roulette wheel.



Random Numbers

- Random numbers are most commonly produced with the help of a random number generators.
- Random numbers have **important applications**, especially in **cryptography** where they act as ingredients in encryption keys.
- One of the most **important prerequisites of a random number** is to be **independent**, as this helps in establishing no correlations between successive numbers.
- It must be ensured that the frequency of the occurrence of these random numbers should be approximately be the same. As a result, theoretically, it is not easy to generate a long random number.

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Random Numbers

- Random numbers are also very important for a simulations.
- All the randomness required by a simulation model is obtained by a random number generator.
- The output of a random number generator is assumed to be a sequence of independent and identically (uniformly) distributed random numbers between 0 and 1.
- These random numbers are transformed into required probability distributions.
- Example: The most common set from which random numbers are derived is the set of single-digit decimal numbers {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}.



Generating Random Numbers

- **Problem:**

Generate sample of a random variable X with a given density f . (The sample is called a random variate)

- **Answer:**

Develop an algorithm such that if one used it repeatedly (and independently) to generate a sequence of samples $X_1, X_2, \dots, X_n$ then as n becomes large, the proportion of samples that fall in any interval $[a, b]$ is close to $P(X \in [a, b])$, i.e

$$\{X_i \in [a, b]\}/n \approx P(X \in [a, b])$$

Generating Random Numbers

- **Solution:** 2-step process
 - Generate a random variate uniformly distributed in $[0, 1]$, also called a random number.
 - Use an appropriate transformation to convert the random number to a random variate of the correct distribution.
- **Why is this approach good ?**

Answer: It focuses on generating samples from ONE distribution only.

Random Number Generators

- A random number generator is a **hardware device or software algorithm** that generates a **number** that is taken from a **limited or unlimited distribution** and outputs it.
- The two main types of random number generators are pseudo random number generators and true random number generators.
- The numbers or sequence of numbers generated must lack any pattern (i.e. must appear random).
- **True random number generator:**
 - It measures some physical phenomenon that is expected to be random and then compensates for possible biases in the measurement process.
 - Example sources include measuring atmospheric noise, thermal noise, and other external electromagnetic and quantum phenomena.

Random Number Generators

- **Pseudo random number generator:**
 - It uses computational algorithms that can produce long sequences of apparently random results.
 - But these results are in fact completely determined by a shorter initial value, known as a **seed value or key**.
 - As a result, the entire seemingly random sequence can be reproduced if the seed value is known.
 - Properties that pseudo-random number generators should possess:
 1. It should be fast and not memory intensive
 2. It must be able to reproduce a given stream of random numbers.
 3. provision for producing several different independent streams of random numbers
- The random numbers generated must meet some statistical tests for randomness intended to ensure that they do not have any easily discernible patterns.

Random Number Seed

- A random seed (or seed state, or just seed) is a **number (or vector) used to initialize a pseudorandom number generator.**
- Computer-based generators use random number seeds for setting the starting point of the random number sequence.
- For a seed to be used in a pseudorandom number generator, it does not need to be random.
- These seeds are often initialized using a computer's real time clock in order to have some external noise.

Pseudo random
number sequence



Random Variate

- A **random variate** is a variable generated from uniformly distributed pseudorandom numbers.
- It is a particular outcome of a random variable.
- The random variates which are other outcomes of the same random variable might have different values.
- Random variates are used when simulating processes are driven by random influences (stochastic processes).
- They are frequently used as the input to simulation models.
- Procedures to generate random variates corresponding to a given distribution are known as procedures for **random variate generation or pseudo-random number sampling**.
- Depending on how they are generated, a **random variate** can be **uniformly or non-uniformly distributed**.
- Examples: Inter-arrival time and service time.

Random Variate Generation

- Random variate generation is a fundamental aspect of simulation modeling and analysis.
- The objective of random variate generation is to produce observations that have the stochastic properties of a given random variable.
- Various methods and algorithms have been developed to generate random variates that are accurate (representative of the target distribution) and computationally efficient.
- The **distribution** from which random variates are generated is assumed to be completely specified.
- We wish to generate samples from this distribution as input to a simulation model.
- Random variate generation relies on generating uniformly distributed random number in the closed interval $[0,1]$.
- Random variate generators use as starting point, random numbers distributed in $U[0,1]$.

Random Variate Generation : Objectives

- The objective of random variate generation is to produce sample observations that have the stochastic properties of a given random variable, X , having distribution function
$$F(x) = \Pr(X \leq x), \text{ where } -\infty < x < \infty$$
- The development of the theory/concepts surrounding random variate generation via computer algorithms is based on the following two key assumptions :
 - **Assumption 1** : There exists a perfect uniform $(0,1)$, $U(0,1)$, random number generator that can produce a sequence of independent random variables uniformly distributed on $(0,1)$.
 - **Assumption 2** : Computers can store and manipulate real numbers.
- Although Assumptions 1 and 2 are used for developing Random Variate Generation theory, the assumptions are violated when implementing Random Variate Generation algorithms on digital computers.

Factors to be considered for random variate generation

1. Exactness:

- Exactness or accuracy refers to how well the generator produces random variates with the characteristics of the desired distribution.
- This refers to the theoretical exactness of the random variate generator itself, as well as the error that is induced by the $U(0,1)$ random number generator and the error induced by digital computer calculations.

2. Speed:

- Speed refers to the computational set-up and execution time required to generate random variates. Contributions to time are:
 - a. Setup time
 - b. Variable generation time

Factors to be considered for random variate generation

3. Space:

- Space refers to computer memory that is required for the generator.
- Although space is not typically a major consideration for modern computers, computer memory was an important consideration in the early days of Random Variate Generation development.

4. Simplicity:

- Simplicity refers to the both the simplicity of the algorithm as well as the simplicity of implementation.
- This includes the number of lines of code, support routines required, number of mathematical operations, as well as portability across platforms and interaction with other simulation methods such as variance reduction techniques.

The importance of each of these factors will vary depending on the particular situation or simulation application.

Random Variate Generation Techniques

- We assume that a pseudo random number generator $RN(0,1)$ producing a sequence of independent values between 0 and 1 is available.
- General methods:
 - Inverse transform method
 - Acceptance-rejection method
 - Composite method
 - Translations and other simple transforms

Acceptance and Rejection Method: Poisson Distribution

Procedure of generating a Poisson random variate N is as follows:

1. Set $n=0$, $P=1$
2. Generate a random number R_{n+1} , and replace P by $P \times R_{n+1}$
3. If $P < \exp(-\alpha)$, then accept $N=n$.

Otherwise, reject the current n , increase n by one, and return to step 2.

Acceptance and Rejection Method: Poisson Distribution example

- Example: Generate three Poisson variates with mean $\alpha=0.2$
 - $\exp(-0.2) = 0.8187$
- Variate 1
 - Step 1: Set $n = 0, P = 1$
 - Step 2: $R1 = 0.4357, P = 1 \times 0.4357$
 - Step 3: Since $P = 0.4357 < \exp(-0.2)$, **accept** $N = 0$
- Variate 2
 - Step 1: Set $n = 0, P = 1$
 - Step 2: $R1 = 0.4146, P = 1 \times 0.4146$
 - Step 3: Since $P = 0.4146 < \exp(-0.2)$, **accept** $N = 0$
- Variate 3
 - Step 1: Set $n = 0, P = 1$
 - Step 2: $R1 = 0.8353, P = 1 \times 0.8353$
 - Step 3: Since $P = 0.8353 > \exp(-0.2)$, reject $n = 0$ and return to Step 2 with $n = 1$
 - Step 2: $R2 = 0.9952, P = 0.8353 \times 0.9952 = 0.8313$
 - Step 3: Since $P = 0.8313 > \exp(-0.2)$, reject $n = 1$ and return to Step 2 with $n = 2$
 - Step 2: $R3 = 0.8004, P = 0.8313 \times 0.8004 = 0.6654$
 - Step 3: Since $P = 0.6654 < \exp(-0.2)$, **accept** $N = 2$

Acceptance and Rejection Method: Poisson Distribution example

- It took five random numbers to generate three Poisson variates
- In long run, the generation of Poisson variates requires some overhead!

N	R_{n+1}	P	Accept/Reject		Result
0	0.4357	0.4357	$P < \exp(-\alpha)$	Accept	$N=0$
0	0.4146	0.4146	$P < \exp(-\alpha)$	Accept	$N=0$
0	0.8353	0.8353	$P \geq \exp(-\alpha)$	Reject	
1	0.9952	0.8313	$P \geq \exp(-\alpha)$	Reject	
2	0.8004	0.6654	$P < \exp(-\alpha)$	Accept	$N=2$

References

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THANK YOU

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